

Master of Business Administration
(For Students admitted from academic year 2023 onwards)
CURRICULUM AND SYLLABUS



SCHOOL OF MANAGEMENT STUDIES

**GANDHI INSTITUTE OF ENGINEERING AND
TECHNOLOGY UNIVERSITY, ODISHA,
GUNUPUR**

MBA AR-23

Programme Outcomes (POs)

Upon successful completion of this MBA Program, students should develop the ability to:

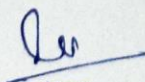
- PO1:** Apply knowledge of management theories and practices to solve business problems.
- PO2:** Foster Analytical and critical thinking abilities for data-based decision making.
- PO3:** To develop Value based Leadership skills.
- PO4:** Ability to understand, analyse and communicate global, economic, legal, and ethical aspects of business.
- PO5:** Ability to lead themselves and others in the achievement of organizational goals, contributing effectively to a team environment.

Program Specific Outcomes (PSOs)

- PSO1:** To imbibe the students with requisite domain knowledge, skills & right attitude necessary to provide effective leadership in a global environment.
- PSO2:** To develop competent management professionals with strong ethical values, capable of assuming a pivotal role in various sectors of the Economy.
- PSO3:** To inculcate proactive thinking to ensure effective performance in the dynamic socioeconomic and business ecosystem.

Program Educational Objectives (PEOs)

- PEO1:** To Impart formal knowledge in all broad areas of business management and leadership and in-depth knowledge in functional areas of business.
- PEO2:** To develop professional skills for employment, entrepreneurship and lifelong learning in management and to adapt to a rapidly changing environment with learned and applied new skills.
- PEO3:** To establish as effective professionals by solving real problems through the use of management science knowledge and with attention to team work, effective communication, critical thinking and problem solving skills.
- PEO4:** To impart in-depth knowledge in all the functional areas of business theories and practices and to develop competence for effective business plans & execution and decision making.
- PEO5:** To produce industry ready graduates having highest regard for Personal & Institutional Integrity, Social Responsibility, Teamwork and Continuous Learning.


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Odisha, Gunupur

MBA AR23

I SEMESTER COURSE STRUCTURE AND SYLLABUS

Abbreviations

AEC	:	Ability Enhancement Course
C	:	Credits
CC	:	Core Course
CE	:	Core Elective
CIA	:	Continuous Internal Assessment
ESE	:	End Semester Examination
L	:	Lecture Hours
MD	:	Multi-Disciplinary
MS	:	Management Studies
P	:	Practice Hours
T	:	Tutorial Hours

S. No.	Course Category	Course Code	Type	Course Title	L	T	P	C	Marks		Total Marks
									CIA	ESE	
1	MS	MB101C	CC01	Principles & Practices of Management	3	0	0	3	40	60	100
2	MD	MB102C	CC02	Managerial Economics	3	0	0	3	40	60	100
3	MD	MB103C	CC03	Managerial Communication	3	0	0	3	40	60	100
4	MD	MB104C	CC04	Legal Aspects of Business	3	0	0	3	40	60	100
5	MS	MB105C	CE01	Quantitative Techniques	4	0	0	4	40	60	100
6	MS	MB106C	CE02	Financial Accounting	4	0	0	4	40	60	100
7	MD	MB107C	CE03	Organisation Behaviour	4	0	0	4	40	60	100
8	MD	MB108C	CE04	IT for Managers	4	0	0	4	40	60	100
I Semester Total					28	0	0	28	320	480	800



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB101C	PRINCIPLES & PRACTICES OF MANAGEMENT	3	0	0	3

Course Description

This course provides a foundational understanding of key management concepts and practices crucial for effective leadership. Covering essential topics such as planning, organizing, leading, and controlling, the course emphasizes practical applications through case studies and real-world scenarios. Students will explore various management theories, develop critical decision-making skills, and learn strategies for ethical leadership and organizational change. Designed for future managers and business professionals, this course combines theoretical knowledge with practical skills to prepare participants for successful careers in management.

Course Objectives

1	To demonstrate an understanding of current and relevant management knowledge.
2	To understand the evolution of management thought and its relevance in decision making.
3	To understand the importance of structure and types and organization in the field of Management.
4	To learn the principles, techniques and concepts needed for managerial planning, process and decision making.
5	To describe the effective management of planning organizing influencing and controlling related to the internal and external environment.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand theoretical aspects, process and principles, scope of Management and its application to Modern Management.
CO2	Analyse how the field of management has evolved and its major contributions.
CO3	Analyse and apply critical role of managers in modern organizational settings
CO4	Illustrate and evaluate the importance of planning, organising, directing and controlling decision making.
CO5	Utilize data and research to support decision-making.
CO6	Evaluate and apply ethical considerations in management practices.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	1	2	3	1	1
CO2	2	2	3	2	3	2	3	2
CO3	2	3	2	2	3	3	2	2
CO4	3	3	2	3	3	3	3	2
CO5	2	3	2	1	2	2	3	3
CO6	1	2	1	3	2	1	2	2
AVG	2.17	2.50	1.83	2.00	2.50	2.33	2.33	2.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Management

Nature, concept, scope and significance; Functions, Management: Art or Science or Professions, Organization VS Administration Vs Management, schools of Management Contributions of F.W. Taylor, Henry Fayol, Elton Mayo; Roles of Managers; Social responsibility and Business Ethics.

Unit-II: Planning

Planning premises, types and planning process, Objectives- Managing by Objective (MBO) strategies, decision making, types of decision, and decision making process

Unit-III: Organising

Structure, types of organizations, principles of organizing, authority and span of control, delegation and decentralization, line and staff relationship. Staffing

Unit-IV: Directing

Introduction, Characteristics of Leader, functions of Leader, Leadership and management, Principle of Leadership, Style of Leadership

Unit-V: Controlling

Introduction, Concept of Controlling, Purpose of Controlling, Types of Control, Steps in Controlling, Techniques in Controlling

Textbook References:

1. Harold Koontz, Heinz Weihrich, A Ramachandra Aryasri, Principles and Practices of Management, Tata McGraw Hill.
2. Pradeep Kumar, R. S Thakur, Management Principles and Practices, Wisdom Publications.
3. Stephen Robbins, Mary Coulter, Neharika Vohra, Management Principles and practices, Pearson Education.
4. Peter F. Drucker, Management: Tasks, Responsibilities, Practices, Harper Business.
5. Thomas S. Bateman and Scott Snell, Management: Leading & Collaborating in a Competitive World, McGraw-Hill Education.
6. Charles W. L. Hill and Steven L. McShane, Principles of Management, McGraw-Hill Education.



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MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB102C	MANAGERIAL ECONOMICS	3	0	0	3

Course Description

In today's competitive business environment, effective managerial decision making requires use of economic concepts and tools. Business efficiency depends on minimization of cost and maximization of production which requires perfect understanding of the economic concepts like demand, supply, production, cost and market conditions. Managerial economics uses economic concepts and principles by emphasizing on demand analysis, production & cost analysis and different market structures which are fundamental for further study. This course also introduces important macroeconomic concepts which are indispensable for understanding the functioning of an economy. Knowledge about those concepts is useful for timely business decisions.

Course Objectives

1	To discuss on the objectives of managerial economics and its role for business management.
2	To comprehend the knowledge of key economic concepts on demand and supply analysis.
3	To make use of the conceptual knowledge on production and cost analysis to determine the economics of scale.
4	To analyse the market structure and determine the price-output under various market conditions.
5	To demonstrate the micro-economics concepts on national income, GDP, NNP and business cycle.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Execute the principles of managerial economics to solve the business problems.
CO2	Analyse the economic concepts of demand and supply to forecast the futuristic market conditions.
CO3	Appraise the production and cost analysis theory to solve the real world business problems.
CO4	Design the different forms of market to determine the price-out for perfect market, monopoly, monopolistic and oligopoly market.
CO5	Develop macroeconomics principles to calculate the GDP, NNP and national income.
CO6	Formulate the managerial economics principles to perform a case study analysis to solve the business problems.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	2	3	3	2	2	2
CO2	3	3	2	3	2	2	1	2
CO3	3	2	3	3	2	1	1	1
CO4	3	2	3	2	1	1	1	1
CO5	2	2	2	2	1	1	1	1
CO6	2	2	2	2	1	1	1	1
AVG	2.67	2.33	2.33	2.50	1.67	1.33	1.17	1.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Basics of Managerial Economics

Nature, scope, Principles of managerial economics – opportunity cost principle, incremental principle, principle of time perspective, discounting principle, Equi-marginal principle- Differences between managerial economics and micro economics-Importance and application of managerial economics concepts in business decision making.

Unit-II: Utility, Demand & Supply Analysis

Utility Concept, TU, MU and DMU. Determinants of demand, Types of demand - Individual demand and Market demand, Industry demand and company demand, producer's goods demand and consumer good's demand, direct demand and indirect demand, derived demand and autonomous demand, short run demand and long run demand. Law of Demand - assumptions, explanation of the law with schedule, graph & mathematical function and exceptions of the law. Determinants of supply, law of supply- assumptions, explanation of the law with schedule, graph & mathematical function and exceptions of the law. Market equilibrium – Price mechanism /Market mechanism with a graphical explanation. Elasticity of demand, types of elasticity, methods to measure elasticity - Point elasticity, total outlay, arc method, gradient method. Demand forecasting - Qualitative Methods - Survey method, Expert opinion method, Delphi Method, sales force opinion method, Focus groups - Quantitative methods - Trend line, Regression, Correlation, Smoothing techniques (Moving Averages), Barometric Demand Forecasting through Economic Indicators, econometric models of demand forecasting.

Unit-III: Production and Cost Analysis

Production function, Laws of Production-Short run production function with one variable input, Short run production function with two variable inputs - Iso- quants, properties of iso-quants, types of iso-quants. Iso-cost line (Budget constraint or factor price line) - change and shifts in iso-cost line, producer's equilibrium, expansion path. Production with all variable inputs - Law of returns to scale – Increasing returns to scale, Constant returns to scale, Decreasing returns to scale. Cost - Cost concepts and classifications, Sunk cost fallacy, Cost output relation- short run cost output relationship, long run cost output relationship, Learning curve, Do firms operate at optimal scale?- Economies of scale, dis-economies of scale and economies of scope.

Unit-IV: Market Structure

Basis for classification of market power, kinds of competitive market, Effect of time on supply - Very short run supply curve, short run supply curve and long run supply curve, Determination of price in short run and long run - price and output decisions in perfect competition under normal profit, super normal profit and loss conditions of firm and shutdown point. Determination of price in short run and long run-price and output decisions in Monopoly–equilibrium at underutilization capacity, over utilization capacity and Optimal Capacity, Monopoly power, Monopsony- Discriminating monopoly - (First degree, second degree and third degree) a Diagrammatic explanation - Monopolistic - Price and output determination in the short run and long run, Chamberlin's Group Equilibrium, Excess Capacity - Oligopoly - collusive and non-collusive Oligopoly, Kinked demand curve, Price and output determination in Oligopoly market, Nash equilibrium, Game theory. Differences between various market structures - Market Failures - public goods, social goods, merit goods, administered prices (ceiling price and floor price) and Externalities– Positive and negative externalities. Fundamentals of Internalization of externalities – Social Cost-Benefit analysis, Pareto optimality criterion, Pigovian taxes, Subsidies and incentives for clean technologies.

Unit-V: Macroeconomics

Macroeconomic indicators- GDP growth rate, consumer price index, interest rate, unemployment, foreign exchange rate, Balance of payments (BOP) - National Income-Concepts of national income (GDP, GNP, NDP, NNP, Personal Income, Personal Savings, Disposable personal Income, Discretionary income) - Methods of calculating national income – Product Method- Final good and Value added method, Income method, Expenditure Method and Social Accounting Matrix, GDP at Purchasing Power Parity (PPP)- Inflation- causes-demand pull and cost push inflation, measures to control inflation, business cycles -phases of business cycles and measures to control business cycles- Stabilization policies - Monetary Policy and Fiscal Policy.

Textbook References:

1. D. N. Dwivedi. Managerial Economics. Vikas Publishing House.
2. D. N. Dwivedi. Managerial Economics, S Chand and Company Ltd.
3. Christopher R. Thomas and S. Charles Maurice. Managerial Economics: Foundations of Business Analysis and Strategy. McGraw Hill.
4. Dominick Salvatore and Siddhartha Rastogi, Managerial Economics: Principles and Worldwide Applications. Oxford University Press.
5. A. Nag, Macroeconomics: Theory, Policy and Application. Atlantic Publishers and Distributors (P) Ltd.
6. R. L. Varshney and K. L. Maheshwari, Managerial Economics. S Chand and Company Ltd.



SCHOOL OF MANAGEMENT STUDIES
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MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB103C	MANAGERIAL COMMUNICATION	3	0	0	3

Course Description

This course focuses on developing effective communication skills essential for successful management. Students will learn how to craft clear and persuasive messages, manage interpersonal interactions, and lead productive meetings. Emphasizing both verbal and written communication, the course covers techniques for improving presentations, negotiating, and handling conflict. Through practical exercises participants will enhance their ability to communicate strategically and build strong professional relationships, equipping them to lead teams and influence stakeholders effectively.

Course Objectives

1	To gain an overall perspective on the nature of human relationships and its impact on communication.
2	To understand the communication-process model to appreciate organizational communication.
3	Analyse the effectiveness of human communication in terms of intrapersonal and interpersonal variables of interpersonal communication.
4	Learn three Models of Interpersonal Communication and apply them to increase their communicative effectiveness by understanding the underlying relationship on which interpersonal communication is based.
5	Learn to attain and develop interpersonal influence and influence others to develop and utilize their talents.

Course Outcomes
(Upon successful completion of this course, students should be able to)

CO1	Analyse how communication functions as a dynamic process and apply strategies to enhance effectiveness in organizational communication.
CO2	Gain insights into the management of interpersonal communication by exploring intrapersonal foundations such as motivation, perception, and emotions.
CO3	Students will be able to apply various models for better understand and navigate interpersonal relationships within a management context.
CO4	Identify common barriers to effective communication and will develop strategies to address these barriers.
CO5	Explore the concept of interpersonal influence for managing resistance to change.
CO6	Acquire skills in non-directive counselling techniques to facilitate open communication and support others in organizational settings.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	2	3	3	3	3	3
CO2	3	3	1	2	2	3	2	2
CO3	3	3	1	2	2	3	2	3
CO4	3	3	1	3	2	3	2	2
CO5	3	3	1	2	3	3	2	2

CO6	3	3	1	3	3	3	2	2
AVG	3.00	3.00	1.17	2.50	2.50	3.00	2.17	2.33
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Communication Basics Communication and Management - The Paradox of Human Communication - The Management Process and Communication - Communication as a process - Achieving effectiveness in Human Communication								
Unit-II: Interpersonal and Intrapersonal Communication Management of Interpersonal Communication - Intrapersonal Foundations for Communication - Motivation; Perception; Emotions.								
Unit-III: Interpersonal Relations Models for Understanding Interpersonal Relationships - Models - Exchange Theory; Johari Window; Transactional Analysis.								
Unit-IV: Barriers Barriers - Power Differences - Language - Defensiveness - Gateways - Interpersonal Trust -Listening - Feedback - Nonverbal Communication - Non Directive Counselling.								
Unit-V: Interpersonal Influence Interpersonal Influence - The Influence Process - Resistance to Change - Organizational Limitations to Interpersonal Influence.								
Textbook References: 1. Wofford, Gerloff and Cummins, Organizational Communication-The Keystone to Managerial Effectiveness, McGraw Hill. 2. Bovee & Thill, Business Communication, Pearson Education. 3. Lesikar & Flatley, Basic Business Communication Skills for Empowering the Internet Generation, McGraw Hill. 4. Monippally. MM., Business Communication Strategies, McGraw Hill. 5. David Irwin, Effective Business Communications, Viva, Thorogood. 6. Rajendra Pal, J S Korlaha HI, Essentials of Business Communication: Sultan Chand & Sons.								



SCHOOL OF MANAGEMENT STUDIES
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Odisha, Gunupur
MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB104C	LEGAL ASPECTS OF BUSINESS	3	0	0	3

Course Description

Ignorance of law is no excuse and everyone is supposed to know the law of the land. Law controls and regulates the external behaviour of human beings. Human conduct must conform to legal rules and regulations. Business law is a branch of general law and it controls and regulates commercial transactions involving businessmen, traders, importers, exporters, consumers, etc. The present business laws consist of the general principles of Contract (Contract Act), Law of Agency, Sales of Goods Act, Partnership and Company Law.

Course Objectives

1	To understand the basic concepts of Contracts, Sale of goods, Agency etc.
2	To acquaint with special legislations dealing with business transactions.
3	To evaluate with special Case laws dealing with business transactions.
4	To analyse the recent amendments dealing with business Legislation.
5	To elucidate the process of formation and winding up of a company.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Analyse the Business Related concepts, foundations etc.,
CO2	Identify Business Law interventions and its impact on Business.
CO3	Define the role of Rules and Regulations framed under the Legal system to run business in judicious way.
CO4	Apply the purposes, relevance and value of specific Jurisdiction of Business Laws related with Contracts, Agency, Sales, Company etc.,
CO5	Apply multidisciplinary approach of the case laws to the context.
CO6	Develop a skill of companies-formation and advantages of incorporation of a company.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	1	2	3	2	3	2
CO2	2	3	1	3	3	3	3	1
CO3	3	3	2	2	3	3	2	2
CO4	3	2	3	2	2	2	1	2
CO5	3	3	1	3	2	2	2	3
CO6	2	3	1	3	2	2	2	3
AVG	2.67	2.83	1.50	2.50	2.50	2.33	2.17	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Indian Contract Act, 1872

Essential Element of valid contract: Offer and Acceptance, Consideration, Capacity to Contract, Free Consent, Legality of object, Void agreements and contracts, Performance of contract, Discharge of contract, Remedies for breach of contract. Special Contract: Contract of Indemnity and Guarantee, Bailment and pledge and contract of agency.

Unit-II: Sales of Goods Act, 1930

Sales of Goods Act, 1930: Basics, Conditions and warranties, Transfer of property, Performance of contract, Right of an unpaid seller. Negotiable Instrument Act, 1881: Negotiable instruments, Notes, Bills and Cheque, Parties, Negotiation, Presentment, Dishonour and Discharge of Negotiable instrument.

Unit-III: Consumer Protection Act, 1986

Consumer Protection Act, 1986: Definitions, Consumer protection Council, Dispute redressal agencies, process to file complain, Appeal, Remedies. Information Technology Act, 2000: Introduction, Objects and scope, Definitions of important terms, Digital signature, Electronic governance, Cybercrime and remedies, Cyber regulation appellate tribunal, Duties of certifying authority, power of central government to make rules.

Unit-IV: Company Act, 2013

Company Act, 2013: Nature of company, Kinds of company, formation of company, Memorandum of Association, Article of Association, Prospectus, Membership in a company, Share Capital, Meeting and proceeding, Company Management

Unit-V: Intellectual Property Rights

Intellectual Property Rights: Introduction, their major types like Patents, Trademarks, Copyrights, Industrial designs etc. Important provisions. With respect to registration, renewal, revocation, remedies in case of infringement.

Textbook References:

1. N.D. Kapoor, Elements of Mercantile Law, Sultan Chand & Sons.
2. N.D. Kapoor, Business Law, Sultan Chand & Sons.
3. Akhileshwar Pathak, Legal Aspects of Business, Tata McGraw Hill.
4. Tejpal Sheth, Business Law, Pearson Education.
5. D. Chandra Bose, Business Law, PHI Learning.
6. MC Kuchhal & Vivek Kuchhal, Business Law, Vikas.



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MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB105C	QUANTITATIVE TECHNIQUES	4	0	0	4

Course Description

This course introduces quantitative techniques essential for data analysis and decision-making in fields such as business, economics, and social sciences. Students will explore key concepts including descriptive and inferential statistics, probability theory, optimization techniques, and time series analysis. Through hands-on exercises and real-world case studies, participants will gain practical skills in analysing data, testing hypotheses, and applying mathematical methods to solve complex problems. Emphasis will be placed on using statistical software for data interpretation and decision support, with the aim of equipping students with the tools to make informed, data-driven decisions.

Course Objectives

1	Utilize fundamental mathematical principles to solve business-related problems and make informed decisions.
2	Understand and apply different probability distributions (e.g., normal, binomial) to model and analyse random variables and their behaviours.
3	Compute and interpret measures of central tendency (mean, median, mode) and dispersion (range, variance, standard deviation) to summarize and describe data sets.
4	Conduct correlation and regression analyses to identify relationships between variables, model dependencies, and make predictions based on empirical data.
5	Utilize forecasting techniques to predict future values and trends based on historical data and quantitative models.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Analyse and interpret financial ratios and proportions to evaluate business performance and make comparative assessments across different entities.
CO2	Calculate and interpret simple and compound interest to assess investment growth, loan repayments, and financial planning scenarios.
CO3	Apply probability distribution models and analyse uncertainty in business contexts.
CO4	Compute and interpret measures of central tendency and dispersion to summarise and describe data sets effectively.
CO5	Examine correlation and regression analyses to identify relationships between variables, model trends, and make data-driven predictions and decisions.
CO6	Apply forecasting techniques to project future business trends and outcomes based on historical data, supporting strategic planning and decision-making processes.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	2	1	3	3
CO2	3	3	0	3	2	1	3	3
CO3	3	3	0	3	2	1	3	3
CO4	3	3	0	3	2	1	3	3
CO5	3	3	0	3	2	1	3	3
CO6	3	3	0	3	2	1	3	3

AVG	3.00	3.00	0.00	3.00	2.00	1.00	3.00	3.00
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Business Mathematics Ratios and Proportions, Simple and Compound interest including application of Annuity, Simultaneous Linear Equations. Differentiation, Derivatives–First order and Second order Derivatives, Maxima & Minima, Integration, Business applications of Derivatives and Integration. (Note: Exclude Trigonometric and Logarithmic functions in derivatives and integration and chain rule in integration).								
Unit-II: Probability Distributions Probability concepts, axioms, Baye's theorem, Random Variables, Mathematical Expectation, Discrete Distributions - Binomial distribution and Poisson distribution, Continuous Distributions - Normal distribution.								
Unit-III: Measures of Central Tendency and Dispersion Introduction, Merits, Demerits, Applications. Mean Geometric Mean, Harmonic Mean, Weighted Mean, Combined Mean, Median, Mode, Standard Deviation, Coefficient of Variance, and Combined Standard Deviation for grouped and ungrouped data, Skewness and Kurtosis.								
Unit-IV: Correlation & Regression Analysis Positive & Negative correlations, Karl Pearson correlation coefficient, Linear regression, Regression Coefficients & Properties, Linear regression model, simple linear regression, coefficient of determination, Testing for significance, estimates through simple regression equation.								
Unit-V: Forecasting Introduction to Time Series Analysis, Cyclic Variation, Seasonal Variations, types of seasonal effects, Smoothing Methods-Moving Averages, weighted moving averages, Exponential smoothing, trend projections Linear Trend.								
Textbook References: 1. S.P. Gupta & M.P. Gupta, Business Statistics, Sultan Chand & Sons. 2. S.C. Gupta & Indra Gupta, Business Statistics, Himalaya Publishing House. 3. J.K Sharma, Business statistics, Pearson Education. 4. David M. Levine, David Stephan Timothy C. Krehbiel, Mark L Berenson, Statistics for managers using Microsoft Excel, Prentice Hall India Pvt. Ltd. 5. Amir D. Aczel, Jayavel Sounder Pandian, Complete Business Statistics, Tata McGraw Hill. 6. N D Vohra, Hitesh Arora, Quantitative Techniques in Management, McGraw Hill.								



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MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB106C	FINANCIAL ACCOUNTING	4	0	0	4

Course Description

This course provides a comprehensive introduction to the principles and practices of financial reporting and analysis. Students will learn how to prepare, interpret, and analyse financial statements, including balance sheets, income statements, and cash flow statements. The course covers key topics such as accounting standards, accrual accounting, and the use of financial data for decision-making. Through hands-on exercises and case studies, participants will gain the skills needed to understand financial performance, ensure accuracy in reporting, and make informed business decisions based on financial information.

Course Objectives

1	Know the accounting frame work to prepare final accounts of trading concerns.
2	Analyse and interpret the accounting information of financial statements for decision making.
3	Understand the cost sheet preparation process and tracing of Activities for the cost object through activity-based costing.
4	Value the concepts of marginal costing and its application in managerial decision making.
5	Develop the budgets and performance reports for planning and control purposes.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Prepare financial statements, including the Profit and Loss Account and Balance Sheet, using the accounting equation and basics of IFRS.
CO2	Analyse financial statements and will prepare funds flow and cash flow statements to assess financial health and performance.
CO3	Identify and categorize elements and types for accurate overhead allocation and cost management.
CO4	Apply marginal costing techniques to make informed decisions about pricing, production, and sales strategies by analysing cost-volume-profit relationships.
CO5	Understand the concepts of budgeting and budgetary control and implement zero-based budgeting to effectively manage financial planning and control.
CO6	Analyse financial data and prepare reports that support strategic decision-making processes.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	3	3	3	3
CO2	2	3	0	2	3	3	2	2
CO3	2	3	0	3	2	3	2	2
CO4	2	2	0	2	3	3	2	1
CO5	2	2	0	3	2	3	3	2
CO6	2	2	0	3	2	3	3	3
AVG	2.17	2.50	0.00	2.67	2.50	3.00	2.50	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Financial Accounting Book-Keeping - Double Entry System - Accounting Concepts and Conventions. Accounting Equation - Preparation of Profit and Loss a/c and Balance Sheet using the accounting equation. Basics of IFRS.
Unit-II: Financial Statement Analysis Concept, objectives and types; Ratio analysis - the study of liquidity, solvency, and profitability ratios. Funds Flow Analysis: Uses and preparation of funds flow statement. Cash Flow Analysis: Uses and preparation of cash flow statements.
Unit-III: Cost Accounting Elements of Cost - Types of Costs - Preparation of Cost Sheet - Special work orders. Activity-Based Costing (ABC): Concept of ABC - Categories in activity-based costing -allocation of overheads under ABC - Benefits and Limitations of Activity Based Costing.
Unit-IV: Marginal Costing Marginal Cost and Marginal Costing - Importance. Break-Even Analysis: Cost Volume Profit Relationship - Application of Marginal Costing Techniques - Fixing Selling Price, Make or Buy, Accepting a Foreign Order, and Deciding Sales Mix
Unit-V: Budgeting and Budgetary Control Definitions of Budget, Budgeting, and Budgetary Control - Need for Budgetary Control - Types of budgets - Preparation of Production Budget, Sales Budget, Cash Budget, and Flexible Budget - Zero-based Budgeting.
Textbook References: 1. Dhanesh K.Khatri, Financial Accounting & Analysis, TMH. 2. PK Jain and K.L. Narang, Financial Accounting, Kalyani Publications. 3. Narayana Swamy, Financial Accounting & Analysis PHI. 4. V.Rajasekharam, Financial Accounting & Analysis, Pearson Education. 5. Ranjan Kumar Bal, Financial Accounting & Analysis, S. Chand. 6. Maheswari, Financial Accounting, IBH.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB107C	ORGANISATIONAL BEHAVIOUR	4	0	0	4

Course Description

This course explores the fundamental principles of human behaviour within organizations, focusing on how individual abilities, attitudes, and perceptions influence job performance and satisfaction. Students will study motivation theories, group dynamics, leadership styles, and conflict management. The course also covers organizational culture and change, providing insights into how culture is developed, sustained, and changed. Additionally, the course addresses resistance to change, the forces driving change, and effective strategies for managing work stress.

Course Objectives

1	Understand how individual abilities, intelligence, attitudes, and job satisfaction influence workplace behaviour and performance.
2	Explore how perception affects decision-making and apply key motivation theories to improve employee motivation.
3	Analyse the stages of group development, group properties, leadership theories, and effectively manage conflicts within groups.
4	Understand the role of organizational culture, how it is created, and the ways employees learn and adopt cultural values.
5	Identify the forces driving organizational change, manage resistance, and develop strategies to address change-related stress in the workplace.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand how individual abilities, intelligence, and attitudes influence behaviour and job satisfaction in organizations.
CO2	Apply perception and motivation theories to improve decision-making and employee motivation.
CO3	Analyse group development, leadership theories, and effectively manage conflicts within teams.
CO4	Understand how organizational culture is created, sustained, and learned by employees.
CO5	Develop strategies for managing organizational change, resistance, and work-related stress.
CO6	Suggest solutions for managing work-related stress associated with organizational change.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	2	3	2
CO2	3	3	2	2	1	3	3	2
CO3	2	2	3	2	2	2	3	3
CO4	2	1	2	3	2	2	3	2
CO5	3	2	2	3	3	3	2	2
CO6	2	1	3	2	2	2	2	3
AVG	2.50	1.83	2.33	2.17	1.83	2.33	2.67	2.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Individual Behavior

The Individual: Nature of Organizational Behavior- Ability–physical ability, intelligence. Attitude–Major Job Attitudes- Job Satisfaction.

Unit-II: Perception and Motivation

Perception-Perception and decision-making. Motivation: Theories of Motivation - Maslow, Herzberg, Vroom, Goal-Setting Theory, and Equity Theory - Applications of Motivation.

Unit-III: Group Dynamics and Leadership

The Group: Stages of Group Development-Group properties. Leadership - Behavioral Theories. Conflict –The Conflict Process.

Unit-IV: Organizational Culture

The Organization System: Organizational Culture-Culture’s functions-Creating a culture-How employees’ learn culture.

Unit-V: Organisational Change

Organizational Change-Forces for change-Planned change-Resistance to change Managing organizational change-Work stress and its management.

Textbook References:

1. Robbins, S. P., Judge, T., & Vohra, N., Organizational Behavior, Pearson Education.
2. Sushma Khanna (editor), Udai Pareek’s Understanding Organizational Behavior, Oxford University Press.
3. Nelson, Quick & Khandelwal, ORGB-An Innovative Approach to Organizational Behavior, A South Asian Perspective, CENGAGE Learning.
4. Mc Shane & VonGlinow, Organizational Behavior, McGraw Hill.
5. Stephen P. Robbins, Timothy A. Judge, Essentials of Organizational Behavior, Pearson Education.
6. Jason Colquitt, Jeffery LePine, Michael Wesson, Organizational Behavior: Improving Performance and Commitment in the Workplace, McGraw Hill.



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GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA I Semester (AR23)

Course Code	Course Title	L	T	P	C
MB108C	IT FOR MANAGERS	4	0	0	4

Course Description

This course is designed to equip management students with essential skills in information systems and advanced MS Excel functionalities. It emphasizes practical applications such as spreadsheet-based decision support systems, data analysis, and business analytics. The course also covers macro programming and introduces students to business analytics, enabling them to leverage technology for informed decision-making and effective management practices.

Course Objectives

1	To understand the role and significance of Information Systems in organizational decision-making.
2	To develop proficiency in using MS Excel for data management and analysis.
3	To learn and apply various functions and tools in Excel for advanced data processing.
4	To explore the application of macros and VBA in automating tasks in Excel.
5	To introduce the concepts and applications of Business Analytics in managerial contexts.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate an understanding of Information Systems and their role in organizations.
CO2	Apply MS Excel functions and tools for data management and decision support.
CO3	Utilize advanced Excel features like Pivot Tables and What-If analysis for complex data analysis.
CO4	Implement and automate tasks using Macros and VBA in Excel.
CO5	Analyse and interpret data using Business Analytics concepts and tools.
CO6	Apply business analytics methods and models for informed decision-making in managerial practices.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	2	1	2
CO2	3	2	2	2	2	3	2	3
CO3	3	3	3	2	3	1	2	1
CO4	1	2	3	3	2	3	2	3
CO5	2	3	1	1	3	1	1	3
CO6	3	3	3	3	1	3	3	1
AVG	2.50	2.50	2.33	2.00	2.00	2.17	1.83	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Information Systems and MS Excel

Information Systems and MS Excel: Information Systems in organization, MS Excel as Spread sheet based DSS, Basic operations in MS Excel: Worksheet Management, Cell referencing, Building formulas, Sorting, Filters, Conditional Formatting, and Working with Charts.

Unit-II: Working with Functions

Working with Functions: Text Functions, Logical Functions, Lookup Functions, Date and Time Functions, Math and Statistical Functions

Unit-III: What - if Analysis and Pivot Tables

What - if Analysis and Pivot Tables: Data Tables, Scenario Manager, Goal Seek, Pivot Tables and Pivot Charts

Unit-IV: Macros: Usage of Macros

Macros: Usage of Macros, Recording Macros, Basics of VBA, Ms G box and Input box functions, Control Structures in VBA

Unit-V: Introduction to Business Analytics

Introduction to Business Analytics: Benefits of Business Analytics, Types of Data - Structured, Semi Structured and Unstructured, Application areas of Business Analytics, Categorization of Analytical methods and models– Descriptive, Predictive and Prescriptive.

Textbook References:

1. Kenneth C. Laudon and Jane P. Laudon, Management Information Systems: Managing the Digital Firm, Pearson.
2. Michael Alexander, Richard Kusleika, and John Walkenbach, Microsoft Excel 2019 Bible, Wiley.
3. Ash Narayan Sah, Data Analysis Using Microsoft Excel, PHI Learning.
4. . Dinesh Kumar, Business Analytics: A Data-Driven Decision Making Approach for Business, Wiley.
5. Jordan Goldmeier, Advanced Excel Essentials, Apress.
6. J. Randolph Kissell and Arne Bathke, Business Analytics: Concepts and Applications, Wiley.



SCHOOL OF MANAGEMENT STUDIES
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MBA AR23

II SEMESTER COURSE STRUCTURE AND SYLLABUS

Abbreviations

AEC	:	Ability Enhancement Course
C	:	Credits
CC	:	Core Course
CE	:	Core Elective
CIA	:	Continuous Internal Assessment
ESE	:	End Semester Examination
L	:	Lecture Hours
MD	:	Multi-Disciplinary
MS	:	Management Studies
P	:	Practice Hours
T	:	Tutorial Hours

S. No.	Course Category	Course Code	Type	Course Title	L	T	P	C	Marks		Total Marks
									CIA	ESE	
1	MD	MB201C	CC05	Entrepreneurship & Small Scale Business	3	0	0	3	40	60	100
2	MS	MB202C	CC06	Business Ethics & Corporate Governance	3	0	0	3	40	60	100
3	MD	MB203C	CE05	Research Methods for Managers	4	0	0	4	40	60	100
4	MS	MB204C	CE06	Marketing Management	4	0	0	4	40	60	100
5	MS	MB205C	CE07	Corporate Finance	4	0	0	4	40	60	100
6	MS	MB206C	CE08	Human Resource Management	4	0	0	4	40	60	100
7	MD	MB207C	CE09	Operations Management	4	0	0	4	40	60	100
8	MD	MB208C	CE10	Business Analytics	4	0	0	4	40	60	100
II Semester Total					30	0	0	30	320	480	800



SCHOOL OF MANAGEMENT STUDIES
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Odisha, Gunupur
MBA II Semester (AR23)

Course Code	Course Title	L	T	P	C
MB201C	ENTREPRENEURSHIP & SMALL SCALE BUSINESS	3	0	0	3

Course Description								
This course provides a comprehensive exploration of the fundamental principles and practices essential for launching and managing a successful small enterprise. Students will delve into key topics such as business planning, market research, financial management, and strategic growth. Emphasis will be placed on practical skills, including developing a business model, navigating legal and regulatory requirements, and leveraging digital tools for marketing and operations. Through case studies, students will gain hands-on experience and strategic insights needed to turn innovative ideas into viable business ventures. This course is ideal for aspiring entrepreneurs and small business owners seeking to refine their skills and drive their ventures to success.								
Course Objectives								
1	To know how the innovations, opportunities, and ideas convert into a new business.							
2	To understand the Entrepreneurship and its role in the society.							
3	To understand the process of Entrepreneurship & preparing business plans.							
4	To gain knowledge about the Entrepreneurship Development Institutions of Government.							
5	To appreciate the issues of Family Business and Social Entrepreneurship.							
Course Outcomes								
<i>(Upon successful completion of this course, students should be able to)</i>								
CO1	Understand various aspects of the entrepreneurial process and trends in entrepreneurship.							
CO2	Identify new business opportunities and outline the legal aspects of Entrepreneurship in practice.							
CO3	Examine the process and prospects of a new venture.							
CO4	Develop a business plan and design a business model.							
CO5	Appraise the entrepreneurship process in the family business and social enterprise.							
CO6	Investigation of small business and preparation of project reports.							
CO-PO & PSO Mapping (Articulation Matrix)								
COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	3	3	3	2	2
CO2	3	2	1	3	2	3	2	3
CO3	3	2	1	3	3	2	3	2
CO4	3	2	3	3	3	2	3	2
CO5	3	2	1	3	2	3	2	3
CO6	3	2	1	3	2	3	2	3
AVG	3.00	2.00	1.33	3.00	2.50	2.67	2.33	2.50
Justification of the Mapping:								
0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to Entrepreneurship								
Importance and concept of Entrepreneurship; Characteristics of Successful Entrepreneur; Entrepreneurship: Classification of entrepreneurs, Myths of entrepreneurship, Entrepreneurial development models. Problems faced by entrepreneurs. Capacity building for entrepreneurship.								

Unit-II: Women Entrepreneurship & SSI

Women Entrepreneurship: Importance of women entrepreneurs, Factors influencing women entrepreneurs, Challenges in the path of women entrepreneurship, Strategies for the development of women entrepreneurs. Modern Small Business Enterprises: Role of small-scale industries, Concepts and definitions of SSI, Government policy and development of the small-scale sector in India.

Unit-III: Setting Up a Small Business Enterprise

Setting up a Small Business Enterprise: Identifying the business opportunity, Formalities for setting up of a small business enterprise, Environment pollution related clearances. Institutions Supporting Small Business Enterprises: Central level institutions, State level institutions, other agencies.

Unit-IV: Sickness in Small Business Enterprises

Sickness in Small Business Enterprises: Sickness of Small-Scale Industries, Causes and symptoms of sickness, Cures of sickness. Break-Even Analysis.

Unit-V: Project Report

Project report preparation and presentation: Introduction; Idea Selection; Selection of the Product / Service; Aspects of a Project; Phases of a Project; Project Report: Contents of a Project Report; Proforma of a Suggested Project Report for a manufacturing and service Organizations.

Textbook References:

1. Ramachandran, K., Entrepreneurship Development: Indian Cases on Change Agents. Tata McGraw-Hill Pub.
2. Rao, N. V. J., Das, S., & Panda, P. R., Entrepreneurship & Small Scale Business. Rights Publications.
3. Kuratko, D. F., Entrepreneurship: Theory, process, and practice. Cengage learning.
4. Osterwalder, A., & Pigneur, Y., Business model generation: a handbook for visionaries, game changers, and challengers. John Wiley & Sons.
5. Neck, H. M., Greene, P. G., & Brush, C. G. (Eds.), Teaching entrepreneurship: A practice-based approach. Edward Elgar Publishing.
6. Landström, H., Pioneers in entrepreneurship and small business research. USA.



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Odisha, Gunupur
MBA II Semester (AR23)

Course Code	Course Title	L	T	P	C
MB202C	BUSINESS ETHICS & CORPORATE GOVERNANCE	3	0	0	3

Course Description								
Ethics and responsibility in business has received critical focus in the wake of the various corporate scams that rocked the global economy. It is believed by many that in the own interest of business, importance be given to ethical functioning. Business decisions often concern complicated situations that are neither totally ethical nor totally unethical. The need for imparting sound ethics and a responsible mind-set in the future leaders is considered as one of the important aspects of higher education. Decision making, when facing ethical dilemmas that arise in a wide range of contemporary business practices, is crucial, and is enabled through moral reasoning and understanding ethical norms of individuals and organizations. Having an insight into the challenges that may be encountered in different functional areas in the business and understanding the impact of decision making on various stakeholders is very crucial for the management professionals.								
Course Objective								
1	Explain the concepts of personal and professional ethics.							
2	Describe the importance of business ethics and understand the need for ethical decision making.							
3	List out various ethical issues in functional areas of business management.							
4	Identify different corporate governance models.							
5	Explain the role of the board and governance committees in corporate governance.							
Course Outcomes								
<i>(Upon successful completion of this course, students should be able to)</i>								
CO1	Appraise the importance of personal and professional ethics.							
CO2	Involves in ethical decision making in building successful business organizations.							
CO3	Differentiate ethical issues in important functional areas of business.							
CO4	Interpret the issues and emerging trends in corporate governance.							
CO5	Appraise importance of the board in corporate governance.							
CO6	Understand the importance of corporate social responsibility in the organizations.							
CO-PO & PSO Mapping (Articulation Matrix)								
COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	2	3	3	3	3
CO2	3	3	2	3	3	3	3	3
CO3	3	3	3	3	2	3	2	3
CO4	3	2	3	3	3	3	3	2
CO5	2	2	3	3	3	2	2	3
CO6	3	1	3	3	3	3	2	2
AVG	2.83	2.17	2.83	2.83	2.83	2.83	2.50	2.67
Justification of the Mapping:								
0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								

Syllabus
Unit-I: Introduction of Business Ethics Definition Business Ethics, Nature Business Ethics, Characteristics, Objective, Type of ethics; Factors Influencing Business Ethics, Sources of Business Ethics, Need & Importance of Business Ethics, Morality Vs Ethics; Ethical Theories, Causes of Unethical Behaviour, Ethical Abuses, Work Ethics, Code of Conduct.
Unit-II: Ethical Decision Making Ethical decision making-personal and professional moral development and moral reasoning, Computer ethics and business: Computer crime, Computers and corporate responsibility Property: information and software, Computer and privacy Professional ethics: Ethics in international business.
Unit-III: Legal Aspects of Ethics Provisions of the Indian constitution pertaining to Business, Political setup: major characteristics and their implications for business, Salient features of Indian culture and values, Economic Environment, Philosophy of economic grow and its implications for business.
Unit-IV: Corporate Governance Concept, need to improve corporate governance standards, features of good governance, Different Approaches to Corporate Governance, Different models of Corporate Governance, Corporate Governance and Other Stakeholders, Board of Directors; Role, Duties and Responsibilities of Auditors, Bank and Corporate Governance, Business Ethics and Corporate Governance.
Unit-V: Corporate Social Responsibility Corporate social responsibility: Meaning, Evolution of corporate social responsibility, Limits of corporate social responsibility, Voluntary responsibility Vs. Legal requirements, Profit maximization vs. social responsibility.
Textbook References: 1. Richard T. De George, "Business Ethics", Pearson, New Delhi. 2. Fernando, A. C., "Business Ethics an Indian Perspective", Pearson, New Delhi. 3. M.G. Velasquez, Business Ethics, Prentice Hall India Limited, New Delhi 4. R.C. Sekhar., Ethical Choices in Business, Response Books, New Delhi 5. Manikutty, S., "Being Ethical – Ethics as the foundation of Business", Random House India, Noida



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Odisha, Gunupur
MBA II Semester (AR23)

Course Code	Course Title	L	T	P	C
MB203C	RESEARCH METHODS FOR MANAGERS	4	0	0	4

Course Description

This course is designed to equip students with essential tools and techniques for conducting rigorous and effective research in a managerial context. This course covers a broad spectrum of research methodologies, including qualitative and quantitative approaches, data collection methods, and statistical analysis. Students will learn how to design research projects, analyse data, and interpret results to make informed managerial decisions. Through hands-on exercises, students will develop practical skills in hypothesis testing, survey design, and data visualization, enabling them to apply research findings to solve real-world business problems. Ideal for current and aspiring managers, this course provides the analytical foundation necessary to drive strategic decision-making and enhance organisational performance.

Course Objectives

1	To understand the formulation of research problem and hypotheses.
2	To learn critical analysis, problem solving and research skills.
3	To enable students to understand the rationale for using a particular qualitative and quantitative research method.
4	To enable students to understand various methods to select appropriate research designs and methods to investigate their chosen research problems.
5	To become familiar with the processes needed to develop, report, and analyze business data.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Formulate of research problems and developments of research outline.
CO2	Develop a scientific knowledge on addressing various management decision Process through business research.
CO3	Demonstrate their skill to apply different research methods in a scientific manner to assist the management for proper decisions on functional aspects.
CO4	Understand research hypothesis and research design for business studies
CO5	Acquire knowledge in generating and handling data with the help of statistical software to draw meaningful conclusions and suitable suggestions.
CO6	Conduct field-based surveys either for problem identification or problem solving in management issues.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3	2	3
CO2	3	2	3	3	3	2	3	3
CO3	3	3	2	3	3	3	2	2
CO4	3	3	2	3	3	3	2	3
CO5	3	2	2	3	3	2	3	3
CO6	3	2	3	2	2	3	2	3
AVG	3.00	2.33	2.50	2.83	2.83	2.67	2.33	2.83

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: An Introduction to Research Methodology

Introduction, Definition, scope, Meaning, Objectives. Business research: Role, Nature, Aim. Types of Research: Pure Research vs. Applied Research, Qualitative Research vs. Quantitative Research, Exploratory Research, Descriptive Research and Experimental Research approaches, Research Process. Research Problem: Definition, selecting the problem, techniques involved in defining the problem. Research Design: Meaning and definition, features, Importance, Types, Developing a research plan.

Unit-II: Data Collection & Sampling

Discussion on primary data and secondary data.

Methods: Primary data - observation method, interview methods through questionnaires, through schedules, and other methods, Processing-editing, coding, classification. Secondary data- Published sources, and Unpublished sources.

Sampling: Steps in a sample design, criteria for selection of sample, characteristics of a good sample design, types of samples.

Unit-III: Measurement

Measurement scales, types of scaling, scaling techniques, scale construction techniques. Tests of sound measurement, test of Validity, test of reliability, test of practicability, techniques of developing measurement tools.

Unit-IV: Hypothesis Testing

Concepts basic to the hypothesis testing procedure; steps in hypothesis testing; Type -I error & Type-II error; Two tailed & one tailed tests of hypothesis. Formulation of Hypothesis. Hypothesis testing of proportions; t - test, Chi square test, F- test, ANOVA of one way & two way classifications.

Unit-V: Report Writing

Report Writing: Steps, layout design, types, mechanics of writing formatting, title page, abstract, body, introduction, methods, sample, measures, design, results, conclusions, references, tables, figures, appendices.

Textbook References:

1. C.R. Kothari, Research methodology Methods and Techniques, New Age International (p) Limited, publishers.
2. Dr Deepak Chawla & Dr Neena Sondhi, Research Methodology Concepts and Cases, Vikas Publishing House Pvt Ltd.
3. R. Panneerselvam, Research Methodology, PHI Learning.
4. Krishna Swamy, Sivakumar, and Mathirajan, Management Research Methodology, Pearson.
5. SL Gupta and Hitesh Gupta, Business Research Methods, Tata McGraw Hill Education.
6. Donald R cooper and Pamela S Schindler, Business research Methods, McGraw-Hill.



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MBA II Semester (AR23)

Course Code	Course Title	L	T	P	C
MB204C	MARKETING MANAGEMENT	4	0	0	4

Course Description

This course is designed to provide students with a thorough understanding of contemporary marketing strategies and practices. The course covers key concepts such as market segmentation, consumer behaviour, branding, digital marketing, and strategic planning. Students will explore the principles of developing effective marketing campaigns, managing product life cycles, and analysing market trends. Through a combination of theoretical frameworks and practical case studies, students will learn how to create and implement marketing strategies that align with business objectives and respond to evolving market dynamics. This course is ideal for aspiring marketers and business professionals looking to enhance their ability to drive growth and build competitive advantage through strategic marketing initiatives.

Course Objectives

1	To explain the conceptual framework of marketing and its applications in "the real world".
2	To apply marketing concepts to make business decisions under various environmental constraints.
3	To illustrate the functionality and application of elements of Marketing Mix.
4	To create a suitable Marketing plan for a product.
5	To assess the range of common strategies used, with each of the various promotional mix tools.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Have an insight into the basic marketing concepts, the role of marketing in the organization.
CO2	Understand issues of marketing with an emphasis on Learning to develop responsive marketing strategies that meet customer needs.
CO3	Get acquainted with the components of marketing mix, stages in new product development.
CO4	Analyse the objectives and methods for pricing products and selecting channel members.
CO5	Evaluate the techniques of promotion mix.
CO6	Understand contemporary marketing topics.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	1	2	2	2	1
CO2	3	3	1	2	2	1	2	2
CO3	2	3	1	2	3	3	1	2
CO4	3	3	1	2	3	3	2	3
CO5	3	2	1	1	3	3	1	2
CO6	3	3	0	2	3	2	1	3
AVG	2.83	2.67	0.83	1.67	2.67	2.33	1.50	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Introduction Definition & Functions of Marketing, Scope of Marketing, Marketing concept, selling versus Marketing, Concept of Marketing Myopia. 80:20 Principle, Introduction to the Concept of Marketing Mix, Bottom of the pyramid concept; Concept of Marketing Environment: Macro and Micro, need for analysing the Marketing Environment.
Unit-II: Segmentation Market segmentation, Bases for market segmentation, Targeting, Positioning; Meaning & importance of consumer behaviour, Comparison between Organizational Buying behavior and consumer buying behaviour, Buying roles, Five steps buyer decision process
Unit-III: Product Product: Classification of consumer products and industrial products, Product Mix, New Product Development Process: Idea Generation to commercialization. Product Life Cycle, Strategies across stages of the PLC. Packaging & Labeling, Basic concept of Branding Pricing Basics: Meaning, Importance of pricing, Factors Influencing pricing decisions. Pricing strategies and approaches.
Unit-IV: Marketing Channels Place: The Role of Marketing Channels: Channel functions & flows, channel levels. Channel conflicts and resolution (Overview only) Channel Options: Introduction to Wholesaling, Retailing, Franchising, Direct marketing, E-Commerce Marketing Practices.
Unit-V: Promotion and Contemporary Topics Promotion: The role of marketing communications, Elements of promotion mix, IMC approach (Overview). Contemporary Topics: Viral Marketing, Guerrilla Marketing, Societal and social Marketing, Relationship Marketing, Green Marketing , Digital Marketing, Network Marketing (Concepts only).
Textbook References: 1. Philip Kotler & Kevin Lane Keller, Marketing Management, Global Edition, Pearson Education Ltd. 2. Baines, Fill and Page, Sinha, Marketing, Oxford. 3. Kotler, Keller, Koshy, Jha, Marketing Management, Pearson Education Ltd. 4. Lamb Hair Sharma, Mc Daniel, Marketing, Cengage Learning. 5. K Karunakaran, Marketing Management, Himalaya Publishing House. 6. Tapan K Panda, Marketing Management - Text and Cases, Excel Books.



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MBA II Semester (AR23)

Course Code	Course Title	L	T	P	C
MB205C	CORPORATE FINANCE	4	0	0	4

Course Description

This course offers a comprehensive examination of the financial principles and practices essential for managing a company's financial health and strategy. The course covers fundamental topics such as financial analysis, capital budgeting, risk management, and capital structure. Students will learn how to evaluate investment opportunities, optimize financing decisions, and assess financial performance using advanced tools and techniques. Emphasizing both theoretical concepts and practical applications, this course equips students with the skills to make informed financial decisions, manage corporate funds effectively, and drive strategic financial planning. Ideal for future finance professionals and business leaders, it provides the knowledge necessary to navigate complex financial environments and contribute to organisational success.

Course Objectives

1	To understand comprehensively how the various principles of financial theory that are being applied for corporate decision making - whether it is strategic, analytical, or simply the routine decisions a financial manager takes every day.
2	To provide clarity on the assumptions and concepts underlying the decision making in Finance.
3	To get familiar with the techniques of Financial Planning and Analysis
4	To give an overview of the problems facing a financial manager in the commercial world.
5	It will introduce you to the concepts and theories of corporate finance that underlie the techniques that are offered as aids for the understanding, evaluation and resolution of financial manager's problems.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understanding of terminology and concepts of financial management.
CO2	Apply & measures cost of capital/solve problems on time value of money.
CO3	Analyze information and construct a statement of cash flows in estimating of capital budgeting and WACC.
CO4	Make use of dividend models, capital structure theories for decision making.
CO5	Evaluate alternative capital budgeting techniques for decision making.
CO6	Understand the importance of working capital requirement in the organization.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	3	1	3	3	2
CO2	3	3	1	3	1	3	3	2
CO3	3	3	1	3	1	3	3	3
CO4	3	3	1	3	1	3	3	3
CO5	3	3	1	3	2	3	3	3
CO6	3	3	1	3	1	3	3	3
AVG	3.00	2.83	1.00	3.00	1.17	3.00	3.00	2.67

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Financial Management

Introduction & Objectives of Financial Management; Introduction to Time value of Money, Basic Valuation Model, Future Value concept, Present Value concept; Concept of Risk, Causes of risk, Classification of Risk and Measurement of Risk; Concept of Return, types of return, Measurement of Returns, Relationship between risk and return, Sources of Finance (Short & Long term).

Unit-II: Capital Budgeting

Introduction to Capital Budgeting - Meaning, Definition, Nature and Need of Capital Budgeting; Investment Evaluation Techniques - Traditional Method and Discounted Method; Dividend Decision - Types of Dividend, Reasons for paying dividend, Factors & Theories of Dividend policy, Walter Model.

Unit-III: Financing Decisions

Financing Decisions - Concept of Cost of Capital, Characteristics and Classification of cost of Capital, WACC; Capital Structure (Meaning, Definition, Features, factors, and theories); Leverage, Types of Leverage.

Unit-IV: Working Capital Management

Meaning, Characteristics, Concept; Classification of Working Capital - Factors, principles, and estimation of WC

Unit-V: Management of Cash, Receivables and Inventory

Cash and Receivable Management; Inventory Management - techniques of Inventory Management, Factors of Inventory Management.

Textbook References:

1. IM Pandey, Financial Management, Vikas.
2. M.Y Khan, P K Jain: "Financial Management-Text and Problems", TMH.
3. Prasanna Chandra, "Financial Management Theory and Practice", TMH.
4. Shashi K. Gupta, R. K. Sharma, "Financial Management" Kalyani Publishers.
5. Rajiv Srivastava, Anil Mishra, Financial Management" Oxford University Press.
6. James C Van Horne, Sanjay Dhamija, "Financial Management and Policy" Pearson Education.



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MBA II Semester (AR23)

Course Code	Course Title	L	T	P	C
MB206C	HUMAN RESOURCE MANAGEMENT	4	0	0	4

Course Description

This course provides an in-depth exploration of the strategies and practices essential for effectively managing an organization's workforce. This course covers key areas such as recruitment and selection, employee development, performance management, compensation and benefits, and labor relations. Students will learn how to design and implement HR policies that align with organizational goals, foster a positive work environment, and drive employee engagement and productivity. Through case studies, and practical exercises, participants will gain hands-on experience in addressing real-world HR challenges and developing solutions that enhance both individual and organizational performance. Ideal for aspiring HR professionals and managers, this course offers the skills and knowledge needed to build and sustain a motivated and high-performing workforce.

Course Objectives

1	Comprehend in depth the theoretical framework and the basic principles of HRM.
2	Comprehend in depth functions of HRM (Job analysis, manpower planning, and recruitment, selection, on boarding, training & development, appraisal, compensation).
3	Apply the principles and techniques of HRM gained through this course to the discussion of major personnel challenges and the solution of typical case problems.
4	To give an understanding that HR Management is more than just accepting employment applications and keeping records; it is a central and strategic organizational activity of increasing complexity and importance.
5	It takes a practical view that integrates the contributions of the behavioural sciences with the technical aspects of implementing the HR function in the real world.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand the fundamentals, evolution & challenges of HRM.
CO2	Explore the role of HRM in procurement of human resources.
CO3	Evaluate training needs, methods of appraisal and perceptual errors.
CO4	Analyse the basic factors in designing the compensation.
CO5	Evaluate the process of integration and separation for quality of work life.
CO6	Understand negating skills and settlement machineries to resolve disputes in industry.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	3	2	3	2	2
CO2	3	3	3	3	3	3	3	2
CO3	3	3	3	3	3	3	3	3
CO4	3	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3
CO6	3	3	3	3	3	3	3	2
AVG	3.00	2.83	2.67	3.00	2.83	3.00	2.83	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Basic Understanding of HRM Human Resources Management - Context and Concept of People Management in a Systems Perspective, HRM Functions and objective, Societal, Organizational, Personal, Functional, Personnel policies and principles; Skills for HR Professionals, Challenges of HRM; HRM Models - The Fombrun, The Harvard, The Guest, The Warwick.
Unit-I: Basic Understanding of HRM Job analysis - Process, Methods of collecting job data; Job design - Factors effecting job design, Job design approaches; Job evaluation - Job Evaluation Process, Methods of Job Evaluation; Workforce planning and forecasting - Forecasting personnel needs (Labour Demand), Forecasting the supply of inside Candidates, Forecasting the supply of outside candidates.
Unit-III: Managing HR Functions Recruitment and Selection - Selection – Induction & Orientation - Performance and Potential Appraisal - Coaching and Mentoring - HRM issues and practices in the context of Outsourcing as a strategy, Human Resources Development - Training and Development Methods - Design & Evaluation of T&D Programmes Human Resource Information System [HRIS], Human Resources Development –Training and Development Methods - Design & Evaluation of T&D Programmes
Unit-IV: Compensation and Integration Compensation Management - Components of Compensation, Theories of Compensation; Incentives - Types of Incentive Schemes, Group Incentive Plans; Fringe benefits, Types of Employee Benefits and Services; Employee Morale and Motivation – Stress Management - Quality of Work Life.
Unit-V: Labour Relations Trade Unions – Managing Conflicts - Disciplinary Process - Collective Bargaining - Workers Participation in Management - Concept, Mechanisms and Experiences. Collective Bargaining- The collective bargaining Process, building negotiating skills, impasses, mediation, and strikes; Resolving Disputes – Settlement machineries
Textbook References: 1. Gary Dessler & Biju Varkkey, "Human Resource Management", Pearson, New Delhi. 2. George W Bohlander, Scott A Snell, "Principles of human Resource Management", Cengage Learning. 3. Edwin B Flippo, "Personnel Management", Tata McGraw Hill Publishing, New Delhi. 4. John H. Bernardin, "Human Resource Management - An Experiential Approach", Tata McGraw Hill, New Delhi. 5. Mirza, Saiyadain, "Human Resource Management", Tata McGraw Hill, New Delhi. 6. Gary Dessler & Biju Varkkey, "Human Resource Management", Pearson, New Delhi.



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MBA II Semester (AR23)

Course Code	Course Title	L	T	P	C
MB207C	OPERATIONS MANAGEMENT	4	0	0	4

Course Description

This course focused on the principles and practices necessary for effectively managing and optimizing an organization's production and operational processes. The course covers essential topics such as process design, supply chain management, quality control, inventory management, and operational strategy. Students will gain insights into techniques for improving efficiency, reducing costs, and enhancing product and service delivery through case studies. By integrating theoretical knowledge with practical applications, students will develop the skills to analyse and refine operational systems, manage resources effectively, and drive continuous improvement. This course is ideal for current and aspiring managers looking to enhance their ability to lead operational teams and contribute to organizational success.

Course Objectives

1	To understand Industry 4.0 and integrate technology in smart production.
2	To apply various production and operations design decisions and relate it to the overall strategies of organizations.
3	To obtain an understanding of quality management practices to enhance the organizational effectiveness.
4	To understand the roles on inventories and basics of managing inventories in various demand settings.
5	To introduce concepts of operations management in an organisation and expose to analytical methods.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand the basics of operations management in manufacturing and service sector and will be able to plan and implement production and service related decisions.
CO2	Apply the planning and control concepts for decision-making.
CO3	Plan production schedules and resources (material and machine) required for production.
CO4	Apply the concepts of purchase, stores and inventory management and analyze and evaluate material requirement decisions.
CO5	Understand the quality management practices in organizations and implements tools (TQM, Six-Sigma, JIT etc.) to facilitate organizational effectiveness.
CO6	Apply analytical skills and problem-solving tools to resolve the operational issues.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	0	2	3	2	1
CO2	3	3	0	0	2	1	2	2
CO3	3	3	0	0	2	2	2	3
CO4	3	3	0	0	3	2	2	3
CO5	3	3	0	0	3	2	2	2
CO6	3	2	0	0	2	2	2	1
AVG	3.00	2.83	0.00	0.00	2.33	2.00	2.00	2.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Introduction to Production and Operation Management**

Concept of Production & Operations, Manufacturing Vs Service Operations; Production Systems - Classification of Production System, Productivity, Factors Improving Productivity; Demand Forecasting - Forecasting System, Qualitative & Quantitative Methods of Forecasting.

Unit-II: Facility Location

Strategic Importance of facility Location, Factors Affecting Plant Location; Location Theories and Models: Factor Rating Method, Weighted Factor Rating Method, Centre of Gravity; Aggregate Planning: Role & Need for Aggregate Planning, Aggregate Planning Strategies.

Unit-III: Production Planning and Control

Objectives & Importance of PPC, Project Management: Concept and characteristics of a project, Project life cycle, Project Management Techniques, Assumption and analysis of CPM and PERT; Sequencing: Johnson's Job Sequencing, 2×M Machines, M×2 Machines, M×N Machines, Gantt Charts.

Unit-IV: Material Handling

Objectives and Principles, Materials Requirement Planning (MRP-I), Manufacturing Resource Planning (MRP – II); Inventory Management: purpose and Costs of Inventories; Techniques of inventory management: P & Q system, ABC analysis, Economic Order Quantity

Unit-V: Quality

Nature & Dimensions of Quality. Determinants of Quality, Costs of Quality; Total Quality Management, Continuous Improvement (KAIZEN), Kanban; Quality Control Techniques: Quality Control, Statistical Quality Control (SQC); Control Chart for Variables, Mean and Range Chart; Control Chart for Attributes, P and C Chart; Six Sigma, ISO – 9000 Series, Just-in-Time and Lean Manufacturing, Enterprise Resource Planning.

Textbook References:

1. Ajay Garg, Production & Operations Management, Tata McGraw Hill India
2. Geither Norman, Production & Operations Management, The Dryden Press, Chicago.
3. K. Aswathappa, K. ShridharaBhat., Production and Operations Management, Himalaya Publishing House
4. Panneerselvam, "Production and Operations Management", PHI.
5. B. Mahadevan, Operations Management: Theory and Practice, Pearson.
6. Kenneth K. Boyer, Rohit Verma, Operations Management: Cengage Learning.



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Course Code	Course Title	L	T	P	C
MB208C	BUSINESS ANALYTICS	4	0	0	4

Course Description

This course explores the critical techniques and tools used to interpret data and make informed business decisions. This course delves into the fundamentals of data analysis, statistical methods, and predictive modelling, equipping students with the skills to analyse complex datasets and extract actionable insights. Key topics include data visualization, decision-making frameworks, and the application of analytics in areas such as marketing, finance, and operations. Through practical exercises, and software tools, students will learn how to transform data into strategic recommendations, enhance business performance, and drive data-driven decision-making. Ideal for professionals and managers aiming to leverage analytics for competitive advantage, this course provides the knowledge needed to harness the power of data in today's business landscape.

Course Objectives

1	To understand the key terms, definitions, and concepts used in the study of Business Analytics.
2	To demonstrate knowledge of statistical data analysis techniques utilized in business decision making.
3	To analyse the correlation between different variables and to predict demand and other estimates.
4	To Understand and explore problems in business and to solve using extensive information.
5	To learn tools such as Excel and open source to interpret data for data interpretation.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Able to recognise the fundamental concepts and best practices of business analytics in modern organizations.
CO2	Able to apply appropriate analytical methods to find solutions to business problems that achieve stated objective.
CO3	Able to analyse the results of business analytic applications into effective courses of action.
CO4	Able to illustrate use of software tools and functions to clean and prepare data for analysis.
CO5	Able to apply best practices for choosing and building data visualization for business data reporting.
CO6	Able to demonstrate technical competencies using leading-edge platforms for managing structured and unstructured big data.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	3	3	3	3
CO2	3	3	0	3	3	3	3	3
CO3	3	3	0	3	3	3	3	3
CO4	3	3	0	1	1	1	3	2
CO5	3	3	0	2	2	2	3	3
CO6	3	3	0	3	3	3	3	3
AVG	3.00	3.00	0.00	2.50	2.50	2.50	3.00	2.83

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit I: Business Analytics Best Practices**

Business Analytics Definition, Meaning, Core Concepts and Definitions, the Competitive Advantage of Business Analytics, Business Analysis vs. Business Analytics, Data Scientist vs. Data Engineer vs. Data Analyst, Advantages of Business Analytics, Challenges of Business Analytics, Establishing Best Practices.

Unit II: Managing Information and Making Decisions

Understand Your Data, Data vs. Information, V's of Data, Data collection methods, Role of competencies. Coping with Information Overload.

Unit III: Decision tools

Introduction to statistics, Simple Probability, Binominal Distribution and Poison Distribution, Components, Methods of Time Series Analysis (freehand-semi averages Straight line least Square-Moving Averages). Introduction to SPSS and R.

Unit IV: Information Management

Creating the Data Architecture, Understanding the Data Value Chain, Data-Management Processes, Capturing the Right Data, Decision Making Structures, Linking Analytics to Value, Reducing Time to Recommendation.

Unit V: Data Visualisation and Working with Ms Excel Workbook

Concept: Importance of Data Visualization - Choosing Appropriate Visual Encodings - Ordering of Items - Number of Distinct Values - Structure of Visualization
Working with MS Excel Workbook: Tabs and Ribbons - Entering Data - Formatting Numbers & Dates - Worksheet Management - Sorting, Filters, Conditional Formatting.

Textbook References:

1. U Dinesh Kumar, Business Analytics: The Science of Data - Driven Decision Making, Wiley
2. David L. Olson and Majid Nabav, Introduction to Business Analytics, Business Expert Press.
3. Christian Albright and Wayne L. Winston, Business Analytics: Data Analysis & Decision Making, Cengage Publisher.
4. Richard Vidgen, Sam Kirshner, Felix Tan, Business Analytics: A Management Approach, Bloomsbury Academic
5. R. N. Prasad and Seema Acharya, Fundamentals of Business Analytics, Wiley India Pvt. Ltd.
6. Dinabandhu, Business Analytics, Routledge



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MBA AR23

III SEMESTER COURSE STRUCTURE AND SYLLABUS

(Dual Specialisation) Students should choose any Two Specialisations

List of Specialisations

- | | |
|--|---|
| 1. Marketing
2. Finance
3. Human Resource
4. Rural Management | 5. Operations & Supply Chain Management
6. Business Analytics
7. Systems & IT
8. Agribusiness Management |
|--|---|

S. No.	Course Category	Course Code	Type	Course Title	L	T	P	C	Marks		Total Marks
									CIA	ESE	
1	MS	MB300P	AEC01	Project Report Presentation & Viva	0	0	0	6	-	100	100
2	MS	MB301C	CC07	Banking & Insurance Management	3	0	0	3	40	60	100
Choose <u>any Three Electives from any One Specialisation</u>											
3	MS	Elective I	DSE01	See Table: III Semester Specialisation Elective Courses	4	0	0	4	40	60	100
4	MS	Elective II	DSE01		4	0	0	4	40	60	100
5	MS	Elective III	DSE01		4	0	0	4	40	60	100
Choose <u>any Three Electives from any One Specialisation</u> excluding the specialisation taken in DSE 01											
6	MS	Elective IV	DSE02	See Table: III Semester Specialisation Elective Courses	4	0	0	4	40	60	100
7	MS	Elective V	DSE02		4	0	0	4	40	60	100
8	MS	Elective VI	DSE02		4	0	0	4	40	60	100
III Semester Total					27	0	0	33	280	520	800

Table: III Semester Specialisation Elective Courses (AR23)

S. No.	Course Code	Course Title
Marketing Specialisation (For Semester III)		
1	MB-E-302MM	Consumer Behaviour
2	MB-E-303MM	Sales & Distribution Management
3	MB-E-304MM	Salesforce Business Analyst-I
4	MB-E-305MM	Marketing Analytics
5	MB-E-306MM	B2B Marketing
Finance Specialisation (For Semester III)		
1	MB-E-302FIN	Strategic Corporate Finance
2	MB-E-303FIN	Security Analysis and Portfolio Management
3	MB-E-304FIN	Financial Institutions, Markets & Services
4	MB-E-305FIN	Financial Analytics
5	MB-E-306FIN	Behavioural Finance
HR Specialisation (For Semester III)		
1	MB-E-302HR	Strategic HRM
2	MB-E-303HR	Industrial & Labour Laws
3	MB-E-304HR	Compensation & Benefits Management
4	MB-E-305HR	HR Analytics
5	MB-E-306HR	Talent & Knowledge Management
Rural Management (For Semester III)		
1	MB-E-302RM	Strategies for Sustainable Rural Development
2	MB-E-303RM	Industry and Agri-Business
3	MB-E-304RM	Rural Infrastructure
4	MB-E-305RM	Financial Inclusion
5	MB-E-306RM	Retail and Rural Marketing
Operations & Supply Chain Management (For Semester III)		
1	MB-E-302OSM	Supply Chain and Logistics Management
2	MB-E-303OSM	Total Quality Management
3	MB-E-304 OSM	Operations & Materials Management
4	MB-E-305 OSM	Supply Chain Planning

5	MB-E-306OSM	Strategic Planning for Supply Chain Management
Business Analytics (For Semester III)		
1	MB-E-302BA	Big Data Analytics with HADOOP
2	MB-E-303BA	Data Analysis with Python
3	MB-E-304BA	Data Analysis with R
4	MB-E-305BA	Data Analysis with SAS
5	MB-E-306BA	Data Visualization for Managers
Systems & IT (For Semester III)		
1	MB-E-302SIT	Data Mining for Business Decisions
2	MB-E-303SIT	Managing Software Projects
3	MB-E-304SIT	Managing Digital Innovation and Transformation
4	MB-E-305SIT	Enterprise Resource Planning
5	MB-E-306SIT	Managing Digital Platforms
Agribusiness Management (For Semester III)		
1	MB-E-302ABM	Agri-economics
2	MB-E-303ABM	Agribusiness Management
3	MB-E-304ABM	Food Retail Management
4	MB-E-305ABM	Management of Agro Business Cooperatives
5	MB-E-306ABM	Agribusiness Financial Management



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MBA III SEMESTER (AR23)

Course Code	Course Title	L	T	P	C
MB300P	PROJECT REPORT PRESENTATION & VIVA-VOCE	0	0	0	6

Course Description

This course is designed to equip students with essential skills for effectively communicating their research findings and defending their work in a formal academic setting. It emphasizes the development of clear, structured, and engaging project reports, alongside honing the ability to deliver compelling presentations that capture the audience's attention. Students will learn to utilize visual aids and multimedia tools to enhance their presentations and will practice responding to questions and critiques with confidence and depth during the viva-voce. The course also focuses on integrating feedback to refine project reports and improve presentation techniques, ensuring students can showcase their expertise and critical thinking skills effectively in both written and oral formats. It also offers students the opportunity to gain exposure to industry projects, interactions with professionals and other summer interns, and improving their presentation, writing, and communication skills. It often acts as a gateway for final placement for many students.

Course Objectives

1	To transfer their analytical, integrative, team skills honed in the classroom to the workplace.
2	To understand the complexities of the corporate world first hand.
3	To focus on a narrow problem within a selected topic/field typically a micro theme and to facilitate an in-depth study using quantitative/qualitative measurement tools to organize and analyse data, intensively focusing on the area to arrive at reasoned conclusions about the issue or problem.
4	To identify opportunities for networking with people in Industry/Organisations/Corporates.
5	To aim for pre-placement offers where feasible/appropriate.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Developing independent thinking on the subject and enables to gain practical experience in research methodology.
CO2	Work effectively as an individual and member of the team to solve managerial problems.
CO3	Apply the theoretical concepts learned in the programme to real-life situations.
CO4	Create a clear and organised project report that effectively communicates research findings and conclusions.
CO5	Apply professional presentation techniques to deliver a compelling and informative project report.
CO6	Compose and present detailed project report of his/ her work and defend effectively.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	2	2	3	1	2	2	3
CO2	3	3	3	3	3	2	3	3
CO3	3	3	3	3	3	3	3	3
CO4	2	3	2	3	3	2	3	3
CO5	3	3	1	3	3	2	3	3
CO6	2	3	1	3	3	2	3	3

AVG	3.00	2.83	2.00	3.00	2.67	2.17	2.83	3.00
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Duration of SIP: SIP is scheduled during the summer between the first and the second year. It generally spans for eight weeks. The duration may be extended beyond eight weeks, but in no case should exceed ten weeks. Any extension for SIP duration would be subject to the decision of the Dean/HoD.								
Initial Stage: 1. Each student will be allotted a faculty member as guide based on the student specialisation. Any problem/assistance related to SIP faced/needed by the student must be report to his/her guide only. 2. Each student must submit an Initial Internship Report (IIR) on second week of post starting of internship to his/her guide.								
During SIP: 1. Students will abide by the applicable policies and norms of the Organisation during the period of internship. 2. The students are expected to first understand the organization and the industry/field in which the organization is operating, in order to make a meaningful difference with the quality of his/her work 3. The organization has the right to terminate students from the internship at any time due to inappropriate behaviour and/or non-cooperation with the internship process and/or continued non-performance in assignment or any other reason the organization deems fit. Therefore, it is strictly advising to all interns/students that do internship with utmost seriousness. 4. The students are expected to concentrate on the specific topic of study, its objectives, its rationale, and adopt a methodology and identify a suitable analysis procedure for the completion of the study. 5. It is expected that all students will adhere to the proper standards of intellectual honesty and professional propriety in their conduct. 6. Students are advised not to do anything directly or indirectly which may create a poor impression about the University. Any student found disregarding any of the norms would be liable for disciplinary action, including rustication depending upon the severity of the offence. 7. The students should abide by the dress code and other professional norms of their Internship organisation. Punctuality is a quality that is appreciated by professionals across all organizations. 8. The students should ensure that the data and other information used in the study report is obtained with the permission of the organisation concerned. 9. Each student must submit an Internship Progress Report (IPR) on every fortnight to his/her guide.								
Completion Stage 1. At the end of the internship, the intern shall prepare Internship Completion Report (ICR). 2. The intern may be required to make a presentation to the Industry Mentor (IM). 3. As proof of completion of the internship, the intern will obtain an Internship Completion Certificate from the organization/company/IM.								
After completion 1. The intern shall submit the SIP Report (including completion certificate) preliminary copy to the guide within one week of commencement of the ensuing semester. 2. The intern must follow five stages before the Project Report Committee (PRC): Stage 1: First presentation at the time of Cycle Test I. Stage 2: Each student is to submit a written project synopsis to his/her respective guides. Stage 3: Second Presentation at the time of Cycle Test II. Stage 4: Project Report Submission (three sets of hardcopy and one softcopy) before								

commencement of External Exams.
 Stage 5: External Viva-voce at the time of External Exams.

Faculty Supervision, Reporting & Evaluation

1. The guides are to ensure that the interns can deliver as expected and explore the possibility of converting the SIP opportunity into a long-lasting association, collaborations for research, executive education, and consultancy projects etc.
2. During SIP, several interactions between the guide and the organizations/companies are expected. These can be conducted through multiple modes such as email, phone, video, or in-person as may be expedient/appropriate. Guide can coordinate with IM and assigned student for the same.
3. During the internship, the intern may seek telephonic/email/online platform interactions with guide to receive support if necessary.
4. Completion of the summer internship is a degree-awarding requirement for the MBA program of School of Management Studies, Gandhi Institute of Engineering & Technology University. For any foreseeable or unforeseeable reasons, if a student is unable to complete the internship in the intended year, he or she should be allowed to complete the internship requirements at the end of the requisite course work of the program he or she has enrolled for. It should be noted that as SIP is a core course, non-completion of SIP would entail no degree.
5. Absenteeism, premature abandonment, non-submission of reports, misconduct at the workplace, unprofessional behaviour, etc. are non-exhaustive examples of serious misconduct during the program. In case the student intern is found to have indulged in serious misconduct, the student intern is liable for any of the punitive actions, including cancellation of internship, withdrawal of final placement assistance, or rustication from the program/university, depending on the severity of inappropriate conduct.

Evaluation of SIP

The internship evaluation will have three components:

1. **Continuous Assessment (During SIP):** Every fortnight student will have to submit a progress report to his/her guide as per template. All these reports will be assessed by guide for content and compliance.
2. **Industry Mentor Assessment (On completion of SIP):** The Industry Mentor will submit in writing/discuss with corresponding guide his/her evaluation of the student in terms of exceeding or at least meeting expectation.
3. **Final Assessment (After completion of SIP):** A student will be required to submit a detailed project report to his/her guide and appear for an oral presentation before a panel to compile his/her learnings from the internship. The guide will share report format with the interns.

Project Report Presentation Marks Summary

Stage	Coverage of Presentation	Marks
I	- Title of the Project - Need for the Study - Objectives of the Study - Scope of the Study - Hypothesis of the Study - Research Methodology & Statistical Tools - Theoretical Framework - Company Profile - Industry Profile - Limitations	20
II	- Data Analysis & Interpretation - Findings, Suggestions and Conclusion	30
III	Project Submission and Viva-voce Examination	50
Total Marks for Evaluation		100

SYNOPSIS FORMAT

Each student is to submit a written project synopsis to his/her respective guides. The project synopsis must comprise the following aspects:

1. **Title of the Project:** Title should be meaningful and should convey the broad aspects that will be covered in the body and the scope of the project.
2. **Organisation/Company:** Mention the name of organization/company and the functional area (e.g. marketing, finance and human resource) where you intend to do your project. Briefly explain the nature of the organization and its business.
3. **Introduction of the Topic:** Briefly introduce the topic on which your project is based.
4. **Objectives, Scope & Hypothesis:** Explain the objectives and the scope of the project along with functional area that will be covered in the study. Also frame the hypothesis under study.
5. **Research Methodology:**
 - a. Research design
 - b. Sampling design
 - c. Sample size
 - d. Sample location
 - e. Data type
 - f. Instrument used
 - g. Analytical Tools
6. **Expected Outcome:** Briefly discuss the expected outcome of the project.
7. **Bibliography:** Source for your synopsis (APA Style or Harvard Style)

Note: Once the Title & the Organization (if applicable) Is Approved, It Cannot Be Changed arbitrarily

FORMAT FOR PROJECT REPORT SUBMISSION AND VIVA VOCE

The final report is comprised of the following:

1. Cover Page
2. Certificates
 - Department Permission Letter
 - Company Permission Letter
 - SIP Completion Letter from Company
3. Declaration
4. Acknowledgements
5. Executive Summary in 250 words
6. Index/Table of Contents
7. List of Tables/Figures/Charts/Symbols
8. List of Abbreviations
9. Chapter 1: Introduction
 - Introduction (in Single Paragraph)
 - Need for the Study (point wise in number format)
 - Objectives of the Study (point wise in number format, Max: 3)
 - Scope of the Study (point wise in number format)
 - Research Methodology
 - Sample Design (in Single Paragraph)
 - Sample Type (in Single Paragraph)

- Sample Size (in Single Paragraph)
 - Statistical Tools used (type wise in bullet format)
 - Data Collection
 - Which type of data collection you chose and why?
 - Limitations (point wise in number format)
10. Chapter 2: Industry and Company Profile
- Industry Profile
 - Brief profile of the Industry including its current status from which the company belongs
 - Company Profile
 - Name of the firm/company (its complete address).
 - Company's vision & mission.
 - Focus on nature of the Organisation and its business (service/production/trading etc.)
 - Size (in terms of manpower & turnover) of organization.
 - Organization structure of the company.
 - Market share & position of the company in the industry.
 - Awards and Recognitions
11. Chapter 3: Theoretical Framework
- Your chosen topic introduction.
 - Importance of the chosen topic.
 - Various strategies/types/categories/etc. of your chosen topic.
 - Mention any other relevant information on your chosen topic.
12. Chapter 4: Data Analysis and Interpretation
- Show your raw data in the form of tables/charts/figures/diagrams/etc.
 - Show the analysis of your collected data using various mathematical and statistical methods.
 - Write a brief interpretation on each table/chart/figure/diagram/etc.
13. Chapter 5: Findings, Suggestions/Recommendations and Conclusion
- Present the facts based on your analysis (point wise in number format)
 - Suggest alternative solutions to the identified problems (point wise in number format)
 - Prepare a brief conclusion (in single paragraph)
14. References/ Bibliography (APA Style or Harvard Style) use the following links
<https://apastyle.apa.org/instructional-aids/reference-examples.pdf>
<https://librarydevelopment.group.shef.ac.uk/Assets/pdfs/referencing/harvard.pdf>
15. Appendices
- Questionnaires
 - Any Supporting Data

FORMAT FOR TABLES/FIGURES/CHARTS/SYMBOLS/ABBREVIATIONS INDEX

List of Tables

S. No.	Table No.	Table Title	Page No.

List of Figures

S. No.	Figure No.	Figure Title	Page No.

List of Charts

S. No.	Chart No.	Chart Title	Page No.

List of Symbols

S. No.	Symbol	Nomenclature & Meaning

List of Abbreviations

S. No.	Abbreviated Name	Full Form of Abbreviation

SPECIFICATIONS FOR BODY OF THE PROJECT REPORT AND VIVA VOCE

Following aspects must be adhered to as given in while compiling the body of report

1. **Page Size:** Good quality white A4 size executive bond paper should be used for typing and duplication.
2. **Chapter/Para Numbering:** The chapters are to be numbered as Chapter-1, Chapter-2 etc. The heading/title of the chapter is to appear below the chapter number in uppercase. Paragraphs are to be numbered as 1,2,3 etc in every chapter separately.
3. **Page Specifications**
 - a. Left Margin : 1.25 inch
 - b. Right Margin : 1.25 inch
 - c. Top Margin : 1 inch
 - d. Bottom Margin : 1 inch
4. **Page Numbers:** All text pages starting from Body of the Project Report as well as program source code listings should be numbered at the bottom centre of the pages.
5. **Normal Body Text**
 - a. Font Size: 12, Times New Roman, 1.5 Spacing, Single Side Writing.
 - b. Paragraphs Heading Font Size: 12, Times New Roman, Bold (Capitalized Each Word)
 - c. Page/Title Font Size: 14
6. **Table/Charts/Diagrams/Figure Number:** Follow standard format
7. **Binding & Colour Code of the Report**
 - a. **Hard Bound Report.**
 - b. **Background of the cover page :** Black/White.
 - c. **Colour of Letters :** Gold in case of Black background/Black in case of white background.

Textbook References:

7. C.R. Kothari, Research methodology Methods and Techniques, New Age International (p) Limited, publishers.
8. Dr Deepak Chawla & Dr Neena Sondhi, Research Methodology Concepts and Cases, Vikas Publishing House Pvt Ltd.
9. R. Panneerselvam, Research Methodology, PHI Learning.
10. Krishna Swamy, Sivakumar, and Mathirajan, Management Research Methodology, Pearson.
11. SL Gupta and Hitesh Gupta, Business Research Methods, Tata McGraw Hill Education.
12. Donald R cooper and Pamela S Schindler, Business research Methods, McGraw-Hill.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)

Course Code	Course Title	L	T	P	C
MB301C	BANKING AND INSURANCE MANAGEMENT	3	0	0	3

Course Description

This course offers a comprehensive exploration of the principles, practices, and regulatory frameworks governing the financial services industry. Students will delve into the fundamentals of banking operations, including risk management, financial analysis, and the role of banking institutions in the economy. The course also covers key concepts in insurance management, such as underwriting, claims processing, and the development of insurance products. Through a combination of theoretical insights and practical cases, students will gain a deep understanding of how banks and insurance companies manage financial risk, optimize performance, and navigate regulatory environments. This course aims to equip students with the knowledge and skills necessary to excel in the dynamic and evolving fields of banking and insurance.

Course Objectives

1	To understand the role and importance of banks and insurance companies in the financial system.
2	To learn about various banking operations, products, and services.
3	To gain insights into the principles and practices of insurance and analyze the regulatory environment governing banking and insurance sectors.
4	To develop an understanding of risk management in banking and insurance.
5	To explore current trends and challenges in the banking and insurance industries.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Adhere to banking operations and insurance principles to make informed financial decisions.
CO2	Implement risk management strategies to mitigate financial risks inherent in banking and insurance activities.
CO3	Evaluate financial products and services offered by banks and insurance companies to meet customer needs.
CO4	Proficient in regulatory compliance requirements and ethical considerations in banking and insurance management.
CO5	Utilize financial analysis skills to assess the performance and stability of banking and insurance institutions.
CO6	Adapt technological advancements and global trends shaping the banking and insurance industries for strategic decision-making.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	2	3	1	2	2	2
CO2	3	2	1	2	3	3	1	2
CO3	1	2	3	2	1	2	3	3
CO4	2	3	3	2	1	2	3	2
CO5	1	2	3	1	2	3	2	1
CO6	2	2	3	2	1	2	2	2
AVG	1.83	2.00	2.50	2.00	1.50	2.33	2.16	2.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit- I: Introduction to Banking**

Definition - Functions of a bank, classification of banks - Evolution of Banking in India - origin - Structure and organization of banks; Reserve Bank of India - Commercial banks - Regional rural banks - Cooperative banks - Development banks - Nationalization, Privatization and Globalization of Banks in India - Banking Sector Reforms.

Unit-II: Commercial Banking in India

Functions - Deposit mobilization by banks – Types of deposits - Opening of accounts - KYC norms - Operation of accounts: pay in slips, withdrawal forms, passbook, and cheques - Closure of bank accounts - Crossing of cheques - Lending of money by banks: cash credit, overdraft, term loans - Principles of lending - Retail credit by banks - NPAs.

Unit-III: Ancillary Services Rendered by Banks

Agency services - General utility services - E-Banking services: ATMs, plastic cards, mobile banking, internet banking, EFTs, CTS - Bank assurance - Marketing of banking services.

Unit-IV: Introduction to Insurance

Definition of insurance - Basic principles of insurance - Segments of insurance - life insurance and non-life insurance - Types of life insurance policies - Pure and Term - Types of non-life insurance: Fire, Marine, Motor, and Miscellaneous - Recent trends in Insurance in India - Role of IRDA in India.

Unit-V: Insurance Marketing

Insurance Marketing - Significance - Products and Pricing - Positioning and Promotion - Distribution channels - Selling Process - Reinsurance: the concept, uses and advantages.

Textbook References:

1. Vasanth Desai, Nature and Problems of Commercial Banking in India, Himalaya Publishing House.
2. Maheshwari and Paul R.R, Banking theory law and practice.
3. Sundaram and Varshney, Banking theory law and practice.
4. Tannans, Banking law and practice in India.
5. M N Mishra, S B Mishra, Insurance Principles & Practice, Sultan Chand and Sons.
6. Mohan Prakash, N.R Banking, Risk and Insurance Management, Vikas Publishing House.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Marketing Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302MM	CONSUMER BEHAVIOUR	4	0	0	4

Course Description

There is an increasing importance of the subject “Consumer Behaviour” in consumer and industrial market in this digitalization era. Consumer behaviour explore the consumer expectations through the attitude formation, cultural differences and personality traits. Now a day, the behaviour of an individual or organization have significant impact on the society. This subject has a higher priority in terms of consumer psychology, sentiment survey, and strategic influence of consumers. Consumer behaviour deserve for employment opportunities in various sectors across the globe.

Course Objectives

1	To discuss on the consumer behaviour theory and practices in this internet age.
2	To investigate the theory of personality, perception, attitude and learning to understand the consumer behaviour.
3	To examine the impact of social media, advertising or publicity to know the consumer communication.
4	To investigate the role of culture in terms of decision-making process.
5	To develop the concepts on innovation and diffusion, ethical behaviour and social responsibility towards business.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate the usage of technology driven consumer behaviour to secure a sustainable business for the society.
CO2	Investigate the concept of personality, perception, attitude and learning to know the consumer behaviour in an individual setting.
CO3	Examine the impact of social media, advertising or publicity to understand the consumer communication for greater transparency.
CO4	Investigate the role of culture in terms of decision-making process for an individual or family.
CO5	Analyse the concepts on innovation and diffusion, ethical behaviour and social responsibility for a sustainable business practice.
CO6	Demonstrate the knowledge on consumer behaviour, personality, perception, culture, decision making or business ethics to design or develop the case studies.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	1	1	0	0	1	1	1
CO2	2	1	1	1	0	3	1	0
CO3	1	1	0	0	2	2	0	1
CO4	1	1	1	1	1	0	1	1
CO5	1	0	1	3	1	1	1	1
CO6	3	3	2	2	1	1	1	1
AVG	1.83	1.17	1.00	1.17	0.83	1.33	0.83	0.83

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit- I: Consumers, Marketers and Technology**

Technology driven Consumer Behaviour: Technology enriches the exchange between consumers and marketers, Customer value, satisfaction and retention; Segmentation, Targeting, and Positioning, Bases for segmentation, Behavioural targeting, Positioning and repositioning.

Unit-II: The Consumer as an Individual

Consumer Motivation and Personality: The dynamics of motivation, Systems of needs & a trio of needs, Measurement of motives, Nature and theories of personality, Personality traits and consumer behaviour, Product and brand personification, Self and self-image; Consumer Perception: Elements of perception, Differential threshold, Subliminal perception, Perceptual selection, Perceptual, organization; Perceptual interpretation: stereotyping, Consumer imagery, perceived quality, Perceived risk; Consumer Learning: Elements of consumer learning, Classical conditioning, Stimulus discrimination and product differentiation, Instrumental conditioning, Information processing, Cognitive learning, Consumer involvement and hemispheric lateralization, Outcomes and measuring of consumer learning; Consumer Attitude Formation and Change: Attitudes and their formation, Tri-component attitude model, Multi-attribute attitude model, Changing the motivational functions of attitudes, Elaboration likelihood model, Cognitive dissonance and resolving conflicting attitude, Assigning causality and attribution theory.

Unit-III: Communication and Consumer Behaviour

Persuading Consumers: Persuading consumers, Designing persuasive messages, Persuasive advertising appeals, Humour in advertising, Measures of message effectiveness; From Print and Broadcast Advertising to Social and Mobile Media: Targeting segments versus eyeballs, Consumers and social media, Consumers and mobile advertising, Measuring media's advertising effectiveness, Traditional media's electronic evolution; Reference Groups and Word-of-Mouth: Source credibility and reference groups, Factors affecting reference group influences, Reference groups, self-concepts and brand associations, Credibility of spokes persons, endorsers and other formal sources, Word-of-mouth and opinion leadership, Strategic applications of word-of-mouth, Diffusion of innovations.

Unit-IV: Consumers in their Social and Cultural Settings

The Family and Its Social Standing: Family as a socialization agent, Family decision-making and consumption-related roles, Family life cycle, Measuring social class, Social class' characteristics and consumer behaviour; Culture's Influence on Consumer Behaviour: Culture's role and dynamics, Learning cultural values, Measuring cultural values, Indian core values, Cultural aspects of emerging markets; Cross-Cultural Consumer Behaviour: An International Perspective, Cross-cultural analysis and acculturation,

Unit-V: Consumer Decision-Making, Marketing Ethics, and Consumer Research

Consumer Decision-Making and Diffusion of Innovations: Consumer decision-making and diffusion of innovations, Diffusion and adoption of innovations; Marketing Ethics and Social Responsibility Societal marketing concept, Exploitive marketing, Provocative marketing

Textbook References:

1. Michael R. Solomon, Consumer Behaviour: Buying, Having and Being Pearson.
2. David Loudon, A. J. Della Bitta, Consumer Behaviour: Concepts and Applications McGraw Hill Latest Edition
3. Majumudar Ramanuj, Consumer Behavior: Insights from Indian Market, PHI Latest Edition
4. Martin M. Evans, Gordon Foxall, Ahmad Jamal, Consumer Behaviour, Wiley.
5. Roger D. Blackwell, Paul W. Miniard, James F. Engel Consumer Behaviour Cengage.
6. Leon G. Schiffman and Joseph Wisenblit & S. Ramesh Kumar, Consumer Behaviour Pearson Education, India.



SCHOOL OF MANAGEMENT STUDIES
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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Marketing Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-303MM	SALES & DISTRIBUTION MANAGEMENT	4	0	0	4

Course Description

This course is intended to make students familiar with the fundamentals of Sales Management and Distribution Management. Since many students with Marketing electives begin their career in sales, business development, retailers, wholesalers and distributors handling, it is essential to study basics of personal selling, sales management, distribution dynamics, various channels of distribution etc. Through this course, students will be equipped with basic skills required in sales and distribution management. The course also enhances the ability on sales force management and distribution strategies through case analysis.

Course Objectives

1	To discuss the conceptual knowledge on theory and practices of sales management.
2	To demonstrate the sales planning, scheduling and budgeting that interface the challenges in sales management.
3	To develop the principle of sales force management and its practicability for a sustainable business management.
4	To design and evaluate the distribution channel management integrating the customer satisfaction.
5	To apply the sustainable practices on managing the distribution channel with conflict resolution.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Execute the theoretical knowledge on sales management for a sustainable development business.
CO2	Implement the theory and practices of sales planning, scheduling and budgeting to provide customer satisfaction.
CO3	Distinguish the application of sales force management, sales force control and evaluation.
CO4	Design the distribution channel to fulfil the customer expectations for a sustainable business.
CO5	Collaborate and direct the channel members to adopt the ethical practices for a sustainable distribution.
CO6	Discuss the relevant case on sales management, sales planning & budgeting, sales force management, distribution management and conflict management.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	1	1	3	2	1	1	1
CO2	2	2	1	1	1	1	1	1
CO3	3	2	2	1	1	2	2	1
CO4	3	2	2	2	1	1	1	1
CO5	2	1	1	2	1	1	1	1
CO6	3	2	3	2	2	2	2	2
AVG	2.67	1.67	1.83	1.83	1.33	1.33	1.33	1.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Introduction to Sales Management**

Introduction to Sales Management: Evolution of Sales Management, Importance of Sales Management, Types of Selling, Difference between Selling and Marketing, Modern Day Sales Activities, Selling Skills, Selling Strategies, Selling Process.

Unit-II: Sales Planning

Sales Planning and Budgeting: Sales planning process, Sales forecasting methods, Sales budgeting process, Methods used for deciding sales budget, Types of quotas and quota setting procedure, Reasons for establishing or revising sales territories, Routing and scheduling sales persons, Market cost analysis.

Unit-III: Sales Force Management

Sales Force Management: Recruitment and selection of the sales force, training the sales force, sales force motivation, sales force compensation, sales force control and evaluation.

Unit-IV: Distribution Management

Basics of Distribution Management: Definition, need for Distribution Channels, designing the Marketing Channels, Motivating and Evaluating Channel Members, Capturing the Customer requirements.

Unit-V: Channel Management

Managing Distribution Channels: Managing Channel Information Systems, reasons for Channel Conflicts, Managing Conflict, Managing, Ethical issues in Sales and Distribution Management.

Textbook References:

1. Havaladar & Cavale, Sales and Distribution Management, McGraw Hill publication.
2. Krishna Havaladar and Vasant M Cavale, Sales and Distribution Management Text and Cases, McGraw Hill Publication.
3. Pradip Mallik, Sales Management, Oxford University Press.
4. Still, Cundiff & Govani, Sales Management, Pearson Education.
5. Panda Tapan K., Sahadev Sunil, Sales & Distribution Management (Latest Edition), Oxford University Press.
6. Krishna K. Havaladar, Vasant M. Cavale, Sales & Distribution Management-Text & Cases, Tata McGraw-Hill.



SCHOOL OF MANAGEMENT STUDIES
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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Marketing Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304MM	SALESFORCE BUSINESS ANALYST-I	4	0	0	4

Course Description

Businesses increasingly rely on data analytics to enhance decision-making. Modern business students must develop skills in analysing data to make informed decisions across various fields like accounting, economics, finance, and marketing. The Salesforce Business Analyst Course is tailored for individuals who understand business needs, gather requirements, and collaborate to create Salesforce solutions that drive improvements. Proficiency in Salesforce best practices, business process mapping, and writing user stories is beneficial.

Course Objectives

1	To Gain a working knowledge of data analysis tools for business decision making using Salesforce
2	To have hands-on experience for processing and analysis of data in business disciplines such as financial portfolio management, supply-chain optimization, healthcare informatics, and social media marketing with the help of Salesforce platform
3	To learn to professionally communicate, analysis, results for the success of the project
4	To maximize the value of a system through architecture best practices and technical specifications.
5	To ensure that business automation and data management are smooth and well established.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Develop Business Analyst Skills: Master key skills in information discovery, communication, and documentation.
CO2	Align Organizations with V2MOM: Create and manage organizational alignment, understanding Salesforce Admin vs. Business Analyst roles.
CO3	Apply Agile and Lifecycle Management: Understand and implement Agile methodologies and release management in Salesforce.
CO4	Utilize Salesforce Customer 360: Leverage Customer 360 for storytelling and organizational change leadership.
CO5	Innovate and Influence: Build trusted relationships, use journey mapping, and innovate customer solutions.
CO6	Enhance Salesforce Adoption: Drive Salesforce adoption and achieve excellence in sales operations.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	1	0	2
CO2	3	0	0	0	0	2	1	0
CO3	2	2	1	2	1	0	1	0
CO4	3	0	0	1	0	2	0	1
CO5	3	1	1	0	0	1	1	0
CO6	2	1	0	2	1	1	0	0

AVG	2.67	1	0.67	1	0.50	1.17	0.50	0.50
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Essential Business Analyst Skills								
Essential Business Analyst Skills: Discover Top Business Analyst Skills, Explore Techniques for Information Discovery, sharpen your communication Skills, Explore Documentation Types. Drucker School—Business Performance Basics: Get to Know Drucker School of Business, Understand the Five Most Important questions to ask about your business, learn best practices from The Financial Times, Identify the Top Five things Salesforce Admin to do Organizational Alignment (V2MOM): Create organizational alignment through the V2MOM, write a V2MOM, Manage the V2MOM Compare the Admin and Business Analyst Roles, Learn about the Salesforce Business Analyst Role. Application Lifecycle and Development Models: Understand what Application Lifecycle Management is, Learn Basics of Release Management, Manage changes in Increasingly complex releases, Use Package Development for more flexible releases Salesforce Agile Basics: Understand why Salesforce Adopted Agile, Learn the basics of Agile, Learn to use Lean Principles								
Unit-II: Salesforce Agile Basics								
Compare the Admin and Business Analyst Roles, Learn about the Salesforce Business Analyst Role. Application Lifecycle and Development Models: Understand what Application Lifecycle Management is, Learn Basics of Release Management, manage changes in Increasingly complex releases, Use Package Development for more flexible releases Salesforce Agile Basics: Understand why Salesforce Adopted Agile, Learn the basics of Agile, Learn to use Lean Principles.								
Unit-III: Salesforce Customer 360								
Discover Salesforce Customer 360, Get to know Customer 360, Tour the customer 360 portfolio. Demo Storytelling: Create your story, Build your storyboard. Drucker School—Organizational Change Leadership: Learn Drucker's Approach to Change, become a Change Leading organization, manage organizational Change, Learn Best practices from American Express, do these Five things to drive organizational change. Content Strategy Development : Learn content strategy Basics, Craft your content vision, Get to know your audience, Evaluate Existing content, Define your content strategy.								
Unit-IV: Trust and Influence, Diagramming Essentials								
Trust and Influence: Develop trusted relationships, Influence others Journey Mapping: Add Journey Mapping to your Solution Toolbox, define intention and Audience, start your Journey Map, run a journey Mapping workshop, Craft your journey map Innovation Customer Discovery: Empathize with your customer, create customer archetypes, Give your customer Future now. Innovation Ideation & Prototyping: Create an Inspiring Environment, Ideate in Teams, Build Prototype and Pitch Ideas.								
Unit-V: Diagramming Essentials								
Explore Diagramming Best Practices, import a Salesforce Diagram into Lucid Chart, design a Business Capability Map Learn How to Whiteboard During a Virtual Meeting Salesforce Adoption Strategies: Accelerate Salesforce Adoption, prepare to build your Salesforce Journey, Create the Ideal Salesforce Org for your company, deploy your Salesforce Technology, Drive Long Term Adoption. Sales Operation Use Case and Best Practices: Deliver Sales Operations Excellence, Collaborate across Teams								
Textbook References:								
7. Srimi Munagavalsa, "The Salesforce Business Analyst Handbook: Proven business analysis techniques and processes for a superior user experience and adoption", Packt Publishing.								
8. Sharif Shaalan and Timothy Royer, "Salesforce for Beginners: A step-by-step guide to optimize sales and marketing and automate business processes with the Salesforce platform", Packt Publishing.								
9. Paul Goodey, "Salesforce CRM - The Definitive Admin Handbook: Build, configure, and customize Salesforce CRM and mobile solutions", Packt Publishing.								
10. Felicia Duarte, Rachelle Hoffman, "Learn Salesforce Lightning: The Visual Guide to the Lightning UI", Wiley Press								

11. Mike Cohn, “Succeeding with Agile: Software Development Using Scrum - Addison Wesley Signature Series”, Pearson.
12. Mike Cohn, “Succeeding with Agile: Software Development using Scrum”, Pearson.
13. Andrew Stellman, Jennifer Greene, “Learning Agile”, O’Reilly Publishers.

Practical / Tutorial Platform / E-learning

Use the Trailhead Platform: <https://www.salesforce.com/blog/what-is-trailhead/>

The Salesforce Business Analyst Trailmix:

<https://trailhead.salesforce.com/users/srebello7/trailmixes/salesforce-business-analyst-professional>



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Marketing Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305MM	MARKETING ANALYTICS	4	0	0	4

Course Description

In the recent development of business, the data science has a considerable influence on the society. With the increasing interest of all the stakeholders of the society, marketing analytics is an important subject for the management graduates. This course is helping the students to solve the business problems through the statistical data through various software such as MS-Excel or R-programming. The marketing analytics will help the business to understand the customer expectation and secure a sustainable business.

Course Objectives

1	To describe the concept on meaning, characteristics, advantages and disadvantages of marketing analytics.
2	To define the pricing policies and practices for a sustainable business.
3	To develop the statistical methods for perfect sales forecast to minimize the wastages.
4	To examine the customer lifetime value and appropriateness of STP.
5	To design a market basket analysis for a sustainable modern business.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Analyse the customer data through various strategies with the application of suitable software.
CO2	Demonstrate the pricing policies and practices to regulate the business smoothly.
CO3	Create a structure data set to find a positive business output.
CO4	Examine the customer lifetime value and interpret the data for a sustainable business.
CO5	Identify the types of market for providing a better solution.
CO6	Develop and design the case analysis in PESTEL, pricing, forecast, customer lifetime value, retailing and advertising analysis.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	1	1	1	1	1	1
CO2	3	2	1	1	1	1	1	0
CO3	2	2	2	2	1	1	0	0
CO4	2	1	1	1	1	0	0	0
CO5	3	1	1	0	0	1	0	1
CO6	3	1	0	0	0	1	1	0
AVG	2.50	1.33	1.00	0.83	0.67	0.83	0.50	0.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to marketing Analytics

Meaning, characteristics, advantages and disadvantages of marketing analytics, Market Data Sources (Primary and Secondary). Market Sizing: Stakeholders, Applications & Approaches (Top-down and

Bottom-up), PESTLE Market Analysis, Porter Five Force Analysis, Data analytics through MS-Excel.

Unit-II: Pricing Analytics

Pricing Policy and Objectives, Estimating Demand: Price Elasticity, Estimating Linear and Power Demand Curves, Optimize Pricing, Incorporating Complementary Products, Pricing using Subjective Demand Curve, Pricing Multiple Products, Price Bundling & Nonlinear Pricing: Pure Bundling & Mixed Bundling, Determine Optimal Bundling Pricing, Profit Maximizing strategies using Nonlinear Pricing Strategies, Price Skimming & Sales, Revenue Management: Markdown Pricing and Handling Uncertainty, R-programming.

Unit-III: Sales Forecasting

Introduction, Simple Linear Regression & Multiple Regression model to forecast sales, Forecasting in Presence of Special Events, Modeling trend and seasonality; Ratio to moving average forecasting method, Using S curves to Forecast Sales of a New Product.

Unit-IV: Customer Analytics

Customer Lifetime Value: Concept, Basic Customer Value, Measuring Customer Lifetime value, Estimating Chance that customer is still active, Using Customer Value to value a business Market Segmentation: The segmentation-targeting-positioning (STP) framework, Segmentation, the concept of market segmentation, managing the segmentation process, deriving market segments and describing the segments using Cluster analysis

Unit-V: Retailing & Advertising Analysis-Market Basket analysis

Computing two way and three-way lift, RFM Analysis, Allocating Retail Space and Sales Resources: Identifying the sales to marketing effort relationship & its modeling, optimizing sales effort Advertising Analysis: Measuring the Effectiveness of Advertising, Pay per Click (PPC) Online Advertising.

Textbook References:

1. Wayne L. Winston, Marketing Analytics: Data-Driven Techniques with Microsoft Excel Wiley India Pvt. Ltd.
2. Moutusy Maity, Pavankumar Gurazada, Marketing Analytics, Oxford University Press.
3. Arunangshu Giri & Pradip Paul, Applied Marketing Analytics Using SPSS: Modeler, Statistics and AMOS Graphics, PHI Learning Pvt. Ltd.
4. Seema Gupta and Avadhoot Jathar, Marketing Analytics, Wiley.
5. Thomas W. Miller, Marketing Data Science: Modelling Techniques in Predictive Analytics with Python and R, Pearson Education.
6. Dr. Anand Vyas, Mrs Jyoti Vyas, Marketing Analytics Software.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Marketing Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306MM	B2B MARKETING	4	0	0	4

Course Description

Business to Business Marketing (B2B) has been one of the most debated subject in marketing. It involves all kinds of industries either manufacturing or service. It covers small, medium and large industries that creates huge employment opportunities. Continuous innovation has become the norm in business organizations since they are always looking to optimize their products and processes to offer better goods and services to other businesses. Its growth has been far faster than others' due to expanding technology, improved transportation options, and cutting-edge research. In addition to improving the learner's understanding of creating effective business plans and initiatives, this course attempts to establish a solid theoretical foundation. In order to give students an actual learning experience, this course will incorporate case studies from all over the world.

Course Objectives

1	To apply the knowledge on B2B (Industrial) marketing, consumer market and business environment at large.
2	To identify the models of B2B buying process for sustainable development of society.
3	To analyse the importance of personal selling, relationship marketing and key account management in B2B marketing perspectives.
4	To evaluate the sales plan and design a constructive idea on demand & forecast.
5	To develop the product, pricing, distribution and promotional strategies for a sustainable B2B market.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Integrate the theoretical perspectives of business markets into consumer centric markets.
CO2	Appraise the organizational buying models for sustainable business marketing.
CO3	Experiment the knowledge on relationship strategies through market segmentation to predict the futuristic demand.
CO4	Formulate the product and pricing techniques for a practicable business to business market condition.
CO5	Develop a structured distribution management and formulating the suitable selling function to administer the business market.
CO6	Discuss the relevant case on B2B market, buying behaviour, relationship marketing, market forecasting, product & pricing techniques and distribution management.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	2	1	1	1	1
CO2	3	2	0	1	1	0	0	1
CO3	3	3	1	0	1	1	1	1

CO4	2	3	0	0	1	1	1	0
CO5	2	3	2	0	1	1	0	0
CO6	3	3	3	3	3	3	2	3
AVG	2.67	2.67	1.17	1.00	1.33	1.17	0.83	1.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Business Marketing

B2B perspective, Classification of business products- Foundation, entering & facilitating, Type of business customers, Commercial, Government & institutional, Differences between consumer and business marketing.

Unit-II: Buying Process

Buying Process-I: Robinson, Faris & Wind Models, Buy- phase model, Buy Class model, buy grid model, Marketing implications of buy phase & buy class models.

Buying Process-II: Webster & Wind model, Buying Centre roles, how buying centre composition changes with buy phase & buying situation, Decision making in Buying Centre- Role of Participants preference & influence, Marketing implication of buying Centre.

Buying Process-III: Selecting suppliers, Price based approach, Cost based approach, Ownership cost based approach, Ownership cost based approach, Changing suppliers, Weighted criteria Vendor rating system, Kraljic Model.

Unit-III: B2B Relationship & Selling Process, Segmentation, Targeting, Demand forecasting & sales planning

Types of relationship & relationship spectrum, understanding customer profitability using whale curve & Net margin/cost to serve matrix, Strategy guidelines for creating collaborative relationships, Enterprise to enterprise bonding, going up business share ladder, Operational & Design Partnering Strategies Switching costs.

Selling Process: Contact strategy, Communication strategy, Presentation strategy-FABV & STD, Buyer Receptivity analysis, Salesman- purchase manager drivers of relationship marketing effectiveness -Relationship breadth, Relationship composition, Relationship Strength, Relationship quality

Segmentation, Targeting, Demand forecasting & sales planning: Macro-segmentation methods, Micro-segmentation methods, Criteria for targeting segments, Company criteria-objective, capability, Segment criteria- segment volume, segment profitability, & segment competition

Demand forecasting & sales planning: Demand Forecasting-Sales force opinion/composite method, Demand Forecasting-Survey of buyer intention method, Demand Forecasting-Growth rate method, Demand Forecasting-End use method, Sales planning using market share method, Customer wise sales planning using business share method, Marketing risk concept

Unit-IV: Product, Solution Selling, Price & Technology adoption life cycle curve

Product: 10 dimensions of product differentiation, Examples of differentiating materials, parts & machines using the following dimensions- Performance & conformance quality, Design, Reliability, Features, Durability.

Solution Selling: Product perspective vs. solution perspective, Examples of solutions with different types of services, Value pricing of solutions

Price: Pricing methods, Mark- up pricing, Target returns pricing, Competitor based pricing Perceived Value (benefit) based pricing, Pricing against imports- Calculating landed cost of imports, Contracts- Spot contract & Long term contract, Discounts & Incentives- (i) Quantity discount, commitment discount & cash discount (ii) Incentives- delivery & service related incentives

Technology adoption life cycle curve: TALC Concept, Strategy guidelines over TALC.

Unit-V: Place, Organizing the personal selling function & Case

Direct & indirect channels, Responsibilities of industrial distributors & MRs, Situations in which to use industrial distributors & MRs, Example of B2B Channel design using distribution cost method.

Organizing the personal selling function: Sales Organization structure- geographic, product & market centered, Determination of size of sales force, Key account management, Sales resource opportunity grid.

Textbook References:

1. Michael D. Hutt, Thomas W. Speh, Business Marketing Management: B2B, Cengage Learning.
2. Ross Brennan, Louise E Canning, Raymond McDowell, Business-to-Business Marketing, Sage Publications India Private Limited.
3. Robert Vitale, Waldemar Pfoertsch, Joseph Giglierano, Business to Business Marketing: Analysis and Practice, Pearson.
4. Michael Miller, B2B Digital Marketing, Pearson Education India.
5. Nick Ellis & Soumya Sarkar, Business-to-Business Marketing, Oxford University Press, India.
6. Michael D. Hutt, Dr. Dheeraj Sharma, Thomas W. Speh, K. Douglas Hoffman, B2B Marketing: A South-Asian Perspective, Cengage Learning.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Finance Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302FIN	STRATEGIC CORPORATE FINANCE	4	0	0	4

Course Description

This course explores the advanced financial strategies and decision-making processes that drive long-term corporate success and value creation. It covers a broad range of topics including capital structure optimization, financial planning, mergers and acquisitions, and risk management. Students will examine strategic financial models and frameworks used to align financial decisions with corporate strategy, assess investment opportunities, and manage financial risks. Through practical applications, students will develop a deep understanding of how financial strategies influence overall business performance and sustainability. This course equips students with the skills necessary to make informed, strategic financial decisions that support organisational objectives and enhance competitive advantage.

Course Objectives

1	To align financial decisions with strategic business goals.
2	To understand the strategic aspects of mergers, acquisitions, and management buy-ins/outs
3	To know analysis of financial statements to support long-term investment decisions.
4	To understand various sources for effective fundraising and due diligence processes.
5	To know various investment opportunities for value creation and strategic fit.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Evaluate competing projects for optimum resource allocation.
CO2	Optimize capital structure, including management buy-outs and buy-ins.
CO3	Communicate and implement financial strategies to support long-term growth.
CO4	Apply valuation techniques to assess company worth and investment potential.
CO5	Facilitate the understanding on the impact of risk and cost of capital on investment appraisal besides their cumulative impact on the value of a capital project.
CO6	Analyse the factors associated with corporate failure in order to predict its likelihood and to provide advice on actions that will prevent its occurrence.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	1	3	3	3	3	3
CO2	3	3	2	3	3	3	3	3
CO3	3	3	2	3	3	3	3	3
CO4	3	3	2	3	3	3	3	3
CO5	3	3	1	3	3	3	3	3
CO6	3	3	1	3	3	3	3	3
AVG	3.00	3.00	1.50	3.00	3.00	3.00	3.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit I: Introduction to Strategic Corporate Finance

Introduction: Strategy Vs Planning, significance of strategy in financial decisions, Different types of financial strategy for Shareholders Wealth Maximisation, overall corporate value addition and Economic Value Addition.

Strategic Cost Management: Traditional costing Vs Strategic Costing, Relevant costs Vs Irrelevant costs, Different types of strategic costing and their relevance- Target Costing, Activity based Costing, Life Cycle Costing, Quality Costing, Zero Based Budgeting, Strategic cost reduction techniques and value chain analysis.

Unit II: Management Buy-outs and Buy-ins

Management Buy-outs: Establishing feasibility of the buy-out, Negotiating the main terms of the transaction with the vendor including price and structure, Developing the business plan and financial forecasts in conjunction with the buy-out team for submission to potential funders, negotiations with potential funders so that the most appropriate funding offers are selected.

Management Buy-ins: Management Buy-in/Buy-outs (“BIMBOs”), Vendor initiated buyouts/buy-ins.

Unit III: Long- term Investment Decisions

Types of long term investment decisions - Phases in long term investment decisions- Different analyses for evaluating an investment proposal - Estimation of project cash flow- Initial cash flow, Sunk cost, Terminal cash flow (problems on estimation of cash flow) - Relevant cost analysis (concept).

Unit IV: Fundraising and Due Diligence

Fundraising: Identification of different sources of development capital, determination of capital structure and factors affecting the capital structure, cost of capital and cost saving strategy, production of a business plan and financial forecasts to enable potential funders to assess the proposition.

Due Diligence: Financial due diligence for both purchasers and financial institutions, good quality “added value” due diligence advice.

Unit V: Valuation of Business Enterprise

Reasons for valuation of business enterprise - Different Approaches to corporate valuation - Discounted Cash Flow valuation - FCFF, NOPLAT, FCFE, EVA - problems on DCF method of valuation- Traditional techniques of evaluating value - ROI, EBIT, EBIDTA, ROCE, RONA - New approaches to shareholder valuation

Textbook References:

7. AswathDamodaran: Corporate finance theory and practice; John Willey Sons, Inc.
8. Jakhotia: Strategic Financial Management (Vikas Publication).
9. Rustagi., R P., Financial Management-Theory, Concepts and Problem, Taxman Publications.
10. Pandey.I.M, Financial Management, Vikas Publications, New Delhi.
11. Jakothiya, Girish, Strategic Financial Management, Vikas Publishing House.
12. Prasanna C, Corporate Valuation and Value creation, TATA McGraw Hill.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Finance Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-302FIN	SECURITY ANALYSIS AND PORTFOLIO MANAGEMENT	4	0	0	4

Course Description

This course provides an in-depth examination of techniques and strategies for evaluating financial securities and managing investment portfolios. Students will explore the methodologies for analysing stocks, bonds, and other financial instruments, including fundamental and technical analysis. The course covers the principles of constructing and managing diversified investment portfolios, focusing on risk assessment, asset allocation, and performance evaluation. Through practical applications, students will learn to apply quantitative and qualitative tools to make informed investment decisions and optimize portfolio returns. Emphasizing both theoretical knowledge and real-world practices, this course prepares students to effectively navigate the complexities of financial markets and achieve strategic investment goals.

Course Objectives

1	To dwells on the investment avenues, and operations of the Indian stock market and the role of various financial institutions in aiding the capital formation.
2	To enable students to discuss the fundamental analysis required to be done for building up an ideal portfolio.
3	To enable students to be acquainted with the various techniques used for doing technical analysis for the purpose of portfolio management.
4	To enable students to discuss the concepts of portfolio revision and portfolio evaluation.
5	To measure the bond and mutual fund performances.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Recall and analyse various portfolio management models.
CO2	Do fundamental analysis in developing and managing a portfolio.
CO3	Acquaint with various technical analysis tools like Charts, Patterns and other mathematical and market indicators.
CO4	Analyse portfolio evaluation and portfolio revision.
CO5	Design and manage the bond as well as equity portfolios in the real world.
CO6	Apply performance measures for evaluating mutual funds.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	1	3	3	3
CO2	3	3	0	3	2	3	3	3
CO3	3	3	0	3	2	3	3	3
CO4	3	3	0	3	1	3	3	3
CO5	3	3	0	3	2	3	3	3
CO6	3	3	0	3	2	3	3	3
AVG	3.00	3.00	0.00	3.00	1.67	3.00	3.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Investment Environment in India and Equity Valuation

Introduction: Overview of Indian Financial System - Securities Trading in Stock Markets - The Investment Management Process.

Investment Alternatives: Negotiable Securities & Non-negotiable Securities. Primary Market: Types of New Issues- Parties to New Issue & Secondary Market.

Equity Analysis & Valuation: Balance Sheet Analysis - Equity Valuation Models - Intrinsic Value & Market Price - The P/E Ratio & Earnings Multiplier Approach.

Unit-II: Security Analysis

Fundamental Analysis: Economic Analysis, Industry Analysis & Company Analysis- Earnings of the Company, Financial Analysis, Growth in Earnings.

Technical Analysis: Dow Theory, Efficient Market Hypothesis, Random Walk Theory, Support and Resistance Levels, Odd Lot Trading Moving Averages.

Data Sources and Contemporary Issues: Fundamental Data Sources - Technical Data Sources.

Unit-III: Portfolio Analysis

Models and Theories: The Returns and Risks from Investing - Markowitz Portfolio Theory - Mean-Variance Approach.

Portfolio Selection: Efficient Portfolios - The Single Index Model - Capital Asset Pricing Model - Arbitrage Pricing Theory.

Unit-IV: Bond Analysis & Valuation & Management

Bond Analysis: Types of Bonds - Interest Rates - Term Structure of Interest Rates - Measuring Bond Yields - Yield to Maturity - Yield to Call - Holding Period Return.

Bond Pricing Theorems: Bond Duration - Active and Passive Bond Management Strategies - Bond Immunization - Bond Volatility - Bond Convexity.

Unit-V: Mutual Funds & Exchange Traded Funds

Mutual Funds: Types of Mutual Funds Schemes - Structure - NAV (Net Asset Value) - Risk and Return.

Performance Evaluation Models: Sharpe Model - Treynor Model - Jensen Model - Fama's Decomposition.

Exchange Traded Funds: Types of ETFs and Trading Process.

Textbook References:

1. William. F. Sharpe, Gordon J Alexander & J V Bailey: Fundamentals of Investments, PHI.
2. Donald E Fischer, Ronald J Jordan: Security Analysis and Portfolio Management, Pearson.
3. Prasanna Chandra, Investment analysis and Portfolio Management, TMH.
4. Punithavathi Pandian: Security Analysis and Portfolio Management, Vikas.
5. M. Ranganatham, R. Madhumathi, Security Analysis and Portfolio Management, Pearson.
6. Kevin, Security Analysis and Portfolio Management, PHI.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Finance Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304FIN	FINANCIAL INSTITUTIONS, MARKETS & SERVICES	4	0	0	4

Course Description

This course offers a comprehensive overview of the structures, functions, and dynamics of the financial sector. Students will explore the roles and operations of various financial institutions, including banks, insurance companies, and investment firms, and their impact on the economy. The course covers the functioning of financial markets, including equity, debt, and derivative markets, as well as the services they provide, such as trading, asset management, and financial advising. Emphasizing both theoretical foundations and practical insights, this course aims to provide students with a robust understanding of how financial institutions and markets interact, the regulatory environment governing them, and the ways in which they contribute to financial stability and economic growth.

Course Objectives

1	To understand the role and function of financial institutions in the economy
2	To develop skills to assess and select appropriate financial products and services for various financial needs.
3	To assess risk management and due diligence practices in venture capital investing.
4	To develop skills in interpreting and using credit ratings for investment and credit decisions.
5	To gain practical skills for managing and reviewing mutual fund investments in the context of overall financial planning.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain various functions and role of varied and multiple constituents of the Indian financial system.
CO2	Lay out a strong and firm theoretical foundation of studies in the field of banking, capital market and financial services
CO3	Examine the structure and terms of venture capital deals, including valuation methods and equity stakes.
CO4	Analyse the methodologies used by credit rating agencies to evaluate creditworthiness.
CO5	Analyse how different economic and market conditions affect mutual fund performance and strategy.
CO6	Equip with the knowledge and skills necessary to become employable in the financial service industry.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	0	3	1	2	1	2
CO2	2	1	0	3	1	2	2	2
CO3	2	1	0	3	2	2	2	2
CO4	3	2	0	3	2	2	3	3
CO5	3	2	0	3	2	2	3	3

CO6	2	1	0	3	1	2	2	3
AVG	2.33	1.33	0.00	3	1.50	2.00	2.17	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Structure of Financial System

Structure of Financial System - Role of Financial System in Economic Development - Financial Markets and Financial Instruments - Money Markets - Bond Markets - Mortgage Markets - Stock Markets - Foreign Exchange Markets - Derivative Securities Markets - Role of SEBI - Secondary Market Operations - Regulation - Functions of Stock Exchanges - Listing - Formalities - Financial Services Sector Problems and Reforms.

Unit-II: Financial Services

Concept, Nature and Scope of Financial Services - Regulatory Frame Work of Financial Services - Growth of Financial Services in India - Merchant Banking - Meaning-Types - Responsibilities of Merchant Bankers - Role of Merchant Bankers in Issue Management - Regulation of Merchant Banking in India. Wealth Management System

Unit-III: Venture Capital

Growth of Venture Capital in India - Financing Pattern under Venture Capital - Legal Aspects and Guidelines for Venture Capital, Leasing - types of Leases - Evaluation of Leasing Option vs. Borrowing. Hire Purchase vs. Leasing (NP in Leasing)

Unit-IV: Credit Rating

Credit Rating - Meaning, Functions - Debt Rating System of CRISIL, ICRA and CARE. Factoring, Forfeiting and Bill Discounting - Types of Factoring Arrangements - Factoring in the Indian Context; (NP in Factoring)

Unit-V: Mutual Funds

Concept and Objectives, Functions and Portfolio Classification, Organization and Management, Guidelines for Mutual Funds, Working of Public and Private Mutual Funds in India. Debt Securitization - Concept and Application - De-mat Services - need and Operations-role of NSDL and CSDL. NAV calculation - Sharpe, Jensen, Treynor models

Textbook References:

1. Clifford Gomez, "Financial Markets, Institutions and Financial Services", PHI learning, New Delhi.
2. Gordon E. and Natarajan K., "Financial Market and Services", Himalaya Publishing House, New Delhi.
3. Avadhani V. A., "Marketing of Financial Services", Himalaya Publishing House, Mumbai.
4. Bharti V. Pathak, "The Indian Financial System", Pearson Education, New Delhi.
5. L.M. Bhole, "Financial Institution and Markets", TMH, New Delhi.
6. Verma J.C., "Management of Financial Services", Anmol Publications, New Delhi.



SCHOOL OF MANAGEMENT STUDIES
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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Finance Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305FIN	FINANCIAL ANALYTICS	4	0	0	4

Course Description

This course equips students with advanced skills and techniques for analysing financial data to drive strategic decision-making. It covers a range of analytical methods, including statistical analysis, forecasting, and data modelling, to evaluate financial performance and assess investment opportunities. Students will learn to utilize various financial analytics tools and software to interpret complex data sets, identify trends, and generate actionable insights. Through hands-on cases, students will develop the ability to apply quantitative analysis to real-world financial problems, enhance their analytical thinking, and support data-driven decisions in financial planning and management. This course is designed to empower students with the expertise needed to excel in the evolving field of financial analytics and contribute to effective financial strategy and decision-making.

Course Objectives

1	Understand the critical role of financial analytics and contemporary workplace situation.
2	Understand the principles of analytical thinking and its application to financial problem-solving and decision-making.
3	Develop skills in cleaning and preparing datasets by handling missing values, removing duplicates, and correcting data inconsistencies.
4	Grasp the concepts and characteristics of time series data in financial markets, including trends, seasonality, and volatility.
5	Explore risk management techniques and their integration into trading strategies to protect against significant losses and manage exposure.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Calculate and interpret financial ratios and metrics to assess the performance and health of companies and investments.
CO2	Identify and analyse trends, patterns, and relationships in financial data using statistical tools and methods.
CO3	Assess financial risk and uncertainty through statistical measures such as variance, standard deviation, and Value at Risk.
CO4	Gain practical skills in analysing, valuing, and managing financial securities in real-world investment scenarios.
CO5	Develop and validate forecasting models to predict future market trends and behaviour based on historical time series data.
CO6	Create and implement various trading models using historical data to evaluate their performance and effectiveness.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	0	2	3	3
CO2	3	3	0	3	2	2	3	3
CO3	3	3	0	3	2	2	3	3

CO4	3	3	0	3	2	2	3	3
CO5	3	3	0	3	2	2	3	3
CO6	3	3	0	3	2	2	3	3
AVG	3.00	3.00	0.00	3.00	1.67	2.00	3.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Financial Analytics and Basic Financial Statements

Introduction: Meaning, Importance of Financial Analytics, Documents used in Financial Analytics. Elements of Financial Health.

Types of Financial Analysis: Cash Flow, Efficiency, Growth, Horizontal, Leverage, Liquidity, Profitability, Rates of Return, Scenario & Sensitivity, Valuation, Variance, Vertical.

Tools: Financial Statement Analysis Tools.

Unit-II: Financial Statistics and Analytical Thinking

Concept and mathematical expectation, Probability, Mean, SD and Variance, Skewness and Kurtosis, Covariance and correlation, Financial Returns, Capital Asset Pricing model. Using Excel Built in functions for financial analysis, Creating graphs and Formatting charts for financial data.

Unit-III: Financial Securities and Datasets

Bond Investments, Stock Investments, Securities Datasets and Visualization, Adjusting for Stock Splits, Adjusting for Mergers, Plotting Multiple Series, Securities Data Importing, Securities Data Cleansing, Securities Quoting

Unit-IV: Time Series Analysis and Market Sentiment

Examining Time Series, Stationary Time Series, Auto-Regressive Moving Average Processes, Markov Regime Switching Model, Reading the Market Data, Bayesian Reasoning, The Beta Distribution, Prior and Posterior Distributions, Examining Log Returns for Correlation, Momentum Graphs

Unit-V: Simulating Trading Strategies

Foreign Exchange Markets, Chart Analytics, Initialization and Finalization, Momentum Indicators, Bayesian Reasoning within Positions, Entries, Exits, Profitability, Short-Term Volatility, The State Machine

Textbook References:

1. Mark J. Bennet, Dirk L. Hugen, Financial Analytics with R, Cambridge: Cambridge University Press.
2. Michael Samonas, Financial Forecasting, Analysis and Modelling: A Framework for Long-Term Forecasting, New Delhi: Wiley Publishing.
3. Chandan Sengupta, Financial Analysis and Modeling using Excel and VBA, New Delhi: Wiley India.
4. Scott Proctor K, Building Financial Models with Microsoft Excel, New Delhi: Wiley India.
5. Pitabas Mohanty, Financial Analytics, Wiley.
6. Timothy Mayes, Financial Analysis with Microsoft Excel, Cengage Publishers.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Finance Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306FIN	BEHAVIOURAL FINANCE	4	0	0	4

Course Description

This course is intended to complement other finance courses that are mainly based on the traditional paradigm which assumes that investors and managers are generally rational. Specifically, this course has three main objectives. First, we aim to examine how the insights of behavioural finance theories shed light on the behaviour of individual investors and finance professionals in investment decision-making and corporate financial decision-making. Second, we explore the possibility to improve investment performance and corporate performance by recognising the cognitive biases and applying appropriate techniques. Finally, we investigate the implications of behavioural finance for the construction of good corporate governance mechanisms.

Course Objectives

1	To introduce the new field of behavioural finance and underline its importance as a driving force in the global markets.
2	To provide the conceptual framework of behavioural finance based on traditional and modern theories.
3	To impart the psychological aspects and challenges underlying the issue of rational and irrational behaviour.
4	To understand some psychological biases which lead to various anomalies.
5	To deepen and broaden students' conceptual understanding of financial decision-making.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Enrich the knowledge with standard finance to evaluate Individual investor psychology on investment decision making process.
CO2	Analyse existing behavioural economics research as well as to present their own research ideas, with a view to empirical application.
CO3	Apply important behavioural biases of beliefs and preferences in financial settings.
CO4	Examine market sentiment and collective behaviour impact market movements and investor actions.
CO5	Evaluate the psychological foundations and the impact of bias and heuristics on financial decision-making.
CO6	Examine liquidity levels and regulatory conditions, illustrating their influence on market efficiency and investor confidence.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	0	1	0	2	3	2
CO2	2	3	0	2	0	2	3	2
CO3	1	3	1	2	1	2	3	2
CO4	3	3	2	3	2	2	3	3

CO5	3	3	0	2	2	2	3	2
CO6	3	3	0	2	1	2	3	2
AVG	2.33	3.00	0.50	2.00	1.00	2.00	3.00	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Basics of Behavioural Finance

Introduction: Nature, scope, objectives and application; Investment Decision Cycle: Judgment under Uncertainty: Cognitive information perception - Peculiarities (biases) of quantitative and numerical information perception - Representativeness – Anchoring - Exponential discounting - Hyperbolic discounting

Unit-II: Behavioural foundation and Market Bubbles

The Behavioural Foundations: Role of behavioural antecedents Role of Emotions, Mood, Sentiments. Emotional Timeline; Emotions and Neuroscience Role of Demographics Role of Psychological Dispositions Limits to Arbitrage, Fundamental risk

Market Bubbles: Introduction and brief history of stock market bubbles, Identification and classification of a stock market bubble, Explaining bubbles through behavioural finance, Investor behaviour during bubbles, Causes of bubbles

Unit-III: Building Block of Behavioural Finance

Cognitive Psychology and limits to arbitrage - Demand by arbitrageurs: Definition of arbitrageur; Long-short trades; Risk vs. Horizon; Transaction costs and short-selling costs; Fundamental risk; Noise-trader risk; Professional arbitrage; Destabilizing informed trading (positive feedback, predation) - Expected utility as a basis for decision-making - The evolution of theories based on expected utility concept

Unit-IV: Overview of Investor Psychology

Formal overview of investor psychology, Behavioural Portfolio theory - Psychographic models - Sound Investment Philosophy different biases such as Overconfidence, Base rate neglect, Anchoring and adjustment, Cognitive Dissonance, Availability, Self-Attribution and Illusion of Control Bias.

Unit-V: External factors and Investor Behaviour

Fear & Greed in Financial Market, emotions and financial markets: geomagnetic storm, Statistical methodology for capturing the effects of external influence onto stock market returns

Textbook References:

1. Chandra, Behavioural Finance, McGraw-Hill Education.
2. Sulphrey M.M., Behavioural Finance, PHI Learning Private Limited.
3. Montier, James, Behavioural Investing, A Practitioner's Guide to Applying Behavioural Finance, Wiley.
4. Baker, K. and Nofsinger, J., Behavioral Finance: Investors, Corporations, and Markets, John Wiley.
5. Shleifer, Andrei, Inefficient Markets: An Introduction to Behavioural Finance, Oxford University Press, Oxford.
6. Ackert, The Psychology of Investing, Pearson Prentice Hall.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(HR Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302HR	STRATEGIC HRM	4	0	0	4

Course Description

Though the concepts of Strategic Human Resource Management students will be able to understand the link between human resources department and strategic goals and objectives of the organization to improve the organization's performance and develop the organization's culture to ensure that there is innovation and flexibility. Strategic Human Resource Management deals with taking care of the organization's human resource and managing the human resource appropriately to help in the generation of the organization's profit and drive the organization towards achieving its goals.

Course Objectives

1	Demonstrate an understanding of strategy and its role in HRM
2	Understand the basic concepts of SHRM
3	Learn various models of SHRM to full fill the business needs
4	Learn the competencies and challenges of HR professionals
5	To understand the formulation of strategies and programs to introduce and sustain competitive HR advantage in organizations.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Identify the relationship of HR strategy with other functional and overall corporate strategy
CO2	Classify HR strategy, apply different models to business needs
CO3	Recognize the application of different models to develop strategies
CO4	Evaluate competencies of HR Professional in a SHRM Scenario
CO5	Implement the Human Resource Evaluation process and methods
CO6	Recognize the development and formulation of strategies and programs to sustain competitive HR advantage in organizations.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	1	1	2	2	1	0	1	1
CO2	3	2	1	3	3	1	1	2
CO3	3	2	2	3	2	1	1	1
CO4	3	3	3	3	3	2	3	2
CO5	3	3	1	1	2	1	2	2
CO6	1	2	2	3	2	2	2	2
AVG	2.33	2.17	1.83	2.50	2.17	1.17	1.67	1.67

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to SHRM

Introduction: Definition of SHRM, Need and Importance of SHRM; Human resources as a source of competitive advantage; Traditional HR Vs. Strategic HR, Barriers to strategic HR, Types of Strategies-Corporate Strategy, Business strategy and HR Strategy.

Unit-II: Integration of HR Strategy

Integration of HR Strategy: Integration of HR Strategy with corporate and business strategies; Different approaches to integration; the 5-P model for linking people with strategic needs of business; Role of HRM in strategy formulation and implementation; HR as a Strategic partner.

Unit-III: HR Environment and Strategic HR Processes

HR Environment and Strategic HR Processes: Impact of Technology, changing nature of work, demographic changes, workforce diversity etc., on HR; HR Planning, Strategic issues in staffing; Performance Management: Meaning and need for performance management, performance appraisal systems and their limitations.

Unit-IV: Training and Development Strategies

Training and Development Strategies: Cross - cultural training, Multi-skilling, Succession Planning; Creating a learning organization; Strategically oriented compensation and reward systems: Skill - based pay broad banding, variable pay, profit sharing, employee stock option plans, executive compensation.

Unit-V: Human Resource Evaluation

Restructuring and SHRM, Corporate Ethics, Values and SHRM, Diversity & SHRM, Competencies of HR Professional in a SHRM Scenario, Human resource Evaluation; Measures of Evaluation; HR Audit.

Textbook References:

1. Jaffrey A Mello, Strategic HRM, Cengage Learning.
2. Das Pulak, Strategic Human Resource management, Cengage Learning.
3. Charles R. Greer, Strategic Human resource management, Pearson.
4. Michael Armstrong, Strategic HRM, Kogan page, London.
5. John Storey, Patrick M. Wright and Dave Ulrich, The Routledge Companion to Strategic Human Resource Management, Routledge.
6. Jeffery Mello, Strategic HRM, Thompson publication, New Delhi.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(HR Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-303HR	INDUSTRIAL & LABOUR LAWS	4	0	0	4

Course Description

This course provides an in-depth exploration of the legal frameworks governing workplace relations, employment practices, and industrial activities. It covers key areas such as employment contracts, labour rights, dispute resolution, and regulatory compliance, focusing on both national and international perspectives. Students will study legislation related to wage standards, working conditions, occupational health and safety, and collective bargaining. Through case studies, practical scenarios, and legal analysis, students will gain a thorough understanding of the laws that impact labour management, employee relations, and organizational responsibilities. This course aims to equip students with the knowledge to navigate complex legal environments, ensure compliance, and effectively address labour-related issues in diverse industrial settings.

Course Objectives

1	To understand the basic concepts of various Employment legislations
2	To acquire knowledge different causes & impact of industrial disputes
3	To understand various machineries for dispute settlement
4	To acknowledge important provisions for labour legislations
5	To gain idea on Social security legislations.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Elaborate the concept of Industrial Relations.
CO2	Outline the important causes & impact of industrial disputes.
CO3	Elaborate Industrial Dispute settlement procedures.
CO4	Summarize the important provisions of Wage Legislations, in reference to Payment of Wages Act 1936, Minimum Wages Act 1948 & Payment of Bonus Act 1965.
CO5	Recognize Protective Legislations.
CO6	Summarize the important provisions of Social Security Legislations.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	2	2	2	2	2	1	1
CO2	2	2	2	2	2	2	2	1
CO3	3	3	3	2	3	2	3	1
CO4	3	2	1	2	2	0	2	1
CO5	3	2	1	2	2	0	2	1
CO6	3	2	1	2	2	0	2	1
AVG	2.67	2.17	1.67	2.00	2.17	1.00	2.00	1.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Labour Legislation Introduction

Concept of Industrial Relations - Factors Affecting Industrial Relations, Importance of Industrial Relations, Collective Bargaining, International Labour Organization: Dimensions, Aims and Objectives, Organs of the International Labour Organization; Role of the Trade Unions in Modern Industrial Society of India - Types of Trade Union in India.

Unit-II: Factories and Contract

Industrial Disputes Act 1947- Objective & Scope, Definitions & Provisions related to Layoff, strike & Lock out, Retrenchment & closures. Strike and Lockout - Legal and Illegal Strikes and Lockouts, Justified and Unjustified Strikes and Lockouts, Distinction between Lockout and Closure, Difference between Lay-Off and Retrenchment, Special Provisions Relating to Lay-Off & Retrenchment

Unit-III: Settlement Mechanism

Industrial Disputes for Settlement - Settlement Machinery for Industrial Disputes, Conciliation Officer, Board of Conciliation, Court of Enquiry, Labour Court, Industrial Tribunal and National Tribunal and Arbitration, their Composition, Powers, and Duties, Different Modes of Settlement, Distinction between Conciliation, Arbitration and Adjudication, their Merits and Demerits

Unit-IV: Compensation and Wages

Factories Act, 1948, The payment of bonus act, 1965, The Sexual Harassment of Women at Workplace, Prevention, Prohibition and Redressal Act-2013, Payment of Wages Act-1936, Minimum Wages Act-1948, Payment of Bonus Act-1965

Unit-V: Miscellaneous Provisions Act

The Workmen's Compensation act-1923, The Employee's State Insurance act-1948, The Employees' Provident Funds and Miscellaneous Provisions Act-1952, The Payment of Gratuity Act-1972

Textbook References:

1. P. SubbaRao, Essentials of Human Resource Management and Industrial Relations, Himalaya Publishers.
2. VenkataRatnam C. S., Industrial Relations, Oxford University Press, New Delhi.
3. T.N.Bhagoliwal Personnel Management and Industrial Relations, Tata McGraw Hill.
4. Singh, B. D, Industrial Relations - Emerging Paradigms, Excel, New Delhi.
5. Memoria & Memoria, Dynamics of Industrial Relations, Himalaya Publishing House, New Delhi.
6. Sinha Shekhar, Industrial Relations, Trade Union & Labour Legislations, Pearson.



SCHOOL OF MANAGEMENT STUDIES
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Odisha, Gunupur
MBA III SEMESTER (AR23)
(HR Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304HR	COMPENSATION & BENEFITS MANAGEMENT	4	0	0	4

Course Description

This course delves into the strategic design and administration of employee compensation and benefits programs within organizations. It explores various compensation structures, including base salary, bonuses, incentives, and equity compensation, and examines benefits such as health insurance, retirement plans, and employee perks. Students will learn to develop competitive compensation strategies aligned with organizational goals, ensure compliance with legal and regulatory requirements, and effectively manage compensation budgets. The course emphasizes both the theoretical aspects of compensation management and practical applications through case studies and industry practices. By the end of the course, students will be equipped with the skills to create equitable and motivating compensation packages that attract, retain, and reward talent while supporting organisational objectives.

Course Objectives

1	To understand the meaning, objectives, and types of compensation, including wages, salaries, and C.T.C.
2	To learn about wage structure components, such as basic pay, dearness allowance, V.D.A., and fringe benefits, and methods for calculating V.D.A.
3	To examine executive remuneration, criteria for fixing executive pay, and new developments in CEO compensation.
4	To study methods of wage fixation, including collective bargaining, wage boards, pay commissions, adjudication, and remuneration preparation for consultants and outsourcing agencies.
5	To assess the role of the HR department in compensation administration, the impact of globalization on compensation, and challenges in managing compensation in multinational corporations, including recommendations from the National Commission on Labour-II.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Able to understand the various wage theories and the concepts related to pay.
CO2	To Analyze and differentiate between the various components of wages and types of wage structure.
CO3	Able to examine the different methods of wage fixation.
CO4	Able to Calculate the DA and other components of wages and salaries and prepare pay roll.
CO5	Able to analyze the wage determination process.
CO6	Able to Present the challenges of compensation management in MNCs.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	3	3	3
CO2	2	3	2	2	2	2	3	2
CO3	2	2	2	3	3	2	2	2

CO4	3	3	3	2	3	3	3	3
CO5	2	2	2	3	2	3	2	3
CO6	2	2	2	2	3	2	3	2
AVG	2.33	2.50	2.33	2.50	2.67	2.50	2.67	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Compensation Management

Meaning and objectives, types of compensation; Scope; Significance; Concepts of Wage- Salary, Earnings, Take-Home pay, C.T.C; Types of Wages- Minimum wages, Fair wages, Living wages, Nominal wages, Real wages, Time Rate and Piece Rate; Wage theories.

Unit-II: Wage Structure

Components of wage structure- Basic; Dearness Allowance; Variable Dearness Allowance (V.D.A.) and Calculation of V.D.A.; Incremental system; Fringe benefits.

Unit-III: Executive Remuneration

Concept; Criteria for fixing executive remuneration Unique features of Executive Remuneration; Why managers should be paid more; CEO's compensation package; New developments in CEO's compensation.

Unit-IV: Wage Fixation

Wage curves, pay grades, broad banding; Wage differentials; Factors influencing wage determination; Wage determination process; Methods of Wage Fixation - Collective Bargaining; Wage Boards; Pay Commissions; Adjudication; Principles & Procedures for Preparation of Remuneration for Consultants and Other Outsourcing Agencies.

Unit-V: Role of HR Department in Compensation Administration

Important recommendations of National Commission on Labour – II; Impact of globalization on compensation. Challenges of compensation management in MNC.

Textbook References:

1. Sarma, A.M and Rao, N.S. Compensation System and Performance Management (First Edition). Himalaya Publishing House Pvt. Ltd.
2. Bhattacharyya, Compensation Management. Oxford University Press Higher Education Division.
3. Singh, B. D. Compensation and Reward Management, Excel Books India.
4. Milkovich, G., Newman, J and Venkataratnam, C. S., Compensation, McGraw Hill Publications
5. Tapomoy Deb, Compensation Management, Text and Cases, Excel Books,
6. Richard.I. Henderson, Compensation Management in a Knowledge Based World, Prentice Hall India.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(HR Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305HR	HR ANALYTICS	4	0	0	4

Course Description

This course leveraging data-driven insights to enhance human resource management and decision-making. Students will explore the use of analytics to address key HR challenges such as talent acquisition, employee performance, retention, and engagement. The course covers various analytical techniques and tools, including data mining, predictive modelling, and workforce metrics, to help HR professionals make informed decisions and drive strategic HR initiatives. Through practical exercises and case studies, students will learn to collect, analyse, and interpret HR data to optimize workforce planning, improve employee experiences, and align HR practices with organizational goals. This course aims to equip students with the skills to effectively utilize HR analytics to create a more data-informed and strategic HR function.

Course Objectives

1	To enable the learner towards critical thinking about data for HR decisions
2	To impart knowledge of conduction of HR Analytics for key HR Processes using MS Excel.
3	To elucidate the methods of capturing, examining & purifying data and to introduce the aspect of HR Metrics in the context of HR Analytics
4	To understand the transformational HR operations with strategic business concepts
5	To understand the data Visualization for Key HR processes.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Gain a clarification of the relevance of HR Analytics in the current business scenario.
CO2	Identify the models of conducting HR Analytics and able to understand the methods of capturing, examining & purifying data for conduction of HR Analytics.
CO3	Analyse a considerate of transformational HR operations in interactions with other strategic business concepts.
CO4	Use MS Excel for conduction of HR Analytics for key HR Processes.
CO5	Recognize of various tools and software technologies used for conduction of Descriptive HR Analytics and Visualization of HR Data.
CO6	Make out how HR results connect to business results.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	2	3	2	0	3	3
CO2	3	3	2	1	2	1	3	3
CO3	2	3	3	2	3	2	3	3
CO4	2	3	1	1	1	1	3	3
CO5	2	3	1	2	1	0	3	3
CO6	3	2	1	1	1	1	3	3
AVG	2.50	2.83	1.67	1.67	1.67	0.83	3.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Framework of HR Measurement

Meaning of HR Analytics; Pitfalls of HR Analytics, Evolution of HR Analytics; HR Metrics and HR Analytics; Analytical Pyramid- Descriptive and Predictive models; Intuition versus analytical thinking; Ethical issues in Analytics; HRMS/HRIS and data sources; HR measurement: Traditional vs. contemporary HR measures; Fundamental analytical concepts from statistics and research design; analytical concepts from economics and finance.

Unit-II: Using HR Metrics for Maximum Impact

Analytics frameworks like LAMP, HCM: 21 Model How to Conduct HR Analytics. Understanding HR Data: Importance of Data, Types and Scales of Data; Methods of Capturing Data, Data Examination & Purification.

Unit-III: Business understanding and Forecasting for HR

HR Metrics, Types of HR Metrics, Applications of HR Metrics and Creating HR Dashboards, Staffing Metrics; Performance and compensation metrics; Learning and developmental metrics. Developing Human Resources Balanced Score Card, Important Excel Formulas Useful for Creating Dashboards, VLOOKUP, HLOOKUP, INDEX, SUMIF, AVERAGEIF and COUNTIF

Unit-IV: Communication HR Data and Processing

Communicating HR data and processing: Data requirements; identifying data needs and gathering data; HR data quality, validity, and consistency; Using historical data; Data exploration; Data visualization; Association between variables; Insights from reports; Root cause analysis of HR issues. (Workshop Mode)

Unit-V: Modelling in HR

Descriptive Analytics in HR: HR Dashboards using MS Excel, Slicing and Dicing of HR Data using MS Excel Pivot Table Applications, Pivot Chart, Data Visualization for Key HR processes.

Textbook References:

1. Rama Shankar Yadav & Sunil Maheshwari, HR Analytics, Wiley.
2. Pratyush Banerjee, Jatin Pandey & Manish Gupta, HR Analytics: Practical Applications of HR Analytics, Sage.
3. Dipak Kumar Bhattacharya, HR Analytics, Sage.
4. Nishant Uppal, Human Resource Analytics, Pearson.
5. Bharti Motwani, HR Analytics: Practical Approach Using Python, Wiley.
6. Ramesh Soundrarajan & Kuldeep Singh, Winning on HR Analytics, Sage.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(HR Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306HR	TALENT & KNOWLEDGE MANAGEMENT	4	0	0	4

Course Description

This course provides a comprehensive exploration of strategies and practices for effectively managing and leveraging organizational talent and intellectual capital. Students will examine key concepts in talent management, including recruitment, development, performance management, and succession planning, as well as strategies for cultivating a high-performance workforce. The course also covers knowledge management techniques, focusing on how to capture, share, and utilize organizational knowledge to enhance innovation and efficiency. Students will develop skills in creating talent development programs, fostering a knowledge-sharing culture, and aligning talent and knowledge management practices with overall business objectives. This course aims to equip students with the tools and insights needed to optimize human capital and drive organizational success.

Course Objectives

1	Understand the meaning, importance, and evolution of talent management, including systems, reservoirs, and competency models.
2	Explore succession and career planning, focusing on both traditional and contemporary approaches, as well as talent acquisition and various coaching methods.
3	Gain insight into knowledge management concepts, driving forces, systems, strategies, and the factors influencing it.
4	Learn about knowledge management infrastructure, encompassing frameworks, the role of knowledge workers, learning organizations, and the challenges in implementation.
5	Assess the impact of knowledge management on organizational effectiveness, examining its effects on people, processes, products, and overall performance.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Work with Talent Management Assessment Tools.
CO2	Aware of Alignment of Recruiting and Talent Management Strategies.
CO3	Identify the Factors influencing Knowledge Management and forces driving knowledge management.
CO4	Recognize the challenges .in the implementation of Knowledge management.
CO5	Analyse the managing mechanism for organizational effectiveness.
CO6	Explore the dependency of Knowledge Management upon Organisational Performance.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	2	2	3	2	3	2
CO2	2	2	2	2	2	3	2	2
CO3	3	2	2	2	2	2	3	3
CO4	3	3	3	3	3	3	3	2
CO5	2	3	2	3	2	2	2	3
CO6	3	2	3	2	3	2	3	3

AVG	2.50	2.50	2.33	2.33	2.50	2.33	2.67	2.50
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Talent Management								
Talent Management: Meaning, importance, Evolution, Talent Management System, Talent Reservoir – Components, Talent Management Grid, Talent Management Assessment Tools, Process of Talent Management, Competence – Competency Assessment, Competency Modeling – Steps in developing a valid competency model.								
Unit-II: Succession and Career Planning								
Succession and Career Planning: Succession Planning – Traditional Approaches, Contemporary Approaches, Talent Acquisition - Talent Identification & Business Alignment, Recruiting & Assessment, Talent Development – Training & Coaching – Action-Oriented Coaching, Remedial Coaching, Attitude Based Coaching; Performance Based Coaching, Executive Coaching, Talent Management Strategies								
Unit-III: Knowledge Management								
Knowledge Management: Concept, Forces Driving Knowledge Management, Knowledge Management Systems, Issues in Knowledge Management, Knowledge Management Strategies, Technologies for Knowledge Management, Factors influencing Knowledge Management.								
Unit-IV: Knowledge Management Infrastructure								
Knowledge Management Infrastructure: Mechanism and system- Knowledge Management Framework and Approaches; Concept of Knowledge Worker; Knowledge management and Learning Organization, Challenges in implementation of knowledge management.								
Unit-V: Managing Knowledge for organizational effectiveness								
Managing Knowledge for organizational effectiveness: Knowledge Management Assessment of Organization- Importance- Types of Timing- Organization Impacts of Knowledge Management on People, Processes, Products & Organization performance.								
Textbook References:								
1. Lance A Berger, Dorothy R Berger, The Talent Management Handbook, McGraw Hill Education. 2. Irma Becerra-Fernandez, Avelino Gonzalez, Rajiv Sabherwal, Knowledge Management: Challenges, Solutions, and Technologies, Pearson. 3. Sudhir Warier, Knowledge Management, Vikas Publications. 4. Ravinder Shukla, Talent Management: Process of Developing and Integrating Skilled Workers, Global India Publications. 5. Marshall Goldsmith, Louis Carter, Best Practices in Talent Management: How the World's Leading Corporations, Wiley. 6. Hugh Scullion, David G. Collings, Global Talent Management, Routledge.								



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Rural Management Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302RM	STRATEGIES FOR SUSTAINABLE RURAL DEVELOPMENT	4	0	0	4

Course Description

This course explores the strategic frameworks and methodologies essential for promoting sustainable rural development. It emphasizes the integration of economic, social, and environmental aspects to enhance the quality of life in rural areas. Through a combination of theoretical insights and practical case studies, students will gain a comprehensive understanding of sustainable practices and policies tailored for rural development.

Course Objectives

1	Explore the fundamental concepts and significance of sustainable rural development.
2	Examine the socio-economic and environmental challenges faced by rural areas
3	Investigate various strategies and policies aimed at achieving sustainable rural development.
4	Assess the roles and impacts of different stakeholders in fostering rural sustainability.
5	Develop practical skills necessary for implementing and managing sustainable development projects in rural regions.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate understanding of sustainable rural development concepts and principles.
CO2	Analyse socio-economic challenges and propose viable solutions for rural areas.
CO3	Evaluate environmental impacts and implement sustainable resource management practices.
CO4	Critically assess policies and institutional frameworks supporting rural development.
CO5	Design and manage rural development projects using innovative strategies.
CO6	Apply theoretical knowledge to real-world rural development challenges through case studies.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	1	2	3	3	3	2	3
CO2	3	3	2	2	3	3	3	0
CO3	3	3	3	0	2	3	1	3
CO4	3	0	3	2	3	0	3	3
CO5	3	1	2	3	2	3	1	0
CO6	0	3	1	3	2	0	3	3
AVG	2.50	1.83	2.17	2.17	2.50	2.00	2.17	2.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Sustainable Rural Development

Definition and Importance, Historical Perspective and Evolution, Key Principles of Sustainability, Global and Local Contexts, Case Studies from Different Regions, Sustainable Development Goals

(SDGs), Rural-Urban Linkages, Community Development, Participatory Development, Integrated Rural Development, Role of Culture in Rural Development

Unit-II: Socio-Economic Dimensions of Rural Development

Rural Demographics and Socio-Economic Profiles, Poverty and Inequality in Rural Areas, Rural Employment and Livelihoods, Education and Health Services in Rural Contexts, Gender Issues in Rural Development, Migration and Urbanization, Social Capital, Rural Infrastructure, Financial Inclusion, Cooperatives and Self-Help Groups (SHGs), Rural Markets and Value Chains, Human Development Index (HDI) in Rural Areas

Unit-III: Environmental and Resource Management

Natural Resources: Land, Water, and Forests, Sustainable Agriculture and Farming Practices, Biodiversity Conservation, Climate Change and Rural Areas, Renewable Energy in Rural Development, Soil and Water Conservation, Agroforestry, Waste Management, Sustainable Fisheries, Environmental Education and Awareness, Impact Assessment, Ecosystem Services

Unit-IV: Policy and Institutional Frameworks

Government Policies and Programs, Role of NGOs and Community-Based Organizations, International Organizations and Rural Development, Rural Development Financing, Monitoring and Evaluation of Rural Development Projects, Decentralization and Governance, Public-Private Partnerships, Policy Analysis and Advocacy, Legal and Regulatory Frameworks, Rural Development Agencies, Capacity Building and Training, Disaster Management and Rural Resilience

Unit-V: Innovative Strategies

ICT for Rural Development, Rural Entrepreneurship and Microfinance, Sustainable Tourism in Rural Areas, Successful Rural Development Projects: Global and Indian Examples, Future Directions and Innovations in Rural Development, Smart Villages, Social Innovation, Technology Transfer and Adoption, Precision Agriculture, Inclusive Development, Green Technologies, Best Practices and Lessons Learned, Participatory Rural Appraisal (PRA), Market Linkages and Access

Textbook References:

1. Katar Singh, Rural Development: Principles, Policies and Management, SAGE Publications.
2. S. Giriappa, Integrated Rural Development and Management: Strategies and Case Studies, Concept Publishing Company.
3. R.B. Singh, Sustainable Rural Livelihoods: A Framework for Analysis, Springer.
4. A.K. Shrivastava, Sustainable Rural Development, CRC Press.
5. George W. Norton and Jeffrey, The New Rural Economy: Change, Dynamism and Government Policy, Routledge.
6. Neil M. Coe and Kevin Ward, Rural Development: Contemporary Issues and Practices, Routledge.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Rural Management Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-303RM	INDUSTRY AND AGRI-BUSINESS	4	0	0	4

Course Description

This course provides a comprehensive understanding of the interplay between industry and agri-business, highlighting the importance of sustainable practices in both sectors. It covers the economic, social, and environmental aspects of industrial and agricultural activities, focusing on strategies to enhance productivity, profitability, and sustainability. Students will explore the latest trends, technologies, and policies impacting the industry and agri-business sectors.

Course Objectives

1	Investigate the key concepts and principles underlying industry and agri-business.
2	Analyse the economic, social, and environmental challenges faced by these sectors.
3	Examine various strategies and technologies to improve efficiency and sustainability in industry and agri-business.
4	Assess the roles and impacts of different stakeholders in promoting sustainable practices.
5	Develop practical skills for managing and leading industry and agri-business enterprises.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate understanding of industry and agri-business concepts and principles.
CO2	Analyse economic challenges and propose viable solutions for industry and agri-business sectors.
CO3	Evaluate technological advancements and implement sustainable practices.
CO4	Critically assess policies and regulatory frameworks impacting industry and agri-business.
CO5	Design and manage projects using innovative strategies in industry and agri-business.
CO6	Apply theoretical knowledge to real-world industry and agri-business challenges through case studies.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3	3	2
CO2	2	3	3	2	3	3	3	2
CO3	3	2	2	2	1	2	3	3
CO4	3	3	1	0	1	1	3	2
CO5	3	2	3	2	2	3	2	1
CO6	3	2	2	3	2	3	3	3
AVG	2.83	2.33	2.33	2.00	2.00	2.50	2.83	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Industry and Agri-Business

Overview of Industry and Agri-Business, Importance and Scope, Historical Development, Key Principles of Industrial and Agricultural Economics, Global and Local Contexts, Case Studies of Industry and Agri-Business Integration

Unit-II: Economic Dimensions of Industry and Agri-Business

Economic Theories and Models, Market Structures and Competition, Supply Chain Management, Trade and Globalization, Economic Impact Assessment, Financing and Investment in Industry and Agri-Business, Rural and Urban Linkages

Unit-III: Technological and Innovation Strategies

Technological Advancements in Industry and Agriculture, Precision Agriculture, Smart Manufacturing, Automation and Robotics, Information and Communication Technologies (ICT), Research and Development (R&D), Innovations in Agri-Business, Sustainable Practices and Green Technologies

Unit-IV: Policy and Regulatory Frameworks

Government Policies and Regulations, International Trade Agreements, Environmental Laws and Regulations, Role of NGOs and International Organizations, Policy Analysis and Advocacy, Rural Development Policies, Agri-Business Support Programs, Compliance and Certification Standards

Unit-V: Sustainable Practices and Case Studies

Sustainable Agriculture Practices, Industrial Ecology and Circular Economy, Corporate Social Responsibility (CSR), Risk Management and Resilience, Successful Case Studies from Industry and Agri-Business, Future Trends and Challenges, Innovation and Best Practices, Sustainable Development Goals (SDGs)

Textbook References:

1. K. Thakur, **Agri-Business Management**, SAGE Publications India Pvt Ltd,
2. B. Singh, **Principles of Agricultural Economics**, Concept Publishing Company Pvt Ltd
3. A. Sharma, **Agricultural and Food Economics**, Springer.
4. A. Kumar, **Industry 4.0: The Industrial Internet of Things**, CRC Press.
5. M. D. Pathak and B. Mishra, **Sustainable Agricultural Development: Recent Approaches in Resources Management and Environmentally-Balanced Production Enhancement**, CRC Press.
6. T. Pal, **Agri-Business: Issues of Sustainable Development**, Routledge.



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GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Rural Management Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304RM	RURAL INFRASTRUCTURE	4	0	0	4

Course Description

This course delves into the critical role of infrastructure development in enhancing the quality of life and economic growth in rural areas. It covers various aspects of rural infrastructure, including transportation systems, water supply and sanitation, energy provision, and communication networks. Students will explore the planning, implementation, and management of infrastructure projects that address the unique challenges and opportunities in rural settings. The course emphasizes sustainable development practices and the integration of local needs and resources into infrastructure solutions. Students will gain practical insights into designing and executing effective infrastructure projects that promote rural development, improve living conditions, and support economic vitality in underserved regions.

Course Objectives

1	To understand the fundamental concepts and types of rural infrastructure
2	To analyse the role of infrastructure in enhancing rural development.
3	To explore planning and implementation strategies for effective rural infrastructure.
4	To evaluate the impact of infrastructure projects on rural communities.
5	To assess sustainable practices and innovations in rural infrastructure development.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain key concepts and types of rural infrastructure and their significance.
CO2	Analyse the role of rural infrastructure in promoting economic growth and social development.
CO3	Develop strategies for the planning and implementation of rural infrastructure projects.
CO4	Assess the impact of infrastructure projects on rural communities and their effectiveness.
CO5	Identify and apply sustainable practices and innovations in rural infrastructure development.
CO6	Propose solutions to challenges faced in rural infrastructure development.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	3	2	1
CO2	3	2	1	2	0	2	2	2
CO3	3	3	2	2	1	3	0	3
CO4	3	3	2	1	3	3	3	0
CO5	3	3	2	3	3	2	2	3
CO6	3	2	3	2	2	2	3	3
AVG	3.00	2.67	2.17	2.00	2.00	2.50	2.00	2.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Rural Infrastructure

covers the definition, scope, and types of rural infrastructure such as transportation, water supply, sanitation, energy, communication, health facilities, education infrastructure, and housing. It highlights the importance of infrastructure in rural development, challenges like funding and maintenance, and the roles of government, NGOs, private sector, and community involvement.

Unit-II: Planning and Implementation of Rural Infrastructure

explores planning processes and methodologies, needs assessment, resource allocation, project design, feasibility studies, cost-benefit analysis, risk assessment, stakeholder analysis, community participation, project management (scheduling, budgeting, quality control), and the use of GIS and other technological tools.

Unit-III: Impact Assessment of Rural Infrastructure

focuses on methods and tools for assessing economic, social, and environmental impacts, including qualitative and quantitative techniques, baseline data collection, comparative analysis, monitoring and evaluation techniques, performance indicators, and case studies.

Unit-IV: Sustainable Practices in Rural Infrastructure Development

principles of sustainability, innovations and technologies for sustainable infrastructure (renewable energy, water conservation, eco-friendly materials, smart technologies), environmental and social impact assessments, and examples of sustainable practices and technologies.

Unit-V: Policy and Regulatory Frameworks for Rural Infrastructure

policies and regulations affecting infrastructure development, the roles of government, NGOs, private sector, and international agencies, funding and financing mechanisms (PPPs, grants, loans, community-based financing), regulatory challenges, compliance issues, governance, and institutional frameworks.

Textbook References:

1. **B.K. Prasad**, *Rural Infrastructure: Issues and Prospects*, Mittal Publications.
2. **M.L. Dantwala**, *Rural Development and Infrastructure*, Sage Publications.
3. **N. K. Raju**, *Sustainable Rural Development*, Concept Publishing Company.
4. **R.S. Deshpande**, *Infrastructure Development in Rural India: Problems and Prospects*, Academic Foundation.
5. **Judith D. Tandler**, *Rural Infrastructure: Strategy, Planning, and Management*, World Bank Publications.
6. **John Hine and J. Rizet**, *Rural Transport and Development: A Case Study in Africa*, World Bank Publications.



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GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
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MBA III SEMESTER (AR23)
(Rural Management Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305RM	FINANCIAL INCLUSION	4	0	0	4

Course Description

The program aims at developing an overall understanding of the concept of financial inclusion, history and evaluation of financial inclusion, rural banks and their roles in financial inclusion, financial inclusion through microfinance, rural credit, and financial inclusion schemes.

Course Objectives

1	To provide students with comprehensive understanding of how financial inclusion impacts in rural people.
2	To understand the key issues linked to the whole discourse on financial inclusion
3	To provide students with insights into current financial inclusion schemes in India.
4	To provide students the knowledge on the role of government and other institutions in developing financial inclusion in rural areas.
5	To know the challenges inherent therein and initiatives taken in the country to promote financial inclusion.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Develop strategies for improving financial literacy and education.
CO2	Explore the impact of financial exclusion on economic and social development.
CO3	Examine government initiatives and schemes designed to support priority sector financing.
CO4	Identify challenges faced by financial institutions in implementing priority sector lending.
CO5	Analyse the challenges and barriers to providing credit in rural communities.
CO6	Analyse the impact of various schemes on financial inclusion.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	3	3	2	3
CO2	2	3	0	3	3	2	2	3
CO3	3	3	0	3	3	2	2	3
CO4	3	3	0	3	3	2	2	3
CO5	3	3	0	3	3	3	2	3
CO6	3	3	0	3	3	3	2	3
AVG	2.83	3.00	0.00	3.00	3.00	2.50	2.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Financial Inclusion: Issues, Challenges and Experiences

Concept of financial Inclusion, exclusion and extent of financial exclusion - challenges of financial inclusion - moral hazard, adverse selection, high transaction cost and information asymmetry - history and evolution of financial inclusion - committee's on financial inclusion - national strategy for financial inclusion - financial literacy and consumer protection - cross country experiences of financial inclusion.

Unit-II: Priority Sector Financing and Govt. initiatives

Components of priority sector - RBI guidelines, government initiatives: poverty alleviation programmes/employment programmes/production oriented programmes - rural housing and urban housing schemes under priority sector - Educational loans.

Unit-III: Priority Sector Financing and Govt. initiatives

Components of priority sector - RBI guidelines, government initiatives: poverty alleviation programmes/employment programmes/production oriented programmes - rural housing and urban housing schemes under priority sector - Educational loans

Unit-IV: Financial Inclusion and Rural Credit Rural Credit

Concept, need, characteristics and sources of rural credit - institutional and non-institutional rural credit - moneylenders, cooperatives, commercial banks, regional rural banks, nationalized banks - NABARD - outline of Govt. programmes for rural credit

Unit-V: Financial Inclusion Schemes in India

Financial inclusion schemes in India - Pradhan Mantri Jan Dhan Yojana (PMJDY) - Atal Pension Yojana (APY) - Pradhan Mantri Mudra Yojana (PMMY) - Sukanya Samriddhi Yojana - Pradhan Mantri Suraksha Bima Yojana - Jeevan Suraksha Bandhan Yojana - Financial inclusion programmes organised by RBI. Modern financial inclusion instruments.

Textbook References:

1. Bhaskaran, R, Microfinance-Perspectives and Operations, Macmillan Education.
2. Khan, M.Y., Indian Financial System, Tata McGraw Hill Publishing Company Ltd.
3. Vasantha Desai, Indian Banking-Nature and Problems, Himalaya Publishing House.
4. Chakrabarty K.C, Financial Inclusion and Banks: Issues and Perspectives, RBI Bulletin, November 2011.
5. Das, Manda & Rath, Micro Finance & Rural Development in India, New Century Publication, N Delhi.
6. Rangarajan C, Report of the Committee on Financial Inclusion.



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MBA III SEMESTER (AR23)
(Rural Management Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306RM	RETAIL AND RURAL MARKETING	4	0	0	4

Course Description

This course explores strategies and techniques for effectively reaching and engaging consumers in both urban and rural markets. It covers key concepts such as consumer behaviour, market segmentation, distribution channels, and promotional tactics tailored to diverse environments. Students will learn how to analyse market needs, develop targeted marketing campaigns, and implement practical solutions to overcome challenges specific to retail and rural settings. Emphasis is placed on understanding local market dynamics, leveraging technology, and creating sustainable marketing practices to drive growth and enhance customer satisfaction in various geographical contexts.

Course Objectives

1	To impart the basic knowledge of retail marketing of goods
2	To make aware of retail marketing location.
3	To update learners on recent trends in marketing..
4	To know the emerging trends of rural marketing.
5	To understand the profile of rural consumers.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Analyse consumer behaviour and preferences in different retail environments.
CO2	Apply best practices in customer relationship management.
CO3	Evaluate the effectiveness of retail strategies and adjust as needed for optimal results.
CO4	Analyse the structure and evolution of retail markets in India and globally.
CO5	Develop required skills to manage rural customers.
CO6	Strategies to overcome the practical difficulties of rural marketing.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	0	3	3	2	2	3
CO2	3	3	0	3	3	2	2	3
CO3	3	3	0	3	3	2	2	3
CO4	2	3	0	3	3	2	2	3
CO5	3	3	2	3	3	2	2	3
CO6	3	3	2	3	3	2	2	3
AVG	2.67	3.00	0.67	3.00	3.00	2.00	2.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Retail Marketing

Retailing Concept, functions, Scope & Trends in Retailing, Issues & Challenges in Organized Retailing, Retail Evolution & Growth in Indian Context, Understanding Retail Consumers. Retail

Model and Theories of structural changes in retailing - Classification of retailers and retail markets - Life cycle and phase in growth of retail markets - Methods of customer interaction.

Unit-II: Retailing Strategy

Process & Development, Types of Retailing Formats. Retail location strategy - Types of location decision and its determining factors - Site selection analysis - Selection of shopping centre or market - Retail location theories - Location assessment procedures, Retail Promotion Strategy.

Unit-III: Retail in India and Global

Retail in India - Evaluation and Size of retail in India - Drivers of retail change in India - Foreign Direct Investment in retail - Challenges to retail developments in India. Strategic planning process for global retailing - Challenges facing global retailers - Challenges and threats in global retailing - Factors affecting the success of a global retailing strategy.

Unit-IV: Introduction to Rural Marketing

Meaning, Concept, Phased Evolution, Significance Factors and Environment. Structure and Characteristics of the Rural Markets. Rural Vs. Urban Marketing. Rural Marketing in India.

Unit-V: Rural Consumer Behaviour

Consumer Behaviour Model in Rural Marketing, Rural Marketing Research-Retail & IT models in Rural Marketing - CSR and Marketing Ethics in Rural Marketing - Source of Financing and credit agencies. Advertising & Media Role in Rural Marketing.

Textbook References:

1. C G Krishnamacharyulu, Lalitha Ramakrishnan, Rural Marketing, Pearson Education.
2. Awadhesh Kumar Singh Satyaprakash Pandey, Rural Marketing: Indian Perspective, New age publishers
3. U.C.Mathur, Rural Marketing, Excel Books.
4. C.S.G. Krishnamacharyulu and Lalitha Ramakrishnan, Rural Marketing Text and Cases, Pearson Education.
5. Chetan Bajaj, Rahnishtuli and Nidhiv.Srivastava, Retail Management, Oxford University Press.
6. J. N.Jain and P.P.Singh, Modern Retail Management-Principles and Techniques, Regal Publications.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Operations & Supply Chain Management Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302OSM	SUPPLY CHAIN AND LOGISTICS MANAGEMENT	4	0	0	4

Course Description

This course allows students to understand both the components of supply chain management and its role within the function and across other functions in an enterprise. The objective of supply chain management is to create value, build a competitive infrastructure, leverage worldwide logistics, synchronize supply with demand, and measure performance globally. Logistics is part of the supply chain, which deals with storage and distribution of goods and services, in the right quantity, right condition, at the right time, and in the right place.

Course Objectives

1	To analyze the supply chain scenario and to make understand the students the insights on supply chain process from sourcing to distribution
2	To understand the importance and dynamics of firm's physical distribution functions and management of its supply chain
3	to ensure efficient and cost-effective production, timely delivery of products and services, effective inventory management, collaborative relationships between suppliers and customers, flexibility, and responsiveness to changes in demand.
4	To impart knowledge and understanding to students on Supply Chain Management and its relevance to today's business decision-making.
5	To gain the knowledge of possibilities of efficient optimization and management of operation in Logistics Management and also the ability to apply them in the enterprise reality

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Recognise the role of network planning and understand basic network planning models.
CO2	Estimate essentials of demand and supply management.
CO3	acquaint with the importance of resource planning and understand the importance of resource scheduling.
CO4	Utilise recent Industry trends in supply chain planning.
CO5	Develop an understanding of the importance of logistics in the formulation of the business strategy and the conduct of supply chain operations.
CO6	Strengthen integrative management analytical and problem-solving skills.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	3	3	2	3	3
CO2	3	2	3	3	3	2	3	3
CO3	3	3	1	3	3	2	3	3
CO4	3	3	2	2	3	2	3	3
CO5	3	3	2	2	3	2	3	2
CO6	3	2	2	1	3	3	2	2
AVG	3.00	2.50	2.17	2.33	3.00	2.17	2.83	2.67

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Supply Chain Foundations**

Supply Chain Foundations: Supply Chain as a network of entities: Role and interactions between the entities. Value Chain Focus of Supply Chain. Impact of Supply Chain Management on Sales, Cost, Profit, Profitability, Balance Sheet, Profit and Loss Account, and Customer Accounts Profitability. Centralized and Decentralized Supply Chains.

Unit-II: Distribution Management

Distribution Management: Distribution Channels: Structure and Operation, Distribution Cost, Pipeline Inventory and Response Considerations, Hub and Spoke Models, Cross docking, Lots streaming, Container Selection, Warehousing: Facility location and Network design, Vehicle Loading and Routing Methods, Lead time, Supply chain sustainability.

Unit-III: Procurement Logistics

Procurement Logistics: Source Identification: Global Vs. Domestic Sourcing, Landed Cost Computation, Vendor Rating, Contract Negotiation, Consolidation, Vendor Management, Individual component Vs. Unit Purchases. Vendor Development and Vendor Relationship Management, Vendor Performance Monitoring.

Unit-IV: Aligning Logistics to Customer Needs

Aligning logistics to customer needs: Quick response logistics, Green logistics, Reverse Logistics, Vendor Managed inventory, cross docking, Packaging Innovations, Third Party Logistic and Service concepts and applications

Unit-V: Managing Global Logistics and Global Supply Chains

Global Logistics, Global supply chain, Global Supply chain business process, Global purchase, Global Strategic alliances, Issues and challenges in Global supply chains, Managing the supply chain for globally integrated products

Textbook References:

1. Sahay B.S, Supply Chain Management for Global Competitiveness, Macmillan India Ltd., New Delhi.
2. Reguram G, Rangaraj N, Logistics and Supply Chain Management Cases and Concepts, Macmillan India Ltd., New Delhi.
3. Bowersox, Logistical Management, Mc-GrawHill.
4. D K Agrawal, 'Distribution and Logistics Management: A Strategic Marketing Approach', Macmillan publishers India.
5. Agarwal, Logistics and Supply Chain Management, McMillan.
6. Ballou & Srivastava, Business Logistics and Supply Chain Management, Pearson.



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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Operations & Supply Chain Management Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-303OSM	TOTAL QUALITY MANAGEMENT	4	0	0	4

Course Description

This course explores comprehensive approaches to improving organizational performance through a focus on quality across all processes and functions. Students will study TQM principles such as customer satisfaction, continuous improvement, and employee involvement, and learn to apply various tools and techniques like statistical process control, root cause analysis, and quality audits. The course covers the development and implementation of quality management systems, fostering a culture of excellence, and aligning quality goals with organizational objectives. Students will gain the skills needed to drive quality initiatives, enhance operational efficiency, and achieve sustainable competitive advantage by embedding quality into every aspect of an organization's operations.

Course Objectives

1	To facilitate the understanding of Quality Management principles and process.
2	To apply various quality improvement techniques.
3	To describe and apply the development and nature of quality control charts.
4	To Understand the cost of quality.
5	To Analyse the Total Quality Management organisation.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Recall and relate the quality concepts as applied to projects of their choice,
CO2	Demonstrate the skills to extend the quality as the strategic decision making process with help of different tools and techniques.
CO3	Exhibit the skills to practical knowhow and use of different quality control tools and techniques
CO4	Critically appraise the organisational, communication and teamwork requirements for effective quality management
CO5	Critically analyse the strategic issues in quality management, including current issues and developments, and to devise and evaluate quality implementation plans
CO6	Analyse and relate TQM with the foundational components of operations management namely, product development, process design & management and supply chain management,

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	2	3	2	2	3	3
CO2	3	2	2	3	3	2	3	1
CO3	3	3	3	3	3	2	2	3
CO4	3	3	3	3	2	2	2	3
CO5	3	3	3	3	3	3	3	3
CO6	3	2	3	3	2	3	1	3
AVG	3.00	2.67	2.67	3.00	2.50	2.33	2.33	2.67

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Quality Management

Introduction to Quality, Dimensions of Quality, Total Quality Management (TQM) Quality Philosophers: Dr. Edwards Deming --W. Edward Deming's 14 points, Philip B. Crosby --Contributions to The Theory of Process Improvement and Six Sigma, Dr. Joseph Juran --The Juran Trilogy, Dr. Genichi Taguchi --Taguchi loss function, The philosophy of off-line quality control, Design of experiments, Dr. Shigeo Shingo -- Poka-yoke, Dr. Kaoru Ishikawa --Cause-and-effect Diagram or Fishbone Diagram, Masaaki Imai

Unit-II: Quality as a Strategic Decision

Meaning of Strategy and Strategic Quality Management, Strategic Planning and Implementation, McKinsey 7s Model, Competitive Analysis, Management Commitment to Quality. Kaizen, Total Productive Maintenance (TPM). The Statistical basis of Six Sigma.

Unit-III: Cost of Quality

Cost of Quality, Categories of Cost of Quality, Models of Cost of Quality, Optimizing Costs, Preventing Cost of Quality

Unit-IV: Quality Control Tools

Check Sheet, Histogram, Shapes of histogram, Drawing a histogram, Pareto Chart, Drawing a Pareto chart, Cause & Effect Diagram, Scatter Diagram, Control charts, Quality Control, Quality Assurance, Product control – acceptance sampling and OC curve.

Unit-V: Quality Management Systems

Quality Management System, Quality Management Principles, ISO 9001 Structure, Quality Audits, ISO Registration, Requirements, Benefits of ISO registration, Examples of ISO Standard Application

Textbook References:

1. J. R. Evans, Total Quality Management, Cengage Learning.
2. Bedi, Quality Management, Oxford Press.
3. Bharat Wakhlu, Total Quality, S. Chand & Company.
4. Joel E. Ross, Total Quality Management (Text, Cases and Readings), Taylor and Francis Group.
5. P. N. Mukherjee, Total Quality Management, PHI.
6. Robert Louis Flood, Beyond TQM, John Wiley & Sons Ltd.



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MBA III SEMESTER (AR23)
(Operations & Supply Chain Management Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304OSM	OPERATIONS & MATERIALS MANAGEMENT	4	0	0	4

Course Description

This course deals with the design and operation of the systems for production of goods and services. It will explore the approaches and analyse strategic decisions in operations management with a focus on designing products and processes, allocating scarce resources to strategic alternatives, and do long-range capacity and facility planning. These operations functions help in achieving the organization's long-range objectives. Subsequent focus will be on medium and short term planning and controlling. Care will be taken to strike a balance between theoretical and practical perspectives in manufacturing and service organizations.

Course Objectives

1	To develop an understanding of how the operations, have strategic importance and can provide a competitive advantage in the workplace.
2	To understand the relationship between operations and other business functions
3	To understand techniques of location and facility planning; line balancing; job designing; and capacity are planning in operations management.
4	To understand the Materials Management function starting from Demand Management through Inventory Management.
5	To Evaluate the Impact of Technology on Operations and Materials Management

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Identify the elements of operations management and various transformation processes to enhance productivity and competitiveness.
CO2	Analyse and evaluate various facility alternatives and their capacity decisions.
CO3	Develop aggregate capacity plans and MPS in operation environments.
CO4	Plan and implement suitable materials handling principles and practices in the operations.
CO5	Plan and implement suitable quality control measures in Quality Circles to TQM.
CO6	Improve Forecasting and Demand Planning Abilities.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	2	3	2	3	3	2
CO2	2	3	3	2	2	3	3	2
CO3	3	3	3	3	2	3	3	2
CO4	3	3	2	3	2	3	3	2
CO5	3	3	3	3	3	3	3	3
CO6	3	3	2	2	2	2	2	3
AVG	2.67	3.00	2.50	2.67	2.17	2.83	2.83	2.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Introduction to Operations and Materials Management**

Introduction to operations and materials management, evolution scope and development stages of om, operations strategy: as a competitive weapon & concept of productivity

Unit-II: Forecasting and Facility Location & Layout

Introduction to forecasting time series introduction, components of time series, types of forecasting, regression method, moving average, exponential method, forecasting errors.

Facility location: parameters & criterion for selection of a location, different types of layouts, their planning and production systems, service layouts & operations.

Unit-III: Assembly Line Balancing & Scheduling

Networking of process flow, assembly line balancing, scheduling of operations. Capacity planning: planning capacity across the organization, planning long-term capacity, capacity timing and sizing strategies.

Unit-IV: Materials Management

Materials management materials handling, role of purchase department, inventory basics, ERP, KANBAN system, Lean operations and JIT

Unit-V: Inventory Planning & Control

EOQ models, inventory control techniques: ABC, VED analysis, EOQ using quantity discount, POQ, quality planning & control, total quality management (TQM), statistical process control, control charts.

Textbook References:

1. James R Evans & David A Collier, Operations Management, Thomson Press Pub.
2. Richard B Chase, F Robert Jacobs, Nicholas J Aquilano, & Nitin K Agarwal, Operations Management for Competitive Advantage, Tata McGraw-Hill.
3. Jay Heizer, Barry Render, and Chuck Munson, Operations Management, Pearson Education.
4. Heizer and Barry Render, Principles of Operations Management, Prentice Hall.
5. Roberta S. Russell and Bernard W. Taylor, Operations and Supply Chain Management, Wiley.
6. J.R. Tony Arnold, Stephen N. Chapman, and Lloyd M. Clive, Introduction to Materials Management, Pearson.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Operations & Supply Chain Management Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305OSM	SUPPLY CHAIN PLANNING	4	0	0	4

Course Description

This course designed to equip students with the skills and knowledge needed to effectively plan and manage supply chain operations. The course covers the essential components of supply chain planning, including demand forecasting, inventory management, and production scheduling. Students will learn how to align supply chain plans with organizational objectives, optimize inventory levels, and enhance the efficiency of supply chain processes. Through a combination of theoretical knowledge and practical case studies, students will explore various planning models and tools, understand their applications in real-world scenarios, and develop strategies to address common challenges. The course emphasizes the importance of integrated planning in achieving operational excellence and maintaining a competitive edge in today's dynamic market environment.

Course Objectives

1	To Understand Supply Chain Planning Fundamentals.
2	To Develop Forecasting and Demand Planning Skills.
3	To Optimize Inventory and Production Planning.
4	To Integrate Supply Chain Planning with Strategic Goals.
5	To Utilize Technology and Tools in Supply Chain Planning.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Proficiency in Supply Chain Planning Concepts.
CO2	Effective Forecasting and Demand Planning.
CO3	Optimization of Inventory and Production.
CO4	Align supply chain planning efforts with organizational strategy.
CO5	Proficient in using supply chain planning tools and technologies.
CO6	Identify and address common supply chain planning challenges, developing strategic solutions to enhance overall supply chain performance.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	3	3	2
CO2	3	3	3	2	2	3	3	3
CO3	3	2	3	3	2	3	3	2
CO4	2	2	3	3	2	3	3	3
CO5	2	3	3	3	3	3	3	2
CO6	3	1	3	3	2	2	3	3
AVG	2.50	2.33	3.00	2.67	2.17	2.83	3.00	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Supply Chain Planning

Supply chain - objectives - importance - decision phases - process view - competitive and supply chain strategies - achieving strategic fit - supply chain drivers - obstacles - framework - facilities - inventory - transportation - information - sourcing - pricing

Unit-II: Designing the Supply Chain Network

Designing the distribution network - role of distribution - factors influencing distribution - design options - e-business and its impact - distribution networks in practice - network design in the supply chain - role of network - factors affecting the network design decisions - modelling for supply chain.

Unit-III: Planning Demand and Supply

Role of forecasting - demand forecasting - approaches - role of IT. Planning and Managing Inventories - Safety inventory and its appropriate level - impact of supply uncertainty, aggregation and replenishment policies.

Unit-IV: Transportation Networks and Sourcing

Role of transportation - modes and their performance - transportation infrastructure and policies - design options and their trade-offs - Tailored transportation. Sourcing - In-house or Outsource - Third and Fourth PLs - supplier scoring and assessment

Unit-V: Coordination in a Supply Chain

Lack of supply chain coordination and the Bullwhip effect - obstacle to coordination - managerial levels - building partnerships and trust - continuous replenishment and vendor-managed inventories - collaborative planning, forecasting and replenishment

Textbook References:

1. Nicolas Vandeput, Demand Forecasting Best Practices, Manning Publications.
2. Keah-Choon Tan, Joel Wisner, G. Leong, Principles of Supply Chain Management: A Balanced Approach, South-Western College Publishing.
3. Edward J. Bardi (Author), C.John Langley (Author), John Joseph Coyle, The Management of Business Logistics: A Supply Chain Perspective, South-Western College Publishing.
4. Jeremy F. Shapiro, Modeling the Supply Chain, South-Western College Publishing.
5. Gerald Feigin, Supply Chain Planning and Analytics: The Right Product in the Right Place at the Right Time, Business Expert Press
6. Colleen Crum and George E. Palmatier, Demand Management Best Practices: Process, Principles, and Collaboration, J Ross Publishing.



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GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Operations & Supply Chain Management Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306OSM	STRATEGIC PLANNING FOR SUPPLY CHAIN MANAGEMENT	4	0	0	4

Course Description

Strategic Planning for Supply Chain Management explores the critical role of strategic planning in optimizing supply chain performance. This course provides an in-depth understanding of how to develop and implement effective supply chain strategies that align with organizational goals. Students will learn to analyse various supply chain models, apply strategic frameworks, and leverage tools and technologies to enhance decision-making. The course emphasizes the integration of strategic planning with operational execution, focusing on creating value through effective supply chain management. By examining real-world case studies and industry best practices, students will gain insights into overcoming challenges and driving supply chain success.

Course Objectives

1	To Develop Comprehensive Supply Chain Strategies:
2	To Analyse and Evaluate Supply Chain Models
3	To Implement Strategic Planning Tools
4	To Integrate Strategy with Operational Execution
5	To Address Challenges and Opportunities in Supply Chain Management

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Develop and present strategic supply chain plans that support business objectives and drive performance.
CO2	Critically evaluate different supply chain models and select the most appropriate ones based on organizational needs and goals.
CO3	Use strategic planning tools and technologies to analyse data and support decision-making in supply chain management
CO4	Integrate strategic planning with operational execution, ensuring that strategies are effectively translated into actionable plans
CO5	Identify key challenges and opportunities in supply chain management and develop innovative strategies to address them.
CO6	Apply theoretical knowledge to real-world case studies, demonstrating their ability to implement strategic planning concepts effectively in practical scenarios.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	3
CO2	3	2	2	3	2	3	2	3
CO3	3	3	3	3	3	1	3	3
CO4	3	3	2	3	2	2	3	3
CO5	3	3	3	3	3	2	3	3
CO6	2	2	3	3	2	2	2	2
AVG	2.83	2.50	2.67	3.00	2.33	2.17	2.50	2.83

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Introduction to Strategic Supply Chain Management**

Introduction to supply chain management, competitive cost analysis, capacity sizing, strategic games and capacity sizing

Unit-II: Capacity Timing and Sizing Strategies

Capacity timing, capacity expansion, capacity types, operational hedging, complementary capacity – operational hedging .

Unit-III: Sales and Operations Planning

Sales and operations planning, product design to hedge demand risk, financial hedging, capacity location and logistical design.

Unit-IV: Logistic Outsourcing and 3PL Challenges

Third Party Logistics, coordination and incentives, coordination of channels of distribution, collaborative planning and forecasting, outsourcing models.

Unit-V: Sustainable Supply Chain Management

Product life cycle and supply chain design, supplier management, sustainability issues in supply chain, eco-efficiency, green supply chain, and reverse logistics.

Textbook References:

1. Jan A Van Mieghem, Operations Strategy: Principles and Practice, Dynamic Ideas.
2. Raví Anupindi, Sunil Chopra, Sudhakar Deshmukh, Jan A. Van Mieghem, and Eitan Zemel, Managing Business Process Flows: Principles of Operations Management, Pearson Education.
3. Sunil Chopra, Peter Meindl, D.V. Kalra, Supply Chain Management. Pearson Education.
4. David Simchi-Levi, Philip Kaminsky, Edith Simchi-Levi, and Ravi Shankar, Designing and Managing the Supply Chain, Tata McGraw Hill.
5. Sridhar Tayur, Ram Ganeshan and Micheal Magazine, Quantitative Models for Supply Chain Management, Springer.
6. Shardul S. Phadnis and Yossi Sheffi, Strategic Planning for Dynamic Supply Chains: Preparing for Uncertainty Using Scenarios, Springer Nature Switzerland AG.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Business Analytics Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302BA	BIG DATA ANALYTICS WITH HADOOP	4	0	0	4

Course Description

This course provides an in-depth understanding of how to manage and analyse large-scale data sets using Hadoop, an open-source framework designed for distributed storage and processing. Students will learn the fundamentals of Hadoop's ecosystem, including its core components such as Hadoop Distributed File System (HDFS), MapReduce, and Hadoop YARN. The course covers data ingestion, storage, and processing techniques, along with advanced analytics and data mining methods. Students will develop practical skills in leveraging Hadoop to perform complex data analyses, derive actionable insights, and address challenges associated with big data. This course aims to equip students with the expertise needed to harness the power of big data technologies and drive data-informed decision-making in various organizational contexts.

Course Objectives

1	Learn the principles of big data and the potential insights from its analysis.
2	Explore HDFS, MapReduce, YARN, Hive, Pig, HBase, and their roles in data processing.
3	Design, implement, and optimize data processing workflows using Hadoop.
4	Use tools like Hive, Pig, and HBase for data analysis on large datasets.
5	Optimize performance, manage resources, and troubleshoot Hadoop clusters.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the concept of big data and analyze the process of big data analytics.
CO2	Develop and implement programs using Hadoop MapReduce for data processing.
CO3	Create and execute data transformation scripts using Apache Pig.
CO4	Execute data analysis and perform complex querying using Apache Hive.
CO5	Configure and demonstrate the components and functionalities of Apache HBase.
CO6	Perform comprehensive big data analytics using Hadoop technologies

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	0	3	2	3	2	2
CO2	3	2	0	1	2	1	2	3
CO3	2	3	0	3	3	2	3	2
CO4	1	2	0	2	3	3	2	3
CO5	3	2	0	3	2	2	3	2
CO6	2	3	0	3	2	3	2	3
AVG	2.17	2.217	0.00	2.50	2.33	2.33	2.33	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Big Data

Concept, Features of big data, big data challenges, Hadoop and its features, Hadoop Ecosystem, Hadoop Components, Hadoop Architecture, Hadoop Cluster, Hadoop Storage: HDFS.

Unit-II: Hadoop Mapreduce

Concept, YARN components, YARN architecture, YARN mapreduce application execution flow, YARN workflow, Anatomy of mapreduce program, distributed cache, MRunit, Reduce Join, Custom input format, sequence input format.

Unit-III: Introduction to Apache PIG

PIG Components & Execution, PIG data types, Data models in PIG, Programming in PIG.

Unit-IV: APACHE HIVE

Introduction, Architecture and components, data types and data models, HIVE partitioning and bucketing, HIVE tables, HIVE QL: joining tables, dynamic partitioning.

Unit-V: APACHE Hbase

Introduction, Architecture and components, Run modes, configuration, data models, HIVE data loading techniques, Introduction to Apache Spark.

Textbook References:

1. Tom White, Hadoop: The Definitive Guide, O'Reilly Media.
2. Alan Gates, Programming Pig, O'Reilly Media.
3. Lars George, HBase: The Definitive Guide, O'Reilly Media.
4. Alex Holmes, Hadoop in Practice, Manning Publications.
5. Edward Capriolo, Dean Wampler, and Jason Rutherglen, Hive: A Petabyte Scale Data Warehouse Using Hadoop, O'Reilly Media.
6. Holden Karau, Andy Konwinski, Patrick Wendell, and Matei Zaharia, Learning Spark: Lightning-Fast Big Data Analysis, O'Reilly Media.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Business Analytics Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-303BA	DATA ANALYSIS WITH PYTHON	4	0	0	4

Course Description

This course offers a thorough introduction to using Python for data manipulation, analysis, and visualization. Students will gain hands-on experience with key Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, learning how to clean, transform, and explore data sets to uncover meaningful insights. The course covers fundamental concepts of data analysis, including statistical analysis, data wrangling, and exploratory data analysis, and emphasizes practical applications and projects. By the end of the course, students will be proficient in leveraging Python's powerful tools to perform comprehensive data analyses, create informative visualizations, and support data-driven decision-making across various domains.

Course Objectives

1	Understand Python syntax, data structures, and programming concepts.
2	Manipulate, clean, and process data using pandas and numpy.
3	Visualize data with matplotlib and seaborn.
4	Apply statistical techniques for data analysis.
5	Implement basic machine learning algorithms with scikit-learn.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Process and transform datasets using pandas and numpy.
CO2	Generate insightful data visualizations using matplotlib and seaborn.
CO3	Perform statistical data analysis and interpret results.
CO4	Develop and evaluate machine learning models using scikit-learn.
CO5	Integrate Python code with other data processing and web development tools.
CO6	Design and execute data analysis projects from start to finish.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	0	2	2	3	2	2
CO2	3	2	0	1	2	1	2	3
CO3	2	3	0	3	3	2	3	2
CO4	1	2	0	2	3	3	2	3
CO5	3	2	0	3	2	2	1	2
CO6	3	3	0	3	3	2	2	1

AVG	2.33	2.50	0.00	2.33	2.50	2.17	2.00	2.17
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to Python Keywords and Identifiers, Statements and Comments, Input-Output and Import, Operators, Python namespace, Data types - Numbers, Strings, Lists, Tuples, Set, Dictionaries, Arrays, Matrix, Flow Control: If – else, for loop, while loop, break and continue, Pass statement, Looping technique								
Unit-II: Functions and OOP Concepts Defining and calling a function, Types of Function, Recursion, Python Modules, Packages, OOP Concepts: OOP concepts in Python – Class, Inheritance, Multiple Inheritance, Operator Overloading.								
Unit-III: IPython, NumPy, and Pandas IPython Basics, code development in IPython, IPython features, NumPy Basics, NumPyArrays, Vectorized Computation, Indexing and sorting arrays, Structured arrays, Pandas Basics, Pandas data structures, Descriptive statistics, Handling missing data, Hierarchical Indexing, Vectorized string operations, working with time series.								
Unit-IV: Working with Data Reading and writing data in text format, binary data formats, interacting with web, interacting with databases, Combining and merging data sets, Reshaping and Pivoting, Data Transformation, Data Aggregation, Pivot tables, and Cross Tabulation.								
Unit-V: Data Visualization Introduction to Matplotlib, line plots, scatter plots, visualizing errors, Density and contour plots, Histograms and Binning's, Text and Annotation, Three-dimensional plotting in Matplotlib								
Textbook References: 1. Wes McKinney, Python for Data Analysis, O'Reilly Media. 2. Jake VanderPlas, Python Data Science Handbook, O'Reilly Media. 3. Al Sweigart, Automate the Boring Stuff with Python, No Starch Press. 4. Mark Lutz, Learning Python, O'Reilly Media. 5. Luciano Ramalho, Fluent Python, O'Reilly Media. 6. Brett Slatkin, Effective Python: 59 Specific Ways to Write Better Python, Addison-Wesley Professional.								



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GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA III SEMESTER (AR23)
(Business Analytics Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304BA	DATA ANALYSIS WITH R	4	0	0	4

Course Description

This course provides a comprehensive introduction to data analysis using R, focusing on equipping students with the skills to manage, analyse, and visualize data effectively. Covering fundamental R programming concepts, data manipulation with packages like `dplyr` and `tidyr`, and statistical analysis including hypothesis testing and regression, students will also learn to create insightful visualizations with `ggplot2`. The course includes hands-on labs and a final project that applies these techniques to real-world datasets. Designed for individuals with basic statistics knowledge, it is ideal for those looking to advance their data analysis skills and pursue careers in data science and related fields.

Course Objectives

1	Learn R programming basics and data structures.
2	Manipulate and clean data using R packages.
3	Create visualizations with ggplot2.
4	Apply statistical methods for data analysis.
5	Implement machine learning algorithms in R.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Write basic R programs using fundamental programming elements.
CO2	Implement control structures in R to manage program flow.
CO3	Create and customize data visualizations using R.
CO4	Conduct statistical analysis with R to interpret data.
CO5	Apply machine learning techniques using R.
CO6	Utilize R for various data science tasks and problem-solving.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	2	2	2	2	2
CO2	1	2	0	1	2	1	2	3
CO3	2	3	0	3	3	3	3	2
CO4	3	2	0	2	3	2	2	3
CO5	2	2	0	3	2	3	1	2
CO6	2	3	0	3	3	3	2	3
AVG	2.17	2.50	0.00	2.33	2.50	2.33	2.00	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Elements of R

Concept of R, IDE of R, Mathematical Operators and Vectors, Assigning Variables, Special Numbers, Logical Vectors, Classes, Different types of numbers, Changing classes, Examining

Variables, The workplace, Elements in R-Vectors, Matrices and Arrays, Lists, Conversion between vectors and lists, Combining lists, Data Frames.

Unit-II: Functions, Strings and Factors and Flow Controls

Environments, Functions, Strings, Factors, Flow Controls - Conditional - if and else, Vectorized if, Multiple Selection, Loops - repeat loops, while loops, for loops, Advanced looping - replication, looping over lists, looping over arrays, Multiple - Input Apply, Instant vectorization, Split-Apply-Combine.

Unit-III: Packages and Visualization

Loading packages, search path, libraries and installed packages, installing packages, maintaining packages, Visualization - The three plotting systems, Scatter plots - base graphics, lattice graphics, ggplots, Line Plots, Histograms, Box Plots, Bar Charts, Other plotting packages and systems.

Unit-IV: Computing Statistics and Exploratory Data Analysis with R

Summarizing data, Calculating relative frequencies, Tabulating Factors and creating contingency tables, Testing categorical variables for independence, Calculating Quantiles of a dataset, Converting data into z-scores, t-test, testing sample proportions, testing normality, comparing means of two samples, testing correlation for significance, Variations, Missing Values, Covariation, Patterns and Models.

Unit-V: Machine Learning and Model Building with R

Types of machine learning algorithm, supervised learning algorithms - Linear regression in R, Logistic Regression in R Unsupervised Learning in R-Clustering with R, Recommendation Algorithms, Steps to generate recommendations in R, Model Building: Model basics, Type of Models, Visualizing models - Predictions, Residuals, Model Building, Communicating results - Basics of R Markdown.

Textbook References:

1. Hadley Wickham and Garrett Golemund, R for Data Science, O'Reilly Media.
2. Roger D. Peng and Elizabeth Matsui, The Art of Data Science, Leanpub.
3. Garrett Golemund, Hands-On Programming with R, O'Reilly Media.
4. Paul Teetor, R Cookbook, O'Reilly Media.
5. Hadley Wickham, Advanced R, CRC Press.
6. Brett Lantz, Machine Learning with R, Packt Publishing.



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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Business Analytics Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305BA	DATA ANALYSIS WITH SAS	4	0	0	4

Course Description								
SAS (Statistical Analysis System) is a powerful suite of business analytics software used for statistical analysis, data mining, predictive modelling, and more. Renowned for its robustness and scalability, SAS is considered a leader in business analytics software and services, widely adopted by organizations across various industries to make data-driven decisions. This course provides an in-depth understanding of SAS and its applications in data analysis. Students will learn how to leverage SAS for performing complex statistical analyses, managing large datasets, and generating meaningful insights from data. The course will cover key components of SAS, including SAS Base, SAS Enterprise Guide, and SAS Visual Analytics.								
Course Objectives								
1	Understand SAS basics and its programming environment.							
2	Manipulate and manage data using SAS.							
3	Apply statistical analysis techniques with SAS.							
4	Use SAS for data mining and pattern discovery.							
5	Generate and interpret reports and visualizations in SAS.							
Course Outcomes								
<i>(Upon successful completion of this course, students should be able to)</i>								
CO1	Use SAS for executing various data analysis tasks.							
CO2	Manage and query databases using Proc SQL in SAS.							
CO3	Develop and implement SAS macros for automation.							
CO4	Apply statistical methods to data within SAS.							
CO5	Explore, analyse, and interpret data using SAS tools.							
CO6	Generate comprehensive reports and visualizations in SAS to effectively communicate findings.							
CO-PO & PSO Mapping (Articulation Matrix)								
COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	2	2	2	2	3
CO2	2	3	0	1	2	1	2	2
CO3	3	2	0	3	3	3	3	1
CO4	2	3	0	2	3	2	2	2
CO5	2	2	0	3	2	3	1	3
CO6	3	3	0	3	3	3	2	2
AVG	2.50	2.50	0.00	2.33	2.50	2.33	2.00	2.17
Justification of the Mapping:								
0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to SAS								
Introduction to SAS interface and library structure, Reading data, importing and exporting datasets, Formats and Informants. Variable attributes and data modification.								

Unit-II: Proc SQL

Introduction to Proc SQL, Retrieving and selecting data from table, Retrieving and selecting data from multiple tables, Concatenating query results.

Unit-III: SAS Macros

Introduction, Macro Functions, SQL clauses for Macros, The %Macro statement, Conditional statement.

Unit-IV: Statistics in SAS

Descriptive statistics, Hypothesis testing, parametric and Non-Parametric tests, Statistical procedures for different statistics.

Unit-V: Data Exploration

Data Preparation, Data Cleaning, Data type conversion, Missing value treatment, Data summarization.

Textbook References:

1. Alan C. Elliott and Wayne A. Woodward, SAS Essentials: Mastering SAS for Data Analytics, Wiley.
2. Lora D. Delwiche and Susan J. Slaughter, The Little SAS Book: A Primer, SAS Institute.
3. Graham Chapman, SAS Programming for Data Science, SAS Institute.
4. J. Walker, Data Analysis Using SAS Enterprise Guide, SAS Institute.
5. SAS Institute, SAS Macro Language 1: Essentials, SAS Institute.
6. R. J. M. Johnson and L. L. Johnson, Applied Statistics and the SAS Programming Language, Pearson.



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MBA III SEMESTER (AR23)
(Business Analytics Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306BA	DATA VISUALIZATION FOR MANAGERS	4	0	0	4

Course Description

This course is designed to equip professionals with the skills to effectively interpret and communicate data through visual means. This course focuses on leveraging data visualization tools and techniques to present complex data insights in a clear and actionable format. Participants will learn to use various visualization tools to create impactful charts, graphs, and dashboards, enhancing their ability to make data-driven decisions and communicate findings to stakeholders. The course will cover fundamental principles of data visualization, best practices, and hands-on experience with popular visualization tools and software. By the end of the course, managers will be able to translate data into compelling visual stories that drive strategic business decisions.

Course Objectives

1	Understand the principles of data visualization.
2	Use various tools to create charts, graphs, and dashboards.
3	Interpret and analyse complex data sets.
4	Design impactful visual representations of data.
5	Enhance data-driven decision-making skills.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Apply key data visualization principles in practice.
CO2	Create charts, graphs, and dashboards using various tools.
CO3	Analyse and interpret complex data sets effectively.
CO4	Design clear and impactful visuals.
CO5	Enhance decision-making through visual data insights.
CO6	Communicate findings effectively to stakeholders using visualizations.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	2	2	2	2	2
CO2	2	1	0	1	2	1	1	2
CO3	1	3	0	3	3	3	3	3
CO4	2	2	0	2	3	2	2	3
CO5	3	3	0	3	2	3	3	2
CO6	2	3	0	3	3	3	3	3
AVG	2.17	2.33	0.00	2.33	2.50	2.33	2.33	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit I: Introduction to Data Visualization

Principles of Data Visualization, Importance of Data Visualization in Business, Overview of Popular Data Visualization Tools (e.g., Tableau, Power BI, Excel), Types of Data Visualizations: Charts, Graphs, Dashboards, Best Practices for Effective Data Visualization.

Unit II: Visualization Tools and Techniques

Getting Started with Visualization Tools (Tableau, Power BI, Excel), Creating Basic Charts and Graphs (Bar, Line, Pie Charts), Customizing Visualizations: Colours, Labels, Legends, Advanced Chart Types: Heat maps, Scatter Plots, Tree maps, Introduction to Dashboards and Interactive Reports.

Unit III: Data Preparation and Analysis

Preparing Data for Visualization, Cleaning and Transforming Data, Aggregating and Summarizing Data, Identifying Key Metrics and KPIs, Techniques for Exploring and Analyzing Data.

Unit IV: Designing Impactful Visualizations

Storytelling with Data: Creating a Narrative, designing for Clarity and Impact, Using Visual Hierarchy and Emphasis, Balancing Aesthetics and Functionality, Case Studies of Effective Data Visualizations.

Unit V: Communicating Insights to Stakeholders

Presenting Data Insights: Tips and Techniques, Crafting Compelling Visual Stories, Tailoring Visualizations for Different Audiences, Effective Use of Dashboards in Decision-Making, Practical Applications and Hands-On Projects.

Textbook References:

1. Cole Nussbaumer Knaflic, Storytelling with Data: A Data Visualization Guide for Business Professionals, Wiley.
2. Steve Wexler, Jeffrey Shaffer, and Andy Cotgreave, The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios, Wiley.
3. Kieran Healy, Data Visualization: A Practical Introduction, Princeton University Press.
4. Stephen Few, Information Dashboard Design: Displaying Data for At-a-Glance Monitoring, Analytics Press.
5. Stephanie D. H. Evergreen, Effective Data Visualization: The Right Chart for the Right Data, SAGE Publications.
6. Ben Fry, Visualizing Data: Exploring and Explaining Data with the Processing Environment, O'Reilly Media.



SCHOOL OF MANAGEMENT STUDIES
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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Systems & IT Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302SIT	DATA MINING FOR BUSINESS DECISIONS	4	0	0	4

Course Description

This course equips students with the skills to apply data mining techniques to derive actionable insights from large datasets. Covering essential concepts like classification, clustering, association analysis, and anomaly detection, the course emphasizes practical experience with tools such as R and Python. Students learn data pre-processing, model building, and evaluation, applying these techniques to real-world business contexts in marketing, finance, operations, and customer relationship management. The course also addresses ethical considerations and best practices, preparing students to make informed, data-driven decisions and effectively communicate findings to stakeholders.

Course Objectives

1	Understand key data mining concepts and methodologies.
2	Gain hands-on experience with data mining tools and software.
3	Learn data preparation, cleaning, and preprocessing techniques.
4	Build, validate, and evaluate predictive data mining models.
5	Apply data mining techniques to real-world business problems.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Apply key data mining concepts and methodologies effectively.
CO2	Utilise popular data mining tools and software for practical data analysis.
CO3	Perform data pre-processing, cleaning, and transformation to prepare high-quality datasets.
CO4	Build, validate, and evaluate predictive models using various data mining algorithms.
CO5	Interpret and communicate data mining results to support business decision-making.
CO6	Address ethical considerations and apply best practices in data mining.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	0	0	0	3	3	3	2
CO2	2	3	0	3	2	2	2	0
CO3	3	2	0	2	1	2	3	3
CO4	3	3	0	2	3	3	2	3
CO5	3	1	0	3	3	3	2	3
CO6	3	3	0	3	3	0	2	2
AVG	2.83	2.00	0.00	2.17	2.50	2.17	2.33	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Data Mining

Kind of data, DM Functionalities, Classification of DM Systems, Issues in DM. Multidimensional data model: Data cubes, Stars, snowflakes and fact constellations Defining schemas, concept hierarchies, CLAP

Unit-II: Data Processing

Data pre-processing Why to pre-process data? Data cleaning: Missing Values, Noisy Data, Data Integration and transformation Data Reduction: Data cube aggregation, Dimensionality reduction. Data Compression, Numerosity Reduction Discretization and Concept Hierarchy Generation

Unit-III: Data Mining Tools Methods and Techniques

Data Mining Primitives, Languages and System Architectures: Task relevant data. Kind of Knowledge to be mined, DM Query languages: Syntax, Designing GUI. Architectures of DM Systems concept of Cluster Analysis- Regression and correlation; Classification- Decision trees; clustering –Neural networks; Market basket analysis- Association rules-Genetic algorithms and link analysis, Support Vector Machine, Ant Colony Optimization

Unit-IV: Basics of Business Decisions

Meaning and significance of Business decisions – Nature of problems requiring business decisions - Recurring/ Repetitive problems, Non-repetitive problems, Structured problems, Non structured problems - Types of Business Decisions – Strategic, Operational and Managerial Decisions – Characteristics of good business decisions - Business decision making process – Reengineering decision making process in a dynamic environment.

Unit-V: Data Mining Application for Business Decisions in Different Sectors

Application and trends in Data mining - Data Mining for Financial data analysis, Data Mining for retail industry, Data mining for telecommunication industry Applications in various sectors – Retailing, CRM, Banking, Stock Pricing, Production and Sales.

Textbook References:

1. David J. Hand, Heikki Mannila, and Padhraic Smyth, Principles of Data Mining, MIT.
2. Michel Berry and Gordon Linoff, Data mining techniques for Marketing, Sales and Customer Support, John Wiley.
3. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, and Vipin Kumar, Introduction to Data Mining, Pearson Education.
4. J. Han, M. Kamber, Morgan, Data Mining Concepts and Techniques, Kaufmann Publishers.
5. M. Kantardzic, Data mining: Concepts, Models, Methods and Algorithms, John Wiley & Sons Inc.
6. Michel Berry and Gordon Linoff, Mastering Data mining, John Wiley and Sons Inc.



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MBA III SEMESTER (AR23)
(Systems & IT Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-303SIT	MANAGING SOFTWARE PROJECTS	4	0	0	4

Course Description

Managing Software Projects is an MBA-level course designed to provide students with a comprehensive understanding of the principles and practices of software project management. This course covers the key aspects of project management including planning, scheduling, monitoring, controlling, and quality assurance, with a special focus on software development projects. Students will learn about various project management methodologies, tools, techniques, and standards essential for successful project execution.

Course Objectives

1	To grasp fundamental project management principles and their significance in the context of software development.
2	To master various techniques and tools for effective project planning and scheduling
3	To acquire skills for monitoring and controlling software projects through metrics and process improvements.
4	To understand and apply different software testing methods and the principles of software re-engineering.
5	To ensure adherence to quality standards through comprehensive software quality assurance practices.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the fundamental concepts and importance of project management in the context of software development.
CO2	Develop effective project plans and schedules using appropriate methodologies and tools.
CO3	Monitor and control software projects using metrics and process improvement techniques.
CO4	Apply various software testing techniques and understand the software re-engineering process.
CO5	Implement software quality assurance practices and adhere to relevant standards.
CO6	Utilize project management tools and techniques to ensure successful project execution and delivery.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	2	1	2	2	2
CO2	3	3	0	2	2	3	2	2
CO3	3	2	0	3	3	3	2	1
CO4	3	3	0	3	3	2	1	2
CO5	3	3	0	2	3	3	3	3
CO6	3	3	0	2	3	3	2	3
AVG	3.00	2.83	0.00	2.33	2.50	2.67	2.00	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Overview of Project Management

Project Management – Definitions; Importance, Factors Influencing Project Management - Project Manager, Project Management Activities, Stakeholders; Project Communication; Project Development Phases; Project Charter; Statement of Work (SoW); Project Management Associations.

Unit-II: Project Planning and Scheduling

Tasks in Project Planning; Work Breakdown Structures (WBS); Planning Methods; Development Life Cycle Models; Estimation and Budgeting of Projects: Software Cost Estimation; COCOMO Model; Budgeting. Project Scheduling: Scheduling Techniques - Program Evaluation and Review Technique (PERT), Gantt chart, Critical Path Method (CPM), Automated Tools.

Unit-III: Project Monitoring and Controlling

Project Status Reporting; Project Metrics; Earned Value Analysis (EVA); Project Communication Plan & Techniques; Steps for Process Improvement. Configuration Management: Software Configuration Management (SCM) - Baselines, Software Configuration Items (SCI); SCM Process; Identification of Objects in the Software Configuration; Version Control; Change Control; Configuration Audit; Status Reporting; Goals of SCM.

Unit-IV: Testing Techniques and Software Re-Engineering

Software Testing Concepts; Types of Software Testing – Manual Testing, Automated Testing; Black Box Testing; White Box Testing Techniques- Software Re-Engineering: Software Maintenance Problems; Redevlopment vs. Reengineering; Business Process Reengineering; Software Reengineering Process Model; Technical Problems of Reengineering.

Unit-V: Software Quality Assurance

Software Quality Assurance Activities; Software Qualities; Software Quality Standards – ISO Standards for Software Organization, Capability Maturity Model (CMM), Comparison between ISO 9001 & SEI CMM, Other Standards.

Textbook References:

1. Bob Hughes, Mike Cotterell, Rajib Mall, Software Project Management, McGraw Hill Education.
2. Robert K. Wysocki, Effective Project Management: Traditional, Agile, Extreme, Wiley.
3. Kathy Schwalbe, Information Technology Project Management, Cengage Learning.
4. Mickey W. Mantle, Ron Lichty, Managing the Unmanageable: Rules, Tools, and Insights for Managing Software People and Teams, Addison-Wesley Professional.
5. Pankaj Jalote, Software Project Management in Practice, Pearson Education.
6. Harold R. Kerzner, Project Management: A Systems Approach to Planning, Scheduling, and Controlling, Wiley.



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MBA III SEMESTER (AR23)
(Systems & IT Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304SIT	MANAGING DIGITAL INNOVATION AND TRANSFORMATION	4	0	0	4

Course Description

It is widely acknowledged that business leaders across all sectors are grappling with the strategic implications of digital transformation taking place in their organizations, industry ecosystems, and in the society. The economic, societal and business implications of digitalization are contested and raising serious questions about the wider impact of this kind of transformation. All private and public operations, including both those internally and externally oriented, are affected by digitalization. These organizations will increasingly acknowledge the need to access students with the right skills to analyse, understand and act upon the challenges of digitalization.

Course Objectives

1	Understand the types and significance of innovation as a core business process and measure its impact within a digital economy.
2	Explore digital infrastructures like the Internet, Cloud Computing, and IoT, and their implications for business model innovation.
3	Analyze the history, environment, and ethical considerations of online business and the evolution of new organizational forms.
4	Evaluate e-business models and strategies, focusing on B2B and B2C systems, supply chain management, and online consumer behavior.
5	Assess the changing patterns of global business operations, international trade, and the management of global information systems and human resources

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Comprehend the significance and measurement of innovation in a digital economy.
CO2	Investigate the impact of digital infrastructures on business model innovation.
CO3	Examine the history and ethical considerations of online business.
CO4	Evaluate e-business models and strategies for B2B and B2C systems.
CO5	Assess the evolving patterns in global business operations and international trade.
CO6	Manage global information systems and human resources in an international context.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	0	2	3	2	2	3
CO2	3	3	0	2	3	3	2	2
CO3	2	2	0	3	1	1	2	1
CO4	3	3	0	3	2	2	3	3
CO5	3	2	0	0	3	3	3	3
CO6	3	2	0	3	3	3	2	3
AVG	2.67	2.50	0.00	2.17	2.50	2.33	2.33	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Innovation

Meaning and need for Innovation - Kinds of Innovation - Innovation as a core business process - Sources of Innovation - Measuring Innovation - System innovation within a digital economy: Opportunities and challenges; identifying and maximizing the benefits of ubiquitous technologies within the business; the changing nature of systems: Web Apps, Mobile Apps.

Unit-II: Digital Infrastructures for Business

Internet - World Wide Web - Cloud Computing - Mobile Services - Social Media - Platforms - Business Model Innovation - Internet-of Things - Key characteristics - Applications challenges confronted by individuals, Companies and Governments.

Unit-III: On-Line Business

History and foundations of online business - E-business environment - legal, ethical and security issues - Lessons from the dot.com boom and bust - New organizational forms – virtual organizations, electronic markets and hierarchies - Social networks and web developments - Innovations involving e-business technologies; the role of open innovation in product and process development

Unit-IV: Managing E-Business Models

Business-to-Business (B2B) and Business-to-Consumer (B2C) business models and strategies for e-business - global supply chain management, electronic markets - B2B systems, intermediation, e-procurement and IT in supply chain management. - B2C strategies - online consumer behaviour, regional and cultural differences and e-marketing.

Unit-V: Innovative Transformation

Changing pattern in Business operations - Business across borders - International trade and investment –patterns, growth of and rationale for foreign direct investment. Global and regional integration and multilateral organizations - Organization of international business - Global sourcing of production and services - Global information systems management - International dimensions of human resources management - Global pay and incentive practices, managing expatriates and multinational workforces.

Textbook References:

1. Thomas M. Siebel, Digital Transformation: Survive and Thrive in an Era of Mass Extinction, Rosetta Books.
2. George Westerman, Didier Bonnet, and Andrew McAfee, Leading Digital: Turning Technology into Business Transformation, Harvard Business Review Press.
3. Paul R. Daugherty and James Wilson, Digital Innovation and Transformation: A Strategic Guide, MIT Press.
4. Dr. Ravi Kalakota and Marcia Robinson, Managing Digital Transformation: A Guide to the Strategic and Practical Aspects of Digital Transformation, Addison-Wesley.
5. Dr. Sanjeev Sinha, Digital Transformation: A Model to Master Digital Disruption, Bloomsbury India.
6. Dr. Anil Gupta, Digital Business Strategy and Value Creation: Big Data, Blockchain, and Competitive Advantage, Cambridge University Press.



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MBA III SEMESTER (AR23)
(Systems & IT Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305SIT	ENTERPRISE RESOURCE PLANNING	4	0	0	4

Course Description

This course provides a technical overview of Enterprise Resource Planning Systems and their impact on organizations. SAP is introduced to illustrate the concepts, fundamentals, framework, general information technology context, the technological infrastructure, and integration of business enterprise-wide applications.

Course Objectives

1	Comprehend the evolution, growth, and justification of ERP systems, with a focus on the Indian scenario.
2	Examine the structure and function of ERP modules, including their role in integrated management information and business modelling.
3	Explore the relationship between ERP and related technologies such as BPR, MIS, EIS, DSS, and SCM.
4	Assess and compare leading ERP vendors and their packages, understanding their market positions and module offerings.
5	Navigate the ERP implementation lifecycle, addressing key phases, challenges, and best practices for successful deployment and maintenance.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the evolution, growth, and significance of ERP systems, particularly in the Indian context.
CO2	Analyse the structure, functionality, and advantages of various ERP modules for integrated management information.
CO3	Examine the relationship and integration of ERP with related technologies such as BPR, MIS, EIS, DSS, and SCM.
CO4	Assess the ERP market by comparing leading vendors and evaluating their packages and modules.
CO5	Develop strategies for ERP implementation, including pre-evaluation screening, project planning, and gap analysis.
CO6	Navigate the ERP implementation lifecycle, focusing on configuration, team training, testing, going live, and post-implementation maintenance.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	2	3	3	2	2
CO2	3	2	0	2	2	3	2	2
CO3	3	3	0	3	2	2	3	2
CO4	2	0	0	2	2	3	3	1
CO5	2	3	0	2	3	3	3	3
CO6	2	2	0	3	3	3	3	3
AVG	2.50	2.00	0.00	2.33	2.50	2.83	2.67	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Introduction to ERP**

Introduction to ERP, Evolution of ERP, Reasons for the growth of ERP, Scenario and Justification of ERP in India, Evaluation of ERP, Various Modules of ERP, Advantage of ERP.

Unit-II: Overview of Enterprise

An overview of Enterprise, Integrated Management Information, Business Modeling, ERP for Small Business, ERP for make to order companies, Business Process Mapping for ERP Module Design, Hardware Environment and its Selection for ERP Implementation.

Unit-III: ERP and Related Technologies

ERP and Related Technologies, Business Process Reengineering (BPR), Management Information System (MIS), Executive Information System (EIS), Decision support System (DSS), Supply Chain Management (SCM).

Unit-IV: ERP Modules

ERP Modules, Introduction, Finance, Plant Maintenance, Quality Management, Materials Management, ERP Market, Introduction, SAP AG, Baan Company, Oracle Corporation, People Soft, JD Edwards World Solutions Company. A Comparative Assessment and Selection of ERP Packages and Modules.

Unit-V: ERP Implementation

ERP implementation lifecycle, issues in implementing ERP packages, pre-evaluation screening, package evaluation, project planning phase, gap analysis, reengineering, configuration, implementation, team training, testing, going live, end-user training, post implementation (Maintenance mode).

Textbook References:

1. Alexis Leon, Enterprise Resource Planning, McGraw Hill Education (India) Private Limited.
2. Rajesh Ray, Enterprise Resource Planning: Text & Cases, Tata McGraw-Hill Education.
3. Simha R. Magal and Jeffrey Word, Integrated Business Processes with ERP Systems, Wiley.
4. Thomas F. Wallace and Michael H. Kremzar, ERP: Making It Happen, Wiley.
5. Marianne Bradford, Modern ERP: Select, Implement, and Use Today's Advanced Business Systems, Lulu.com.
6. Mary Sumner, Enterprise Resource Planning, Pearson.



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MBA III SEMESTER (AR23)
(Systems & IT Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306SIT	MANAGING DIGITAL PLATFORMS	4	0	0	4

Course Description

The course provides a comprehensive overview of strategies and best practices for overseeing and optimizing digital platforms in today's tech-driven marketplace. Students will explore the key components of digital platforms, including their design, architecture, and functionality, as well as their role in enabling innovation and business growth. The course covers topics such as platform business models, user experience design, data management, and performance metrics. Students will learn to manage and scale digital platforms effectively, address challenges related to security and compliance, and leverage analytics to drive engagement and value creation. Students will develop the skills needed to navigate the complexities of digital platform management and harness their potential to achieve organizational objectives and competitive advantage.

Course Objectives

1	Understand the fundamentals and evolution of digital platforms
2	Explore the strategic and design aspects of managing digital platforms
3	Analyse the technological infrastructure and related technologies supporting digital platforms.
4	Gain insights into the operational management and module functionalities of digital platforms.
5	Learn about the lifecycle and challenges associated with digital platform implementation.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the basic concepts and historical evolution of digital platforms.
CO2	Analyse the strategic considerations and design principles of digital platforms.
CO3	Evaluate the role of related technologies such as BPR, MIS, EIS, and DSS in digital platforms.
CO4	Assess the functionalities of various digital platform modules like finance, plant maintenance, and quality management.
CO5	Develop a comprehensive plan for the implementation and management of digital platforms.
CO6	Identify and address the operational challenges and opportunities in managing digital platforms.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	2	1	2	1	2
CO2	3	3	0	2	2	3	2	2
CO3	3	2	0	3	3	3	2	2
CO4	3	3	0	3	3	2	2	2
CO5	3	3	0	2	3	3	3	2
CO6	2	2	0	2	3	3	2	3
AVG	2.83	2.67	0.00	2.33	2.50	2.67	2.00	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Digital Platforms

Introduction to Digital Platforms- Definition and significance of digital platforms in modern business environments- Overview of how digital platforms facilitate connectivity and transactions- Evolution of Digital Platforms- How technological advancements have shaped the growth of digital platforms- Reasons for the Growth of Digital Platforms- Scenario and Justification of Digital Platforms in India- Evaluation of Digital Platforms- Various Types of Digital Platforms- Advantages of Digital Platforms

Unit-II: Overview of Digital Platforms

An Overview of Digital Platforms- core components and structure of digital platforms- Integrated Platform Management Information- Business Modeling for Digital Platforms- Digital Platforms for Small Businesses- How digital platforms can be leveraged by small businesses- How digital platforms can be leveraged by small businesses- Benefits and challenges of using digital platforms- Hardware Environment and its Selection for Digital Platform Implementation-

Unit-III: Digital Platforms and Related Technologies

Digital Platforms and Related Technologies- Business Process Reengineering (BPR)- Principles and practices of business process reengineering in the context of digital platforms- Management Information System (MIS)- Role of MIS in supporting digital platform operations- Integration of MIS with digital platforms for enhanced data management- Executive Information System (EIS)- Decision Support System (DSS)- Supply Chain Management (SCM)

Unit-IV: Digital Platform Modules

Introduction to Digital Platform Modules- common modules in digital platforms- Importance of modular design in platform development- Finance Module- features and functions of the finance module in digital platforms- Plant Maintenance Module- Quality Management Module- Key components of the quality management module- Materials Management Module- Digital Platform Market Overview- Comparative Assessment and Selection of Digital Platform Packages and Modules

Unit-V: Digital Platform Implementation

Digital Platform Implementation Lifecycle- Stages of the digital platform implementation lifecycle, from planning to execution- Issues in Implementing Digital Platform Packages- Pre-Evaluation Screening- Package Evaluation- Project Planning Phase- Gap Analysis- Reengineering- Configuration and Implementation- Team Training and Testing- Going Live and End-User Training- Post-Implementation (Maintenance Mode)

Textbook References:

1. Geoffrey G. Parker, Marshall W. Van Alstyne, and Sangeet Paul Choudary, Platform Revolution: How Networked Markets Are Transforming the Economy - and How to Make Them Work for You, W.W. Norton & Company.
2. Michael A. Cusumano, Annabelle Gawer, and David B. Yoffie, The Business of Platforms: Strategy in the Age of Digital Competition, Innovation, and Power, Harper Business.
3. Amrit Tiwana, Platform Ecosystems: Aligning Architecture, Governance, and Strategy, Morgan Kaufmann.
4. Thomas M. Siebel, Digital Transformation: Survive and Thrive in an Era of Mass Extinction, Rosetta Books.
5. P. S. Ravindra, Digital India: Understanding Information Technology Policies and Strategies, Sage Publications India Pvt. Ltd.
6. R. K. V. Raghavan, Managing Digital Enterprise, McGraw-Hill Education.



SCHOOL OF MANAGEMENT STUDIES
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MBA III SEMESTER (AR23)
(Agribusiness Management Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-302ABM	AGRI-ECONOMICS	4	0	0	4

Course Description

This course offers a comprehensive examination of the economic principles and practices pertinent to the agriculture sector. Students will explore key topics such as agricultural production, market dynamics, farm management, and rural development, with a focus on how economic theories apply to agricultural systems and policies. The course covers issues like supply and demand in agricultural markets, pricing mechanisms, resource allocation, and the impact of government policies on farming and rural economies. Students will develop a nuanced understanding of the economic challenges and opportunities in agriculture, equipping them to make informed decisions and contribute to the advancement of sustainable and efficient agricultural practices.

Course Objectives

1	To understand the economic principles underlying agricultural production, consumption, and distribution.
2	To analyse the role of markets and market structures in agricultural systems.
3	To examine the impact of government policies and interventions on agriculture.
4	To explore the relationship between agriculture and the environment, including issues related to sustainable practices and resource management.
5	To develop analytical skills for evaluating economic problems and making informed decisions in agricultural contexts.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Applied economic principles to agriculture for effective decision-making.
CO2	Analyse agricultural markets and policies to assess their impact on stakeholders.
CO3	Evaluate trade dynamics and global implications on agricultural sectors.
CO4	Address sustainability concerns, integrating environmental and economic perspectives.
CO5	Develop strategies to manage risks associated with agricultural production and market volatility.
CO6	Evaluate economic analyses persuasively to diverse stakeholders.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	2	3
CO2	3	3	3	3	3	3	2	3
CO3	3	3	2	3	2	3	2	3
CO4	3	3	3	3	2	3	2	3
CO5	3	3	3	3	3	3	2	3
CO6	3	3	2	3	1	3	2	3
AVG	3.00	3.00	2.67	3.00	2.00	3.00	2.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Agriculture and Economic Development Nature and scope of agricultural and rural economics; Traditional agriculture and its modernization; Role of agriculture in economic development; Interdependence between agriculture and industry.
Unit-II: Diversification of Rural Economic Activities Livestock economics- Livestock resources and their productivity; White revolution; Fishery and poultry development; Forestry, horticulture and floriculture; Issues and problems in rural industrialization and development of agro-based industries.
Unit-III: Agricultural Production and Productivity Agricultural production- Resource use and efficiency; production function analysis in agriculture, factor combination and resource substitution; Cost and supply curves; Size of firm and laws of returns.
Unit-IV: Agricultural Prices Marketing and state policy; Agricultural markets and market efficiency Marketing functions and costs; Market structure and imperfections; regulated markets; marketed and marketable surplus; Crop insurance; Terms of trade between agricultural and non-agricultural prices; Need for state interventions; Objectives of agricultural policy- Instruments and evaluation; Food security in India and Public Distribution System.
Unit-V: Rural Savings and Capital Formation Characteristics and sources of rural credit institutional and non-institutional; Reorganization of rural credit- cooperatives, commercial banks, regional rural banks, Role of NABARD.
Textbook References: 1. R.G. Desai. Agricultural Economics: (Models, Problems & Policy Issues). Himalaya Publishing House. 2. Reddy S.S., P. Raghuram, T.V. Nilakantha Sastry., I. Bhavanidevi. Agricultural Economics, Oxford & IBH Publishing. 3. Nain S. S., Vinay Mehala. Introduction to Agricultural Economics, CBS Publishers and Distributors. 4. R.K. Lekhi & Joginder Singh, Agricultural Economics an Indian Perspective, Kalyani Publishers. 5. S K Nag & P K Verma, Fundamentals of Agricultural Economics, Science Technology; Jaya Publishing House. 6. Rajendra Kumar Rout, Basic Agricultural Economics, Kalyani Publishers.



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MBA III SEMESTER (AR23)
(Agribusiness Management Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-303ABM	AGRI-BUSINESS MANAGEMENT	4	0	0	4

Course Description

This course provides an overview of the principles and practices of managing agribusiness enterprises. It covers topics such as strategic management, marketing, finance, operations, human resource management, and risk management specific to the agricultural industry. Students will learn how to apply management concepts to various sectors within agribusiness, including production, processing, distribution, and retail.

Course Objectives

1	To understand the unique characteristics of agribusiness and its management challenges.
2	To learn the strategic planning process and its application to agribusiness enterprises.
3	To explore marketing strategies tailored to agricultural products and services.
4	To know financial management techniques to assess the financial performance of agribusiness firms.
5	To learn managerial operational aspects of agribusiness, including supply chain management and logistics.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Analyse strategic management principles for effective decision-making in agribusiness enterprises.
CO2	Devise marketing strategies tailored to agricultural products and services, considering value chain dynamics.
CO3	Apply financial management techniques to assess the financial performance and viability of agribusiness firms.
CO4	Manage operational aspects such as supply chain logistics and human resources within agricultural contexts.
CO5	Identify and mitigate risks associated with agribusiness operations, integrating sustainability principles.
CO6	Leverage technology and innovation to enhance productivity and competitiveness in agribusiness ventures.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	2	3
CO2	3	3	3	3	2	3	2	3
CO3	3	3	1	3	1	3	1	3
CO4	3	3	2	3	3	3	2	3
CO5	3	3	2	3	2	3	1	3
CO6	3	3	2	3	2	3	2	3
AVG	3.00	3.00	2.17	3.00	2.00	3.00	1.67	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Agriculture in Indian Economy

Role of agriculture in Indian economy; problems and policy changes relating to farm supplies, farm production, agro processing, economic reforms and Indian agriculture; impact of liberalization, privatization and globalization on Agri-business sector; Emerging trends in production, processing, marketing and exports

Unit-II: Agricultural Marketing

Agricultural marketing and economic development; Agricultural market structure – meaning, components and dynamics of market structure; marketing strategy - meaning & significance, formulation of marketing strategy; agribusiness marketing environment, design of marketing mix, market segmentation and targeting, determinants of consumer's behaviour.

Unit-III: Agricultural Finance

Importance, need and scope of financial management; classification and credit need in changing agriculture scenario; finance functions, investment financing; Working capital management – concept and components of working capital, need for working capital in agribusiness, Capital budgeting - steps and concept of capital budgeting; Agri-business financing system in India.

Unit-IV: Agricultural Information Systems

Scope of ICT in Indian agriculture; Agricultural information systems: design, testing, implementation and maintenance for developing agricultural information systems; current situation, challenges and opportunities in India.

Unit-V: Agricultural Input Marketing

Meaning and importance - Management of distribution channels for agricultural input marketing; Agricultural Inputs and their types - farm and non-farm; Role of cooperative, public and private sectors in agri-input marketing; Role of NSC and State Seed Corporation

Textbook References:

1. Acharya SS & Agarwal NL. Agricultural Marketing in India. Oxford & IBH.
2. Kohls RL & Uhj JN, Marketing of Agricultural Products., Prentice Hall.
3. James G. Beierlein, Kenneth C. Schneeberger and Donald D. Osburn, Principles of Agribusiness Management, Waveland Press.
4. Broadway, A. C. and Broadway, A Text Book of Agri-Business Management. Kalyani Publications.
5. Singh, A. K. and Pandey, S., Rural Marketing. New Age.
6. Singh, S., Rural Marketing- Focus on Agricultural Inputs. Vikas Publ. House.



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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Agribusiness Management Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-304ABM	FOOD RETAIL MANAGEMENT	4	0	0	4

Course Description

This course provides an in-depth examination of the principles and practices involved in managing food retail businesses. It covers various aspects of food retail management, including retail strategy, merchandising, inventory management, customer service, pricing strategies, and store operations. Students will explore the unique challenges and opportunities within the food retail industry and develop strategies for effective management in this competitive market.

Course Objectives

1	To understand the dynamics of the food retail industry, including market trends, consumer behaviour, and competitive forces.
2	To analyse retail strategies and concepts tailored to food retail businesses, including format selection, location planning, and store design.
3	To develop effective merchandising strategies to optimize product assortment, layout, and presentation in food retail stores.
4	To know various pricing strategies and promotional tactics to drive sales and profitability in food retail operations.
5	To Identify emerging trends and innovations in food retail management and adapt strategies accordingly.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Proficient in inventory control and management techniques specific to food retail environments.
CO2	Apply marketing and merchandising principles to create appealing displays and enhance customer engagement.
CO3	Implement operational strategies to optimize efficiency, minimize waste, and enhance overall productivity.
CO4	Adhere food safety regulations and best practices to maintain product quality and ensure consumer safety.
CO5	Utilize technology and data analytics tools to analyse consumer behaviour, forecast demand, and make informed business decisions.
CO6	Develop strategies for sustainability and ethical sourcing practices in food retail operations to meet evolving consumer preferences and environmental concerns.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	3	2	3
CO2	3	2	2	2	3	3	2	3
CO3	3	3	1	3	3	3	2	3
CO4	3	2	1	2	1	3	2	3
CO5	3	2	1	2	3	3	2	3
CO6	3	3	1	3	3	3	2	3

AVG	3.00	2.33	1.50	2.50	2.50	3.00	2.00	3.00
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to Food Retailing Introduction to international food market, India's competitive position in world food trade, Foreign investment in global food industry, Retail management and food retailing, the nature of change in retailing, organized retailing in India E-Retailing and understanding food preference of Indian consumer, food consumption and expenditure pattern, Demographic and psychographic factors affecting food pattern of Indian consumer.								
Unit-II: Value Chain Analysis in Food Retailing Value chain in food retailing, principal trends in food wholesaling and retailing, the changing nature of food stores, Various retailing formats, competition and pricing in food retailing, market implications of new retail developments, Value chain and value additions across the chain in food retail, food service marketing.								
Unit-III: Pricing and Distributing System in Food Retailing 4 P's in food retail management, brand management in retailing, merchandise pricing, Pricing strategies used in conventional and non-conventional food retailing, Public distribution system, promotion mix for food retailing.								
Unit-IV: Promotional Operations in Food Retailing Management of sales promotion and publicity, advertisement strategies for food retailers, Managing retail operations, managing retailers' finances, Merchandise buying and handling, merchandise pricing, logistics, procurement of food products and handling, transportation of food products.								
Unit-V: Retail Sales, CRM and Ethical Issues Retail sales management, types of retail selling, salesperson selection, training, evaluation and monitoring, Customer relationship management, managing human resources in retailing, Legal and ethical issues in retailing.								
Textbook References: 1. Berman & Evans, Retail Management: A Strategic Approach, Prentice Hall of India. 2. Cox, Retailing: An Introduction, Pearson Edu. 3. Levy, M. & Weitz, B. W., Retailing Management, McGraw Hill. 4. Samarendra Mahapatra, Food Retail Management, Kalyani Publications 5. H. M. Chandrashekar, Food Retail Management, LAP Lambert Academic Publishing 6. Russell J. Zwanka, Food Retail Management Strategic Cases, Createspace Independent Publicaions.								



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Odisha, Gunupur
MBA III SEMESTER (AR23)
(Agribusiness Management Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-305ABM	MANAGEMENT OF AGRO BUSINESS COOPERATIVES	4	0	0	4

Course Description

This course offers a detailed examination of the principles and practices essential for the effective management of cooperatives within the agricultural sector. Students will explore the structure, governance, and operational dynamics of agro-business cooperatives, focusing on how they can enhance productivity, profitability, and sustainability for member farmers. The course covers key topics such as cooperative formation, member engagement, financial management, strategic planning, and conflict resolution. Students will learn to address the unique challenges faced by agricultural cooperatives, implement best practices for efficient management, and leverage cooperative models to achieve collective goals and foster rural development.

Course Objectives

1	To understand the concept and significance of agro-business cooperatives in the agricultural sector.
2	To analyse the organizational structure, governance, and legal frameworks governing agricultural cooperatives.
3	To explore various management functions including planning, organizing, staffing, directing, and controlling within agro-business cooperatives.
4	To identify and assess the challenges and opportunities faced by agro-business cooperatives in today's dynamic business environment.
5	To develop practical skills in cooperative management through case studies, simulations, and real-world projects.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Adhere principles and governance structures specific to agro-business cooperatives.
CO2	Apply strategic planning skills tailored to the unique challenges of agricultural cooperatives.
CO3	Navigate legal and regulatory frameworks governing agro-business operations.
CO4	Manage financial aspects including budgeting, analysis, and risk mitigation within agricultural cooperatives.
CO5	Develop marketing strategies and distribution channels optimized for agricultural products and cooperatives.
CO6	Cultivate ethical leadership and teamwork abilities essential for sustainable cooperative management.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	2	3	1	3	1	3
CO2	3	1	2	3	2	3	3	3
CO3	3	1	2	1	2	2	2	3
CO4	3	1	0	2	1	3	3	3
CO5	3	2	1	2	1	3	3	3

CO6	2	0	3	0	3	3	3	3
AVG	2.67	1.00	1.67	1.83	1.67	3.00	2.50	3.00
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to Cooperation Importance and principles of cooperation, Cooperative movement in India, Cooperative administration- a global perspective, Ecology of cooperative administration, cooperative sector and economic development.								
Unit-II: Cooperative Management Meaning and Definition of Cooperative management, Nature, functions and purpose of cooperatives, Procurement, storage, processing cooperatives, Role of leadership in cooperative management								
Unit-III: Cooperative Marketing Cooperative Marketing meaning and definition, Marketing Mix Applied To Cooperatives, Identification of Areas of Improvement, Cooperative Marketing Redevelopment Proposal Writing, The state and cooperative movement, Effects of cooperative law in management								
Unit-IV: HRM in Agricultural Cooperatives Long range planning for cooperative expansion, policy making Human Resource Management (HRM) - Placement and role of board of directors in cooperative management								
Unit-V: Agribusiness Cooperative Overview of agribusiness cooperative- credit cooperatives, cooperative marketing, dairy cooperatives and financing agribusiness cooperative								
Textbook References: 1. S. Nakkiran, and M. Karthikeyan, Cooperatives and agribusiness, Discovery Publishing. 2. Armstrong Emmanuel Ogidi, Agribusiness and Cooperative Management Paperback, LAP LAMBERT Academic Publishing. 3. Akmat, J. S., New Dimensions of Cooperative Management. Himalaya Publishing House. 4. Ansari, A. A., Cooperative Management Patterns. Anmol Publications 5. Sah, A. K., Professional Management for the Cooperatives. Vikas Publishing House. 6. Freddie Barnard, Jay Akridge, Frank Dooley and John Foltz, Agribusiness Management, Routledge.								



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MBA III SEMESTER (AR23)
(Agribusiness Management Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-306ABM	AGRIBUSINESS FINANCIAL MANAGEMENT	4	0	0	4

Course Description

This course provides an in-depth exploration of financial principles and practices tailored specifically for the agribusiness sector. Students will learn to manage financial resources effectively within agricultural enterprises, focusing on key areas such as budgeting, financial forecasting, investment analysis, and risk management. The course covers financial strategies for optimizing farm operations, evaluating capital projects, and securing financing, with a particular emphasis on the unique financial challenges and opportunities in agriculture. Students will gain the skills to analyse financial statements, make data-driven decisions, and develop strategies that enhance the financial health and sustainability of agribusinesses.

Course Objectives

1	To understand the financial landscape of agribusiness, including the influence of market dynamics, government policies, and environmental factors.
2	To analyse financial statements and performance indicators to assess the financial health and profitability of agribusiness enterprises.
3	To develop skills in budgeting, forecasting, and financial planning to optimize resource allocation and mitigate financial risks.
4	To explore investment analysis techniques and decision-making processes related to capital investments in agricultural assets and projects.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Evaluate the financial health and profitability of agribusiness enterprises through the analysis of financial statements and performance indicators.
CO2	Develop effective budgeting, forecasting, and financial planning strategies to optimize resource allocation and mitigate financial risks in agribusiness operations.
CO3	Maintain proper financial reporting system for effectiveness of business.
CO4	Prepare summary sheets of overall financial performance of business.
CO5	Prepare cash flow statement for effective utilisation of cash.
CO6	Identify and assess various sources of financing available to agribusinesses, and analyse their suitability and implications for business sustainability and growth.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	3	2	3	3
CO2	3	3	0	3	3	2	3	3
CO3	3	3	0	3	3	2	3	3
CO4	3	3	0	3	3	2	3	3
CO5	3	3	0	3	3	2	3	3
CO6	3	3	0	3	3	2	3	3
AVG	3.00	3.00	0.00	3.00	3.00	2.00	3.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Financial Aspects of Business

Managing an agribusiness, Business planning, Importance, need and scope of financial management, Classification and credit, need in changing agriculture scenario- finance functions, investment financing.

Unit-II: Financial Recording System

Assets, liabilities and owner's equity - Income and expenditure - Meeting tax obligations - Management control and analysis - Setting up a cash record - The cash record format - Column headings - Recording transactions - Bank transactions - Loan accounts - Overdraft - Term loan - Off-farm investments - Debtors accounts - Creditors accounts - Totalling and checking the cash records - Bank reconciliation.

Unit-III: End-of-year Summaries

Debtors (amounts owed to the business) - Creditors (amounts owed by the business) - Produce, supplies and feed inventories - Livestock inventories - Improvements - Plant and machinery - Statement of assets and liabilities.

Unit-IV: Cash Flow Management, Monitoring and Budget

Cash flow statement - Monitoring cash flow performance - Budget analysis - End-of-year cash flow statement - Cash flow budget - Preparation of the cash flow budget - Budget revision.

Unit-V: Assessment of Business Equity

Statement of assets and liabilities - Valuation of land and fixed improvements - Calculation of owner's equity - Provision for taxation - Owner's equity and net worth - The purpose of the statement of assets and liabilities.

Textbook References:

1. WJ Obst, R Graham, G Christie, Financial Management for Agribusiness, CSIRO Publishing
2. Nelson, A. G & Murrey, W. G., Agricultural Finance. Kalyani Publ.
3. Chandra, P., Financial Management. Tata McGraw Hill.
4. Khan, M. Y. & Jain, P. K., Management Accounting. Tata McGraw Hill.
5. Pandey, I. M., Financial Management. Vikas Publ. House.
6. Verma/Banafar, Agri Business and Finance Management, Narendra Publishing House



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Odisha, Gunupur

MBA AR23

IV SEMESTER COURSE STRUCTURE AND SYLLABUS

(Dual Specialisation) Students should choose any Two Specialisations

List of Specialisations

- | | |
|---|---|
| 9. Marketing
10. Finance
11. Human Resource
12. Rural Management | 13. Operations & Supply Chain Management
14. Business Analytics
15. Systems & IT
16. Agribusiness Management |
|---|---|

S. No.	Course Category	Course Code	Type	Course Title	L	T	P	C	Marks		Total Marks
									CIA	ESE	
1	MS	MB400D	AEC02	Dissertation	0	0	0	3	-	100	100
2	MS	MB401C	CC18	Strategic Management	3	0	0	3	40	60	100
Choose <u>any Three Electives from the Specialisation</u> as chosen earlier in the third Semester											
3	MS	Elective I	DSE03	See Table: IV for Semester Specialisation Elective Courses	4	0	0	4	40	60	100
4	MS	Elective II	DSE03		4	0	0	4	40	60	100
5	MS	Elective III	DSE03		4	0	0	4	40	60	100
Choose <u>any Three Electives from the Specialisation</u> as chosen earlier in the third Semester											
6	MS	Elective IV	DSE04	See Table: IV for Semester Specialisation Elective Courses	4	0	0	4	40	60	100
7	MS	Elective V	DSE04		4	0	0	4	40	60	100
8	MS	Elective VI	DSE04		4	0	0	4	40	60	100
IV Semester Total					27	0	0	30	280	520	800

Table: IV Semester Specialisation Elective Courses

S. No.	Course Code	Course Title
Marketing Specialisation (For Semester IV)		
1	MB-E-402MM	Salesforce Business Analyst-II
2	MB-E-403MM	Digital Marketing & Social Media
3	MB-E-404MM	Advertising & Brand Management
4	MB-E-405MM	Integrated Marketing Communication
5	MB-E-406MM	Service Marketing
Finance Specialisation (For Semester IV)		
1	MB-E-402FIN	Financial Risk Management
2	MB-E-403FIN	Project Finance
3	MB-E-404FIN	Direct Taxation
4	MB-E-405FIN	Financial Derivatives
5	MB-E-406FIN	Mergers & Acquisitions
HR Specialisation (For Semester IV)		
1	MB-E-402HR	Advanced Manpower Planning
2	MB-E-403HR	Organisational Development and Change
3	MB-E-404HR	Performance Management System
4	MB-E-405HR	Leadership & Change Management
5	MB-E-406HR	Training and Development
Rural Management (For Semester IV)		
1	MB-E-402RM	Agripreneurship
2	MB-E-403RM	Agro Processing, Packing and Quality Control
3	MB-E-404RM	Financial Management in Rural Perspective
4	MB-E-405RM	Operations Management in Rural Perspective
5	MB-E-406RM	Human Resource Management in Rural Perspective
Operations & Supply Chain Management (For Semester IV)		
1	MB-E-402OSM	Service Operations Management
2	MB-E-403OSM	Green Supply Chain Management
3	MB-E-404OSM	Supply Chain Analytics
4	MB-E-405OSM	Supply Chain Information System

5	MB-E-406OSM	Management of Manufacturing Systems
Business Analytics (For Semester IV)		
1	MB-E-402BA	Finance and Risk Analytics
2	MB-E-403BA	Machine Learning and Applications
3	MB-E-404BA	Marketing & Retail Analytics
4	MB-E-405BA	Web and Social Media Analytics
5	MB-E-406BA	Data Mining
Systems & IT (For Semester IV)		
1	MB-E-402SIT	Cyber Security
2	MB-E-403SIT	Business Intelligence
3	MB-E-404SIT	Decision Support System
4	MB-E-405SIT	Database Management Systems
5	MB-E-406SIT	Information System Control and Audit
Agribusiness Management (For Semester IV)		
1	MB-E-402ABM	Food Technology and Processing Management
2	MB-E-403ABM	Agri Import and Export Management
3	MB-E-404ABM	E-Commerce and Agri-Business
4	MB-E-405ABM	Agri-Supply Chain Management
5	MB-E-406ABM	International Trade in Agriculture



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MBA IV SEMESTER (AR23)
Ability Enhancement Course-2

Course Code	Course Title	L	T	P	C
MB400D	DISSERTATION	0	0	0	3

Course Description

There is an increasing importance of ability enhancement courses in this two years Master of Business Administration programme (MBA). In this course, a student will show his/her ability to prepare a 10,000-15,000 words dissertation report on specific topic pertaining to the domain area of knowledge. The dissertation report will be comprising of theoretical data, analysis, findings and conclusion on specific problem of a business or industry. In this course, the student will be able to interpret the theoretical knowledge about the business or industry and prepare a case study. This dissertation report will enhance the subject awareness and reasonableness on the chosen topic.

Course Objectives

1	To discuss on the various business problems across the globe.
2	To collect the information from various sources of data.
3	To examine importance of the problems and try to solve through proper investigation.
4	To analyse the sources of information using various statistical tools.
5	To develop the theoretical knowledge on the business problem and suggest the measures for reformation.

Course Outcomes
(Upon successful completion of this course, students should be able to)

CO1	Demonstrate the usage of the dissertation to solve the business problems and society at large.
CO2	Investigate the literature gap from the existing academic research articles, conference proceedings, books, periodicals etc.
CO3	Examine the hypothetical relevance of business problems through statistical tools.
CO4	Investigate the primary and secondary source of information.
CO5	Analyse the data and provide some suggestion to solve the business problem.
CO6	Describe the findings and conclusion for the future researcher.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	1	2	2
CO2	2	3	3	2	3	3	2	2
CO3	3	3	3	2	3	2	0	3
CO4	3	3	2	2	3	3	1	3
CO5	3	2	2	2	3	1	2	3
CO6	3	3	2	2	1	1	1	3
AVG	2.83	2.67	2.33	2.00	2.50	1.83	1.33	2.67

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

About dissertation:

The dissertation is a specific project report that solve the business problem. It comprises of primary and secondary information on domain knowledge. The dissertation particularly covers a

business/industry related problem that can be resolved through proper investigation or examination. Initially the student researchers should select a topic in his/her interest reciprocating with the business/industry related problems. The student researcher should collect all the theoretical information from various secondary sources such as journals, periodicals or books. After this literary information, the researcher should define the literature gap to understand the relevance of the study. Based on the gap analysis, the student researcher should define the objectives of the study, hypothesis formulation, research methodology or findings to solve the case.

Stages of dissertation:

1. Introduction
2. Review of literature
3. Theoretical framework
4. Research Methodology
5. Findings, Suggestions, Conclusion
6. Bibliography
7. Questionnaire

The overall dissertation should have a clear methodological approach that help the student researcher for further investigation. The dissertation will help the student on case study analysis, statistical knowledge and justification of the study. The dissertation should contain all the chapters includes the references. The entire dissertation should be around 10,000-15,000 words content. The formatting style, line space and other relevant information are instructed at the beginning of the regulation.

Textbook References:

7. C. R. Kothari and Gaurav Garg, Research Methodology: Methods and Techniques. New Age International Publishers.
8. Ranjit Kumar, Research Methodology: A Step-By-Step Guide for Beginner. Sage Publications Pvt. Ltd.
9. Panneerselvam, R., Research methodology. PHI Learning Pvt. Ltd.
10. Verma, R., Verma, S., & Abhishek, K., Research methodology. Books clinic Publishing.
11. Chawla, D., & Sodhi, N., Research methodology: Concepts and cases. Vikas Publishing House.
12. U. Flick, introducing research methodology: A beginner's guide to doing a research project. Sage Publications.



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MBA IV SEMESTER (AR23)

Course Code	Course Title	L	T	P	C
MB401C	STRATEGIC MANAGEMENT	3	0	0	3

Course Description

The Strategic Management course delves into the formulation, implementation, and evaluation of cross-functional decisions that enable an organization to achieve its long-term objectives. This course provides a comprehensive framework for understanding how firms gain and sustain competitive advantage in dynamic environments, emphasizing strategic planning, competitive analysis, and strategic decision-making processes. Students will explore case studies, engage in strategic simulations, and apply theoretical models to real-world scenarios, equipping them with the skills to navigate complex business landscapes, anticipate market trends, and drive organizational success.

Course Objectives

1	Be familiar with the concepts of Strategic management and describe the basic model of strategic management.
2	Know the Industry analysis, Competitor Analysis and Organizational Analysis to gain Competitive Advantage.
3	Realize the Strategy Formulation framework to arrive at strategic choice/formulation.
4	Comprehend the formulation of Corporate, Business and Functional Strategies.
5	Appreciate the various aspects of Strategy Implementation and Control for accomplishing desired results

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Describe strategic management and present its model through the examination of vision, mission and objectives of an organisation.
CO2	Illustrate industry & competitive analysis as well as resource-based view of the firm to assess the competitive position of the firm.
CO3	Generate alternative strategies by aligning key external and internal factors using appropriate techniques.
CO4	Evaluate the directional strategies in a typical organization and build Competitive advantage through business and functional level strategies.
CO5	Generate strategic decisions need to be made on sound judgment.
CO6	Explain various issues of strategy implementation and evaluate Strategic performance using strategic measures.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	2	2	3	2	3	3	2
CO2	3	2	2	2	2	2	2	3
CO3	2	3	2	3	3	2	3	2
CO4	3	2	3	2	3	3	2	2
CO5	2	3	2	3	2	2	3	2
CO6	2	3	2	3	2	3	2	3
AVG	2.3	2.5	2.1	2.6	2.3	2.5	2.5	2.3

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Basic Concepts of Strategic Management

Strategy, Strategic Decisions, Strategic Management, Evolutionary Phases of Strategic Management, Benefits of Strategic Management, Basic Model of Strategic Management, Pitfalls in Strategic Planning, Levels of Strategic Management, Vision, Mission and Objectives-Strategy competitive advantage.

Unit-II: Strategic Analysis

Environmental scanning: Components of the external environmental analysis, Identifying External Environmental Variables. Industry analysis: Industry Life Cycle Analysis, Porters Five Force Model, Strategic Groups. Competitor Analysis: Analysing Market Commonality and Resource Similarity, Competitive Intelligence, Monitoring Competitors for Strategic Planning. Organizational Analysis and Competitive Advantage: Resources, Capabilities, and Core Competencies, Building Core Competencies through VRIO framework and Value Chain Model.

Unit-III: Strategy Formulation

Strategy Formulation framework, SWOT Matrix, Boston Consulting Group (BCG) Matrix, GE Matrix, The Grand Strategy Matrix-Blue ocean strategy, Resource based strategy formulation.

Unit-IV: Formulation of Corporate, Business and Functional Strategies

Corporate Level Strategies: Stability Strategies, Growth Strategies: Intensive Strategies - Integration Strategies - Diversification Strategies, Defensive Strategies: Turnaround -Divestment - Liquidation. Business Level Strategies: Michael Porter's Generic Strategies: Cost Leadership - Differentiation - Focus. Functional Level Strategies: Marketing, Finance, Production, R&D, Human Resources, Logistics, Information Technology.

Unit-V: Strategy Implementation and Control

Issues of Strategy Implementation, Strategy – Structure relationship, Types of Organizational Structure for Strategy Implementation: Functional, Divisional, SBU, Matrix and Network/Virtual structures. Strategic leadership, Strategy supportive culture, Strategic change management, Strategy Control, Balanced Score card approach.

Textbook References:

1. Thomas L. Wheelen, J.David Hunger, Alan N. Hoffman and Charles E. Bamford Concepts in Strategic Management and Business Policy, Pearson Education.
2. Fred R. David and Forest R. David, Strategic Management: Concepts and Cases-A Competitive Advantage Approach, Pearson Education
3. Collis, D. J. & Montgomery, C. A. Creating corporate advantage, Harvard Business Review
4. Dr. F.C. Sharma, Strategic Management for M.Com & MBA, Mahavir Publications.
5. [Frank T. Rothaermel](#), Strategic Management, McGraw Hill India
6. Dr. Ravi Kumar Sharma and Dr. Smita Verma Principles of Strategic Management, Sahitya Bhawan Publications.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Marketing Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402MM	SALESFORCE BUSINESS ANALYST-II	4	0	0	4

Course Description

This course equips students with the skills and knowledge necessary to effectively leverage Salesforce for business analysis and decision-making. Students will explore key concepts such as Salesforce CRM functionality, data analysis, and reporting, focusing on how to gather and interpret business requirements to drive system configurations and enhancements. The course covers topics including workflow automation, custom object creation, and user experience optimization, with practical training on using Salesforce tools and features to analyse business processes and improve operational efficiency. Through hands-on exercises and real-world case studies, students will develop the expertise to serve as a bridge between technical teams and business stakeholders, ensuring that Salesforce implementations align with organizational goals and deliver measurable value.

Course Objectives

1	Learn to visualize and analyse business processes using process mapping techniques and tools.
2	Acquire skills to plan, test, and deploy changes in Salesforce using version control and collaboration tools.
3	Understand UX design principles, create key UX artefacts, and use feedback to design effective interfaces.
4	Develop strategies for managing Salesforce releases, including feature prioritization and communication.
5	Gain proficiency in Salesforce data management, including data import/export and handling large data volumes.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Create detailed business process maps to analyse and improve organizational processes.
CO2	Plan and execute Salesforce changes from development through deployment, ensuring smooth transitions and minimal disruptions.
CO3	Design and implement user experience solutions by developing UX artefacts and integrating user feedback for enhanced usability.
CO4	Prioritize and test Salesforce features effectively, and communicate new features and changes to stakeholders and executives.
CO5	Optimize data management in Salesforce by designing robust data models and handling data import/export and large data volumes efficiently.
CO6	Utilize Git and GitHub to manage version control and facilitate team collaboration throughout the development lifecycle.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	3	2	2	2	2
CO2	3	3	3	2	2	3	3	3

CO3	2	2	2	2	2	2	2	2
CO4	3	2	2	3	3	2	2	2
CO5	3	3	3	2	3	3	3	3
CO6	2	2	1	2	2	2	3	2
AVG	2.67	2.33	2.17	2.33	2.33	2.33	2.50	2.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Relationship Building & Change Set Developmental Model

Relationship Building: Respond to People with Empathy and build skilled relationships Process Mapping for Business Analyst: Get started with Process mapping, Speak the process mapping language, Use Process Mapping tools and Software.

Business Process Mapping: Understand the value of Business Analysis, Understand Universal Process notation, Create a Business Process map, See Business Process Mapping in Practice. User Experience Basics: Learn about User Experience Design, Define Project Scope and Requirements, Create UX Artifacts, Create Tools and Request Feedback.

Unit-II: Change Set Developmental Model

Plan for Changes to your Org, Develop and Test Changes locally, Test the Integration environment, and deploy changes.

Git and GitHub Basics: Learn why version control is important for Term Based Development, Learn where GitHub fits in the Development Cycle, Discover Collaboration within GitHub, Work with GitHub workflow, Work with Teams in GitHub, Work with your history in Git User

Story Creation: Learn about user stories, Construct a user story Salesforce Devops with Copado: Get Started with DevOps, Create value Stream maps, Learn about version control and agile planning, support collaboration with continuous integration, Let innovation flow with continuous delivery; Use Acceptance Testing.

Unit-III: Advanced Salesforce Release Readiness Strategies

Master the release readiness process, Select and prioritize features, Test new features before a release, and Communicate new features to users

Strategies to Present Executive Audiences: Outline your presentation, Craft your project presentation, present to executives.

Project Management Essentials: Get to know Project Management fundamentals, Explore Project management methodologies, Use Agile for non-software projects, Deliver value through project management.

Data Modelling: Optimize customer data with standard and custom objects, create Object Relationships, Work with Schema Builder Data Management: Import Data, Export Data **Large Data Volumes:** Design your Data Model, Connect Data Queries and searches, Load our Data, Perform Data Deletes and Extracts Duplicate Management: Improve Data Quality in Salesforce, Resolve and Prevent duplicate data in Salesforce

Unit-IV: Salesforce Platform Basics

Salesforce Platform Basics: Unlock Business with Salesforce Data Platform, Discover use cases for the Platform, Understand the Salesforce Architecture, Navigate setup, Powerup with AppExchange CRM Analytics

Data Integration Basics: Kick off the Data Journey, Extract External Data into CRM analytics, Extract Salesforce Data into CRM Analytics, Prepare your data.

Unit-V: Project

Import and Export with Data Management Tools: Use the Data Import Wizard, Use Dataloader.io to Export Data, Use Dataloader.io to update data Project : Customize an Org to Support a New Business Unit: Manage User Access, Manage Chatter, Modify your Data Model, Configure an Email Letterhead and Template, Automate your Business Process

Textbook References:

1. **Benioff, Marc, and Carlye Adler,** Behind the Cloud: The Untold Story of How Salesforce.com Went from Idea to Billion-Dollar Company-and How You Can, Too, Wiley.

2. **Greenberg, Paul**, CRM at the Speed of Light: Essential Customer Strategies for the 21st Century, McGraw-Hill Education.
3. **Harris, David**, Salesforce.com for Dummies, Wiley.
4. **Powers, Tim**, Mastering Salesforce DevOps: A Practical Guide to Achieving Continuous Delivery, Apress.
5. **Fitzgerald, Brian, and James K. McHugh**, Business Process Management: Practical Guidelines to Successful Implementations, Springer.
6. **Siddiqui, A. N.**, Effective Salesforce Development: A Complete Guide to Building and Deploying Custom Applications, Packt Publishing.



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Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Marketing Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403MM	DIGITAL MARKETING & SOCIAL MEDIA	4	0	0	4

Course Description

The era of digital environment and digital convergence has opened up new opportunities for marketing. The shift in the consumer behaviour from brick to click is posing new challenges for the marketers. The digital revolution has created opportunity for innovation and empowered customer to digital convergence. The goal of digital marketing is to understand the accessibility of customers to digital media and how it created avenues to investigate the mind-set of online consumers.

Course Objectives

1	To discuss on the comprehensive digital marketing strategy.
2	To investigate the use of search engines for optimizing the visibility and search results.
3	To examine and implement Social Media Marketing Strategies.
4	To create, analyse and send promotional content through E-mails and Mobiles.
5	To develop and apply the measurement techniques for evaluating the digital marketing efforts.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate the knowledge about the designing, developing and promoting through Blogs.
CO2	Investigate the Use the search engines for better visibility in the online world.
CO3	Examine the creativity, novelty and functions of social networking sites.
CO4	Investigate the use of YouTube/e-mail / Mobile marketing for promotional activities.
CO5	Analyse the use of Google analytics for identifying, filtering and tracking the day to day business activities in the organisation.
CO6	Demonstrate the knowledge on digital marketing, social media, networking and their applications to develop the case studies for solving the business problems.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	2	2	2
CO2	2	2	3	2	3	3	1	2
CO3	3	2	3	2	2	2	2	1
CO4	3	2	1	1	2	2	1	1
CO5	2	2	1	3	2	1	1	1
CO6	3	3	2	2	1	1	1	1
AVG	2.67	2.33	2.17	2.17	2.17	1.83	1.33	1.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Online Marketing Foundations

Digital marketing strategy - exploring digital marketing - starting with a website - user experience design (UXD) - user interface design (UI), Content marketing - foundations of content marketing - creation of content plan - content creation and promotion - measuring content effectiveness -

designing and developing blogs - creating, promoting and measuring blog content - using newsletter in content - using photos in content marketing.

Unit-II: Search Engine Optimization

An introduction to search engine marketing (SEM) - Enhancing the organic search preferences, keyword allocation for improving website's search ability - Pay per click marketing - Online Reputation Management to improve SEM - Enhancing the organic search preferences - Keyword allocation for improving website's search ability - Pay per click marketing - Online reputation; Management to improve search engine marketing - keywords as the foundation of SEO - managing keyword data - link building strategies - measuring SEO effectiveness - local search - Paid search - creating an advertisement - pay per click advertising - researching keywords - creating a campaign.

Unit-III: Social Media Marketing

Understanding social media marketing - building online community - growing an online community - getting started with twitter - tweeting on twitter - building presence on twitter - getting started with face book - marketing on face book - building your presence on face book - measuring your efforts - understanding integral and advertising on integral - developing LinkedIn's business strategy - finding audience - engaging audience.

Unit-IV: Marketing YouTube, E-Mail, Mobile

Understanding YouTube - monetizing with YouTube partner program - building an audience - customizing the channel - advanced video optimization on YouTube - using YouTube analytics, Email Marketing - Basic components of Email marketing - Designing emails to enhance Email Marketing - Viral email marketing campaigns - email marketing tools and setup - understanding of how email fits into a marketing plan - managing the audience - audience engagement, Mobile Marketing - understanding mobile marketing - Alternative mobile access technologies: platforms, services, devices and speeds - Consumer and corporate applications: content, marketing, sales promotion, ticketing, participation/interaction - Location-based services (Proximity marketing), SMS/ MMS, GPRS based services and marketing - Mobile messaging based Marketing - mobilizing web presence - SMS campaign and building mobile applications.

Unit-V: Web Analytics

Usage of Google Analytics - Implementation of Google Analytics - understanding analytics core concepts - setting up goals - using different types of reports, Affiliate marketing foundations - fundamental affiliate marketing principles - preparations for affiliate marketing - rolling out affiliate efforts.

Textbook References:

1. Godfrey Parkin, Digital Marketing: Strategies for Online Success by, New Holland Publishers.
2. Ian Dodson, The art of Digital Marketing: The Definitive Guide to Creating Strategic, Targeted, and Measurable Online Campaigns, Wiley.
3. Chris Tread away and Mari Smith. Face book Marketing: An Hour a Day, Sybex.
4. Puneet Singh Bhatia, Fundamentals of Digital Marketing. Pearson.
5. A. Charlesworth, An introduction to social media marketing. Routledge.
6. L. M. Mahoney & T. Tang, Strategic social media: From marketing to social change. John Wiley & Sons.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Marketing Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404MM	ADVERTISING & BRAND MANAGEMENT	4	0	0	4

Course Description

Advertising is the dissemination of information concerning an idea, product or service that induces the need in customer, leading to the sale of the product. Advertising is used for communicating business information to the present and prospective customers. Advertising Management is the process of employing various media to sell a product or service. This process begins quite early from marketing research and encompasses the media campaign that helps to sell the product. In the modern age of large scale production, producers cannot think of pushing sale of their products without advertising. Brand plays a very important role in influencing the minds of the customer. Brand not only creates a distinction from the other competitor brands but also adds value to customer by possessing it. Branding and maintaining brand equity play significant role in marketing of products.

Course Objectives

1	To discuss on concepts of Advertising and the role of Ad Agencies.
2	To investigate the significance of creativity in development of advertisements.
3	To examine and understand the planning and evaluation of media strategy.
4	To Identify and Establish Brand Positioning.
5	To develop and design marketing programmes to enhance brand equity and manage brands over time.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate the concepts and trends in Advertising, examine the methods of budgeting and specify the services of Ad agencies.
CO2	Investigate on the various styles of execution strategies of creative Ad appeals to target customers and develop major selling ideas.
CO3	Analyse and develop right media strategy to achieve advertising objectives.
CO4	Discuss the basics of branding, brand positioning strategies and investigate the in depth importance of marketing mix strategy to leverage brand equity.
CO5	Analyse the influence of different aspects of branding strategies to grow and sustain brand equity.
CO6	Demonstrate the knowledge on advertising management, brand management, brand equity and brand portfolios to analyse and discuss the case study to establish a sustainable communication.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	2	2	2
CO2	3	3	3	2	3	3	1	2
CO3	3	2	3	2	2	2	2	1
CO4	3	2	2	2	3	2	2	2
CO5	2	2	2	3	2	2	2	2
CO6	3	3	2	2	1	1	1	1

AVG	2.83	2.50	2.50	2.33	2.33	2.00	1.67	1.67
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit- I: Advertising Concepts								
Advertising: Concept, types and objectives of advertising, social, ethical and economic aspects of advertising; Advertising budgets: Factors influencing budgeting decisions, methods of budgeting Advertising Agencies: types, services of Ad Agencies; Trends in Advertising: Social media, Mobile, Banner & Video, Search and Display Advertising.								
Unit-II: Creativity in Advertising								
Creativity: Importance of creativity, creativity process; Creativity Implementation and Evaluation: Advertising appeals, Execution styles; Creative Strategy Development: Ad campaigns, Creative brief, major selling ideas Creative tactics: In print and electronic media.								
Unit-III: Market Analysis & Developing Media Strategies								
Market analysis: Analysis of market, Company's marketing strategy, Creative strategy Establishing Media objectives; Developing and Implementing Media strategies: The Media Mix, Target Market Coverage, Geographic Coverage, Scheduling, creative aspects, Reach and Frequency, flexibility, budget considerations; Evaluation of Media: Merits and demerits of Broadcast, Print and Support Media.								
Unit-IV: Branding & Brand Equity								
Definition, Benefits of branding, Inputs for branding, Model (David Aker's) to develop brand strategy, Brand management process. Brand Positioning: Definition and importance of positioning. Positioning Strategy: Target market, defining competition, Establishing Points of difference and Points of parity, Updating positioning overtime. Brand Equity: Meaning, Relevance of brand equity to business, Brand elements to build brand equity. Designing marketing programme (Product, Price, Distribution channel and IMC strategy) to build brand equity.								
Unit-V: Brand Portfolio								
Brand Portfolio: Meaning and Objectives of portfolio, developing portfolio, Challenges in brand portfolio management Brand Architecture: Models of brand architecture, Developing brand architecture, Indicators of brand architecture issues. Product and Brand Extensions: Advantages of Extensions, Disadvantages of Brand Extensions, Brand Extension Guidelines Leveraging Secondary Brand Associations: Company, Country of origin, Channels of Distribution, Co-Branding, Licensing, Celebrity Endorsement, and Events.								
Textbook References:								
1. Prof. Rajpal Yadav, Advertising and Brand Management, Laxmi Publications Pvt. Limited. 2. Larry D. Kelley, Kim Bartel Sheehan, Advertising Media Planning: A Brand Management Approach, Routledge 3. Vibha Mathur and Saloni Arora, Brand Management. PHI Learning. 4. Kenneth Clowand and Donald Black, Integrated Advertisements, Promotion and Marketing Communication. Prentice Hall of India. 5. George Belch and Michael Belch, Advertising and Promotion: An Integrated Marketing Communications Perspective, McGraw-Hill College. 6. George E. Belch, Michael A. Belch and Keyoor Purani. Advertising and Promotion, McGraw-Hill College.								



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Marketing Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405MM	INTEGRATED MARKETING COMMUNICATION	4	0	0	4

Course Description

This course covers key aspects of marketing communication and advertising, focusing on Integrated Marketing Communication (IMC), advertising agencies, and media planning. Students will explore IMC planning models, agency types and selection, media strategies, and direct marketing. The course also examines international advertising and internet advertising, including B2B communication and industrial advertising. By integrating theory with practical insights, students will understand how to develop effective marketing communication strategies.

Course Objectives

1	Understand Integrated Marketing Communication (IMC) principles and their role in marketing.
2	Identify and assess different types of advertising agencies and their services.
3	Develop and implement effective media plans across various channels.
4	Analyze and apply direct marketing strategies and tools.
5	Examine international advertising and internet advertising impacts.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Design Integrated Marketing Communication (IMC) strategies tailored to various business needs.
CO2	Select and evaluate advertising agencies based on their services and alignment with campaign goals.
CO3	Develop and execute comprehensive media plans across print, broadcast, and digital platforms.
CO4	Implement direct marketing tactics effectively and assess their impact on target audiences.
CO5	Analyse international advertising strategies and adapt them to diverse global markets.
CO6	Create and manage internet advertising campaigns to optimize digital marketing efforts.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	2	3	2	3	2	2
CO2	3	2	2	1	2	1	2	3
CO3	2	3	1	3	3	2	3	2
CO4	1	2	2	2	3	3	2	3
CO5	3	2	3	3	2	2	3	2
CO6	2	3	2	3	2	3	2	3
AVG	2.17	2.17	2.00	2.50	2.33	2.33	2.33	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Role of IMC in marketing process

Role of IMC in marketing process, IMC planning model, Marketing and promotion process model. Communication process, steps involved in developing IMC programme, Effectiveness of marketing communications Purpose, Role, Functions, Types, Advertising Vs Marketing mix, Advertising appeal in various stages of PLC.

Unit-II: Advertising Agency

Type of advertising agencies, Services offered by various agencies, Criteria for selecting the agencies and evaluation. Advertising objectives and Budgeting: Goal setting - DAGMAR approach, various budgeting methods used.

Unit-III: Media planning

Developing Media plan, Problems encountered, Media Evaluation Print, Broadcast media, Support media in advertising. Media strategy: Creativity, Elements of creative strategies and its implementation, Importance of Headline and body copy.

Unit-IV: Direct Marketing

Features, Functions, Growth, Advantages/ Disadvantages, and Direct Marketing Strategies. Promotion: Meaning, Importance, tools used, Conventional/unconventional, drawbacks, push pull strategies, Co-operative advertising, Integration with advertising and publicity Public relation/ Publicity: Meaning, Objectives, tools of public relations, Public relation strategies, Goals of publicity, Corporate Advertising - Role, Types, Limitations, PR Vs Publicity.

Unit-V: International Advertising

Global environment in advertising, Decision areas in international advertising Internet advertising: Meaning, Components, Advantages and Limitations, Types of Internet advertising Industrial advertising: B2B Communication, Special issues in Industrial selling.

Textbook References:

1. Belch, G. E., & Belch, M. A., Advertising and Promotion: An Integrated Marketing Communications Perspective, McGraw-Hill Education.
2. Keller, K. L., Strategic Brand Management: Building, Measuring, and Managing Brand Equity, Pearson.
3. Shimp, T. A., Advertising, Promotion, and Other Aspects of Integrated Marketing Communications, Cengage Learning.
4. Clow, K. E., & Baack, D., Integrated Advertising, Promotion, and Marketing Communications, Pearson.
5. Jobber, D., & Ellis-Chadwick, F., Principles and Practice of Marketing, McGraw-Hill Education.
6. Baker, M. J., & Hart, S. J., The Marketing Book, Routledge.



SCHOOL OF MANAGEMENT STUDIES
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MBA IV SEMESTER (AR23)
(Marketing Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406MM	SERVICE MARKETING	4	0	0	4

Course Description

This course provides an in-depth exploration of strategies and techniques tailored specifically for marketing services rather than tangible products. Students will study key concepts such as the intangibility, inseparability, variability, and perishability of services, and how these characteristics influence marketing strategies. The course covers topics including service design, service quality management, customer experience enhancement, and relationship marketing. Students will learn to develop effective marketing plans, create compelling service propositions, and leverage customer insights to build strong brand loyalty. This course aims to equip students with the tools and strategies necessary to successfully market services in competitive environments, drive customer satisfaction, and achieve business growth.

Course Objectives

1	Develop service design and blueprinting strategies using technology.
2	Apply the seven Ps of service marketing to create effective service strategies.
3	Manage service demand and capacity through segmentation and internal marketing.
4	Identify and resolve service quality gaps to enhance customer satisfaction.
5	Implement marketing strategies for various service sectors including financial, health, and hospitality services.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Design and implement service blueprints to enhance service delivery processes.
CO2	Apply the seven Ps framework to develop comprehensive service marketing strategies.
CO3	Optimize service capacity and demand through effective segmentation and internal marketing techniques.
CO4	Identify and address service quality gaps to improve customer satisfaction and service performance.
CO5	Develop targeted marketing strategies for financial, health, hospitality, professional, public utility, and educational services.
CO6	Evaluate and improve service communication to bridge the promise versus delivery gap.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	2	3	2	3	2	2
CO2	3	2	2	3	2	2	2	3
CO3	2	3	1	3	3	2	3	2
CO4	3	3	2	2	3	3	2	1
CO5	3	2	3	3	2	2	3	2
CO6	2	3	2	3	2	3	2	3
AVG	2.50	2.33	2.00	2.83	2.33	2.50	2.33	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Marketing Services Introduction growth of the service sector. The concept of service. Characteristics of service - classification of service designing of the service, blueprinting using technology, developing human resources, building service aspirations.
Unit-II: Marketing Mix in Service Marketing The seven Ps: product decision, pricing strategies and tactics, promotion of service and distribution methods for services. Additional dimension in services marketing- people, physical evidence and process.
Unit-III: Effective Management of Service Marketing Marketing demand and supply through capacity planning and segmentation - internal marketing of services - external versus internal Orientation of service strategy.
Unit-IV: Delivering Quality Service Causes of service - quality gaps. The customer expectations versus perceived service gap. Factors and techniques to resolve this gap. Customer relationship management. Gaps in services - quality standards, factors and solutions – the service performance gap - key factors and strategies for closing the gap. External communication to the customers- the promise versus delivery gap - developing appropriate and effective communication about service quality.
Unit-V: Marketing of Service with Special Reference to Financial services, Health services, Hospitality services including travel, hotels and tourism, Professional service, Public utility service, Educational services.
Textbook References: 1. Zeithaml, V. A., Bitner, M. J., & Gremler, D. D., Services Marketing: Integrating Customer Focus Across the Firm, McGraw-Hill Education. 2. Lovelock, C., & Wirtz, J., Services Marketing: People, Technology, Strategy, Pearson. 3. Parasuraman, A., Zeithaml, V. A., & Berry, L. L., Delivering Quality Service: Balancing Customer Perceptions and Expectations, Free Press. 4. Berry, L. L., & Parasuraman, A., Service Quality Management in Service Sector, John Wiley & Sons. 5. Hoffman, K. D., & Bateson, J. E., Services Marketing: Concepts, Strategies, & Cases, Cengage Learning. 6. Shostack, L., Designing Services That Deliver, Harvard Business Review Press.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
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MBA IV SEMESTER (AR23)
(Finance Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402FIN	FINANCIAL RISK MANAGEMENT	4	0	0	4

Course Description

This course delves into the identification, analysis, and mitigation of financial risks within organizations. Students will explore various types of financial risks, including market, credit, liquidity, and operational risks, and study methods for measuring and managing these risks through advanced analytical techniques. The course covers key concepts such as risk assessment frameworks, hedging strategies, and the use of financial instruments like derivatives to manage exposure. Students will develop the skills to implement effective risk management practices, make informed decisions to safeguard financial stability, and create strategies that align with organisational goals. This course aims to equip students with the knowledge and tools necessary to navigate the complexities of financial risk and ensure sound financial decision-making.

Course Objectives

- 1 Understand the nature and types of financial risks faced by organizations.
- 2 Apply quantitative and qualitative tools to measure and manage financial risks.
- 3 Utilize derivatives and other financial instruments for hedging risk.
- 4 Develop risk management strategies that align with organizational goals.
- 5 Conduct stress testing and scenario analysis to evaluate financial risk exposure.

Course Outcomes

(Upon successful completion of this course, students should be able to)

- | | |
|-----|---|
| CO1 | Evaluate various financial risks and their impact on organizational performance and stability. |
| CO2 | Apply quantitative tools and models to measure and manage market, credit, liquidity, and operational risks. |
| CO3 | Design effective risk management strategies using financial instruments, including derivatives. |
| CO4 | Implement regulatory frameworks and compliance measures within financial institutions. |
| CO5 | Analyse case studies to identify key risk factors and management failures during financial crises. |
| CO6 | Develop integrated risk management plans incorporating emerging technologies like AI and machine learning. |

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	3	2	2
CO2	3	2	2	2	2	2	2	3
CO3	2	3	1	3	3	2	3	2
CO4	1	2	2	2	3	3	2	3
CO5	3	2	3	3	2	2	1	2
CO6	3	3	2	3	3	2	2	3
AVG	2.33	2.50	2.17	2.50	2.50	2.33	2.00	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit I: Introduction to Financial Risk Management

Overview of Financial Risks: Market, Credit, Liquidity, and Operational Risks, Importance of Risk Management in Financial Institutions, The Risk Management Process: Identification, Measurement, Mitigation, Monitoring, and Reporting, Case Studies on Major Financial Crises and the Role of Risk Management.

Unit II: Market Risk

Understanding Market Risk: Definition and Sources, Measuring Market Risk: Value at Risk (VaR), Expected Shortfall, and other Quantitative Methods, Hedging Market Risk: Use of Derivatives (Futures, Options, Swaps), Case Studies: Managing Market Risk in Practice.

Unit III: Credit Risk

Introduction to Credit Risk: Key Concepts and Types, Measuring Credit Risk: Credit Scoring Models, Credit Ratings, and Credit VaR, Managing Credit Risk: Credit Derivatives, Collateral Management, and Credit Default Swaps (CDS), Basel III Framework: Credit Risk Capital Requirements, Case Studies: Credit Risk Management in Financial Institutions.

Unit IV: Liquidity and Operational Risk

Liquidity Risk: Causes, Measurement, and Management, Stress Testing for Liquidity Risk: Approaches and Applications, Operational Risk: Identifying and Managing Non-Financial Risks, Regulatory Requirements: Basel II and Basel III Operational Risk Management Guidelines, Case Studies: Liquidity and Operational Risk Management in Practice.

Unit V: Advanced Risk Management Strategies

Integrated Risk Management: Enterprise Risk Management (ERM), Capital Allocation and Risk-Adjusted Performance Measures (RAPM), Scenario Analysis and Stress Testing: Tools and Techniques, Regulatory Environment: Dodd-Frank Act, Basel Accords, and Other Regulations, Emerging Trends in Financial Risk Management: Fintech, AI, and Machine Learning Applications, Case Studies: Holistic Risk Management Strategies in Leading Firms.

Textbook References:

1. Hull, J. C., Risk Management and Financial Institutions, Wiley.
2. Jorion, P., Value at Risk: The New Benchmark for Managing Financial Risk, McGraw-Hill.
3. Allen, S. L., Financial Risk Management: A Practitioner's Guide to Managing Market and Credit Risk, Wiley.
4. Crouhy, M., Galai, D., & Mark, R., The Essentials of Risk Management, McGraw-Hill.
5. Basel Committee on Banking Supervision., Basel III: A global regulatory framework for more resilient banks and banking systems. Bank for International Settlements.
6. Tarantino, A., & Cernauskas, D., Risk Management in Finance: Six Sigma and Other Next-Generation Techniques. Wiley.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Finance Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403FIN	PROJECT FINANCE	4	0	0	4

Course Description

This course provides a comprehensive understanding of the financial structuring and management of large-scale projects. Students will explore key concepts such as project valuation, risk assessment, financial modelling, and funding strategies, focusing on how to secure and manage financing for capital-intensive projects. The course covers various financing mechanisms, including debt, equity, and public-private partnerships, and examines the financial, legal, and regulatory considerations involved in project finance. Students will gain the skills to analyse financial feasibility, develop detailed financial plans, and effectively manage project financials to ensure successful project completion and long-term sustainability. This course is designed to equip students with the expertise needed to navigate the complexities of project finance and drive successful outcomes in diverse project environments.

Course Objectives

1	Understand the core concepts, including the distinction between project finance and corporate finance, and the role of special purpose vehicles.
2	Learn about the structuring of financing deals, including debt and equity components, and the implications of various financing structures on project risk and return.
3	Delve into risk identification, assessment, and mitigation strategies, focusing on financial, operational, and political risks.
4	Gain practical skills in creating detailed financial models that project cash flows, profitability, and financial viability of projects.
5	Examine the legal and regulatory considerations that influence project finance, including contract law, taxation, and environmental regulations.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Describe the stages of the project life cycle, from initiation to closure, and understand the progression through these phases.
CO2	Assess the financial viability and profitability of a project, using various evaluation techniques to determine potential returns and risks.
CO3	Comprehend different types of contracts, the 3R's of contracting (Roles, Responsibilities, and Relationships), and the process of tendering and contract selection.
CO4	Develop and use project procedure manuals to standardize processes and ensure consistency in project execution.
CO5	Develop and manage project reporting systems to communicate progress, issues, and results to stakeholders effectively.
CO6	Outline the roles and responsibilities of a project manager and assess their impact on project success.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	2	3	3	3	3

CO2	2	3	1	3	3	3	3	2
CO3	2	2	1	3	3	3	3	3
CO4	2	2	1	3	2	3	2	2
CO5	2	3	2	3	2	3	3	2
CO6	2	3	2	2	3	3	3	2
AVG	2.17	2.50	1.17	2.67	2.67	3.00	2.83	2.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Concepts of Project Management

Project Management Concept, Categories of Projects, Project Life cycle, Tools and Techniques of Project Management, Roles and Responsibilities of Project Manager.

Unit-II: Establishing the Project

Feasibility Report - Financial aspects of Project Preparation, Technical aspects, Finalization of Project Implementation Schedule, Evaluation of Project Profitability.

Unit-III: Contracting

Accountability in Project Execution, Contracts, 3'R's of Contracting, Tendering and Selection of Contracts, Team Building.

Unit-IV: Project Implementation

Project Execution Plan, Project Procedure Manual, Project Control System, Project Scheduling and Monitoring, Monitoring Contracts.

Unit-V: Project Evaluation & Reporting

Project Evaluation, Project Review Meetings, Project Reporting Systems, Closing the Contract, Project Extensions, Role of Project Management Professional.

Textbook References:

1. S. Choudhury, Project Management, Tata McGraw Hill Publishing Company Ltd.
2. Harvey Maylor, Project Management, Pearson Education Ltd.
3. Jeffrey K.Pinto, Project Management Achieving Competitive Advantage, Pearson Education.
4. Dr. Pradip Kumar Sinha, Project Appraisal, Planning and Control, Excel Books.
5. Rajendra Mishap, Project Management, Excel Books.
6. Stefano Gatti, Project Finance in Theory and Practice: Designing, Structuring, and Financing Private and Public Projects, Academic Press.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Finance Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404FIN	DIRECT TAXATION	4	0	0	4

Course Description

This course provides a comprehensive overview of the key principles, structures, and applications of taxation systems. Participants will gain a deep understanding of direct taxes, such as income tax and corporate tax, including their impact on individuals and businesses, as well as the methodologies for tax planning and compliance. This course also covers GST filing and focusing on their implementation, administration, and effects on economic transactions. Through detailed lectures, case studies, and practical exercises, students will develop the skills to navigate complex tax regulations, optimize tax strategies, and ensure compliance with evolving tax laws.

Course Objectives

1	Gain a clear grasp of the essential definitions, concepts, and regulations outlined in the Income-tax Act of 1961.
2	Learn to accurately compute taxable income across various heads and taxable calculations.
3	Develop skills to assess the taxable income of companies and apply tax planning strategies for corporate restructuring events such as amalgamations, mergers, and demergers.
4	Explore the concepts and principles behind the GST and gain insights into GST registration, tax rates, and administrative procedures both in India and globally.
5	Analyse the contributions of direct and indirect taxation to the Indian economy, and role of taxation as a governmental tool for economic and social governance.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Elucidate the concept of residential status and its impact on tax incidence, and outline the important dates and forms related to income tax filing.
CO2	Identify and explain various heads of income, including salaries, income from house property, business or profession, capital gains, and other sources.
CO3	Perform calculations for taxable income, including determining surcharges, marginal relief, and deductions.
CO4	Explicate Minimum Alternative Tax (MAT), Dividend Distribution Tax (DDT), and tax implications for amalgamations, mergers, and demergers.
CO5	Explain the processes for GST registration, filing, and the tax rates applicable in India and internationally.
CO6	Analyse the role of tax as a government levy and the principles of General Anti-Avoidance Rules (GAAR) in preventing tax avoidance.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	0	2	3	3
CO2	3	3	0	3	0	2	3	3
CO3	3	3	0	3	0	2	3	3
CO4	3	3	0	3	0	2	3	3
CO5	3	3	0	3	0	2	3	3

CO6	3	3	0	3	0	2	3	3
AVG	3.00	3.00	0.00	3.00	0.00	2.00	3.00	3.00
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction								
Income-Tax Law: An introduction - Important definitions in the Income-tax Act, 1961 - Concept of previous year and assessment year - Basis of Charge and Rates of Tax - Residential Status and Incidence of Tax - Income Tax Important Dates and Forms. Individual Income Exempted from Tax.								
Unit-II: Heads of Income								
Heads of Income: Salaries - Income from House Property - Profits & Gains from Business or Profession - Capital Gains - Income from Other sources - Clubbing of Incomes. Calculation of Tax: Calculation of Taxable Income - Tax Calculation including Surcharge and Marginal relief - Deduction - Rebate, Relief - Set-off & Carry Forward of Losses - Principles - Meaning - Inter-sources & Inter-head Set-off.								
Unit-III: Corporate Tax								
Assessment of Companies: Computation of taxable income - Carry-forward and set-off of losses for companies - Minimum Alternative Tax (MAT) - Dividend Distribution Tax (DDT) - Set-off and Carry-forward of Amalgamation Losses - Tax Planning for Amalgamation - Merger and Demerger of Companies - Tax Provisions for Venture Capital Funds - Other special provisions relating to companies								
Unit-IV: Introduction to GST								
GST Concepts: GST as the preferred Tax Structure - Model of GST - Need for Tax Reforms - GST Principles - Single GST - Dual GST - Transactions covered under GST - Impact of GST. Registration and Filing: Rates of Tax - Rates in Foreign Countries - In India - Assessment and Administration of GST.								
Unit-V: Taxation & Governance								
Contribution: Contribution of direct and indirect taxation to the Indian economy, tax evasion and tax avoidance, tax as a levy for the Government's enabling role and GAAR								
Textbook References: 1. Dr. Vinod K. Singhania & Dr. Monica Singhania Students Guide to Income Tax, Taxmann Publication, Latest Edition according to assessment year. 2. Dr. B. K. Agarwal & Dr. Rajeev Agarwal Tax Planning and Managemen, Nirupam Publication, Latest Edition according to assessment year. 3. Paolo M. Panteghini Corporate Taxation in a Dynamic World, Springer. 4. A. Srinivas Corporate Tax Planning, Tata McGraw Hill. 5. Lal B. B., Direct Taxes, Pearson Education. 6. Datey V.S., Indirect Taxes-Law & Practice, Taxmann Publication								



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Finance Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405FIN	FINANCIAL DERIVATIVES	4	0	0	4

Course Description

This course provides an in-depth exploration of financial instruments used to manage risk and speculate in the markets. Students will delve into the fundamentals of derivatives, including forwards, futures, options, and swaps, examining their pricing, valuation, and strategic applications. The course covers essential concepts such as hedging, arbitrage, and risk management techniques, to illustrate the use of derivatives in various financial contexts. By the end of the course, participants will have a solid understanding of how to leverage derivatives for financial strategy and risk control, equipping them with the skills to navigate complex financial environments effectively.

Course Objectives

1	Gain a comprehensive knowledge of various financial derivatives, including forwards, futures, options, and swaps, and their fundamental characteristics, uses, and market mechanics.
2	Learn to accurately price and value derivative contracts using key models and methods, such as the Black-Scholes model for options and the concept of no-arbitrage pricing for futures and swaps.
3	Develop practical skills in designing and implementing hedging strategies to manage financial risks and protect portfolios against adverse price movements using derivatives.
4	Explore advanced risk management techniques involving derivatives, including the assessment of risk exposures, the use of derivatives for portfolio optimization, and the application of strategies to mitigate financial risks.
5	Analyse real-world case studies and scenarios to understand the application of derivatives in various financial contexts, including speculation, arbitrage, and the impact of market conditions on derivative performance.

Course Outcomes
(Upon successful completion of this course, students should be able to)

CO1	Describe the development and growth of derivative markets, including the fundamental linkages between spot and derivative markets.
CO2	Analyse the factors influencing the determination of forward and future prices, understanding their impact on trading strategies.
CO3	Apply option pricing models, including the Binomial Model and the Black-Scholes Merton Model, to evaluate options.
CO4	Develop and apply trading strategies using options to capitalize on market movements and manage risk.
CO5	Design and execute hedging strategies using currency options to mitigate risk in foreign exchange exposures.
CO6	Use swaps to manage various types of financial risks, including interest rate, currency, and commodity risks.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3

CO1	3	3	0	3	0	1	3	2
CO2	3	3	0	3	0	1	3	2
CO3	3	3	0	3	0	1	3	2
CO4	3	3	0	3	0	1	3	2
CO5	3	3	0	3	0	1	3	2
CO6	3	3	0	3	0	1	3	2
AVG	3.00	3.00	0.00	3.00	0.00	1.00	3.00	2.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Derivatives

Introduction: Development and Growth of Derivative Markets - Types of Derivatives Fundamental Linkages between Spot & Derivative Markets - The Role of Derivatives Market - Uses & Misuses of Derivatives.

Commodity Exchanges: Role - Functions - Trading - National and Regional Exchanges.

Unit-II: Future and Forward Market

Structure of Forward and Future Markets: Mechanics of Future Markets - Hedging Strategies Using Futures - Determination of Forward and Future Prices.

Types of Futures: Interest rate Futures - Currency Futures and Forwards.

Unit-III: Options

Options Market: Distinguish between Options and Futures - Structure of Options Market - Principles of Option Pricing.

Option Pricing Models: The Binomial Model - The Black-Scholes Merton Model.

Unit-IV: Option Strategies

Strategies: Basic Strategies - Advanced Strategies - Trading with Options - Hedging with Options - Currency Options.

Unit-V: Swaps

SWAP: Concept - Nature - Features - Evolution of Swap Market.

Major Types of Swaps: Interest Rate Swaps - Currency Swaps - Commodity Swaps - Equity Index Swaps -Credit Risk in Swaps - Credit Swaps.

Managing Risk: Using Swaps to Manage Risk - Pricing and Valuing Swaps.

Textbook References:

1. John C Hull: Options, Futures and other derivatives, Pearson.
2. M. Ranganatham & R. Madhumathi : Derivatives and Risk Management, Pearson
3. S. L. Gupta: Financial Derivatives, PHI.
4. Dubofsky, Miller: Derivatives Valuations and Risk Management, Oxford.
5. Don M. Chance, Robert Brooks: Derivatives and Risk Management Basic, Cengage.
6. Sundaram Das, Derivatives – Principles and Practice, McGraw Hill.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Finance Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406FIN	MERGERS & ACQUISITIONS	4	0	0	4

Course Description

This course which explores the core concepts of mergers & acquisitions and corporate restructuring and the challenges encountered in implementing them. Beginning with the conceptual framework of corporate restructuring, the course goes on to discuss takeovers and M&A, the concept and process of due diligence and legal issues in M&As. The key issues relating to valuation and accounting will be explained. This course will also be discussing the post-merger issues, the human aspects of M&As and cross-border acquisitions

Course Objectives

1	Gain a comprehensive grasp of the principles, processes, and terminology related to mergers and acquisitions, including strategic motivations, types of deals, and regulatory considerations.
2	Learn to evaluate and determine the fair value of target companies using various valuation methods such as discounted cash flow (DCF), comparable company analysis, and precedent transactions.
3	Develop skills in conducting thorough due diligence, including financial, operational, and legal assessments, to identify risks and opportunities associated with potential M&A transactions.
4	Explore effective strategies for post-merger integration, focusing on aligning organizational structures, cultures, and systems to achieve the synergies and goals of the acquisition.
5	Assess the success of mergers and acquisitions by analysing performance metrics, measuring integration effectiveness, and understanding the impact on shareholder value and strategic objectives.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the processes involved in mergers and acquisitions, including strategic planning, due diligence, and integration.
CO2	Identify and analyse barriers to successful corporate restructuring, including organizational and operational challenges.
CO3	Evaluate different expressions of corporate value, including their interrelationships and applications in valuation processes.
CO4	Formulate and analyse strategies for entering new markets, using tools like SWOT analysis and other strategic frameworks.
CO5	Develop and apply strategies for mergers and acquisitions, including considerations for cross-border acquisitions and their associated challenges and benefits.
CO6	Apply concepts of financial leverage and risk to leveraged buyouts, including understanding associated theories and exit strategies.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	0	3	2	3	3	3

CO2	3	3	0	3	2	3	3	3
CO3	3	3	0	3	2	3	3	3
CO4	3	3	0	3	2	3	3	3
CO5	3	3	0	3	2	3	3	3
CO6	3	3	0	3	2	3	3	3
AVG	2.83	3.00	0.00	3.00	2.00	3.00	3.00	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Understanding Mergers & Acquisitions

Introduction, Meaning of Merger, amalgamation, acquisition, takeover. Types of Mergers, reverse merger, Motives and Benefits of Mergers and Acquisitions, divestiture, de merger, Diversification etc., Reasons for failure of M & As. Process of M&A

Unit-II: Corporate Restructuring

Introduction, Corporate Restructuring - Meaning, types. Causes of Corporate Restructuring. Barriers of Restructuring, Key elements of Restructuring Process and Strategies for restructuring, Implications of Corporate Restructuring

Unit-III: Takeovers and Corporate Valuation

Introduction, Forms of Takeover, Takeover Defences, Benefits and disadvantages of Takeovers. Buyback of Shares and its process. Basics of Value, Various Expressions of Value. Relationship among different types of value

Unit-IV: Corporate Restructuring Strategies & Strategic Alliance

Strategies for entering a New Market; Tools for Strategy Analysis - SWOT etc. Framework for M&A Strategies, Formulating Strategies for M&A. Cross-border acquisitions, Needs, Benefits and difficulties in Cross Border Acquisitions. Strategic alliances as an

Unit-V: Leveraged Buyouts & Integration for M&A Success

Leveraged buyouts (LBO) & LBO Sponsors and Mode of LBO, Criteria for Selecting LBO Candidate. Concept of Financial Leverage and Risk, Theories of LBO, Exit Strategies for LBO. Meeting the challenges of M&As. Post-Merger Growth Strategies, Strategies for Post-merger Success

Textbook References:

1. Sudi Sudarsanam, Value Creation from Mergers And Acquisitions, Pearson Education.
2. Fred Weston, Kwang S Chung, Susan E Hoag, Mergers, Restructuring and Corporate Control, Pearson Education
3. Ravindhar Vadapalli , Mergers acquisitions and Business valuation, Excel books.
4. Arzak, Valuation for mergers Buyouts & Restructuring, Wiley India (P) Ltd.
5. Chandrashekar Krishna Murthy & Vishwanath S. R., Merger Acquisitions & Corporate Restructuring, Sage Publication.
6. Sheeba Kapil, Kanwal N. Kapil, Mergers and Acquisitions–Strategy, Valuation, Leveraged Buyouts, and Financing, Wiley India Pvt. Ltd.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(HR Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402HR	ADVANCED MANPOWER PLANNING	4	0	0	4

Course Description

The course provides a comprehensive examination of sophisticated strategies and techniques for optimizing human resource allocation and management within organizations. Students will delve into advanced topics such as workforce forecasting, strategic talent management, and succession planning, with a focus on aligning manpower planning with long-term organizational goals. The course covers methodologies for analysing workforce data, predicting future staffing needs, and developing integrated plans to address skills gaps and support organizational growth. Students will gain the expertise to design and implement effective manpower planning strategies, ensure optimal resource utilization, and enhance overall organizational performance. This course aims to equip students with the advanced skills required to navigate complex workforce challenges and drive strategic human resource initiatives.

Course Objectives

1	Analyse the theory and concepts of human resource planning.
2	Identify the evolution of HRP throughout the organization.
3	Relate and apply models and methods used in forecasting.
4	Describe the applications of a Human Resources Information System.
5	Evaluate the organization's planning program

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Identify the purpose, process, and applications of human resource planning in the context of different organizational strategies.
CO2	Create a critical appreciation and knowledge of understanding the determinants of human resource requirements and recent trends in HRP.
CO3	Acknowledge the problems of HRP and analyse the job.
CO4	Create awareness about the HR forecasting techniques.
CO5	Acquire knowledge and skill necessary for preparing the manpower plan of the business enterprise.
CO6	Comprehend the stages of career development process and succession management process and Management Development Program.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	1	0	1	2	0	0	2
CO2	2	2	1	2	1	1	1	0
CO3	1	3	1	2	1	0	1	2
CO4	2	2	0	1	2	0	0	0
CO5	2	3	2	2	2	0	2	1
CO6	1	3	2	2	2	0	3	1
AVG	1.83	2.33	1.00	1.67	1.67	0.17	1.17	1.00

Justification of the Mapping:

0=NoRelevance;1=Low Relevance;2=Medium Relevance;3=High Relevance

Syllabus

Unit-I: Introduction to Human Resource Planning

Introduction: definition and concept of HRP, benefits, process, HRP components, Strategic HRP, Methods and tools for Strategic HRP, Strategic HRP model, Key Points for Successful HR Planning

Unit-II: HR planning and Corporate Strategies

HR planning and corporate strategies: HR planning as a strategic process-employees as resources-goal attainment, linking HR process to strategy, involvement in strategic planning process, strategic HR Planning model, staffing system, Recent trends in HRP

Unit-III: Job Analysis

Job analysis: meaning and definition, job analysis process, techniques of job analysis, methods and practice of job analysis, competency-based approach. Job description, Problems in HRP

Unit-IV: HR Forecasting

HR Forecasting: Forecasting Manpower Needs, the Forecasting Process, inventorying available talent, Projecting Future Talent Supply, forecasting Staffing Requirements. Index analysis-expert forecasts-Delphi technique-nominal group technique-HR budget and staffing table, scenario forecasting and regression analysis.

Unit-V: Career Planning and Succession Management

Career planning and succession management: definitions, concepts, stages of career development process, Career Anchors – Stages of growth and career, career processes; Succession management process and Management development programmes, objectives of MDPs, Job rotation, management development methods, challenges of succession management

Textbook References:

1. S. K Acharya & S. K Tripathy, Manpower Planning and Strategic Change, Himalaya Publishing House.
2. D. K Bhattacharyya, Human resource planning, Excel Books India.
3. Thomas Henry Patten, Manpower Planning and Development of Human resources, Wiley-Interscience.
4. R. Krishnamoorthy, Nitin Kamat & Chinmay Nitin Kamat, Human Resource Planning and Information System, Himalaya Publishing House.
5. Monica Belcourt, Mark Podolsky, Strategic Human Resources Planning, Cengage.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(HR Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403HR	ORGANISATIONAL DEVELOPMENT AND CHANGE	4	0	0	4

Course Description

This course explores the theories, strategies, and practices essential for managing and facilitating effective organizational transformations. Students will examine the principles of organizational development (OD), including change management models, organizational culture assessment, and the role of leadership in driving change. The course covers methods for diagnosing organizational issues, designing and implementing change initiatives, and evaluating the impact of these changes on overall performance. Students will develop the skills to lead and support organizational change, foster a culture of continuous improvement, and achieve sustainable organisational growth. This course aims to equip students with the tools and knowledge needed to navigate complex change processes and enhance organisational effectiveness in dynamic environments.

Course Objectives

1	Understand the concept and forces of Organisational Change.
2	Analyse various models of change and their characteristics.
3	Assess the relationship between organisational culture and change.
4	Understand the principles and processes of Organisational Development (OD).
5	Evaluate and implement OD interventions.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand the concept of organizational change and the forces driving it, both internally and externally.
CO2	Evaluate various models of change, including Lewin's Force Field Analysis and the Continuous Change Process Model, and understand their impact on organizations.
CO3	Analyse the influence of corporate culture on organizational change
CO4	Identify the aligning culture with change initiatives
CO5	Understand the principles and processes of Organisational Development, including the Action Research model.
CO6	Identify and implement appropriate Organisational Development interventions to enhance organisational performance and manage change effectively.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	0	1	2	0	1
CO2	3	2	2	1	1	2	1	2
CO3	2	3	2	0	1	3	2	1
CO4	3	2	2	1	2	3	1	3
CO5	3	3	3	2	2	3	2	3
CO6	3	2	2	1	2	3	1	2
AVG	2.83	2.33	2.00	0.83	1.50	2.67	1.17	2.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Organizational Change

Organisational Change: Introduction, forces of change, planned internal change, planned external change, types of change

Unit-II: Models of Change

Models of Change: Characteristics, Systems Model, Lewin's Force Field Analysis, Continuous Change Process Model, change and its impact, resistance to change.

Unit-III: Organisational Culture and Change

Organisational Culture and Change: Corporate culture, a systematic approach to making change, ten keys to effective Change Management.

Unit-IV: Organisational Development

Organisational Development: Characteristics, assumptions of OD, Model of OD Action Research.

Unit-V: OD Interventions

OD Interventions: Selecting an OD intervention, classification of OD interventions, OD interpersonal interventions, OD team development interventions, OD intergroup development interventions.

Textbook References:

14. Kavitha Seth, Organization Change and Development, Excel Books.
15. Radha R. Sharma, Change Management, Tata McGraw-Hill.
16. Nilakantand Ramanarayan, Change Management, Response Books.
17. Kirpatrick, D. L., Managing Change Effectively, Butter worth.
18. Donald L. Anderson, Organization Development: The Process of Leading Organizational Change, SAGE Publications.
19. Thomas G. Cummings and Christopher G. Worley, Organization Development and Change, Cengage Learning.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(HR Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404HR	PERFORMANCE MANAGEMENT SYSTEM	4	0	0	4

Course Description

This course provides a comprehensive overview of designing, implementing, and managing systems that enhance employee performance and align individual goals with organisational objectives. Students will explore key components such as goal setting, performance appraisal, feedback mechanisms, and development planning. The course covers various performance management models and best practices, including how to create effective performance metrics, conduct constructive evaluations, and use performance data to drive continuous improvement. Students will gain hands-on experience in developing and refining performance management systems that foster employee growth, motivation, and organisational success. This course aims to equip students with the skills to build robust performance management frameworks that contribute to achieving strategic business goals and enhancing overall organizational effectiveness.

Course Objectives

1	Understand the fundamental concepts and objectives of Performance Management and the role of HR professionals in implementing PM.
2	Analyse the characteristics and functions of a Performance Management System (PMS) and understand its integration with goal-setting theories.
3	Understand the importance of Performance Planning, including the process and methodologies used to ensure effective performance outcomes.
4	Explore the objectives and methods of Performance Appraisal, including the ethical considerations in performance management.
5	Learn about the process of Performance Monitoring, including performance management documentation, audits, and leading high-performance teams.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand and articulate the fundamental concepts, objectives, and role of HR professionals in Performance Management.
CO2	Analyse the characteristics and functions of Performance Management Systems, integrating theories of goal-setting with corporate and individual performance.
CO3	Evaluate the importance of Performance Planning, including the methodologies and processes that drive effective performance outcomes.
CO4	Assess the objectives, methods, and ethical considerations involved in Performance Appraisal, and their impact on organisational performance.
CO5	Develop an understanding of the process of Performance Monitoring, including documentation, audits, and leading high-performance teams.
CO6	Identify and overcome barriers to effective performance planning, ensuring alignment between organizational goals and individual performance objectives.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	2	1	1

CO2	3	3	2	1	2	3	2	2
CO3	3	2	3	2	2	3	2	2
CO4	3	2	3	2	2	3	2	3
CO5	2	3	3	2	3	3	3	3
CO6	3	2	2	2	2	3	2	2
AVG	2.83	2.33	2.50	1.67	2.00	2.83	2.00	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Performance Management

Performance Management: Introduction, Definition of PM - Objectives, Characteristics and Prerequisites of PM - Dimensions of PM - Factors affecting of Performance Management - Linkage of performance management with other HR Sub Systems - Role of HR Professionals in Performance Management.

Unit-II: Performance Management System

Performance Management System (PMS): Introduction to PMS - Objectives and Functions of PMS - Conceptual model of Performance Management System - Characteristics of PMS - Performance Management Process - Theories of Goal Setting and Corporate and Individual levels of goal setting - Expectancy Theory.

Unit-III: Performance Planning and Managing

Performance Planning: Definition and Importance of Performance Planning Objectives and Characteristics of performance planning - Methodologies of Performance Planning - Process of Performance Planning - Barriers to performance planning.

Performance Managing: Meaning and Importance - Objectives and Characteristics - Performance Managing Process

Unit-IV: Performance Appraisal

Performance Appraisal: Meaning and Characteristics of appraisal - Objectives and Importance of appraisal - Process of Performance Appraisal - Methods of Performance Appraisal - Ethical Issues in Performance management.

Unit-V: Performance Monitoring

Performance Monitoring: Definition of Performance Monitoring and Characteristics - Objectives of monitoring - Process of Performance Monitoring - performance management Documentation - Annual Stocktaking - Performance Management Audit - Leading high-performance teams - Integrated Performance Management - Maturity Alignment.

Textbook References:

1. Kohil A. S., & Deb, T., Performance Management, Oxford University Press.
2. Herman Aguinis, Performance Management, Pearson.
3. Michael Armstrong and Angela Baron, Performance Management, Jaico Publishing House.
4. Rao, T. V., Performance Management and Appraisal Systems, Response Books.
5. Aguinis, Herman, Performance Management, Pearson Education.
6. Cascio, Wayne F, Managing Human Resources: Productivity, Quality of Work Life, Profits, McGraw-Hill Education.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(HR Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405HR	LEADERSHIP & CHANGE MANAGEMENT	4	0	0	4

Course Description

This course is about understanding why and how leaders influence change. The focus of the unit is increasing the effectiveness of leadership and change management skills at an individual and organisation levels. Leadership and change management are increasingly seen to be key factors affecting the performance of contemporary organisations and are important areas of study in the fields of management and organisational behaviour. Effective change management requires strong diagnosis skills linked to the history, context, challenges, and opportunities faced by an organisation. This content will integrate theoretical and practical aspects of change management to enhance capabilities in the design, development, and implementation of change management strategies. It will examine theoretical aspects from both traditional and critical perspectives.

Course Objectives

1	Analyse the theory and concept of organizational change with models of change.
2	Identify different organizational development intervention.
3	Describe the individual, group and inter-group based intervention.
4	Analyse how leadership is associated with techno-structural interventions.
5	Give knowledge future and research in organizational development.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate ability to apply leadership and change management principles.
CO2	Recognize how a result-oriented work environment can be created.
CO3	Be familiar with how to manage stakeholders and their expectations.
CO4	Comprehend how the role of leadership can be the part of change process through a systematic approach.
CO5	Be acquainted with how to lead and manage cross functional teams.
CO6	Demonstrate knowledge in strengthening the organization, empowering individuals and promoting accountability and ownership principles.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	1	3	1	1	2	0	0
CO2	1	2	2	1	2	3	2	2
CO3	2	3	2	2	2	2	1	1
CO4	3	3	3	3	3	2	2	2
CO5	2	2	3	2	3	3	2	2
CO6	2	2	2	3	2	2	2	1
AVG	2.17	2.17	2.50	2.00	2.17	2.33	1.50	1.33

Justification of the Mapping:

0=NoRelevance;1=Low Relevance;2=Medium Relevance;3=High Relevance

Syllabus**Unit-I: Organizational Change**

Introduction, nature of change, Internal & External changes, types of change, Models of change- Lewis's Force field, Systems Model, Action research model, Resistance to change- reasons for the resistance, overcoming resistance for the change, systematic approach to making change- factors for effective change, skills of leaders in change management, designing the change

Unit-II: Organization Development

Introduction, evolution of OD, OD interventions: Definition, factors to be considered, classification of OD interventions, results of OD, typology of interventions based on target groups. Process of Organization Development: Entering into OD relationship

Unit-III: Human Process Interventions

Human process interventions (individual, group and inter-group human relations): Individual based: coaching, counselling, training, behavioural modelling, delegating, leading, morale boosting, mentoring, motivation, etc., Group based: conflict management, dialoguing, group facilitation, group learning, self-directed work teams, large scale interventions, team building, and virtual teams. Inter-group based: Organization mirroring, third party peace-making interventions.

Unit-IV: Techno-structural Interventions

Intervention Strategy, Structural, technological and process factors in intervention strategies. Advantages / limitations of change technologies and associated leadership models. Role of leadership in change process. Leadership and emotional knowledge strategies to achieve congruence of personnel, structure and culture.

Unit-V: The Future of OD

The changing environment, Fundamental strengths of OD, Implications of OD for the client, ethical standards in OD, OD's future. OD Consultant's role, issues in consultant-client relationship, Power, Politics & OD, Research on OD.

Textbook References:

1. Kavitha Singh, Organization Change & Development, Excel Books.
2. Nilakant V and Ramnarayan, Managing Organizational Change, Response Books.
3. Joan V. Gallos, Edgar H. Schein Organization development: a Jossey-Bass reader, Wiley Publications.
4. Adrain Thornhill, Phil Lewis, Mike Millmore, Mark Saunders, Managing Change – A Human Resource Strategy Approach, Pearson Education.
5. Wendell French and Cecil Bell Organization Development: Behavioral Science Interventions for organization, PHI.



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MBA IV SEMESTER (AR23)
(HR Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406HR	TRAINING AND DEVELOPMENT	4	0	0	4

Course Description

This course focuses on the strategies and practices essential for enhancing employee skills and fostering continuous professional growth within organizations. Students will explore the principles of effective training design, including needs assessment, curriculum development, instructional methods, and evaluation techniques. The course covers various training delivery methods, such as workshops, e-learning, and on-the-job training, and emphasizes the importance of aligning training programs with organisational goals and employee career development. Students will gain the skills to create impactful training programs, assess their effectiveness, and implement development initiatives that drive employee performance and organizational success. This course aims to equip students with the tools and knowledge necessary to develop and manage comprehensive training and development strategies that support a skilled and motivated workforce.

Course Objectives

1	Understand the training process model, important concepts, and trends in training, and align training design with the learning process.
2	Analyse training needs using the Training Need Analysis (TNA) model, and develop objectives and strategies for effective training design and transfer.
3	Evaluate and select appropriate training methods based on criteria and outcomes, including lectures, demonstrations, games, simulations, and e-learning.
4	Implement training programs effectively, address challenges, and evaluate training programs, including understanding resistance and various evaluation types.
5	Explore and apply training strategies in specific areas such as orientation, diversity, sexual harassment, team training, and emotional intelligence.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the training process model and align training design with learning principles.
CO2	Conduct a Training Need Analysis (TNA) and design effective training programs.
CO3	Select and apply appropriate training methods based on training objectives.
CO4	Implement and evaluate training programs, addressing challenges and resistance.
CO5	Develop training programs for specific areas like diversity, sexual harassment, and emotional intelligence.
CO6	Integrate various training strategies to foster continuous learning and innovation.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	3	2	1
CO2	3	3	2	2	2	3	2	2
CO3	2	2	3	2	2	2	2	2
CO4	2	3	3	3	2	2	3	3
CO5	2	2	3	2	3	2	3	3
CO6	3	2	2	2	2	3	2	2

AVG	2.50	2.33	2.50	2.00	2.00	2.50	2.33	2.17
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Training in Organisations								
Overview of Training: Training process model - Important concepts and meanings - Open system model - Trends in training - Challenges and Career opportunities in training - Principles of learning in training - Aligning training design with learning process.								
Unit-II: Need Analysis and Training Design								
Need Analysis: Meaning and concepts - Need Assessment Process - The Training Need Analysis (TNA) Model - TNA and Design - Organizational constraints - Developing objectives - Facilitation of learning and training transfers to the job.								
Unit-III: Training Methods								
Methods: Criteria of selecting appropriate training methods - Basic Training Methods - Matching methods with outcomes - Lectures and Demonstrations - Games and Simulations - JIT - Computer Based Training (CBT) - e-Learning.								
Unit-IV: Implementation & Evaluation of Training								
Implementation of Training Program: Approaches to Implementation of training program - Development of training - Implementation - Transfer of training - Major players in training & development.								
Evaluation of Training Program: Resistance to training evaluation - Types of evaluation.								
Unit-V: Areas of organizational Training								
Organizational Training: Orientation training - Diversity training - Sexual harassment training - Team training - Cross cultural training - Training for talent management and competency mapping - Training in Emotional Intelligence - Innovation and Continuous learning.								
Textbook References:								
1. P.Nick Blanchard, James W. Thacker, A. Anand Ram, Effective Training, Pearson. 2. Anjali Ghansekar, Training & Development, Everest. 3. Raymond A Noe, Amitabh Deo Kodwani, Employee Training and Development, McGraw Hill. 4. Kirkpatrick, Donald L., and James D. Kirkpatrick., Evaluating Training Programs: The Four Levels, Berrett-Koehler Publishers.								
Journals								
1. Indian Journal of Industrial Relation. 2. NHRD Journal of Career Management. 3. Management and Labour Studies. 4. Personnel today. 5. Leadership excellence. 6. Indian Journal of Training & Development.								



SCHOOL OF MANAGEMENT STUDIES
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MBA IV SEMESTER (AR23)
(Rural Management Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402RM	AGRIPRENEURSHIP	4	0	0	4

Course Description

This course equips students with the skills and knowledge necessary to innovate and succeed in the agricultural business sector. Covering key aspects of agricultural entrepreneurship, the course explores how to identify and capitalize on opportunities within the agribusiness industry, from crop and livestock management to agri-tech solutions. Students will learn about business planning, market analysis, and the development of sustainable practices, while also gaining insights into financing options, risk management, and regulatory considerations.

Course Objectives

1	Promote Sustainable Agricultural Practices.
2	Empower Rural Entrepreneurs.
3	Enhance Market Access for Farmers.
4	Foster Innovation in Agribusiness.
5	Contribute to Rural Development.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Identify key opportunities and challenges within agripreneurship, including the dynamics of the agribusiness environment.
CO2	Describe the overview of the agricultural sector, agricultural value chains, and government policies and schemes affecting agripreneurship.
CO3	Perform feasibility analyses covering market, technical, and financial aspects to assess the viability of agribusiness ideas.
CO4	Analyse trends in agri-tech start-ups and other emerging areas in the agricultural sector.
CO5	Implement financial management techniques to ensure effective use of funds and financial stability in agribusiness ventures.
CO6	Engage in workshops and practical training sessions focused on business plan development, agribusiness software, sustainable farming techniques, and networking with industry professionals.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	2	1	2	1	3
CO2	3	3	3	2	3	1	2	3
CO3	2	3	3	2	2	2	2	3
CO4	3	3	2	2	2	3	2	1
CO5	3	3	2	2	2	2	3	3
CO6	2	2	2	3	2	3	2	1
AVG	2.67	2.67	2.50	2.17	2.00	2.17	2.00	2.33

Justification of the Mapping:

0=NoRelevance;1=Low Relevance;2=Medium Relevance;3=High Relevance

Syllabus

Unit-I: Introduction to Agripreneurship

Definition and Scope of Agripreneurship. Role of Agripreneurs in Rural Development. Opportunities and Challenges in Agripreneurship.

Agribusiness Environment: Overview of the Agricultural Sector. Agricultural Value Chains. Government Policies and Schemes for Agripreneurship. Global and National Agricultural Markets

Unit-II: Entrepreneurial Process

Idea Generation and Opportunity Identification. Business Plan Development. Feasibility Analysis (Market, Technical, Financial). Legal Aspects of Starting an Agribusiness.

Agribusiness Management: Supply Chain Management in Agribusiness. Marketing of Agricultural Products. Financial Management for Agripreneurs. Risk Management in Agribusiness. Quality Management and Standards.

Unit-III: Innovation in Agriculture

Technological Innovations in Agriculture. Sustainable Agriculture Practices. Agri-Tech Startups and Emerging Trends. Precision Farming and Digital Agriculture.

Agripreneurship Development: Incubation and Support for Agripreneurs. Role of Agri-Business Incubators and Accelerators. Case Studies of Successful Agripreneurs. Networking and Collaboration in Agribusiness.

Unit-IV: Funding and Finance

Sources of Funding for Agripreneurs. Venture Capital, Angel Investors, and Crowdfunding. Rural and Agri-Finance Institutions. Government Grants and Subsidies for Agripreneurs.

Social and Environmental Responsibility: Ethical Issues in Agribusiness. Corporate Social Responsibility (CSR) in Agriculture. Impact of Agripreneurship on Rural Communities. Environmental Sustainability in Agribusiness

Unit-V: Project Work and Case Studies

Real-life Case Studies of Agripreneurs. Field Visits to Successful Agribusinesses. Group Projects on Agribusiness Models. Development of a Business Plan for an Agripreneurial Venture.

Workshops and Practical Training: Workshops on Business Plan Writing. Training on Agribusiness Software and Tools. Networking Events with Agribusiness Professionals. Practical Sessions on Sustainable Farming Techniques

Textbook References:

1. Dr. B.S. Rathore and Dr. J.S. Patel, Agripreneurship Development, Agrotech Publishing Academy.
2. Dr. L. Lalitha, Dr. K. Sivaramane, and Dr. C. Aruna, Entrepreneurship Development and Agribusiness Management, New India Publishing Agency.
3. Dr. Anjani Kumar and Dr. Ramesh Chand, Agribusiness Management: Challenges, Opportunities & Strategies, Springer.
4. Bruce L. Gardner and Gordon C. Rauser, Handbook of Agricultural Economics. Elsevier.
5. David Wilson, Kathleen Wood, and Jim Woodhill, Small Farmers, Big Change: Scaling Up Impact in Smallholder Agriculture, Practical Action Publishing.
6. Kristin E. Davis, Sophie Theis, Innovation and Entrepreneurship in Agriculture: Case Studies from Developing Countries, International Food Policy Research Institute (IFPRI).



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MBA IV SEMESTER (AR23)
(Rural Management Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403RM	AGRO PROCESSING, PACKING AND QUALITY CONTROL	4	0	0	4

Course Description

This course provides an in-depth understanding of agro-processing, including the techniques and technologies used to process agricultural products, ensure quality, and manage packaging. It aims to equip students with the skills necessary to improve the efficiency and effectiveness of agro-processing operations, ensuring that products meet quality standards and are packaged appropriately for the market.

Course Objectives

1	Understand key agro-processing technologies to enhance product value and quality.
2	Learn effective packaging techniques to preserve agricultural products and meet market standards.
3	Gain knowledge of quality control standards to ensure compliance in agro-processing.
4	Explore sustainable practices in agro-processing and packaging to minimize environmental impact.
5	Develop strategies to improve the market competitiveness of agro-processed products.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Apply advanced agro-processing techniques to add value to agricultural products.
CO2	Select and implement appropriate packaging solutions to maintain product integrity.
CO3	Ensure product compliance with quality control standards and regulatory requirements.
CO4	Incorporate sustainable practices in agro-processing and packaging operations.
CO5	Enhance the market readiness and competitiveness of agro-processed goods.
CO6	Analyse and improve the overall efficiency and effectiveness of agro-processing systems.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	2	3	3	2	3
CO2	3	3	3	2	2	2	2	3
CO3	2	3	3	2	2	2	2	3
CO4	2	2	2	3	2	2	2	1
CO5	3	2	2	2	3	3	2	3
CO6	3	3	3	2	2	3	3	1
AVG	2.67	2.50	2.50	2.17	2.33	2.50	2.17	2.33

Justification of the Mapping:

0=NoRelevance;1=Low Relevance;2=Medium Relevance;3=High Relevance

Syllabus

Unit-I: Introduction to Agro-Processing

Overview of Agro-Processing Industries. Importance of Agro-Processing in Rural Development. Types of Agro-Processed Products (e.g., cereals, fruits, vegetables, dairy, meat). Role of Agro-Processing in Value Addition and Employment Generation.

Processing Techniques for Agricultural Products: Primary and Secondary Processing Methods. Techniques for Processing Cereals, Pulses, Fruits, and Vegetables. Dairy and Meat Processing Techniques. Preservation Methods (e.g., drying, freezing, canning, fermenting)

Unit-II: Packaging of Agro-Processed Products

Importance of Packaging in Agro-Processing. Types of Packaging Materials (e.g., plastics, glass, metal, paper). Advanced Packaging Techniques (e.g., vacuum packing, modified atmosphere packaging). Packaging Design and Labeling Requirements. Environmental Impact and Sustainability in Packaging.

Quality Control in Agro-Processing: Introduction to Quality Control and Assurance. Standards and Regulations for Agro-Processed Products (e.g., FSSAI, ISO, HACCP). Techniques for Quality Testing (e.g., microbiological, chemical, sensory evaluation). Food Safety Management Systems. Traceability in Agro-Processing.

Unit-III: Storage and Distribution of Agro-Processed Products

Post-Harvest Management and Storage Techniques. Cold Chain Management for Perishables. Transportation and Logistics in Agro-Processing. Shelf Life Extension and Inventory Management. **Value Addition through Agro-Processing:** Economic and Social Benefits of Value Addition. Opportunities for Small-Scale Agro-Processing Units. Case Studies of Successful Agro-Processing Ventures. Role of Cooperatives and Farmer Producer Organizations (FPOs) in Agro-Processing.

Unit-IV: Sustainability and Environmental Considerations

Waste Management in Agro-Processing Industries. Eco-Friendly Packaging Alternatives. Sustainable Processing Practices. Impact of Agro-Processing on Rural Ecosystems. **Entrepreneurship in Agro-Processing:** Identifying Opportunities in Agro-Processing. Business Plan Development for Agro-Processing Units. Funding and Support for Agro-Processing Ventures. Challenges and Solutions in Agro-Processing Entrepreneurship.

Unit-V: Regulatory Framework and Compliance

Overview of National and International Food Safety Standards. Licensing and Certification Requirements. Compliance with Environmental and Labour Laws. Intellectual Property Rights in Agro-Processing. **Practical Components:** Hands-on Training in Agro-Processing Techniques. Workshops on Packaging Design and Quality Control. Field Visits to Agro-Processing Units. Case Studies and Project Work on Quality Control and Packaging Strategies.

Textbook References:

1. **Andrew J. W. O'Connor**, Postharvest Technology of Horticultural Crops, University of California Agriculture and Natural Resources (UCANR) Publications
2. **Stephanie Clark, Stephanie Jung, and David P. Murphy**, Food Processing: Principles and Applications, Wiley-Blackwell.
3. **Denise R. Warner**, Handbook of Food Processing: Food Safety, Quality, and Manufacturing Processes, CRC Press.
4. **Gordon L. Robertson**, Food Packaging: Principles and Practice, CRC Press.
5. **Norman G. Marriott and Robert V. Gravani**, Quality Control in the Food Industry, Springer.
6. **Amarjit S. Basra and R. K. Gupta**, Sustainable Food Processing and Packaging, CRC Press.



SCHOOL OF MANAGEMENT STUDIES
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MBA IV SEMESTER (AR23)
(Rural Management Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404RM	FINANCIAL MANAGEMENT IN RURAL PERSPECTIVE	4	0	0	4

Course Description

This course offers a specialised exploration of financial strategies and practices tailored to rural management contexts. Students will investigate the unique financial challenges faced by rural organizations and enterprises, such as managing seasonal income variability, accessing capital, and optimizing resource allocation. The course covers critical topics including budgeting, financial forecasting, risk management, and investment strategies specific to rural settings. Emphasis is placed on developing practical financial solutions that support sustainable rural development and improve economic outcomes for rural communities. Students will acquire the skills needed to effectively manage financial operations, drive rural enterprise growth, and contribute to the economic resilience of rural areas.

Course Objectives

1	Develop skills to manage financial resources effectively in rural enterprises.
2	Understand financial planning and budgeting tailored to rural contexts.
3	Apply financial analysis techniques to evaluate rural investment opportunities.
4	Learn to design and implement cost control measures in rural businesses.
5	Explore financing options and strategies for rural development projects.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Apply financial management techniques to optimize resource allocation in rural enterprises.
CO2	Create comprehensive financial plans and budgets suited to rural business needs.
CO3	Evaluate investment opportunities using financial analysis tailored to rural contexts.
CO4	Implement effective cost control measures to enhance profitability in rural businesses.
CO5	Identify and utilize financing options for rural development and growth projects.
CO6	Assess the financial health and performance of rural enterprises using key metrics

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	2	1	2	1	3
CO2	3	3	3	2	3	1	2	3
CO3	2	3	3	2	2	2	2	3
CO4	2	3	2	2	2	3	2	1
CO5	2	3	2	2	2	2	3	3
CO6	2	2	2	3	2	3	2	1
AVG	2.33	2.67	2.50	2.17	2.00	2.17	2.00	2.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Financial Management

Introduction, Meaning, Objectives, Functions of Financial Management. Role of Finance Manager in Rural Enterprises. Take interview of Finance Manager of any business enterprise to Understand his functions and role in the organization.

Unit-II: Sources of Finance

Long Term, Medium Term and Short Term Sources; Rural Financial Intermediaries – NABARD, DCBs, LDBs, RRBs, Cooperative Banks and Credit Societies and other agencies; Innovative Weapons of Finance for Rural Enterprises

Unit-III: Introduction to Hire Purchase and Leasing

Hire Purchase, Leasing, Micro Finance, Seed Capital, Venture Capital Financing
Visit any of the NABARD, DCB's, LDB's, RRB's and Co-Operative Bank and Societies to identify Sources of finance and loan disbursement system.

Unit-IV: Financial Statement Analysis

Comparative Financial Statements, Trend Analysis, Common Size Statements, Ratio Analysis
Profitability, Liquidity, Activity and Leverage Ratios
Download annual report of any company from its website and Analyze its financial performance by the techniques of ratio analysis.
Prepare comparative financial statement and common-size financial Statement.

Unit-V: Working Capital and Capital Budgeting for Rural Enterprises

Working Capital: Concept, Types, Factors, Estimation and Financing of Working Capital
Capital Budgeting: Concept, Nature, Significance and Techniques of Capital Budgeting - Pay Back Period, Net Present Value, Profitability Index, Accounting Rate of Return, Internal Rate of Return
Take interview of finance manager or entrepreneur regarding their need of working capital and its finance.

Textbook References:

1. **Khan and Jain: Financial Management**, McGraw Hill Education.
2. **Prasanna Chandra: Financial Management**, McGraw Hill Education.
3. **I.M. Pandey: Financial Management**, Vikas Publishing House.
4. **R.M. Srivastava: Financial Management and Policy**, Himalaya Publishing House.
5. **Vasatn Desai: Indian Financial System**, Himalaya Publishing House.
6. **L.M. Bhole: Indian Financial System**, McGraw Hill Education.

Web sites:

1. www.rbi.org.in
2. www.nabard.org



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MBA IV SEMESTER (AR23)
(Rural Management Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405RM	OPERATIONS MANAGEMENT IN RURAL PERSPECTIVE	4	0	0	4

Course Description

This course focuses on the principles and practices of operations management with a specific emphasis on rural enterprises and agricultural operations. It aims to provide students with the skills needed to optimize operational processes, improve efficiency, and manage resources effectively in rural settings. The course addresses the unique challenges and opportunities that rural businesses face, offering strategies to enhance operational performance and support sustainable development.

Course Objectives

1	Optimise operational processes to enhance efficiency in rural businesses.
2	Develop effective supply chain strategies tailored to rural environments.
3	Implement inventory management techniques suited to rural contexts.
4	Apply quality control measures to improve product standards in rural operations.
5	Utilise operational planning tools to address challenges in rural enterprise management.

Course Outcomes
(Upon successful completion of this course, students should be able to)

CO1	Improve operational efficiency through optimized processes in rural businesses.
CO2	Design and implement supply chain strategies that address rural logistics challenges.
CO3	Manage inventory effectively to meet demand and reduce costs in rural settings.
CO4	Apply quality control techniques to enhance product quality in rural operations.
CO5	Use operational planning tools to overcome challenges and enhance rural enterprise management.
CO6	Analyse and resolve operational issues to improve overall performance in rural enterprises

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	2	2	2	2	3	2
CO2	2	3	3	2	2	2	2	3
CO3	2	2	2	2	2	2	2	2
CO4	2	2	2	3	2	2	2	2
CO5	3	2	2	3	3	2	3	2
CO6	3	2	2	2	3	3	3	2
AVG	2.50	2.33	2.17	2.33	2.33	2.17	2.50	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Production and Operation

The Production/Operation Function, Operations Management Decisions, Operations Strategy, Significance of Operations Management in Manufacturing and Service Environment, Interface of Operations Function with Other Functional Areas, Productivity - Meaning, Measures of Productivity, Work Study. Study of manufacturing plant in nearby area and its operations.

Unit-II: Manufacturing Processes

Facility Location - Factors for selection, Facility Layout - Visit an organization and enlist various factors considered for finalizing its location. Study layout of any one organization.

Unit-III: Introduction to Production Planning and Control

Types and features, Production Planning and Control - Need, Objectives, Functions; Materials Management - Objectives, Types and Costs associated.

Unit-IV: Proprietorship and Cooperative Farming

Peasants, Proprietorship, Cooperative Farming, Capital Farming, Corporate Farming, Land Tenure Systems and Agricultural Production Management - Rural Godowns, Agriculture Supply Chain. Study supply chain of any one agri-commodity in detail.

Unit-V: Technology Management

Effects on consumers and producers, Advantages and disadvantages of technology; Emerging technologies in agriculture - Green houses, Automation; Agro processing industries. Visit an agro processing industry / green house. Discuss various new technologies used by them.

Textbook References:

1. Chunawala Patel, Production And Operations Management, Himalaya Publishing House.
2. R. Pannerselevam, Production and Operations Management, Eastern Economy Edition.
3. **E. S. Buffa, Operations Management, John Wiley & Sons.**
4. Robertson C.A., ABN Introduction to Agricultural Production Economics and Farm Management, Tata McGraw Hill.
5. Heady, Earl. O and Jensen Herald R., Farm Management Economics, Prentice Hall.
6. **Barnard C.S. and Nix J.S., Farm Planning and Control, Cambridge University Press.**



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Rural Management Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406RM	HUMAN RESOURCE MANAGEMENT IN RURAL PERSPECTIVE	4	0	0	4

Course Description

This course explores the principles and practices of HRM with a focus on rural organizations. Students will gain insights into the unique challenges and opportunities of managing human resources in rural settings. This course emphasizes practical learning through field visits and case studies, enabling students to connect theoretical knowledge with real-world applications in the rural context.

Course Objectives

1	To introduce the fundamental concepts and importance of Human Resource Management (HRM) in rural contexts.
2	To understand the processes involved in procurement, placement, and socialization of human resources in rural organizations.
3	To explore the methods and practices of performance appraisal, training, and development specific to rural settings.
4	To analyse wage and salary administration practices and employee benefits in rural organizations.
5	To develop the ability to apply HRM concepts through practical visits and research in rural organizations.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand the scope, objectives, and importance of HRM in rural contexts and distinguish it from personnel management.
CO2	Analyse the processes involved in human resource procurement, job analysis, recruitment, and selection in rural organizations.
CO3	Apply HR planning, recruitment, selection, and induction processes through field visits and report preparation.
CO4	Evaluate training and development methods and performance appraisal practices in rural organizations.
CO5	Investigate wage and salary administration, employee benefits, and recent HRM trends in rural contexts.
CO6	Synthesise knowledge gained from field visits to develop presentations and reports on HRM practices in rural organizations.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	2	1	2
CO2	3	3	2	2	2	3	2	3
CO3	3	3	2	2	3	3	2	1
CO4	2	2	3	3	2	2	2	3
CO5	3	2	2	3	3	3	1	1
CO6	3	3	3	3	3	1	3	3

AVG	2.83	2.50	2.33	2.33	2.33	2.33	1.83	2.17
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to Human Resource Management Introduction to Human resource management - Definition, Scope, Objectives, Importance, HRM versus Personnel Management, Changing role of Human Resource Management, HR Min Indian context. Visit any organization Practicing Human Resource Management and Prepare a Presentation on it.								
Unit-II: Procurement and Placement of Human Resources Procurement and Placement: Concept of HRP, Job Analysis, Job Description, Job Specification, Recruitment: Objective, Factors affecting Recruitment, Sources of Recruitment, Process of Recruitment. Selection, Placement, Induction, Socialization.								
Unit-III: Human Resource Planning, Recruitment, and Selection Visit any manufacturing or service organization to study a process of Human Resource Planning, recruitment, and selection, Induction. Prepare a report and submit.								
Unit-IV: Development and Maintenance of Human Resources Development and Maintenance of Human Resource: Performance Appraisal: Meaning, Need, Problems of Performance Appraisal, Training and development: Difference between training and Development, Methods of Training & Development Visit any manufacturing or service organization and study Training and Development and submit a Report.								
Unit-V: Wage and Salary Administration and Employee Benefits Wage and Salary administration: Factors affecting wage/salary, objective of wage and salary administration, Employee Benefits, Principle of employee benefit programme, Employee Service. Recent Trends in HRM Visit/Research organizational Practices of Wage and salary administration and submit a Report								
Textbook References: 1. B.B. Mahapatro, Human Resource Management , New Age International Publishers. 2. Dipak Kumar Bhattacharyya, Human Resource Management , Oxford University Press. 3. Garry Dessler, Human Resource Management , Pearson. 4. Edwin Flippo, Personnel & Human Resource Management , McGraw Hill Education. 5. S. Seetaraman & B. Venkateswara Prasad, Personnel Management , Scitech Publications. 6. P. Subba Rao, Human Resource Management , Himalaya Publishing House.								



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Operations & Supply Chain Management Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402OSM	SERVICE OPERATIONS MANAGEMENT	4	0	0	4

Course Description

This course focuses on the strategies and techniques required to efficiently manage and optimize service delivery within organizations. This course delves into the core principles of service operations, including process design, quality management, and resource allocation, while emphasizing the unique challenges of service environments compared to traditional manufacturing settings. Students will explore topics such as service blueprinting, capacity planning, and customer experience management, with practical case studies highlighting real-world applications. By the end of the course, students will gain a comprehensive understanding of how to enhance operational performance, improve customer satisfaction, and drive organizational success through effective service management.

Course Objectives

1	To Develop a comprehensive understanding of the fundamental concepts and principles of service operations management.
2	to design efficient service processes and apply methodologies such as Lean and Six Sigma to identify and implement process improvements.
3	To Develop techniques for forecasting, scheduling, and resource allocation to ensure optimal service delivery and customer satisfaction.
4	To Implement strategies for designing and managing the customer experience to meet or exceed customer expectations.
5	To Explore how technology and human resource management impact service operations. Learn to utilize information technology, such as CRM systems, and manage HR aspects like recruitment, training, and employee performance to support effective service operations.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Demonstrate a thorough understanding of the key principles and concepts of service operations management, including service design, process management, and quality assurance.
CO2	Design efficient service processes and applying process improvement methodologies such as Lean and Six Sigma to enhance operational performance and customer satisfaction
CO3	Skilled in forecasting demand, planning capacity, and managing resources to balance service delivery with operational efficiency, ensuring the ability to handle varying levels of demand effectively.
CO4	Capable of developing and executing strategies to manage and improve service quality and customer experience.
CO5	Integrate technology solutions and human resource management practices into service operations.
CO6	Identify common challenges in service operations (e.g., variability, demand fluctuations) and develop strategies to address them.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs	PSOs
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	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	2	3	2	2	3	3
CO2	3	2	2	3	3	2	3	1
CO3	3	3	3	3	3	2	2	3
CO4	3	3	3	3	2	2	2	3
CO5	3	3	3	3	3	3	3	3
CO6	3	2	3	3	2	3	1	3
AVG	3.00	2.67	2.67	3.00	2.50	2.33	2.33	2.67

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Service Operations Management

Definition, scope, and significance of service operations management. Intangibility, perishability, heterogeneity, and inseparability. Differences between service and product management. Service Industry Sectors: Exploration of various service sectors such as healthcare, hospitality,

Unit-II: Service Design and Process Management

Service Blueprinting: Techniques for mapping and analysing service processes. Service Process Design: Designing effective service delivery processes. Service Facility Layout: Planning and organizing physical and virtual service environments. Capacity Planning and Management: Techniques for managing service capacity and demand.

Unit-III: Unit 3: Service Quality and Improvement

Service Quality Models: Introduction to models such as SERVQUAL and the GAP Model. Quality Measurement and Assurance: Methods for measuring and ensuring service quality. Continuous Improvement: Tools and techniques for service improvement, including Six Sigma, Lean, and Total Quality Management (TQM). Customer Feedback and Complaint Management: Systems for collecting and responding to customer feedback and complaints.

Unit-IV: Managing Service Operations and Resources

Human Resource Management in Services: Recruitment, training, and management of service personnel. Technology in Service Operations: Role of technology and digital tools in enhancing service operations. Resource Allocation and Scheduling: Techniques for optimal allocation and scheduling of resources. Service Performance Metrics: Key performance indicators (KPIs) and methods for evaluating service performance.

Unit-V: Unit 5: Strategic Issues in Service Operations

Service Strategy and Competitive Advantage: Developing strategies to gain a competitive edge in service operations. Service Innovation: Exploring new service concepts and innovations. Global and Ethical Considerations: Managing service operations in a global context and addressing ethical issues. Change Management: Strategies for implementing and managing changes in service operations.

Textbook References:

1. Robert Johnston and Graham Clark, Service Operations Management: Improving Service Delivery, Pearson.
2. James A. Fitzsimmons and Mona J. Fitzsimmons, Services Management: Operations, Strategy, Information Technology, McGraw-Hill Education.
3. Christian Grönroos, Service Management and Marketing: Managing the Service Value Chain, Wiley.
4. James L. Heskett, W. Earl Sasser, and Leonard A. Schlesinger, The Service Profit Chain: How Leading Companies Link Profit and Growth to Loyalty, Free Press.
5. L. J. Krajewski, Larry P. Ritzman, and Manoj K. Malhotra, Managing Service Operations: Design and Implementation, Pearson.
6. John T. Mentzer and Michael B. Myers, Introduction to Operations and Supply Chain Management, Sage Publications.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Operations & Supply Chain Management Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403OSM	GREEN SUPPLY MANAGEMENT	4	0	0	4

Course Description

This course explores the principles and practices of green supply management, focusing on strategies and tools for creating environmentally sustainable supply chains. Students will gain a comprehensive understanding of how to integrate sustainability into supply chain operations, minimize environmental impact, and enhance overall supply chain performance. The course covers the theoretical foundations of green supply management, practical approaches to implementing green practices, and the role of various stakeholders in fostering sustainability.

Course Objectives

1	Define green supply management and its importance in contemporary supply chains.
2	Assess the environmental impacts of supply chain activities and processes.
3	Examine practices for sustainable sourcing, including eco-friendly materials and ethical suppliers.
4	Explore options for green transportation, packaging, and reverse logistics.
5	Review key environmental regulations, standards, and certifications relevant to green supply management

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Able to design and propose effective green supply chain strategies that integrate sustainability principles into procurement, logistics, and production processes.
CO2	able to assess and select suppliers based on their environmental performance, sustainability certifications, and eco-friendly practices
CO3	able to apply strategies for reducing the environmental impact of logistics and transportation, including optimizing routes, reducing emissions, and adopting green packaging.
CO4	to use sustainability metrics and key performance indicators (KPIs) to measure and report on the environmental performance of supply chain activities and initiatives.
CO5	able to identify common challenges in implementing green supply management practices and develop strategies to address and overcome these obstacles.
CO6	able to understand and apply relevant environmental regulations, standards, and certifications (such as ISO 14001) to ensure compliance and enhance supply chain sustainability.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	2	3	2	3	3	2
CO2	2	3	3	2	2	3	3	2
CO3	3	3	3	3	2	3	3	2
CO4	3	3	2	3	2	3	3	2
CO5	3	3	3	3	3	3	3	3
CO6	3	3	2	2	2	2	2	3
AVG	2.67	3.00	2.50	2.67	2.17	2.83	2.83	2.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Introduction to Green Supply Management**

Definition of green supply management and its role in contemporary supply chains. Key principles of sustainability and their application in supply chains. Overview of the environmental impacts of supply chain activities. Introduction to key environmental regulations and sustainability standards (e.g., ISO 14001).

Unit-II: Sustainable Sourcing and Procurement

Strategies for selecting eco-friendly materials and suppliers. Criteria and tools for evaluating supplier sustainability and performance. Addressing ethical and social responsibility issues in procurement.

Unit-III: Green Logistics and Transportation

Analysing the environmental impacts of logistics and transportation. Techniques for reducing emissions and improving efficiency in logistics. Methods for reducing packaging waste and using sustainable materials.

Unit-IV: Measuring and Reporting Sustainability Performance

Key performance indicators (KPIs) for assessing environmental performance. Techniques for measuring and analysing sustainability metrics. Best practices for sustainability reporting and communication.

Unit-V: Challenges and Future Trends in Green Supply Management

Common challenges in adopting green supply management practices. Strategies for overcoming obstacles and driving sustainable practices. Emerging trends and innovations in green supply management.

Textbook References:

1. S. J. Zhang, K. Xu, Green Supply Chain Management: Product Life Cycle Approach, Springer.
2. Joerg Seidel and Jodie Clarke, Sustainable Supply Chain Management: Practical Ideas for Moving Towards Best Practice, Routledge.
3. John P. Sutton, Green Logistics and Transportation, Routledge.
4. Jonathan S. Scott, Managing Green Supply Chains, Springer.
5. B. W. Y. Lee, A. P. T. Chan, Sustainable Operations and Supply Chain Management, Wiley.
6. G. T. R. Marsh, Handbook of Green Supply Chain Management: Operations, Sustainability, and Supply Chain Management, CRC Press.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Operations & Supply Chain Management Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404OSM	SUPPLY CHAIN ANALYTICS	4	0	0	4

Course Description

This course delves into the application of data-driven methodologies and analytical tools to optimize supply chain operations. This course aims to provide students with a comprehensive understanding of how analytics can be used to enhance decision-making processes in various supply chain functions, including demand forecasting, inventory management, procurement, and logistics. By integrating theoretical concepts with practical applications, students will learn to address complex supply chain challenges and drive efficiency, agility, and resilience in supply chains. The course also covers emerging trends in supply chain analytics, such as the use of big data, machine learning, and artificial intelligence, to prepare students for future industry demands.

Course Objectives

1	Equip students with the ability to apply analytical techniques to solve supply chain problems.
2	Enable students to understand the role of data in supply chain decision-making processes.
3	Develop students' skills in interpreting and visualizing supply chain data.
4	Guide students in evaluating and implementing supply chain strategies using analytics.
5	Prepare students to understand and leverage emerging trends and technologies in supply chain analytics

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Apply quantitative and qualitative analytical techniques to optimize supply chain performance.
CO2	Develop the ability to interpret and visualize data to support supply chain decision-making processes.
CO3	Proficient in using advanced tools and technologies, such as machine learning and artificial intelligence, in supply chain analytics.
CO4	Integrate and evaluate supply chain strategies using data-driven approaches.
CO5	Prepare complex supply chain challenges with innovative analytical solutions.
CO6	Demonstrate an understanding of emerging trends in supply chain analytics and their impact on global supply chains.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	1	0	3	0	0
CO2	2	3	2	2	0	2	3	0
CO3	0	1	1	3	2	0	0	3
CO4	2	0	3	1	2	3	2	0
CO5	3	0	3	2	0	2	0	3
CO6	0	3	0	0	2	0	3	2
AVG	1.67	1.50	1.50	1.50	1.00	1.67	1.33	1.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Introduction to Supply Chain Analytics Basics of Supply Chain Management (SCM), Key Components of SCM (Procurement, Production, Distribution, Logistics), Supply Chain Stages (Suppliers, Manufacturers, Distributors, Retailers), Supply Chain Models (Traditional, Agile, Lean, Hybrid), Role of Analytics in SCM (Efficiency, Decision Support, Competitive Advantage), Analytics Tools Overview (Excel, SQL, Python, R, Tableau, Power BI)
Unit-II: Data Analysis Techniques in Supply Chain Descriptive Statistics (Mean, Median, Mode, Variance, Standard Deviation, Data Distribution), Regression Analysis (Simple Linear Regression, Multiple Linear Regression, Model Diagnostics, Advanced Regression Techniques), Time-Series Analysis (Trend Analysis, Seasonality, Cyclic Patterns, Forecasting Methods, ARIMA), Data Visualization Techniques (Basic Charts, Advanced Visualizations, Dashboards, Reporting Tools)
Unit-III: Optimization Models in Supply Chain Management Optimization Principles (Objective Functions, Constraints, Sensitivity Analysis), Linear Programming (Formulation, Solution Methods, Duality Theory), Integer Programming (Binary Integer Programming, Mixed Integer Programming, Heuristic Methods), Network Optimization (Shortest Path Problems, Maximum Flow Problems, Minimum Cost Flow Problems), Inventory Management Techniques (Economic Order Quantity, Safety Stock, Reorder Point Analysis)
Unit-IV: Predictive Analytics and Machine Learning Predictive Modelling (Time-Series Forecasting, Regression-Based Forecasting, Model Selection, Model Validation), Machine Learning Algorithms (Decision Trees, Random Forests, Support Vector Machines, Clustering Algorithms, Principal Component Analysis), Model Evaluation (Confusion Matrix, ROC Curve, Precision-Recall Curve, Performance Metrics), Applications in Demand Forecasting (Seasonal Trends, Promotion Effects, Advanced Techniques), Risk Management (Predictive Risk Analysis, Scenario Planning, Risk Mitigation Strategies)
Unit-V: Case Studies and Practical Applications Case Studies from Different Industries (Retail: Inventory Management, Demand Forecasting; Manufacturing: Production Scheduling, Optimization; Logistics: Route Optimization, Warehouse Management; Healthcare: Medical Supplies, Pharmaceuticals), Analysis of Analytics Tools in Practice (Tool Selection Criteria, Implementation Experiences, Custom Solutions), Application of Optimization Models (Real-World Problem Solving, Optimization Software, Custom Algorithms), Implementation Challenges (Data Quality Issues, System Integration, Change Management), Solutions and Best Practices (Data Management, Effective Communication of Results)
Textbook References: 1. Peter W. Robertson, Supply Chain Analytics: Using Data to Optimize Supply Chain Processes, Wiley. 2. G. R. Kanet, Introduction to Supply Chain Management Technologies, CRC Press 3. J. A. Burkett, Supply Chain Analytics: A Comprehensive Guide to Optimization, Forecasting, and Analytics, Springer. 4. Nicolas Vandeput, Data Science for Supply Chain Forecasting, Wiley. 5. Prof. G. Raghuram, Supply Chain Management: Theory and Practice, Pearson India. 6. Prof. M. K. Gupta, Supply Chain Analytics: A Practical Guide to Solving Real-World Problems, Sage Publications.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Operations & Supply Chain Management Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405OSM	SUPPLY CHAIN INFORMATION SYSTEM	4	0	0	4

Course Description

This course delves into the role of information systems in optimizing supply chain operations. It covers the strategic integration of supply chain processes through technology, focusing on the tools and systems that enable efficient data sharing, inventory management, procurement, and logistics. Students will explore advanced systems like Enterprise Resource Planning (ERP) and Warehouse Management Systems (WMS) while also examining emerging technologies such as blockchain and IoT. The course equips students with the skills to design, implement, and manage robust supply chain information systems, crucial for enhancing global competitiveness.

Course Objectives

1	To understand the fundamentals of Supply Chain Information Systems and their impact on organizational efficiency.
2	To analyse the various components and functionalities of ERP, WMS, and TMS in a supply chain context.
3	To explore the application of emerging technologies like blockchain and IoT in supply chain management.
4	To develop the ability to assess and select appropriate information systems for different supply chain functions.
5	To gain practical knowledge on the implementation and management of supply chain information systems for optimizing performance.

Course Outcomes
(Upon successful completion of this course, students should be able to)

CO1	Understand and explain the core concepts of Supply Chain Information Systems and their role in supply chain integration.
CO2	Analyse the functionalities of ERP, WMS, and TMS in streamlining supply chain processes.
CO3	Evaluate the potential impact of emerging technologies on supply chain information systems.
CO4	Develop strategies for the selection and implementation of appropriate information systems in supply chains.
CO5	Apply knowledge of information systems to optimize inventory management, procurement, and logistics.
CO6	Demonstrate the ability to manage and troubleshoot supply chain information systems in real-world scenarios.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	2	3	2	1
CO2	3	3	2	2	2	3	2	2
CO3	2	2	3	2	2	2	3	3
CO4	3	3	2	3	3	3	3	2
CO5	3	3	2	3	3	3	3	2

CO6	3	3	2	3	3	3	3	3
AVG	2.83	2.67	2.17	2.33	2.50	2.83	2.67	2.17
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to Supply Chain Information Systems Overview of Supply Chain Information Systems, Role of Information Systems in Supply Chain Management, Types of Supply Chain Information Systems (SCIS), Architecture of SCIS, Key Components of SCIS, Data Integration and Management, Information Flow in Supply Chains, SCIS and Business Strategy Alignment, Technology Trends in SCIS, Challenges in SCIS Implementation								
Unit-II: Enterprise Resource Planning (ERP) Systems Introduction to ERP Systems, ERP Modules and Functionalities, ERP Implementation Strategies, Benefits and Challenges of ERP Systems, ERP Integration with Supply Chain Processes, ERP in Manufacturing and Retail, Case Studies of ERP Implementations, ERP Software Evaluation and Selection, Customization and Configuration, Data Migration and System Testing								
Unit-III: Supply Chain Analytics and Decision Support Systems Introduction to Supply Chain Analytics, Types of Analytics (Descriptive, Predictive, Prescriptive), Decision Support Systems (DSS) in Supply Chain, Data Mining and Big Data Analytics, Predictive Modelling and Forecasting, Optimization Techniques, Visualization Tools and Techniques, Role of Analytics in Decision Making, Case Studies in Supply Chain Analytics, Advanced Analytics Applications								
Unit-IV: E-Supply Chain and E-Procurement Systems E-Supply Chain Management, E-Procurement Systems and Tools, Online Marketplaces and Platforms, Electronic Data Interchange (EDI), Integration of E-Procurement with Supply Chain, Benefits and Challenges of E-Supply Chains, E-Supply Chain Strategies, Security and Privacy Issues, Impact of E-Commerce on Supply Chains, Case Studies in E-Supply Chain and E-Procurement								
Unit-V: Emerging Technologies and Innovations in SCIS Emerging Technologies in Supply Chain Information Systems, Blockchain Technology in Supply Chains, Internet of Things (IoT) and Supply Chain Integration, Artificial Intelligence (AI) and Machine Learning Applications, Cloud Computing in SCIS, Cybersecurity in Supply Chains, Digital Twins and Simulation, Augmented Reality (AR) in Supply Chain Operations, Trends and Future Directions, Case Studies of Innovations in SCIS								
Textbook References: 1. Sunil Chopra, Peter Meindl, Supply Chain Management: Strategy, Planning, and Operation, Pearson. 2. Martin Christopher, Logistics and Supply Chain Management, Pearson. 3. Vinod Kumar Garg and N.K. Venkitakrishnan, Enterprise Resource Planning: Concepts and Practice, PHI Learning. 4. David L. Olson and Subodha Kumar, Supply Chain Information Technology, Business Expert Press. 5. Alexis Leon, ERP Demystified, Tata McGraw-Hill. 6. David Simchi-Levi, Philip Kaminsky, and Edith Simchi-Levi, Designing and Managing the Supply Chain, McGraw-Hill.								



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Operations & Supply Chain Management Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406OSM	MANAGEMENT OF MANUFACTURING SYSTEMS	4	0	0	4

Course Description

This course explores the principles and practices of managing manufacturing operations. It provides a thorough understanding of manufacturing processes, production planning, and control systems. Students will learn about various manufacturing strategies, process improvements, and quality management techniques. The course emphasizes the integration of technology and management practices to optimize manufacturing performance. Key topics include supply chain management, lean manufacturing, and advanced manufacturing technologies.

Course Objectives

1	To understand the fundamental principles of manufacturing systems and their impact on business operations.
2	To analyse various manufacturing processes and technologies for effective production management.
3	To develop skills in production planning and control, focusing on efficiency and cost management.
4	To explore quality management practices and their application in manufacturing systems.
5	To evaluate and implement strategies for continuous improvement and innovation in manufacturing operations.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand the core concepts and functions of manufacturing systems.
CO2	Analyse and apply various manufacturing processes and technologies.
CO3	Develop and implement production planning and control systems.
CO4	Evaluate and apply quality management techniques in manufacturing.
CO5	Assess and integrate continuous improvement strategies in manufacturing systems.
CO6	Use advanced manufacturing technologies to enhance production efficiency.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	1	2	1	2	2	2	1
CO2	3	2	2	2	3	3	2	2
CO3	2	2	3	2	3	2	3	2
CO4	2	2	3	3	2	2	3	2
CO5	3	3	2	3	3	3	2	3
CO6	3	3	3	2	3	3	2	3
AVG	2.50	2.17	2.50	2.17	2.67	2.50	2.33	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus**Unit-I: Introduction to Manufacturing Systems**

Fundamentals of Manufacturing Systems, Manufacturing Process Types, Manufacturing System Design, Production System Types, Role of Manufacturing in Supply Chain, Production Flow Analysis, Facility Layout Planning, Work Study and Ergonomics, Manufacturing Metrics and Performance Indicators, Production Strategy Development, Technology Integration in Manufacturing, Product Life Cycle Management, Benchmarking in Manufacturing

Unit-II: Production Planning and Control

Production Planning Strategies, Forecasting Methods, Scheduling Techniques, Capacity Planning, Production Control Methods (JIT, MRP, ERP), Optimization of Production Efficiency, Inventory Management, Production Order Management, Demand Management, Material Planning and Control, Lead Time Reduction, Supply Chain Coordination, Safety Stock Management, Resource Allocation

Unit-III: Quality Management in Manufacturing

Quality Management Principles, Total Quality Management (TQM), Six Sigma, Statistical Process Control (SPC), Quality Assurance Techniques, Quality Management Systems (QMS), ISO Standards, Process Capability Analysis, Root Cause Analysis, Quality Improvement Tools, Design of Experiments (DOE), Failure Modes and Effects Analysis (FMEA), Reliability Engineering, Supplier Quality Management

Unit-IV: Lean Manufacturing and Process Improvement

Lean Manufacturing Principles, Value Stream Mapping, 5S Methodology, Kaizen and Continuous Improvement, Kanban System, Theory of Constraints (TOC), Waste Reduction Techniques, Just-in-Time (JIT) Production, Cellular Manufacturing, Lean Metrics and Performance Measurement, Root Cause Analysis in Lean, Standard Work Procedures, Lean Six Sigma Integration, Continuous Flow Manufacturing

Unit-V: Advanced Manufacturing Technologies

Automation and Robotics, Additive Manufacturing (3D Printing), Smart Manufacturing, Impact of Advanced Technologies on Production, Enhancing Production Capabilities, Innovation in Manufacturing Operations, Internet of Things (IoT) in Manufacturing, Cyber-Physical Systems, Advanced Data Analytics for Manufacturing, Industry 4.0 Concepts, Augmented Reality (AR) and Virtual Reality (VR) in Manufacturing, Digital Twins, Advanced Sensors and Actuators, Cloud Computing in Manufacturing

Textbook References:

1. F. Robert Jacobs, William Berry, **Manufacturing Planning and Control for Supply Chain Management**, McGraw-Hill Education.
2. Jay Heizer, Barry Render, **Operations Management: Sustainability and Supply Chain Management**, Pearson.
3. **Prof. M. Mahajan**, **Production and Operations Management: Manufacturing and Services**, PHI Learning.
4. Rob Thompson, **Manufacturing Processes for Design Professionals**, Laurence King Publishing.
5. **Prof. S. Kumar**, **Principles of Manufacturing: A Modern Approach**, Tata McGraw-Hill.
6. William M. Feld, **Lean Manufacturing: Tools, Techniques, and How to Use Them**, CRC Press.



SCHOOL OF MANAGEMENT STUDIES
GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY
Odisha, Gunupur
MBA IV SEMESTER (AR23)
(Business Analytics Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402BA	FINANCE AND RISK ANALYTICS	4	0	0	4

Course Description

This course explores how banks and other financial institutions leverage predictive analytics to model and manage risk. Building on a solid foundation of statistical and financial principles, participants will delve into comprehensive modules, advanced techniques, and real-world case studies focused on Market Risk and Credit Risk. In the aftermath of financial crises, the importance of risk analytics has grown significantly. This course will address the students to How to estimate aggregate default rates. What analytical tools are most effective in these pursuits? Through this course, students will gain the skills necessary to navigate the complexities of risk analytics in today's financial landscape.

Course Objectives

1	Understand key concepts in financial risk analytics.
2	Learn credit risk modeling techniques, including logistic regression.
3	Perform market risk estimation using statistical tools.
4	Apply advanced quantitative methods to risk management.
5	Analyze real-world case studies to enhance decision-making in risk management.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Analyze financial risks using predictive analytics and statistical tools.
CO2	Develop credit risk models to predict loan default probabilities.
CO3	Implement logistic regression and discriminant analysis for risk assessment.
CO4	Evaluate market risk through advanced quantitative techniques.
CO5	Apply risk management strategies to real-world financial scenarios.
CO6	Create comprehensive risk reports to support decision-making in financial institutions.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	0	2	2	3	2	2
CO2	3	2	0	1	2	2	2	3
CO3	2	3	0	3	3	2	3	2
CO4	1	2	0	2	3	3	2	3
CO5	3	2	0	3	2	2	1	2
CO6	3	3	0	3	3	2	2	1
AVG	2.33	2.50	0.00	2.33	2.50	2.33	2.00	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Finance and Risk Analytics

Overview of Finance and Risk Analytics, **Financial Markets and Instruments**, Kinds of Financial Risk, Basic Financial Statements Analysis, Role of data analytics in finance, And analytical tools and software.

Unit-II: Credit Risk Modelling

Introduction to Credit Risk Modelling, Importance of credit risk modelling in financial institutions, Overview of key components: Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD).

Unit-III: Discriminant Model for Credit Risk

Introduction to Discriminant Analysis, Linear Discriminant Analysis (LDA), Application in Credit Scoring, Logistic Model for Credit Risk, Logistic Model for Credit Risk.

Unit-IV: Default Risk Estimation

Introduction to Default Risk, Estimation Techniques, Data Requirements and Sources, Market Risk - Returns and Risk, Understanding Market Returns, Volatility and Risk Measurement, and Market Risk Optimization.

Unit-V: Market Risk (ML in Consumer Finance)

Introduction to Market Risk in Consumer Finance, Machine Learning in Market Risk Management, Operational Risk - Basics, Operational Risk Mitigation Strategies, Fraud & Conclusion of Financial Risk Analytics (FRA).

Textbook References:

1. Saunders, A., & Cornett, M. M., Financial Institutions Management: A Risk Management Approach. McGraw-Hill Education.
2. Hull, J. C., Risk Management and Financial Institutions, Wiley.
3. Loffler, G., & Posch, P. N., Credit Risk Modeling using Excel and VBA. Wiley.
4. Jorion, P., Value at Risk: The New Benchmark for Managing Financial Risk, McGraw-Hill.
5. Tsay, R. S., Analysis of Financial Time Series, Wiley.
6. Crouhy, M., Galai, D., & Mark, R., The Essentials of Risk Management, McGraw-Hill.



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MBA IV SEMESTER (AR23)
(Business Analytics Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403BA	MACHINE LEARNING AND APPLICATIONS	4	0	0	4

Course Description

This Machine Learning course will guide you through the entire process of analysing data from a machine-learning perspective. You'll not only gain a foundational understanding of selected machine learning techniques but also delve into the details of when, why, and how to apply them effectively. The course emphasizes training and evaluating machine learning models, with a strong focus on achieving an accurate and reliable final model.

Course Objectives

1	Understand Machine Learning Basics: Grasp fundamental concepts and algorithms.
2	Preprocess Data: Clean and prepare structured and unstructured data.
3	Train Models: Learn to train and optimize machine learning models.
4	Evaluate Performance: Use metrics to assess model accuracy.
5	Apply Techniques: Solve real-world problems with machine learning.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain machine learning concepts and algorithms.
CO2	Preprocess and clean structured and unstructured data.
CO3	Train and optimize various machine learning models.
CO4	Evaluate model performance using appropriate metrics.
CO5	Apply machine learning techniques to solve real-world problems.
CO6	Develop and implement practical machine learning solutions.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	2	2	2	2	2
CO2	2	1	0	1	2	1	1	2
CO3	1	3	0	3	3	3	3	3
CO4	2	2	0	2	3	2	2	3
CO5	3	3	0	3	2	3	3	2
CO6	2	3	0	3	3	3	3	3
AVG	2.17	2.33	0.00	2.33	2.50	2.33	2.33	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to machine learning
 Overview, History. Machine learning in other fields, Theories in machine learning: Approaches and models. Limitations and ethics. Software tools for machine learning practice.

Unit-II: Supervised Learning 1
 linear models, linear and quadratic discriminant, kernel ridge regression, support vector machines,

stochastic gradient descent, nearest neighbors, Gaussian processes, and cross decomposition.

Unit-III: Supervised Learning 2

Naive Bayes, decision trees. Ensemble methods, multiclass and multi-label algorithms, feature selection, isotonic regression, probability calibration, neural network models (supervised)

Unit-IV: Unsupervised learning

Gaussian mixture models, manifold learning, clustering, biclustering, covariance estimation, novelty and outlier detection, density estimation, neural network models (unsupervised).

Unit-V: Practical Applications and Deployment

Practice of supervised and unsupervised learning methods on live data sets. Practice developing and deploying applications of machine learning using free hosting services.

Textbook References:

1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer.
2. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press.
3. Ethem Alpaydin, Introduction to Machine Learning, MIT Press.
4. Tom M. Mitchell, Machine Learning, McGraw-Hill Education.
5. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press.
6. Francis Chollet, Deep Learning with Python, Manning Publications.



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MBA IV SEMESTER (AR23)
(Business Analytics Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404BA	MARKETING & RETAIL ANALYTICS	4	0	0	4

Course Description

The course emphasizes the application of Analytics within a business context, particularly in the Retail domain, with a strong focus on Marketing Analytics. It prioritizes understanding and applying concepts over specific techniques or programming tools (such as SAS, R, SPSS, etc.). The course is delivered in an interactive format, incorporating ideas and concepts from various functional areas to enhance participants' understanding.

Course Objectives

1	Address retail business challenges with analytics.
2	Drive marketing decisions and strategies.
3	Understand key concepts and their applications.
4	Combine concepts from various business functions.
5	Enhance learning through interactive methods.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Identify marketing problems that analytics can address.
CO2	Outline the analytics landscape within the retail sector.
CO3	Apply predictive modelling to influence customer behaviour.
CO4	Analyse technology trends in marketing analytics.
CO5	Plan and execute analytics projects effectively.
CO6	Integrate analytical solutions to solve marketing challenges.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	2	2	2	2	2
CO2	2	2	0	2	2	1	3	2
CO3	1	3	0	3	3	3	3	3
CO4	2	2	0	2	3	2	2	3
CO5	3	3	0	3	2	3	3	2
CO6	2	3	0	3	3	3	3	3
AVG	2.17	2.50	0.00	2.50	2.50	2.33	2.67	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: RFM Analysis
 Overview of Marketing, Marketing Terminologies, RFM Analysis Using Tableau, RFM Analysis Using KNIME.

Unit-II: Advanced Analytics Techniques
 Percentiles and Auto Binner Function in KNIME, Cluster Analysis, K-Means Segmentation.

Unit-III: Churn Prediction Techniques

Churn Rate Prediction, Churn Prediction - Logistic Regression Hands-On.

Unit-IV: Customer Churn Analysis

Churn Prediction - Decision Tree & Random Forest Hands-On, Customer Churn Rates.

Unit-V: Advanced Customer Analytics

Market Basket Analysis, Customer Lifetime Value (CLV) Model.

Textbook References:

1. James B. Freeman, Marketing Analytics: A Practical Guide to Real Marketing Science, Kogan Page.
2. Michael Berry and Gordon Linoff, Data Mining Techniques: For Marketing, Sales, and Customer Relationship Management, Wiley.
3. Kerry Bailey and C.J. McKinney, Data-Driven Marketing: The 15 Metrics Everyone in Marketing Should Know, Wiley.
4. Wayne L. Winston, Marketing Analytics: A Practical Guide to Real Marketing Science, Wiley.
5. Paul R. Niven, Balanced Scorecard Step-by-Step: Maximizing Performance and Maintaining Results, Wiley.
6. Gordon S. Linoff and Michael J.A. Berry, Data Mining for the Masses: With Implementations in RapidMiner, Wiley.



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(Business Analytics Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405BA	WEB AND SOCIAL MEDIA ANALYTICS	4	0	0	4

Course Description

Marketing has transitioned from traditional methods to digital platforms, including social media, mobile, and emerging technologies like wearable devices. The challenge for marketing departments is to justify expenditures with measurable returns, especially as budgets are split across various online and offline channels. This course, designed for marketing professionals familiar with digital marketing basics, uses WSMA to streamline efforts and provide meaningful metrics for evaluating marketing spend in both digital and traditional contexts.

Course Objectives

1	Analyse Digital Marketing Trends: Assess the impact of evolving digital technologies and platforms on marketing strategies.
2	Justify Marketing Spend: Develop methods to evaluate and justify marketing expenditures with measurable returns on investment.
3	Utilize WSMA Tools: Apply WSMA techniques to streamline marketing efforts and optimize performance.
4	Integrate Offline and Online Metrics: Combine metrics from both offline and online channels to provide a comprehensive view of marketing effectiveness.
5	Apply Statistical Concepts: Employ statistical concepts to make informed decisions and enhance marketing strategies.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Address marketing problems using analytical solutions.
CO2	Outline the analytics landscape within the retail sector.
CO3	Apply predictive modelling to shape customer behaviour.
CO4	Analyse technology trends in marketing analytics.
CO5	Execute analytics projects from planning to implementation.
CO6	Integrate solutions to resolve marketing challenges.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	1	2	3	2	2
CO2	2	2	0	2	2	1	3	2
CO3	1	3	0	3	3	3	2	3
CO4	2	2	0	2	3	2	2	3
CO5	3	3	0	3	2	3	3	2
CO6	2	3	0	3	3	3	3	3
AVG	2.17	2.50	0.00	2.33	2.50	2.50	2.50	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Digital Data Understanding Digital Data, Understanding Consumer Behavior, Digital Marketing - Biggest Business Use Case, Data Types and Data Generation.
Unit-II: Google Metrics Google trends and analysis, Google Ads, Understanding the Web Metrics - Basic, Understand the web metrics - Inferential, Understanding the web metrics - Advanced.
Unit-III: Google Analytics Understanding Google Analytics, Understanding Google Analytics - Google Merchandise Platform Understanding Google Analytics - Setting Goals.
Unit-IV: Google Merchandise Business Case - Google Merchandise Store, WSMA - Business Case - Google Merchandise Store Evolution and Measurement.
Unit-V: Text Mining Overview, Text Mining: Process Flow, Text Mining & Sentiment Analysis Process Flow, and Text Mining: Executing Hands-on, Text & Sentiment Analysis Hands-on.
Textbook References: 1. Avinash Kaushik, Web Analytics 2.0: The Art of Online Accountability & Science of Customer Centricity, Wiley. 2. Jules J. Berman, Principles of Data Mining, CRC Press. 3. Nathaniel J. Scharf, Google Analytics: The Missing Manual, O'Reilly Media. 4. Claudia Perlich, Foster Provost, and John H. Miller, Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking, O'Reilly Media. 5. Stephen Few, Now You See It: Simple Visualization Techniques for Quantitative Analysis, Analytics Press. 6. Matthew A. Johnson, Text Mining and Analysis: Practical Methods, Examples, and Case Studies Using SAS, SAS Institute.



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(Business Analytics Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406BA	DATA MINING	4	0	0	4

Course Description

This course introduces MBA students to the principles and applications of data mining in business contexts. It covers key concepts, methodologies, and tools used to extract valuable insights from large datasets. The focus is on practical applications in areas such as marketing, finance, operations, and customer relationship management. Students will learn how to leverage data mining techniques to make informed business decisions and gain a competitive advantage.

Course Objectives

1	Understand the role of data mining in business decision-making.
2	Apply data mining techniques to real-world business problems.
3	Use data mining tools to analyse large datasets.
4	Interpret and present the results of data mining analyses.
5	Evaluate the ethical implications of data mining in business.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Apply data mining techniques to address business problems.
CO2	Pre-process and clean large datasets for accurate analysis.
CO3	Develop predictive models to support business decision-making.
CO4	Utilize data mining tools like R, Python, and KNIME in practical scenarios.
CO5	Assess ethical considerations in the application of data mining.
CO6	Communicate data-driven insights effectively to stakeholders.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	0	3	2	3	2	2
CO2	2	2	0	2	2	2	3	2
CO3	3	3	0	3	3	3	2	3
CO4	2	2	0	2	3	2	2	3
CO5	3	3	0	3	2	3	3	2
CO6	2	3	0	3	3	3	3	3
AVG	2.50	2.50	0.00	2.67	2.50	2.67	2.50	2.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit I: Introduction to Data Mining

Overview of Data Mining: Definition, scope, and significance, Data Mining vs. Data Analytics, Data Mining Process: CRISP-DM framework, Applications of Data Mining in business, Ethical considerations in data mining.

Unit-II: Data Pre-processing

Data Cleaning and Preparation, Data Transformation and Reduction, Handling Missing Data, Feature Selection and Extraction, Data Sampling and Balancing.

Unit-III: Data Mining Techniques

Classification: Decision Trees, Naive Bayes, and k-Nearest Neighbours, Regression: Linear and Logistic Regression, Clustering: K-Means, Hierarchical Clustering, and DBSCAN, Association Rule Mining: Apriori and FP-Growth, Anomaly Detection.

Unit IV: Advanced Data Mining Methods

Text Mining and Sentiment Analysis, Time Series Analysis, Web Mining, Big Data Analytics: Techniques and Tools, Introduction to Machine Learning in Data Mining.

Unit V: Practical Applications and Tools

Data Mining with R and Python, Data Mining with KNIME and RapidMiner, Case Studies in Marketing Analytics, Financial Modelling, and Customer Segmentation, Developing a Data Mining Project: From Problem Definition to Model Deployment, Presentation and Interpretation of Data Mining Results.

Textbook References:

1. Jiawei Han, Micheline Kamber, and Jian Pei, Data Mining: Concepts and Techniques, Morgan Kaufmann.
2. Ian H. Witten, Eibe Frank, and Mark A. Hall, Data Mining: Practical Machine Learning Tools and Techniques, Morgan Kaufmann.
3. Pang-Ning Tan, Michael Steinbach, and Vipin Kumar, Introduction to Data Mining, Pearson.
4. Foster Provost and Tom Fawcett, Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking, O'Reilly Media.
5. Anil Maheshwari, Data Analytics Made Accessible, Amazon Digital Services LLC.
6. Matthew A. Johnson, Data Mining and Predictive Analytics, SAS Institute.



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MBA IV SEMESTER (AR23)
(Systems & IT Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402SIT	CYBER SECURITY	4	0	0	4

Course Description

This course designed to equip students with the knowledge and skills necessary to protect digital assets from a broad spectrum of cyber threats. This course covers fundamental concepts such as risk management, threat analysis, and defensive strategies, alongside practical skills in areas like network security, cryptography, and incident response. Through hands-on labs and real-world case studies, students will learn to identify vulnerabilities, implement security measures, and respond to cyber-attacks effectively. By the end of the course, participants will have a solid understanding of how to safeguard information systems, ensure data integrity, and maintain the confidentiality and availability of critical information in an increasingly complex digital landscape.

Course Objectives

1	Grasp fundamental cyber security principles, including risk management and threat identification, to build a strong foundation in securing digital environments.
2	Learn to design and deploy robust security solutions and protocols to protect information systems from various cyber threats and vulnerabilities.
3	Develop skills to perform threat and vulnerability assessments, enabling the identification of potential security risks and the evaluation of existing controls.
4	Gain practical experience in managing and responding to cyber incidents, including detection, analysis, and mitigation strategies.
5	Understand and apply relevant cyber security regulations and standards to ensure compliance and protect data privacy and integrity.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Analyse the security implications of mobile technology for organizations and develop strategies for effective mobile security management.
CO2	Identify and describe various tools and methods used in cybercrimes, including proxy servers, anonymisers, phishing, and password cracking.
CO3	Design and establish a computer forensic laboratory, understanding the role of the OSI 7 Layer model in forensic analysis.
CO4	Evaluate computer forensics from a compliance perspective, understanding legal and regulatory requirements for forensic investigations.
CO5	Conduct forensic investigations on hand-held devices, applying appropriate techniques and tools for data extraction and analysis.
CO6	Develop and implement forensic best practices for organizations, ensuring effective handling of cybercrime incidents and evidence.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	3	1	2	3
CO2	3	3	0	3	3	1	2	3
CO3	3	3	0	3	3	1	2	3

CO4	3	3	0	3	3	2	3	3
CO5	3	3	0	3	3	2	3	3
CO6	3	3	0	3	3	1	3	3
AVG	3.00	3.00	0.00	3.00	3.00	1.33	2.50	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Cyber Crime

Introduction: Mobile and Wireless devices - Trend mobility - Authentication service security - Attacks on mobile phones - Mobile phone security Implications for organizations - Organizational measurement for Handling mobile - Security policies and measures in mobile computing era.

Unit-II: Tools and Methods

Tools and Methods used in Cyber Crime: Proxy servers and Anonymizers – Phishing - Password cracking - Key loggers and Spy wares - Virus and worms - Trojan horse and Backdoors Steganography - SQL Injection - Buffer overflow - Attacks on wireless network.

Unit-III: Computer Forensic Analysis

Understanding Computer Forensic: Historical background of cyber forensic - Forensic analysis of e-mail - Digital forensic life cycle - Network forensic - Setting up a computer forensic Laboratory - Relevance of the OSI 7 Layer model to computer Forensic - Computer forensic from compliance perspectives.

Unit-IV: Hand-Held Devices Forensic Analysis

Forensic of Hand: Held Devices - Understanding cell phone working characteristics - Hand-Held devices and digital forensic - Toolkits for Hand-Held device - Forensic of i-pod and digital music devices - Techno legal Challenges with evidence from hand-held Devices.

Unit-V: Implications

Organizational Implications: Cost of cybercrimes and IPR issues

Web Threats for Organizations: The evils and Perils - Social media marketing - Security and privacy Implications - Protecting people privacy in the organizations - Forensic best practices for organizations.

Textbook References:

1. Nina Godbole & Sunit Belapure, “Cyber Security”, Wiley India.
2. Harish Chander, “cyber laws & IT protection”, PHI learning pvt.ltd.
3. Dhiren R Patel, “Information security theory & practice”, PHI learning pvt ltd.
4. Ms. M. K. Geetha & Ms. Swapne Raman “Cyber Crimes and Fraud Management” Macmillan.
5. Pankaj Agarwal: Information Security & Cyber Laws (Acme Learning), Excel.
6. Vivek Sood, Cyber Law Simplified, TMH.



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(Systems & IT Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403SIT	BUSINESS INTELLIGENCE	4	0	0	4

Course Description

In this course, students will explore the tools, techniques, and methodologies used to transform raw data into actionable insights that drive strategic decision-making. The curriculum covers data collection, storage, and analysis, emphasizing the use of BI software and platforms to visualize trends, identify opportunities, and enhance organizational performance. Participants will gain hands-on experience with key BI tools such as Tableau, Power BI, and SQL, and will learn to apply these skills to real-world scenarios through case studies and projects. By the end of the course, students will be equipped to leverage data-driven insights to influence business strategies and operational improvements.

Course Objectives

1	Gain a comprehensive understanding of business intelligence concepts, including data warehousing, data mining, and analytics.
2	Acquire skills in data visualization, dashboard creation, and report generation.
3	Learn to extract, clean, and analyse data from various sources to identify trends and insights.
4	Create and present actionable business reports that convey complex data insights in a clear and impactful manner.
5	Demonstrate the ability to solve business problems and drive decision-making through data-driven insights.

Course Outcomes
(Upon successful completion of this course, students should be able to)

CO1	Perform metrics and key performance indicators (KPIs) to measure and enhance business performance.
CO2	Formulate and implement strategic plans and performance measurement frameworks within a BPM context.
CO3	Employ text mining processes and tools, including natural language processing, to analyse and derive value from unstructured textual data.
CO4	Implement business rules approach and manage business rule systems, identifying sources and management strategies for effective rule application.
CO5	Analyse legal, privacy, and ethical issues related to Business Intelligence and propose solutions for compliance.
CO6	Evaluate the impact of social networking on Business Intelligence and integrate social data into BI systems for enhanced decision-making.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	2	3	1	2
CO2	3	3	0	2	2	2	3	2
CO3	3	3	0	0	0	0	0	0
CO4	3	3	0	3	2	1	2	2

CO5	3	3	0	3	2	1	1	2
CO6	3	3	0	3	2	1	1	1
AVG	3.00	3.00	0.00	2.33	1.67	1.33	1.33	1.50

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to Business Intelligence

Introduction: The Business pressure - Responses and support model.

Business Intelligence: Definition of BI - Architecture of BI - Styles of BI - vent-Driven alerts - A cyclic process of Intelligence Creation - The value of Business intelligence - Value driven and Information use - Performance metrics and key performance indicators - Horizontal use cases for BI.

Unit-II: Data Ware Housing

Data Ware Housing Process: An Innovation - Data Warehousing Implementation - Data Warehousing Administration - Security Issues and future trends.

Business Performance Management: Overview - Strategic Plan - Monitor - Performance measurement - BPM methodologies - BPM Techniques - Performance dashboard and scorecards.

Unit-III: Data mining for Business Intelligence

Data Mining: Concepts and definitions - Data mining applications - Artificial Neural Networks for data mining.

Text Mining: Natural language processing - Text mining applications - Text mining process - Tools.

Web Mining: Overview - Web content overview - Web structure mining - Web usage mining.

Unit-IV: Business Rules

Business Rules: The value preposition of business rules - Business rules approach - Business rule system sources of business rules and management approach.

Unit-V: Business Intelligence Implementation

BI Issues: Business intelligence implementation - Business Intelligence and integration implementation - connecting in BI systems - Issues of legality - Privacy and ethics - Social networking and BI.

Textbook References:

1. Efraim Turban et al. "Business Intelligence", Pearson Education.
2. David Loshin "Business Intelligence", Elsevier.
3. Rajiv Sabherwal "Business Intelligence" Wiley Publications.
4. Philo Janus, Stacia Misner, Building Integrated Business Intelligence Solutions with SQL Server, 2008 R2 & Office 2010, TMH.
5. Nina Godbole & Sunit Belapure "Cyber Security" Wiley India.
6. Jawadekar, MIS Text and Cases, TMH.



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MBA IV SEMESTER (AR23)
(Systems & IT Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404SIT	DECISION SUPPORT SYSTEMS	4	0	0	4

Course Description

This course explores the design, implementation, and use of systems that aid in making informed and effective decisions within organizations. Students will delve into the theoretical foundations of DSS, including data management, model building, and the integration of decision-making processes with technology. The course emphasizes practical applications, teaching students to develop and use DSS tools to analyse complex data, simulate scenarios, and evaluate outcomes. Through hands-on projects and case studies, participants will gain experience in creating systems that enhance strategic and operational decision-making, ultimately improving organizational efficiency and effectiveness.

Course Objectives

1	Gain a solid understanding of the core principles and components of Decision Support Systems, including data management, models, and user interfaces.
2	Develop the skills to create and implement various decision models, such as optimization, simulation, and forecasting models.
3	Acquire hands-on experience with DSS software and tools, focusing on their application for data analysis, scenario planning, and decision simulation.
4	Learn to evaluate and enhance decision-making processes using DSS by analysing complex data and generating actionable insights.
5	Demonstrate the ability to use DSS for strategic planning and operational improvements.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Identify and differentiate between various types and hierarchies of DSS, understanding their general categories and applications.
CO2	Apply internet and client/server computing technologies to enhance DSS functionality and performance.
CO3	Identify strategies to overcome resistance to change when implementing DSS, ensuring smooth adoption and integration.
CO4	Apply various models in DSS to solve complex decision-making problems and enhance system capabilities.
CO5	Assess how media richness and task types influence the effectiveness of Group DSS.
CO6	Design and implement data warehouse database structures, ensuring efficient data storage and retrieval.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	0	3	0	0	2	1	0	3
CO2	0	3	0	0	2	1	0	3
CO3	0	3	0	0	2	1	0	3
CO4	0	3	0	0	2	1	0	3
CO5	0	3	0	0	2	1	0	3
CO6	0	3	0	0	2	1	0	3

AVG	0.00	3.00	0.00	0.00	2.00	1.00	0.00	3.00
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Introduction to Decision Support System (DSS)								
Introduction: How DSS evolved - DSS Benefits-Systems, Information Quality and Models - Dataflow diagram Information quality - Types of DSS - DSS hierarchy - Generalizing DSS categories.								
Unit-II: Building and implementing DSS								
System Implementation: DSS on the centre corporate systems - Internet and client/ Server computing in DSS - DSS on standalone system -Choosing a DSS Hardware environment.								
Unit-III: DSS Software Tools								
S/W Tools: Standard Package - Specialized tools and Generators - Programming language for DSS - Building DSS - DSS development process - Project development participants - Overcoming Resistance of Change - Models in DSS.								
Unit-IV: Group Decision Support System								
Group DSS: Why group DSS? - Group Vs Individual activities - Media richness and Task Types - Types of Group DSS - Group DSS in use today-Group ware products.								
Unit-V: Data Warehousing								
Data Warehousing: DW concepts - The DW database - Database structure - Getting the data into the data warehouse - Metadata.								
Textbook References:								
1. Efreem G.Mallach” Decision Support and Data Warehouse Systems” TMH.								
2. Efraim Turban “Decision Support and Busines Intelligence Systms” Pearson.								
3. Dhiraj Sharma: Foundations of I T, Excel.								
4. Nina Godbole & Sunit Belapure, “Cyber Security” Wiley India.								
5. Jawadekar, MIS Text and Cases, TMH.								
6. Dr Milind M Oka “Cases in Management Information system ‘Everest.								



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MBA IV SEMESTER (AR23)
(Systems & IT Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405SIT	DATABASE MANAGEMENT SYSTEMS	4	0	0	4

Course Description

This course offers a comprehensive exploration of the principles and practices essential for designing, implementing, and managing relational databases. Students will delve into database design concepts, including normalization, entity-relationship modelling, and schema development, as well as learn SQL for querying and manipulating data. Through hands-on exercises and projects, participants will gain practical experience in database administration, ensuring data integrity, optimizing performance, and addressing real-world challenges in data management. By the end of the course, students will be equipped with the skills needed to effectively manage and utilize databases in various organizational contexts.

Course Objectives

1	Develop skills in designing and modelling relational databases using normalization techniques and entity-relationship diagrams.
2	Gain proficiency in SQL for querying, updating, and managing data within a database.
3	Learn to apply database security practices to protect data integrity and privacy.
4	Understand and apply techniques for optimizing database performance and managing transactions.
5	Acquire practical experience in database administration tasks, including backup, recovery, and user management.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Evaluate different database technologies and their effectiveness in managing organisational data.
CO2	Identify and explain the roles of hardware, software, data, and people in database systems and their interrelationships.
CO3	Outline the phases of the Database Development Life Cycle (DDLC) and apply them to manage database projects effectively.
CO4	Analyse the costs associated with database projects, including overhead costs and the need for specialized personnel.
CO5	Define what a data warehouse is, who uses it, and the applications and need for data warehousing in organisational contexts.
CO6	Develop a project plan for constructing a data warehouse, including planning stages, justifying the need, obtaining user buy-in, and overcoming resistance.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	2	3	0	3	3	1	1	3
CO2	2	3	0	3	3	1	1	3
CO3	2	3	0	3	3	1	1	3
CO4	2	3	0	2	3	1	1	3
CO5	2	3	0	2	3	1	1	3

CO6	2	3	0	2	3	1	1	3
AVG	2.00	3.00	0.00	2.50	3.00	1.00	1.00	3.00
Justification of the Mapping: 0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance								
Syllabus								
Unit-I: Managing Data								
Data Management: Individual Data Management - Organizational Data Management - Components of Organizational Memory - Evaluation of Database Technology. File-Oriented Systems: Meeting the Need for Random Access Processing Information as Resource - Other Limitations of Traditional File Systems - Data Base Systems - Hierarchical Network Model Systems - Relational Database Systems. Database Systems: Hardware - Software - Data - People and Relationship of the four System Components.								
Unit-II: Database Systems in the Organizations								
Data Sharing: Sharing Data between Functional Units - Sharing Data between Different Levels of Users - Sharing Data between Different Locations. Data Base: The Role of the Data Base - Strategic Data Base Planning - The Need for Data Base Planning - The Data Base Project - The Data Base Development Life Cycle (DDLC).								
Unit-III: Risks and Costs of Database								
Project Risks and Costs: Organizational Conflicts - Development Project Failures - System Failure - Overhead costs - Need for Sophisticated Personnel - Separating Physical and Logical Data Representation - Client/Server Data Base Architecture - Three-Level Data Base Architecture.								
Unit-IV: Data Warehousing								
Data Warehouse Introduction: What is a Data warehouse - Who uses Data warehouse - Need for Data warehouse - Applications of Data Warehouse concepts. The Data Warehouse Data Base: Context of Data Warehouse Data Base - Data Base Structures - Organizing Relational Data warehouse - Multi-Dimensional Data Structures - Choosing a Structure. Meta Data: Human Meta Data - Computer Based Meta Data for people to use - Computer-based Meta Data for the computer to use.								
Unit-V Analyzing the Contexts of the Data warehouse								
Analysis: Active Analysis - User Queries. Constructing A Data Warehouse System: Stages of the Project - The Planning Stage - Justifying the Data warehouse - Obtaining User Buy-in - Overcoming Resistance to the Data warehouse - Developing a Project Plan. Data Warehousing Design Approaches: The Architecture Stage - The Data warehouse Data Base - The Analysis Architecture - Data warehouse Hard Ware.								
Textbook References: 1. Richard T Watson: Data Management Data Bases and Organisations, Wiley-India. 2. Berson Smith, Data Warehousing, Data Mining 7 OLAP, TMH. 3. Marakas: Modern Data Warehousing, Mining, and Visualization Core Concepts, Pearson 4. Sivanandam, Data Mining Techniques and Tasks, Thomson. 5. Gary W Hansen, James V Hansen: Data Base Management and Design, PHI. 6. Michael J.A. Berry, Gordon S. Linoff, Data Mining Techniques, Wiley-India.								



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MBA IV SEMESTER (AR23)
(Systems & IT Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406SIT	INFORMATION SYSTEM CONTROL AND AUDIT	4	0	0	4

Course Description

This course delves into the essential principles and practices for ensuring the integrity, security, and efficiency of information systems. Students will explore key topics such as risk assessment, internal controls, audit methodologies, and compliance with regulatory standards. The course covers the design and implementation of control frameworks to protect against fraud and data breaches, and emphasizes the role of audits in evaluating system performance and regulatory adherence. Through case studies and practical exercises, participants will develop skills to conduct thorough audits, identify vulnerabilities, and recommend improvements, ultimately enhancing the reliability and accountability of information systems within organizations.

Course Objectives

1	Assess and identify risks associated with information systems and their impact on organizational operations.
2	Design and implement effective internal controls to safeguard data integrity and system security.
3	Perform comprehensive audits of information systems to ensure compliance with regulations and internal policies.
4	Analyse audit results to identify vulnerabilities and recommend corrective actions for improvement.
5	Understand and apply relevant legal and regulatory requirements to ensure adherence in information systems management.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Assess the need for IT audits in various organizational contexts and explain how IT auditors contribute to effective risk management and compliance.
CO2	Develop an audit budget and set clear objectives and context for IT audits, ensuring alignment with organisational goals.
CO3	Execute fieldwork and apply audit methodologies, documenting results and ensuring thoroughness in the audit process.
CO4	Evaluate and generalize the use of audit software to streamline audit procedures and improve accuracy.
CO5	Apply codes of ethics and professional standards in IT auditing, understanding their importance in maintaining integrity and professionalism.
CO6	Manage audit demand and resource allocation, ensuring efficient use of resources and timely completion of audit activities.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	0	3	1	1	2	3
CO2	3	3	0	3	1	1	1	3
CO3	3	3	0	3	1	1	1	3

CO4	3	3	0	3	1	1	1	3
CO5	3	3	0	3	1	1	1	3
CO6	3	3	0	3	1	1	1	3
AVG	3.00	3.00	0.00	3.00	1.00	1.00	1.17	3.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: IT Audit

Audit and review the role of Information Technology: Audit standards - Importance of audit standard independence - AICPA preannouncement - The need for IT audit function - role of the IT auditor - Legal implications.

Unit-II: Audit Process

Audit process in an information Technology Environment: Risk Assessment - Audit budget - Objectives and context - preliminary review - Field-work and implementing audit Methodology - Documenting results.

Unit-III: Audit Tools

Computer Assistance Audit Tools: Auditor productivity tools - Flowcharting Techniques - Flowcharting an Analytical tool - Defining critical data - Flowcharting technique - Generalizing audit software - Computer forensic.

Unit-IV: Audit Management

Managing IT Audits: Establish a career development and planning - Evaluating IT audit quality-criteria for IT audit quality - IT auditing in new millennium - Code of ethics and professional standards - Private industry-Management consultancy.

Unit-V: Audit Standards

Strategy and standards for auditing: Strategic planning - Communication - Demand management - Architecture and standards - Business architecture - application and information architecture - Architecture functions.

Textbook References:

1. Sandra Senft & Fredrick "Information Technology Control and Audit "CRC Press.
2. D P Dube, V P Gulati, Information System Audit and Assurance – Includes case studies and chelists from the banking industry, TMH.
3. Nina Godbole & Sunit Belapure "Cyber Security", Wiley India.
4. Jawadekar, MIS Text and Cases, TMH.
5. Dr. Milind M Oka "Cases in Management Information Systems, Everest.
6. Subash Chandra Das-Management Control Systems: Principles and Practices-PHI.



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(Agribusiness Management Elective-I)

Course Code	Course Title	L	T	P	C
MB-E-402ABM	FOOD TECHNOLOGY AND PROCESSING MANAGEMENT	4	0	0	4

Course Description

This course explores the technological and managerial aspects of food production and processing. It covers the principles and practices involved in food processing, including methods and technologies used to enhance food quality, safety, and shelf life. Students will gain insights into the management of food processing operations, from raw material handling to product development and distribution. The course also addresses regulatory and sustainability issues in the food industry, preparing students for roles in managing and innovating within the food technology sector.

Course Objectives

1	To understand the fundamental principles of food processing and the technologies used in the industry.
2	To analyse the management practices involved in food production, including quality control, safety, and regulatory compliance.
3	To explore the various methods of food preservation and their impact on food quality and shelf life.
4	To develop skills in managing food processing operations and optimizing production efficiency.
5	To address sustainability issues and innovations in food technology and processing.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Describe the core principles and technologies used in food processing.
CO2	Assess and apply management practices in food production to ensure quality and safety.
CO3	Evaluate different food preservation methods and their effects on product quality and shelf life.
CO4	Develop strategies to manage and optimize food processing operations effectively.
CO5	Analyse sustainability challenges and innovations in the food technology sector.
CO6	Implement regulatory standards and compliance measures in food processing operations.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	3	2	1
CO2	3	3	2	2	2	3	2	2
CO3	2	2	3	2	2	2	3	3
CO4	3	3	2	3	3	3	3	2
CO5	3	3	2	3	3	3	3	2
CO6	3	3	2	3	3	3	3	3
AVG	2.83	2.67	2.00	2.50	2.50	2.83	2.67	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Introduction to Food Technology and Processing Overview of Food Technology and Processing, Importance of Food Processing, Types of Food Processing Techniques, Principles of Food Preservation, Food Safety and Quality, Food Processing Equipment and Technologies, Process Flow in Food Processing, Emerging Trends in Food Technology, Regulatory Frameworks and Standards, Challenges in Food Technology.
Unit-II: Food Processing Techniques Heat Processing Methods, Cold Processing Methods, Chemical Preservation Techniques, Mechanical Processing Techniques, Fermentation Technology, Dehydration and Drying Methods, Extrusion and Texturization, Packaging Technologies, Quality Control in Food Processing, Case Studies of Food Processing Techniques.
Unit-III: Food Quality and Safety Management Principles of Food Quality Management, Food Safety Standards and Regulations, Hazard Analysis and Critical Control Points (HACCP), Food Quality Assurance Systems, Sensory Evaluation of Food Products, Microbiological Safety in Food Processing, Contamination Control and Prevention, Traceability and Recall Systems, Food Fraud Prevention, Case Studies on Food Safety Management.
Unit-IV: Food Product Development and Innovation Product Development Process in Food Industry, Research and Development in Food Technology, Ingredient Functionality and Formulation, Sensory Analysis and Consumer Testing, Innovations in Food Product Design, Health and Nutritional Aspects of Food Products, Packaging Design and Development, Market Trends and Consumer Preferences, Product Launch Strategies, Case Studies in Food Product Development.
Unit-V: Sustainable Food Processing and Supply Chain Management Sustainable Practices in Food Processing, Waste Management and By-Product Utilization, Energy Efficiency in Food Processing, Sustainable Packaging Solutions, Supply Chain Management in Food Industry, Logistics and Distribution in Food Processing, Food Traceability and Certification, Circular Economy in Food Industry, Impact of Climate Change on Food Processing, Case Studies in Sustainable Food Processing.
Textbook References: 1. Richard W. Hartel, Food Processing: Principles and Applications, CRC Press. 2. R. Paul Singh & Dennis R. Heldman, Introduction to Food Engineering, Academic Press. 3. Inglett, G. E. & D. M. W. D. Narayanasamy, Handbook of Food Processing: Food Safety, Quality, and Manufacturing Processes, CRC Press. 4. Inteaz Alli, Food Quality Assurance: Principles and Practices, CRC Press. 5. S. M. H. K. Begum & N. K. Gupta, Food Processing and Preservation, New Age International Publishers. 6. Paul Singh & Dennis R. Heldman, Food Processing Technology: Principles and Practice, CRC Press.



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(Agribusiness Management Elective-II)

Course Code	Course Title	L	T	P	C
MB-E-403ABM	AGRI IMPORT AND EXPORT MANAGEMENT	4	0	0	4

Course Description

This course provides a comprehensive overview of the principles, practices, and strategies involved in managing the import and export of agricultural products. It is designed for individuals aiming to understand the global agricultural trade dynamics and to develop the skills necessary for successful international trade operations in the agri-sector.

Course Objectives

1	To explore various concepts of Agricultural practices in India
2	To apply fundamental concepts and terminology on Indian Agricultural Trade
3	To analyse the systems of import on Agricultural Product
4	To explain the terminology of exports of Agricultural products in India
5	To distinguish various types IT application on Agricultural Field

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand the fundamentals of agriculture, including different forms, production methods, and the need for import and export.
CO2	Analyse marketable and marketed surpluses and evaluate the implications of import and export on agricultural products.
CO3	Examine statistics of agricultural products in India and assess macroeconomic forces impacting agriculture.
CO4	Explore the global agricultural trade landscape, international issues, and agricultural policies affecting trade.
CO5	Evaluate technology advancements in agriculture and their impact on agricultural production and trade.
CO6	Identify the requirements and processes for importing agricultural products, including quality standards and documentation.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	2	3	2	2	1
CO2	2	2	3	2	2	1	1	2
CO3	2	2	2	2	2	1	1	0
CO4	2	2	2	2	2	1	0	1
CO5	2	2	2	2	2	1	1	1
CO6	2	2	1	2	1	0	1	1
AVG	2.17	2.00	2.17	2.00	2.00	1.00	1.00	1.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Introduction to Agriculture Introduction to agriculture, forms of agriculture, Production , Need for import and export, Analysis of marketable surplus and marketed surplus , Analysis of Import and Export, Statistics of Agricultural products in India, Macro Economic Forces.
Unit-II: Agricultural Trade Overview of world agricultural trade, issues impacting, international agricultural trade, Agricultural Policy, technology advancement on agricultural product.
Unit-III: Import of Agricultural Product Import requirement- causes, identifying agri-products for import, Import substitution, regulation of imports, Quality standards, Scanning the countries for importing the required Agri-products , Import planning & documentation required.
Unit-IV: Agricultural Product Export Identifying foreign markets for agri export, Marketing plan for exports, Exports documents and procedure, terms of payment and export finance, legal dimensions.
Unit-V: Application of IT in SCM Institutional infrastructure for export promotion in India, India Export Assistance, State trading import and exports, Working of State trading Organizations of India
Textbook References: 1. Singh, D., & Gautam, A., Export Management, Himalaya Publishing House. 2. Sheth, R.B., Thoke, N., & Kulkarni, A.V. Management of Agri business and Agri Exports, Nirali Prakashan. 3. Pillai, R. S. N & Bagavathi, V. Marketing Management, S. Chand & Co. 4. Cherunilam, F. International Marketing, Himalaya Publishing House. 5. Ram Singh, Export and Import Management: Text and Cases, Wiley. 6. Parul Gupta, Export Import Management, McGraw Hill.



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(Agribusiness Management Elective-III)

Course Code	Course Title	L	T	P	C
MB-E-404ABM	E-COMMERCE AND AGRI-BUSINESS	4	0	0	4

Course Description

This course provides a comprehensive understanding of e-commerce, focusing on its features, importance, models, and specific relevance to India. Students will explore various e-commerce technologies, practices, and strategies, including internet trading, electronic data interchange, and electronic payment systems. The course will also delve into the adoption of e-commerce by different stakeholders and offer practical guidelines for successful e-commerce implementation. Special emphasis will be placed on e-agribusiness, including case studies, online transactions, marketing strategies, and future trends in the sector.

Course Objectives

1	To explore various concepts of E-commerce and its application in agribusiness.
2	To apply fundamental terminology of e-commerce technology.
3	To analyse the process of E-commerce implementation.
4	To explain the present practices of e-commerce in India.
5	To distinguish various successful stories of e-commerce in agricultural and other sector.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Understand the fundamental concepts of e-commerce, its features, and its significance to the global and Indian economies.
CO2	Analyse various models of e-commerce and their applications in different business contexts.
CO3	Evaluate the current status of e-commerce in India, including key success factors, advantages, limitations, and challenges.
CO4	Explore the technologies and systems integral to e-commerce, such as internet trading, electronic data interchange, and electronic payment systems.
CO5	Examine the roles of various stakeholders in e-commerce, including employees, customers, service providers, suppliers, channel partners, and regulatory authorities.
CO6	Develop practical skills for designing and implementing e-commerce strategies, including website design, search engine optimization, and online advertising.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	2	2	1
CO2	3	3	3	3	3	1	1	2
CO3	2	1	1	2	2	1	1	0
CO4	2	3	2	2	3	1	0	1
CO5	3	1	2	2	2	1	1	1
CO6	2	3	1	2	1	0	1	1
AVG	2.50	2.17	2.00	2.17	2.17	1.00	1.00	1.00

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Introduction to E-Commerce Ecommerce over view: Introduction features and importance to economy, Models of e-commerce, E-commerce status in India, Key success factors, advantages, limitations and challenges in e-commerce with special reference to India.
Unit-II: E-Commerce and Technology Technology for E-Commerce, internet trading, electronic data interchange, electronic payment system, adoption of e-commerce by employees, customers, service provider, supplier, channel partner regulatory authority.
Unit-III: Implementation of E-Commerce Guidelines for effective implementation of e-commerce in India, Important decision for E-commerce: Designing website, designing web address, presence in search engines and directories, advertising the website, monitoring and updating.
Unit-IV: Present Practices of E-Commerce Present practices of e-commerce in agribusiness in agribusiness, online transactions, marketing, delivery of information intensive product through online , business trust through branding , online presentation through website, customer service and value addition through e-CRM and e-SRM.
Unit-V: Success Stories of E-commerce Success stories of organizations involved in e-agribusiness case studies and future vision of e-agribusiness in India
Textbook References: 1. Kenneth C. Laudon and Carol Guercio Traver, E-Commerce 2024: Business, Technology, Society, 2. Brad Stone, The Everything Store: Jeff Bezos and the Age of Amazon, 3. Frederick T. Barlow and M. John McLoughlin, Agribusiness Management, 4. William M. Lineberry and David R. Kohl, Introduction to Agribusiness, 5. M. D. L. Wiskerke and J. L. S. J. P. de Wit, Sustainable Agribusiness: A Practical Guide to a New Era, 6. "Digital Commerce 360 Guide to E-Commerce" by Digital Commerce 360



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(Agribusiness Management Elective-IV)

Course Code	Course Title	L	T	P	C
MB-E-405ABM	AGRI-SUPPLY CHAIN MANAGEMENT	4	0	0	4

Course Description

This course offers an in-depth exploration of the processes and strategies involved in managing the flow of agricultural products from farm to table. This course covers key areas such as logistics, procurement, inventory management, and demand forecasting, emphasizing the unique challenges and opportunities within the agricultural sector. Students will gain insights into optimizing supply chains to enhance efficiency, reduce waste, and improve sustainability. Through real-world applications, learners will develop practical skills to address issues like supply chain disruptions, regulatory compliance, and market dynamics, preparing them to effectively manage and innovate within the agri-supply chain.

Course Objectives

1	To explore various concepts of Supply chain Management.
2	To apply fundamental concepts and terminology on demand and manufacturing management.
3	To analyse the systems of procurement management applied in agricultural supply chain.
4	To explain the concepts and terminology of Logistic management.
5	To distinguish various types of derivatives for investors.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the evolution of Supply Chain Management and its conceptual model.
CO2	Distinguish between traditional and modern approaches to SCM in agriculture.
CO3	Identify and describe the key elements in SCM.
CO4	Analyze different types of demand and implement demand planning and forecasting techniques.
CO5	Apply basic principles of manufacturing management in the context of SCM.
CO6	Describe the procurement management cycle, types of purchases, and concepts of corporate farming.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	2	1	2
CO2	2	2	2	2	3	2	1	1
CO3	2	2	3	2	2	2	1	1
CO4	2	2	2	2	2	2	2	1
CO5	2	2	3	1	2	1	1	2
CO6	2	1	1	1	1	1	1	1
AVG	2.17	1.83	2.33	1.83	2.00	1.67	1.17	1.33

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus
Unit-I: Introduction to SCM Supply Chain: Changing Business Environment; SCM: Present Need; Conceptual Model of Supply Chain Management; Evolution of SCM; SCM Approach; Traditional Agri. Supply Chain Management Approach ; Modern Supply Chain Management Approach; Elements in SCM.
Unit-II: Demand & Manufacturing Management Demand Management in Supply Chain: Types of Demand, Demand Planning and Forecasting; Operations Management in Supply Chain, Basic Principles of Manufacturing Management.
Unit-III: Procurement Management Procurement Management in Agri. Supply Chain: Purchasing Cycle, Types of Purchases, Contract/Corporate Farming, Classification of Purchases, Goods or Services, Traditional Inventory Management, Material Requirements Planning, Just in Time (JIT), Vendor Managed Inventory (VMI).
Unit-IV: Logistic Management Logistics management; History and Evolution of Logistics; Elements of Logistics Management; Distribution Management, Distribution Strategies; Pool Distribution; Transportation Management; Fleet Management; Service Innovation; Warehousing; Packaging for Logistics, Third-Party Logistics (TPL/3PL); GPS Technology.
Unit-V: Application of IT in SCM Concept of Information Technology: IT Application in SCM; Advanced Planning and Scheduling; SCM in Electronic Business; Role of Knowledge in SCM; Performance Measurement and Controls in Agri-Supply Chain Management - Benchmarking: introduction, concept and forms of Benchmarking.
Textbook References: 1. Altekar R V., Supply Chain Management: Concepts and Cases, Prentice Hall of India. 2. Monczka R, Trent R & Handfield R., Purchasing and Supply Chain Management. Thomas Asia. 3. Van Weele AJ., Purchasing and Supply Chain Management Analysis, Planning and Practice. Vikas Publ. House. 4. Deepak Bhagat, U R Dhar, Supply Chain Management in Agribusiness, Botech (Astral). 5. G. Raghuram, N. Chandrasekaran, Agribusiness Supply Chain Management, CRC Press. 6. Prof. T. Justin, Dr. S Amrutha, Dr. M. Muthukamu, Dr. C. Mayilsamy, Dr. D. Suja, Agri-Logistics & Supply Chain Management, Shanlax Publications.



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(Agribusiness Management Elective-V)

Course Code	Course Title	L	T	P	C
MB-E-406ABM	INTERNATIONAL TRADE IN AGRICULTURE	4	0	0	4

Course Description

This course focuses on the dynamics of international trade within the agricultural sector. It explores global trade policies, market trends, and the impact of international trade agreements on agricultural commodities. Students will examine the roles of different stakeholders, including governments, trade organizations, and multinational companies. The course also covers the challenges and opportunities in international agricultural trade, including sustainability issues, trade barriers, and the influence of global economic factors on agricultural markets.

Course Objectives

1	To understand the fundamentals of international trade in agriculture and its global significance.
2	To analyse trade policies, agreements, and their impact on agricultural commodities and markets.
3	To explore the roles of various stakeholders in international agricultural trade.
4	To assess the challenges and opportunities in global agricultural trade, including sustainability and trade barriers.
5	To develop strategies for navigating and optimizing international trade practices in agriculture.

Course Outcomes

(Upon successful completion of this course, students should be able to)

CO1	Explain the key concepts and importance of international trade in the agricultural sector.
CO2	Analyse the impact of trade policies and agreements on agricultural trade and markets.
CO3	Evaluate the roles and influences of different stakeholders in international agricultural trade.
CO4	Identify and assess the challenges and opportunities associated with international agricultural trade.
CO5	Develop strategies for effective participation and optimization in global agricultural trade.
CO6	Address sustainability issues and trade barriers impacting the international agricultural trade sector.

CO-PO & PSO Mapping (Articulation Matrix)

COs	POs					PSOs		
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	2	2	1	2	3	2	1
CO2	3	3	2	2	2	3	2	2
CO3	2	2	3	2	2	2	3	3
CO4	3	3	2	3	3	3	3	2
CO5	3	3	2	3	3	3	3	2
CO6	2	2	3	2	2	2	3	3
AVG	2.67	2.50	2.33	2.17	2.33	2.67	2.67	2.17

Justification of the Mapping:

0=No Relevance; 1=Low Relevance; 2=Medium Relevance; 3=High Relevance

Syllabus

Unit-I: Introduction to International Trade in Agriculture

Overview of International Trade in Agriculture, Importance of Agriculture in Global Trade, Historical Perspectives on Agricultural Trade, Key Theories of International Trade, Trade Policies and Agreements, Global Agricultural Trade Patterns, Trade Barriers and Facilitators, Role of International Organizations (WTO, FAO), Impact of Trade on Agricultural Development, Current Trends in Agricultural Trade.

Unit-II: Agricultural Trade Policies and Agreements

Trade Policy Frameworks and Objectives, Agricultural Trade Policies in Developed vs. Developing Countries, Bilateral and Multilateral Trade Agreements, Trade Liberalization and Protectionism, Tariffs and Non-Tariff Barriers, Subsidies and Support Mechanisms, Regional Trade Agreements (EU, NAFTA, ASEAN), Impact of Trade Agreements on Agriculture, Case Studies of Trade Policies, Trade Disputes and Resolutions.

Unit-III: Market Structures and International Trade

Market Structures in Agriculture, Commodity Markets and Price Determination, Supply and Demand Analysis, Market Integration and Segmentation, Role of Futures and Spot Markets, Impact of Exchange Rates on Agricultural Trade, Global Agricultural Market Players, Trade and Market Efficiency, Market Access and Export Promotion, Case Studies of Agricultural Markets.

Unit-IV: Trade and Sustainable Agriculture

Sustainability in Agricultural Trade, Environmental Impacts of Agricultural Trade, Sustainable Farming Practices and Trade, Certification and Standards for Sustainable Agriculture, Trade Policies Supporting Sustainability, Role of Trade in Food Security, Ethical Considerations in Agricultural Trade, Impacts of Climate Change on Trade, International Trade and Resource Management, Case Studies on Sustainable Agricultural Practices.

Unit-V: Emerging Issues and Future Directions in Agricultural Trade

Emerging Trends in Agricultural Trade, Technological Innovations and Their Impact, Role of Digital Platforms in Agricultural Trade, Trade and Food Security Challenges, Global Supply Chain Disruptions, Impact of Geopolitical Tensions on Agricultural Trade, Trade Policy Innovations, Future Directions and Prospects for Agricultural Trade, Strategic Planning for Agricultural Exporters, Case Studies on Emerging Issues in Agricultural Trade.

Textbook References:

1. Christopher L. Gilbert & Alan A. Krueger, **International Trade in Agricultural Products**, Cambridge University Press.
2. William A. Masters & R. E. Evenson, **Global Agricultural Trade and Developing Countries**, Routledge.
3. P. L. P. H. B. Chinnappa, **The Economics of International Agricultural Trade**, Springer.
4. T. L. Schmitz, **Agricultural Trade Policy: Principles and Practice**, Routledge.
5. S. K. R. Mahajan, **International Agricultural Trade: A Review of the Literature**, Oxford University Press.
6. Prabhu Pingali & Mark Rosegrant, **Trade Policy and Food Security: An Overview of the Issues**, Academic Press.